AOS-CX 10.18.xxxx IP Routing Guide (4100i, 5420, 6000, 6100, 6200 Switch series)

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Table of contents

About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing HPE Aruba Networking switches on a network.

Subtopics

Applicable products

This document applies to the following products:

- HPE Aruba Networking 4100i Switch Series (JL817A, JL818A)
- HPE Aruba Networking 5420 Switch Series (S0U67A, S0U55A, S0U63A, S0U64A, S0U65A, S0U75A, S0U72A, S0U78A, S0U58A, S0U73A, S0U74A, S0U71A, S0U76A, S0U70A, S0U77A, S0U60A, S0U61A, S0U62A, S0U66A, S0U68A)
- HPE Aruba Networking 6000 Switch Series (R8N85A, R8N86A, R8N87A, R8N88A, R8N89A, R9Y03A, S4R20A, S4R21A, S4R22A, S4R23A, S4R24A, S4R25A, S4R26A, S4R27A, S4R28, S4R29A)
- HPE Aruba Networking 6100 Switch Series (JL675A, JL676A, JL677A, JL678A, JL679A)
- HPE Aruba Networking 6200 Switch Series (JL724A, JL725A, JL726A, JL727A, JL728A, R8Q67A, R8Q68A, R8Q69A, R8Q70A, R8Q71A, R8V08A, R8V09A, R8V10A, R8V11A, R8V12A, R8Q72A, JL724B, JL725B, JL726B, JL727B, JL728B, S0M81A, S0M82A, S0M83A, S0M84A, S0M85A, S0M86A, S0M87A, S0M88A, S0M89A, S0M90A, S0G13A, S0G14A, S0G15A, S0G16A, S0G17A)

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Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in Support and Other Resources.

Command syntax notation conventions

Convention	Usage
example - text	Identifies commands and their options and operands , code examples, filenames, pathnames, and output d isplayed in a command window. Items that appear li ke the example text in the previous column are to be entered exactly as shown and are required unless en closed in brackets (**[ ]**).
example - text	In code and screen examples, indicates text entered by a user.
Any of the following: • <example - text> • <example - text> • example - text • example - text	Identifies a placeholder—such as a parameter or a va riable—that you must substitute with an actual valu e in a command or in code: • For output formats where italic text cannot be di splayed, variables are enclosed in angle brackets (**< >**). Substitute the text—including the enclosin g angle brackets—with an actual value. • For output formats where italic text can be displ ayed, variables might or might not be enclosed i n angle brackets. Substitute the text including th e enclosing angle brackets, if any, with an actual value.
|	Vertical bar. A logical **OR** that separates multiple ite ms from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required pa rt of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed ite ms is required.
[ ]	Brackets. Indicates that the enclosed item or items a re optional.
… or ...	Ellipsis: • In code and screen examples, a vertical or horizo ntal ellipsis indicates an omission of information. • In syntax using brackets and braces, an ellipsis i ndicates items that can be repeated. When an ite

Convention	Usage
	m followed by ellipses is enclosed in brackets, ze ro or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term **switch**, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME)#
```

Indicates the configuration context for a feature. For example:

```
switch(config-if)#
```

Identifies the **interface** context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100)#
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>)#
```

Where <VLAN-ID> is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format: **member/slot/port.**

On the HPE Aruba Networking 4100i Switch Series

- member: Always 1. VSF is not supported on this switch.
- slot: Always 1. This is not a modular switch, so there are no slots.
- port: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on the switch.

On the HPE Aruba Networking 6000 and 6100 Switch Series

- member: Always 1. VSF is not supported on this switch.
- slot: Always 1. This is not a modular switch, so there are no slots.
- port: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on the switch.

On the HPE Aruba Networking 6200 Switch Series

- member: Member number of the switch in a Virtual Switching Framework (VSF) stack. Range: 1 to 8. The primary switch is always member 1. If the switch is not a member of a VSF stack, then member is 1.
- slot: Always 1. This is not a modular switch, so there are no slots.
- port: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 in slot 1 on member 1.

On the HPE Aruba Networking 6400 and 5420 Switch Series

- member: Always 1. VSF is not supported on this switch.
- slot: Specifies physical location of a module in the switch chassis.
  - Management modules are on the front of the switch in slots 1/1 and 1/2.
  - Line modules are on the front of the switch starting in slot 1/3.
- port: Physical number of a port on a line module.

For example, the logical interface **1/3/4** in software is associated with physical port 4 in slot 3 on member 1.

Virtual Routing and Forwarding (VRF)

VRF is a technology that allows multiple instances of a routing table to co-exist within the same router. Because the routing instances are independent, the same or overlapping IP addresses can be used without conflicting with each other. Network functionality is improved because network paths can be segmented without requiring multiple routers.

Subtopics

VRF support

The 4100i, 6000, 6100, 5420 and 6200 Switch Series support predefined VRFs:

- mgmt: Assigned to the management port and is used to isolate management traffic (5420 and 6200 Switch Series only).
- default: All other interfaces are automatically assigned to the VRF **default** (6000, 6100, 5420 and 6200 Switch Series).

> **NOTE**
> User-defined VRFs are not supported on the 4100i, 6000, 6100, and 6200 Switch Series.

Loopback

> **NOTE**
> Loopback interfaces are not supported on the HPE Aruba Networking 6000 and 6100 Switch Series.

> **NOTE**
> Loopback interfaces are not supported on the HPE Aruba Networking 4100i Switch Series.

A loopback interface is a virtual interface supporting IPv4/IPv6 address configuration. The loopback interface can be considered stable because once created, it remains up. The loopback interface can then be configured with an address to use as a reference or identifier independent of the physical interfaces.

- **Device identification**: As long as the router is operational, the state of the loopback interface is always up. Even if only one link to the router is active, the loopback interface can be reached. This functionality makes it possible to identify an active device in the network using the IP address configured on the loopback interface.
- **Routing**: Since the loopback interface is always active, a routing session (such as a BGP session) can continue on an alternate path even if the outbound interface fails. In OSPF, a loopback interface address is advertised as an interface route into the network. This functionality increases reliability by allowing traffic to take alternate paths if there is a link failure. In OSPF and BGP, the router ID can be set to the loopback address to avoid reassignment of the router ID when physical interfaces are added or removed.
- **Device management**: Loopback interface is always reachable and can be used for sending and receiving management information such as logs and SNMP traps without interruption.

Subtopics

Loopback commands

Loopback commands

Select a command from the list in the left navigation menu.

Subtopics

interface loopback
ip address
ipv6 address
show interface loopback

`interface loopback`

Syntax

```
interface loopback <INSTANCE>
no interface loopback <INSTANCE>
```

Description

Creates a loopback interface and enters loopback configuration mode.

The **no** form of this command deletes a loopback interface.

Parameter	Description
`<INSTANCE>`	Selects the loopback interface ID. Range: 0 to 4

Examples

```
switch(config)# interface loopback
switch(config-loopback-if)#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`ip address`

Syntax

```
ip address <IPV4-ADDR/MASK> [secondary]
no ip address <IPV4-ADDR/MASK> [secondary]
```

Description

Sets the IPv4 address for a loopback interface.

The **no** form of this command reverses the set of the IPv4 address for a loopback interface.

Parameter	Description
<IPV4-ADDR>	Specifies an IP address in IPv4 format (**x.x.x.x**), where **x** is a de cimal number from 0 to 255.
<MASK>	Specifies the number of bits in the address mask in CIDR format (**x**), where **x** is a decimal number from 0 to 128.
secondary	Indicates that the IPv4 address is a secondary address.

Examples

```
switch(config)# interface loopback 1
switch(config-loopback-if)# ip address 16.93.50.2/24
switch(config-loopback-if)# ip address 20.1.1.1/24 secondary
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`ipv6 address`

Syntax

```
ipv6 address <IPV6-ADDR/MASK>
```

Description

Sets the IPv6 address for a loopback interface.

Parameter	Description
`<IPV6-ADDR>`	Specifies an IP address in IPv6 format (**xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx**), where **x** is a hexadecimal number from 0 to F.
`<MASK>`	Specifies the number of bits in the address mask in CIDR format (**x**), where **x** is a decimal number from 0 to 128.

Examples

switch(config)# **interface loopback 1**

switch(config-loopback-if)# **ipv6 address fd00:5708::f02d:4df6/64**

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show interface loopback`

Syntax

```
show interface loopback [brief | instance <ID>]
```

Description

This command displays the configuration and status of loopback interfaces.

Parameter	Description
`brief`	Displays brief information about all configured loopback interfaces.
`instance <ID>`	Displays the configuration and status of a loopback interface ID. Range: 1 - 255

Examples

```
switch# show interface loopback
 Interface loopback1 is up
 IPv4 address 192.168.1.1/24
 Interface loopback2 is up
 IPv4 address 182.168.1.1/24
switch# show interface loopback brief
----------------------------------------------------------------------
Loopback      IP Address                                    Status
Interface
----------------------------------------------------------------------
loopback1    10.1.1.1/24                                       up
loopback1    1111:2222:3333:4444::6666/128                     up
switch# show interface loopback 1
 Interface loopback1 is up
 IPv4 address 192.168.1.1/24
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Static routing

Static routes are manually configured. If the network topology is simple enough, you can use static routes for the network's routing requirements. The proper configuration and usage of static routes can improve network performance and ensure bandwidth for important network applications.

The disadvantage of using static routes is that they cannot adapt to network topology changes. If a fault or topological change occurs in the network, the relevant routes will be unreachable and a network administrator must modify the static routes manually.

Static routes can now be assigned descriptive names using the optional name parameter. This feature enhances route management by allowing administrators to identify routes by their purpose rather than just by their network addresses. Names can be up to 64 characters long and help improve readability and maintainability of network configurations.

NOTE

These 'long prefix' IPv6 routes are programmed in a switching ASIC's TCAM table instead of the usual route table.

The access switch series are comprised of multiple ASICs of different capabilities with a mixture of line cards with a varying number of capable ASICs. For consistency across platforms, and to support a fair usage scheme amongst TCAM users, there is an imposed maximum long prefix IPv6 route capacity of 510 for all access switch series. Once this hardware limit is exhausted, long prefix IPv6 routes can still be configured/learned but will only be programmed in the software (CPU) path.

Once TCAM/long prefix IPv6 resources are freed up there is no automatic migration of a software-only long prefix IPv6 route to hardware. The route must be removed and recreated, at which point it consumes the freed hardware resource.

Note that since the TCAM table is shared between multiple applications, the 510 long prefix IPv6 route capacity is not guaranteed to be available, depending on current switch configuration/conditions. Use the 'show' resources' command to display current TCAM resource usage & availability.

Each configured/learned long prefix IPv6 route consumes four TCAM entries and these are noted in the 'Long Prefix IPv6 Unicast Route Lookup' row. The maximum 510 TCAM programmed long prefix IPv6 routes would therefore consume 2040 TCAM entries. The remaining Reserved space is necessary for internal TCAM management.

Below is an example of resource usage:

```
show resources
Resource Usage:
Mod  Description
Resource                                              Used    Reserved
Free
----------------------------------------------------------------------------
--
1/3  Long Prefix IPv6 Unicast Route Lookup
          Ingress TCAM Entries                          16        2048
     Total
          Ingress TCAM Entries                          16        2048
18432
          Egress TCAM Entries                            0           0
8192
          Ingress Lookups                                1
8
          Ingress Flex Lookups                           0
1
          Egress Lookups                                 0
4
          Ingress Policers                               0
2047
          Egress Policers                                0
2047
1/4  Long Prefix IPv6 Unicast Route Lookup
          Ingress TCAM Entries                          16        2048
     Total
          Ingress TCAM Entries                          16        2048
18432
          Egress TCAM Entries                            0           0
8192
          Ingress Lookups                                1
8
          Ingress Flex Lookups                           0
1
          Egress Lookups                                 0
4
          Ingress Policers                               0
2047
          Egress Policers                                0
2047
1/5  Long Prefix IPv6 Unicast Route Lookup
```

```
          Ingress TCAM Entries                             16         8192
     Total
          Ingress TCAM Entries                             16         8192
57344
          Egress TCAM Entries                               0            0
8192
          Ingress Lookups                                   1
8
          Ingress Flex Lookups                              0
1
          Egress Lookups                                    0
4
          Ingress Policers                                  0
8191
          Egress Policers                                   0
4095
```

Subtopics

- **Default route**
- **Recursive static routes**
- **Route manager**
- **Configuration concepts**
- **Configuration example procedure**
- **Static routing commands**

Default route

Without a default route, a packet that does not match any routing entries is discarded and an ICMP destination-unreachable packet is sent to the source. A default route is used to forward packets that do not match any routing entry.

The network administrator can configure a default route with the destination as 0.0.0.0 and the mask as 0. The router forwards any packet whose destination address fails to match any entry in the routing table to the next hop of the default static route.

Recursive static routes

A recursive static route is a route for which the next hop is learned from another routing look up (dynamic protocol or another static route, with or without an explicit outgoing interface). For example, the following commands create an unsupported recursive static route:

- `ip route 99.0.0.0/24 1/1/51 30.0.0.2` - Main static route on explicit outgoing interface with gateway 30.0.0.2.
- `ip route 30.0.0.0/24 20.0.0.2` - Static route to reach the next-hop of the previously configured route, without explicit outgoing interface.

When a static route is configured without an explicit outgoing interface, the router resolves the next-hop IP address through recursive lookup. In this process, the routing table is consulted to determine how to reach the specified next hop.

As a result, if the next-hop IP address becomes reachable through an alternate path discovered during recursive resolution, the static route may remain installed in the routing table, even if the original or preferred outgoing interface transitions to a down state.

To ensure that a static route is removed from the routing table when a specific interface becomes unavailable, it is recommended to configure the route with an explicit outgoing interface. This approach tightly couples the validity of the route to the operational state of the interface, preventing unintended forwarding behavior through alternate paths.

Route manager

Route Manager (or Routing Table Manager) is the central repository for all routing information. It maintains the best routes selected by each routing protocol along with various route attributes in the Routing Information Base (RIB). Route Manager also manages routes from various routing protocols and determines the best route to each destination network which needs to be programmed into Forwarding Information Base (FIB).

> **CAUTION**
>
> If the number of routes is larger than 120,000, displaying all routes will require a significant amount of time to display output. A warning message is displayed to suggest using filters to display routes for prefixes for faster results. Use Ctrl+C to interrupt and return to prompt.

Configuration concepts

Before configuring a static route, you must understand the following concepts:

- **Destination address and mask:** In the `ip route` command, an IPv4 address is in dotted-decimal format. A mask can be in the form of mask length - the number of consecutive 1s in the mask.

- **Output interface and next hop address:** When configuring a static route, specify the output interface or next hop address. The next hop address cannot be a local interface IP address or the route configuration will not take effect.
- **Other attributes**: You can configure different priorities and administrative distance for different static routes to make route management policies more flexible. For example, specifying the same priority for different routes to the same destination enables load sharing, but specifying different priorities for these routes enables route backup.

Configuration example procedure

Before configuring a static route, complete the following tasks:

- Configure the physical parameters for related interfaces.
- Configure the link-layer attributes for related interfaces.
- Configure the IP addresses for related interfaces.

> **NOTE**
>
> **6000 and 6100 Switch Series:** The number of IPv4 and IPv6 static routes that can be configured is limited to 512 combined.
>
> **5420 and 6200 Switch Series:** The number of IPv4 and IPv6 static routes that can be configured is limited to 1024 combined. 16K routes are not supported.
>
> 6000, 6100, 5420 and 6200 Series Switches do not support IPv6 addresses with prefixes greater than 64. Prefixes between 65 and 127 will be software-forwarded.

Subtopics

[Basic static route configuration example]

Basic static route configuration example

The IP addresses and masks of the switches and hosts are displayed here. Static routes are required for interconnection between any two hosts.

Procedure

1. Configure IP addresses for interfaces (details not shown).

2. Configure static routes.

   a. Configure a default route on Switch A.
      6300# config 6300(config)# <SwitchA> config [SwitchA] ip route 0.0.0.0/0 1.1.4.2

   b. Configure two static routes on Switch B.
      <SwitchB> config [SwitchB] ip route 1.1.2.0/24 1.1.4.1 [SwitchB] ip route 1.1.3.0/24 1.1.5.6

   c. Configure a default route on Switch C.
      6300# config 6300(config)# [SwitchC] ip route 0.0.0.0/0 1.1.5.5

3. Configure the hosts. The default gateways for hosts A, B, and C are `1.1.2.3`, `1.1.6.1`, and `1.1.3.1`. The configuration procedure is not shown.

4. Display the configuration.

   a. Display the IP routing table of Switch A.

   b. Display the IP routing table of Switch B.

   c. Use the `ping` command on Host B to check the reachability of Host A, assuming Windows XP runs on the two hosts.
      C:\Documents and Settings\Administrator>ping 1.1.2.2 Pinging 1.1.2.2 with 32 bytes of data: Reply from 1.1.2.2: bytes=32 time=1ms TTL=255 Reply from 1.1.2.2: bytes=32 time=1ms TTL=255 Reply from 1.1.2.2: bytes=32 time=1ms TTL=255 Reply from 1.1.2.2: bytes=32 time=1ms TTL=255 15 Ping statistics for 1.1.2.2: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 1ms, Average = 1ms

   d. Use the traceroute command on Host B to check the reachability of Host A.
      8320# traceroute traceroute Trace the IPv4 route to a device on the network traceroute6 Trace the IPv6 route to a device on the network 8320# traceroute A.B.C.D Enter IPv4 address of the device to traceroute WORD Enter host name oo fthe device to traceroute 8320# traceroute

Static routing commands

Select a command from the list in the left navigation menu.

Subtopics

ip route
ipv6 route
show ip rib
show ipv6 rib

`ip route`

Syntax

```
ip route <DEST-IPV4-ADDR>/<NETMASK> [<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-
VLAN> |
nullroute | reject]
   {distance | name <WORD> | tag <VALUE> | track <WORD> | use-forwarding-
address | vrf <WORD>]
no ip route <DEST-IPV4-ADDR>/<NETMASK> [<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-
VLAN> |
nullroute | reject]
   {distance | name <WORD> | tag <VALUE> | track <WORD> | use-forwarding-
address | vrf <WORD>]
ip route <DEST-IPV4-ADDR>
            <NETMASK> [<NEXTHOP-PORT-LAG-VLAN> | nullroute | reject]
   {distance | name <WORD> | tag <VALUE> | vrf <WORD>]
no ip route <DEST-IPV4-ADDR>
            <NETMASK> [<NEXTHOP-PORT-LAG-VLAN> | nullroute | reject]
   {distance | name <WORD> | tag <VALUE> | vrf <WORD>]
```

> **NOTE**
> The **ip route distance** command is not supported on the HPE Aruba Networking 6200 Switch Series.

Description

Adds an IPv4 static route on the default VRF.

The **no** form of this command deletes a IPv4 static route.

> **NOTE**
> You can configure a maximum of 32 next hops per route.

Parameter	Description
`<DEST-IPV4-ADDR>/<NETMASK>`	Specifies the IPv4 route destination.
	Specifies the number of bits in the address mask in CIDR format (**x**), where **x** is a decimal number from 0 to 32.

Parameter	Description
<NETMASK>	
<NEXTHOP-ADDR>	Specifies the next hop address for reaching the destination in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255.
<NEXTHOP-PORT-LAG-VLAN>	Specifies the next hop as an outgoing interface.
`nullroute`	Specifies that packets matching the destination route are silently discarded and no ICMP error notification is sent to the sender.
`reject`	Specifies that packets matching the destination route are discarded and an ICMP error notification is sent to the sender.
`distance` *<VALUE>*	Specifies the administrative distance to associate with this static route. Range: 1 to 255. Default: 1.
`name` *<WORD>*	Specifies the name for the destination route. Range: 1 to 64 characters.
`tag` *<VALUE>*	Specifies and assigns tag for the route. Range: 1 to 4294967295.
`track` *<WORD>*	Specifies a name for the track object.
`use-forwarding-address`	Specifies the nexthop address as forwarding address.
`vrf` *<WORD>*	Specifies the VRF route leak destination.

Examples

```
switch(config)# ip route 10.0.0.0/24 nullroute
switch(config)# ip route 10.0.1.0/24 reject
switch(config)# ip route 10.0.2.0/24 20.0.0.2
switch(config)# ip route 10.0.3.0/24 1/1/1
switch(config)# ip route 10.0.3.0/24 1/1/1.110
switch(config)# ip route 10.0.2.0/24 20.0.0.2 distance 4
switch(config)# ip route 10.0.4.0/24 20.0.0.2 name AdminRoute
switch(config)# ip route 10.1.1.5/32 1/1/1 tag 20
switch(config)# ip route 1.1.2.0 255.255.255.0 2.2.2.2 track track1
```

```
switch(config)# ip route 10.1.1.6 255.255.255.0 20.1.1.2 use-forwarding-
address
switch(config)# ip route 20.0.3.0/24 reject vrf test
switch(config)# ip route 10.0.4.0/24 20.0.0.2 distance 5 name AdminRoute
switch(config)# ip route 1.1.2.0 255.255.255.0 2.2.2.2 track track1 name
Tracked-Critical-Path
switch(config)# ip route 10.1.1.6 255.255.255.0 20.1.1.2 use-forwarding-
address name OSPF-Forward
switch(config)# ip route 10.0.4.0/24 20.0.0.2 tag 15 name TaggedRoute
switch(config)# ip route 20.0.4.0/24 10.0.0.2 vrf test name VRF-Management
```

Command History

Release	Modification
10.17	The **name** parameter was introduced.
10.10	Inclusive language update.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`ipv6 route`

Syntax

```
ipv6 route <DEST-IPV6-ADDR>/<NETMASK> [<NEXTHOP-ADDR> | <NEXTHOP-PORT-LAG-
VLAN> | nullroute | reject]
   {distance | name <WORD> | source <IFNAME> | tag <VALUE> | track <WORD> |
vrf <WORD>}
no ipv6 route <DEST-IPV6-ADDR>/<NETMASK> [<NEXTHOP-ADDR> | <NEXTHOP-PORT-
LAG-VLAN> | nullroute | reject]
```

```
   {distance | name <WORD> | source <IFNAME> | tag <VALUE> | track <WORD> |
vrf <WORD>}
```

NOTE

The **ipv6 route tag** command is not supported on the HPE Aruba Networking 6200 Switch Series.

Description

Adds an IPv6 static route.

The **no** form of this command deletes an IPv6 static route on the default VRF.

Parameter	Description
`<DEST-IPV6-ADDR>`	Specifies the route destination in IPv6 format (**xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx**), where **x** is a hexadecimal number from 0 to F.
`<NETMASK>`	Specifies the number of bits in the address mask in CIDR format (**x**), where **x** is a decimal number from 0 to 128.
`<NEXTHOP-ADDR>`	Specifies the next hop in IPv6 format (**xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx**), where **x** is a hexadecimal number from 0 to F.
`<NEXTHOP-PORT-LAG-VLAN>`	Specifies the next hop as an outgoing interface.
`nullroute`	Specifies that packets matching the destination route are silently discarded and no ICMP error notification is sent to the sender.
`reject`	Specifies that packets matching the destination route are discarded and an ICMP error notification is sent to the sender.
`distance <VALUE>`	Specifies the administrative distance to associate with this static route. Range: 1 to 255. Default: 1.
`name <WORD>`	Specifies the name for the destination route. Range: 1 to 64 characters.
`source <IFNAME>`	Specifies the name for the outgoing interface.

Parameter	Description
`tag <VALUE>`	Specifies and assigns tag for the route. Range: 1 to 429496729 5.
`track <WORD>`	Specifies the object monitor. Range: 1 to 64 characters.
`vrf <WORD>`	Specifies the VRF route leak destination.

Examples

```
switch(config)# ipv6 route 120::/124 nullroute
switch(config)# ipv6 route 122::/124 1/1/1
switch(config)# ipv6 route 122::/124 1/1/1.110
switch(config)# ipv6 route 123::/124 120::1
switch(config)# ipv6 route 122::/124 nullroute distance 5
switch(config)# ipv6 route 3001::1/128 nullroute tag 10
switch(config)# ipv6 route 122::/124 reject distance 5
switch(config)# ipv6 route 3001::1/128 reject tag 10
switch(config)# ipv6 route 123::/124 120::1 distance 6
switch(config)# ipv6 route 124::/124 120::1 name IPv6ToHost
switch(config)# ipv6 route 3002::1/128 1000::2 tag 20
switch(config)# ipv6 route 2000::/64 1000::2 track track1
switch(config)# ipv6 route 123::/124 reject vrf test
switch(config)# ipv6 route 123::/124 nullroute distance 6 name IPv6-Block
switch(config)# ipv6 route 3002::1/128 nullroute tag 11 name IPv6-Block
switch(config)# ipv6 route 123::/124 reject distance 6 name IPv6-Reject-
Route
switch(config)# ipv6 route 3002::1/128 reject tag 11 name IPv6-ICMP-Error
switch(config)# ipv6 route 125::/124 120::1 distance 7 name V6ToServer
switch(config)# ipv6 route 3004::1/128 1000::3 tag 40 name Backup-Link
switch(config)# ipv6 route 2001::/64 1000::2 track track1 name IPv6-
Monitored-Link
switch(config)# ipv6 route 124::/124 121::3 vrf test name IPv6-VRF-Link
```

Command History

Release	Modification
10.17	The **name** parameter was introduced.
10.10	Inclusive language update.

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config`	Administrators or local user group members with execution righ ts for this command.

`show ip rib`

Syntax

```
show ip rib <FILTER> [vrf <VRF-NAME>]
```

Description

Shows the IPv4 Routing Information Base (RIB) of VRF with name (**<VRF-NAME>**). If VRF name is not specified, default VRF routes are displayed.

Parameter	Description
`<FILTER>`	Selects filter, see Usage section.
`vrf <VRF-NAME>`	Specifies the VRF name.

Usage

There are sub-options available within this command:

- **A.B.C.D:** Shows longest prefix match.
- **A.B.C.D/M:** Shows exact route match.
- **all-vrfs:** Shows all VRF information.
- **bgp:** Shows BGP routes only.

- **connected:** Shows connected routes only.
- **connected:** Shows connected routes only.
- **local:** Shows local routes only.
- **ospf:** Shows OSPF routes only.
- **rip:** Shows RIP routes only.
- **static:** Shows static routes only.
- **summary:** Shows aggregate count of routes per routing protocol.
- **vrf:** Specifies the VRF name.
- **selected:** Shows routes selected for forwarding only.
- **non-selected:** Shows routes not selected for forwarding only.

The output of the **show ip rib** commands are not available in information generated by the **show tech files** command. This information is now available in the file ipv4_rib_dump.gz, which can be generated using the command **sudo ovs-appctl -t hpe-routing hpe-metaswitch/show_ip_rib ipv4 all-vrfs**.

Examples

Showing IPv4 routes in RIB for HPE Aruba Networking 8325, 10000, 8360, and 9300/9300S Switch series:

```
switch# show ip rib
Origin Codes: R - RIP, O - OSPFv2, B - BGP
              C - connected, S - static, H - host-routes
Type Codes:   E - External BGP, I - Internal BGP, IA - OSPF inter area
              E1 - OSPF external type 1, E2 - OSPF external type 2
* indicates selected for forwarding
VRF: default
Prefix          Nexthop       Interface   VRF       Origin/ Distance/
Age
Type    Metric
----------------------------------------------------------------------------
--------
*1.1.1.1/32     -             1/1/33      -         H       [1/0]
00h:00m:07s
*10.0.0.0/30    -             1/1/1       -         S       [20/0]
0d:10h:01m:41s
*10.0.1.0/30    -             1/1/1       -         B/I     [200/0]
2d:20h:01m:42s
*10.1.64.0/18   -             loopback2   -         C       [0/0]       -
*10.2.64.0/18   10.0.0.3      lag1        -         O/E1    [110/25]
1d:05h:03m:43s
```

```
*10.2.64.0/18   20.10.0.1    vlan100      -          O/E1    [110/25]
0d:05h:03m:43s
*20.1.2.3/32    2.2.2.2      1/1/4        vrf_red    B/E     [20/0]
2d:10h:01m:45s
*30.1.3.0/24    -            reject       -          S       [1/0]
33d:10h:01m:43s
*50.10.13.0/24  -            reject       -          S       [1/0]
12d:10h:01m:44s
*61.1.1.2/32    4.4.4.4      1/1/5        -          B/I     [200/0]
1d:11h:01m:45s
*62.1.1.3/32    5.5.5.5      1/1/6        -          B/I     [200/0]
0d:12h:01m:45s
*193.0.0.2/32   50.0.0.2     1/1/2        -          S       [1/0]
0d:04h:01m:43s
 193.0.0.2/32   56.0.0.3     1/1/3        -          O/E1    [110/25]
0d:04h:03m:43s
Total Route Count : 14
```

Showing IPv4 routes in RIB for all other switches:

```
switch# show ip rib
Origin Codes: R - RIP, O - OSPFv2, B - BGP
              C - connected, S - static, D - DHCP
Type Codes:   E - External BGP, I - Internal BGP, IA - OSPF inter area
              E1 - OSPF external type 1, E2 - OSPF external type 2
* indicates selected for forwarding
VRF: default
Prefix          Nexthop      Interface    VRF        Origin/ Distance/
Age
Type    Metric
----------------------------------------------------------------------------
--------
*1.1.1.1/32     -            1/1/33       -          H       [1/0]
00h:00m:07s
*10.0.0.0/30    -            1/1/1        -          S       [20/0]
0d:10h:01m:41s
*10.0.1.0/30    -            1/1/1        -          B/I     [200/0]
2d:20h:01m:42s
*10.1.64.0/18   -            loopback2    -          C       [0/0]       -
*10.2.64.0/18   10.0.0.3     lag1         -          O/E1    [110/25]
1d:05h:03m:43s
*10.2.64.0/18   20.10.0.1    vlan100      -          O/E1    [110/25]
0d:05h:03m:43s
*20.1.2.3/32    2.2.2.2      1/1/4        vrf_red    B/E     [20/0]
2d:10h:01m:45s
```

```
*30.1.3.0/24     -            reject       -          S       [1/0]
33d:10h:01m:43s
*50.10.13.0/24   -            reject       -          S       [1/0]
12d:10h:01m:44s
*61.1.1.2/32     4.4.4.4      1/1/5        -          B/I     [200/0]
1d:11h:01m:45s
*62.1.1.3/32     5.5.5.5      1/1/6        -          B/I     [200/0]
0d:12h:01m:45s
*193.0.0.2/32    50.0.0.2     1/1/2        -          S       [1/0]
0d:04h:01m:43s
 193.0.0.2/32    56.0.0.3     1/1/3        -          O/E1    [110/25]
0d:04h:03m:43s
Total Route Count : 14
```

Showing IPv4 exact route match in RIB:

```
switch# show ip rib 10.0.0.0/30
VRF : default
Prefix          : 10.0.0.0/30              VRF(egress)        :  -
Nexthop         :  -                       Interface          :  1/1/1
Origin          : Connected                Type               :  -
Distance        : 0                        Metric             :  0
Age             : -                        Tag                :  0
Selected        : Yes                      Recursive Nexthop  :  No
```

Showing IPv4 RIB summary:

```
switch# show ip rib summary
IPv4 RIB Table Summary
VRF name :       default
  Protocol       RIB Routes
-------------- -------------
connected      1010
local          1011
static         4
ospfv2         509
bgp            9014
selected       10008
non-selected   1518
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`) or Manager (`#`)	Administrators or local user group members with execution rights for this command.

`show ipv6 rib`

Syntax

```
show ipv6 rib <FILTER> [vrf <VRF-NAME>]
```

Description

Shows the IPv6 Routing Information Base (RIB) of VRF with name (**<VRF-NAME>**). If VRF name is not specified, default VRF routes are displayed.

Parameter	Description
`<FILTER>`	Selects filter, see usage section.
`vrf <VRF-NAME>`	Shows routes in the VRF and specifies VRF name.

Usage

There are sub-options available within this command:

- **X:X: :X:X:** Shows longest prefix match.
- **X:X: :X:X/M:** Shows exact route match.
- **all-vrfs:** Shows all VRF information.
- **bgp:** Shows BGP routes only.
- **connected:** Shows connected routes only.
- **local:** Shows local routes only.

- **ospf:** Shows OSPF routes only.
- **rip:** Shows RIP routes only.
- **static:** Shows static routes only.
- **summary:** Shows aggregate count of routes per routing protocol.
- **vrf:** Specifies the VRF name.
- **selected:** Shows routes selected for forwarding only.
- **non-selected:** Shows routes not selected for forwarding only.

Examples

Showing IPv6 routes in RIB for HPE Aruba Networking 8325, 10000, 8360, and 9300/9300S Switch series:

```
switch# show ipv6 rib
Origin Codes: R - RIPng, O - OSPFv3, B - BGP
              C - connected, S - static, H - host-routes
Type Codes:   E - External BGP, I - Internal BGP, IA - OSPF inter area
              E1 - OSPF external type 1, E2 - OSPF external type 2
* indicates selected for forwarding
VRF: default
Prefix                Nexthop      Interface   VRF    Origin/ Distance/
Age
Type    Metric
-----------------------------------------------------------------------------
--------
*1::2/128             1::2         1/1/1       -      H       [1/0]
00h:00m:06s
*1000::/64            -            1/1/1       -      C       [0/0]      -
*1000::8/128          -            1/1/1       -      L       [0/0]      -
*1001:db8::/32        1000::10     1/1/1       -      B/I     [200/0]
1d:20h:01m:42s
*2000::/64            fe80::3182   vlan100     -      S       [1/0]
2d:05h:03m:43s
 2000::/64            fe80::1241   1/1/1       -      O/E1    [110/25]
0d:05h:03m:43s
*2000::2000:0:0:0/67 fe80::1111   lag1        Green  B/E     [20/0]
1d:10h:01m:45s
*3001::0/64           -            vlan100     -      C       [0/0]      -
*3001::1/128          -            vlan100     -      L       [0/0]      -
*6101::0/64           -            nullroute   -      S       [1/0]
12d:10h:01m:43s
Total Route Count : 10
```

Showing IPv6 routes in RIB for the other switch series:

```
switch# show ipv6 rib
Origin Codes: R - RIPng, O - OSPFv3, B - BGP
              C - connected, S - static, D- DHCP
Type Codes:   E - External BGP, I - Internal BGP, IA - OSPF inter area
              E1 - OSPF external type 1, E2 - OSPF external type 2
* indicates selected for forwarding
VRF: default
Prefix                 Nexthop     Interface  VRF    Origin/ Distance/
Age
Type    Metric
----------------------------------------------------------------------------
--------
*1000::/64             -           1/1/1      -      C       [0/0]     -
*1000::8/128           -           1/1/1      -      L       [0/0]     -
*1001:db8::/32         1000::10    1/1/1      -      B/I     [200/0]
1d:20h:01m:42s
*2000::/64             fe80::3182  vlan100    -      S       [1/0]
2d:05h:03m:43s
 2000::/64             fe80::1241  1/1/1      -      O/E1    [110/25]
0d:05h:03m:43s
*2000::2000:0:0:0/67 fe80::1111  lag1       Green  B/E     [20/0]
1d:10h:01m:45s
*3001::0/64            -           vlan100    -      C       [0/0]     -
*3001::1/128           -           vlan100    -      L       [0/0]     -
*6101::0/64            -           nullroute  -      S       [1/0]
12d:10h:01m:43s
Total Route Count : 10
```

Showing IPv6 exact route match in RIB:

```
switch# show ipv6 rib 2000::2000:0:0:0
VRF : default
Prefix           : 2000::2000:0:0:0/67        VRF(egress)       :  Green
Nexthop          : fe80::1111                 Interface         :  lag1
Origin           : BGP                        Type              :
External
Distance         : 20                         Metric            :  0
Age              : 1d:10h:01m:45s             Tag               :  20
Selected         : Yes                        Recursive Nexthop :  Yes
```

Showing IPv6 RIB summary:

```
switch# show ipv6 rib summary
IPv6 RIB Table Summary
VRF name :  default
```

```
  Protocol        RIB Routes
-------------- -------------
connected       1009
local           1010
static          3
ospfv3          508
bgp             1013
selected        10004
non-selected    1527
```

Command History

Release	Modification
10.10	Inclusive language update.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

Open Shortest Path First version 2 (OSPFv2)

NOTE

OSPFv2 is not supported on the 6000 and 6100 Switch Series.

NOTE

OSPFv2 is not supported on the 4100i Switch Series.

Open Shortest Path First version 2 (OSPFv2) is a routing protocol described in RFC2328. It is a Link State-based IGP (Interior Gateway Protocol) routing protocol applied to routers grouped into OSPF areas identified by the routing configuration on each routing switch. It is widely used in medium to large-sized enterprise networks.

Subtopics

Overview

The characteristics of OSPFv2 are:

- Provides a loop-free topology using SPF algorithm.
- Supports load balancing with equal cost routes for the same destination.
- OSPFv2 is a classless protocol and allows for a hierarchical design with VLSM (Variable Length Subnet Masking) and route summarization.
- Provides authentication of routing messages.
- Provides fast convergence with triggered, incremental updates via LSAs.
- Supports OSPFv2 operation over GRE IPv4 tunnel interfaces (platform-specific).

Some OSPFv2 configuration is done in the global configuration context, others in the router ospf context, or in the interface configuration context, or in the vlink context. OSPFv2 can be configured on L3 ports, VLAN interfaces, LAG interfaces, GRE tunnel interfaces, and loopback interfaces. All such configurations work in the mentioned interfaces context. OSPFv2 mandates the associated interface to be a routed interface.

Supported features

- OSPFv2 neighbor adjacency, Hello protocol, multiple areas, Inter-area routing
- AS external routes, stub areas, totally stubby areas, NSSA, ASBR

- Designated Router/Backup Designated Router
- Point-to-point interfaces/broadcast interfaces
- Point-to-multipoint network types in both non-broadcast mode (static neighbor configuration) and broadcast mode (dynamic neighbor discovery using the **dynamic** option)
- Equal-cost multipath
- Null authentication, Simple password authentication, MD5 authentication
- Graceful Restart - un-planned, restart interval, Graceful Restart Helper
- Stub router advertisement
- SPF throttling
- LSA Throttling
- SPF throttling
- Configuration of interface parameters such as priority, cost, hello-interval, dead-interval, retransmit-interval, transit-delay, etc
- Configuration of virtual link parameters such as hello-interval, dead-interval, retransmit-interval, transit-delay, etc.
- Route redistribution
- Passive interfaces
- Multi VRF support with each VRF having up to eight OSPF process instances
- Congestion control (prioritizing hello packets, inactivity timer reset, and adjacency throttling)
- IP unnumbered interfaces

Listed below are a few recommendations while configuring IP unnumbered interfaces:

- Enable OSPF on the lending interface in the same area as the borrowing interface.
- If different unnumbered interfaces are used for different areas, then configure them to use different loop backs.
- Configure the address on the lending loopback as /32.
- OSPFv3 can also be enabled on an IP unnumbered interface. OSPFv3 neighborship can be established on IP unnumbered interfaces as it works over IPv6 link-local address.

How OSPFv2 protocol works

The protocol uses Link State Advertisements (LSAs) transmitted by each router to update neighboring routers regarding its interfaces and the routes available through those interfaces. Each routing switch in an area also maintains a link-state database (LSDB) that describes the area topology. (All routers in a given OSPF area have identical LSDBs.) The routing switches used to connect areas to each other flood summary link LSAs and external link LSAs to neighboring OSPF areas to update them regarding available routes. Through this means, each OSPF router determines the shortest path between itself and a desired destination router in the same OSPF domain (Autonomous System (AS)).

OSPFv2 concepts

The following sections describe OSPFv2 Link-State types, router types, and area types.

Subtopics

OSPFv2 Link-state advertisement (LSA) types
OSPFv2 router types
OSPFv2 area types

OSPFv2 Link-state advertisement (LSA) types

OSPFv2 sends routing information in Link-State Advertisements (LSAs). The switches support the following types of LSAs.

- **Router LSA—Type-1 LSA.** Describes the states of the router interfaces to an area, and is flooded throughout a single area only.
- **Network LSA—Type-2 LSA.** Describes the list of routers connected to the network, and is flooded throughout a single area only.
- **Summary LSA—Type-3 LSA.** Describes the route to networks in another OSPF area of the same Autonomous System (AS). Propagated through backbone area to other areas.
- **ASBR summary LSA—Type-4 LSA.** Describes the route to an ASBR in an OSPF normal or backbone area of the same AS. Propagated through backbone area to other areas.
- **AS external LSA—Type-5 LSA.** Describes the route to a destination in another AS (external route.) Originated by ASBR in normal or backbone areas of an AS and propagates through backbone area to other normal areas.

- **NSSA LSA—Type-7 LSA.** Describes the route to a destination in another AS (external route.) Originated by ASBR in NSSA. ABR converts type-7 LSAs to type-5 LSAs for injection into the backbone area.

OSPFv2 router types

The following router types are supported

Internal routers

Internal OSPFv2 routers belong to only one area. Internal routers flood type-1 LSAs to all routers in the same area and maintain identical LSDBs.

Area border routers (ABRs)

Area border routers have membership in multiple areas. ABRs are used to connect the various areas in an AS to the backbone area for that AS. Multiple ABRs can be used to connect a given area to the backbone, and a given ABR can belong to multiple areas other than the backbone.

An ABR maintains a separate LSDB for each area to which it belongs. (All routers within the same area have identical LSDBs.) The ABR is responsible for flooding summary LSAs between its border areas. You can reduce summary LSA flooding by configuring area ranges. An area range enables you to assign an aggregate address to a range of IP addresses. This aggregate address is advertised instead of all the individual addresses it represents.

Autonomous system boundary router (ASBR)

Autonomous system boundary routers run multiple interior gateway protocols and serve as a gateway to other autonomous systems operating with interior gateway protocols. The ASBR imports and translates different protocol routes into OSPF through redistribution. ASBRs can be used in backbone areas, normal areas, and NSSAs, but not in stub areas.

Designated routers (DRs)

In an OSPF network having two or more routers, one router is elected to serve as the DR and another router to act as the Backup Designated Router (BDR). All other routers in the area forward their routing information to the DR and BDR, and the DR forwards this information to all routers in the network. This action minimizes the amount of repetitive information that is forwarded on the network by eliminating the need for each individual router in the area to forward its routing information to all other routers in the network. If the area includes multiple networks, each network elects its own DR and BDR.

In an OSPF network with no DR and no BDR, the neighboring router with the highest priority is elected the DR, and the router with the next highest priority is elected the BDR. If the DR goes off-line, the BDR automatically becomes the DR, and the router with the next highest priority then becomes the new BDR. If multiple routing switches on the same OSPF network are declaring themselves DRs, both priority and router ID are used to select the DR and BDRs.

Priority is configurable using the `ip ospf priority` command at the interface level. If two neighbors share the same priority, the router with the highest router ID is elected as the DR. The router with the next highest router ID is elected as the BDR.

OSPFv2 area types

OSPFv2 is built upon a hierarchy of network areas. All areas for a given OSPF domain reside in the same AS. An AS is defined as a number of contiguous networks that share the same interior gateway routing protocol.

An AS can be divided into multiple areas. Each area represents a collection of contiguous networks and hosts, and the topology of a given area is not known by the internal routers in any other area. Areas define the boundaries to which types 1 and 2 LSAs are broadcast, which limits the amount of LSA flooding that occurs within the AS and also helps to control the size of the LSDBs maintained in OSPF routers. An area is represented in OSPF by either an IP address or a number. Area types include: Backbone, Normal, Not-so-stubby (NSSA), and stub.

Backbone area

Every AS must have one (and only one) backbone area (identified as area 0 or 0.0.0.0.) The ABRs of all other areas in the same AS connect to the backbone area, either physically through an ABR or through a configured virtual link. The backbone is a transit area that carries the type-3 summary LSAs, type-5 AS external link LSAs and routed traffic between non-backbone areas, as well as the type-1 and type-2 LSAs and routed traffic internal to the area. ASBRs are allowed in backbone areas.

Normal area

Normal area connects to the backbone area through one or more ABRs (physically or through a virtual link) and supports type-3 summary LSAs and type-5 external link LSAs to and from the backbone area. ASBRs are allowed in normal areas.

Stub area

Stub area connects to the AS backbone through one or more ABRs. It does not allow an internal ASBR, and does not allow external (type 5) LSAs. A stub area supports these actions:

- Advertise the area summary routes to the backbone area.
- Advertise summary routes from other areas.
- Use the default summary (type-3) route to advertise both of the following:
    - Summary routes to other areas in the AS
    - External routes to other ASs

You can configure the stub area ABR to do the following:

- Suppress advertising from some or all area summarized internal routes into the backbone area.
- Suppress LSA traffic from other areas in the AS by replacing type-3 summary LSAs and the default external route from the backbone area with the default summary route (0.0.0.0/0.)

Virtual links are not allowed for stub areas.

Not-so-stubby (NSSA) area

NSSA area connects to the backbone area through one or more ABRs. NSSAs are intended for use where an ASBR exists in an area where you want to control the following:

- Advertising the ASBR external route paths to the backbone area
- Allowing LSAs from the backbone area to advertise in the NSSA:
  - Summary routes (type-3 LSAs) from other areas
  - External routes (type-5 LSAs) from other areas as a default external route (type-7 LSAs)

Virtual links are not allowed for NSSAs.

OSPFv2 configuration task list

Tasks at a glance

- Configuring OSPF on the routing switch
- Configuring external route redistribution and control
- Influencing route choice by changing the administrative distance
- Configuring graceful restart of OSPF routing
- Configuring OSPF interface settings
- Configuring OSPF interface authentication
- Configuring all OSPF interfaces as passive
- Clearing OSPF statistics on a switch

Configuring OSPF on the routing switch

Create the OSPF instance and enter the OSPF router configuration context. From this, you can proceed with other OSPF configuration tasks.

Prerequisites

- You must be in the global configuration context, as indicated by the `switch(config)#` prompt to create the OSPF instance and enter the OSPF router configuration context.

- To configure a router ID, create OSPF network areas, or adjust other global OSPF configuration items, you must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

Procedure

1. Create the OSPF instance and enter the OSPF router configuration context using the following command. For command details, see router ospf.
   router ospf <PROCESS-ID>

   For example, the following command creates OSPF instance 1.

   ```
   switch(config)# router ospf 1
   switch(config-ospf-1)#
   ```

2. Configure a global router ID using the following command. For command details, see router-id.
   router-id <ROUTER-ADDRESS>

   For example, the following command sets the router ID to 1.1.1.1.

   ```
   switch(config-ospf-1)# router-id 1.1.1.1
   ```

3. Optionally, if the OSPF process was disabled (it is enabled by default), enable the OSPF process using the following command.
   enable

   For command details, see enable. (Refer to disable for disabling the OSPF process).

Setting OSPF network for the area

After you define an OSPF area, you can assign one or more networks to it. The OSPF protocol will run on the interface with the configured IPv4 address. The interfaces which have an IP address configured in this network or in a subset of this network, will participate in the OSPF protocol.

Prerequisites

You must be in the interface configuration context, as indicated by the `switch(config-if)#` prompt.

Procedure

1. Set an OSPF network for the area using the following command. For command details, see ip ospf area.
   ip ospf <PROCESS-ID> area <AREA-ID>

   For example, use the following command to assign interface 1/1/1 to OSPF area 1. The area can alternatively be entered as an IPv4 address.

```
switch(config)# interface 1/1/1
switch(config-if)# ip ospf 1 area 1
```

2. Optionally, you can disable OSPF on the interface using the following command. For command details, see ip ospf shutdown.
   ip ospf shutdown

```
switch(config)# interface 1/1/1
switch(config-if)# ip ospf shutdown
```

Configuring external route redistribution and control

About this task

Configuring route redistribution for OSPFv2 establishes the routing switch as an ASBR for importing and translating different protocol routes into OSPFv2. When you configure redistribution for OSPFv2, you can specify that routes external to the OSPFv2 domain are imported as OSPFv2 routes.

Procedure

1. Enable route redistribution using the `redistribute` command.

```
   redistribute {bgp | connected | local loopback | static | rip | ospf
<PROCESS-ID>}
                 [route-map <ROUTE-MAP-NAME>]
```

   For example, setting redistribution of connected routes as OSPFv2 routes:

```
switch(config-ospf-1)# redistribute connected
```

   If a route map is specified, then only the routes that pass the match clause specified in the route map are redistributed to OSPFv2. For example, setting redistribution of local loopback routes, that only match the routes specified by the route map:

```
switch(config-ospf-1)# redistribute local loopback route-map local_routes
```

2. Optionally, modify the default metric for redistribution using the `default-metric` command.

```
   default-metric <METRIC-VALUE>
```

   For example, setting the default metric for redistribution to 37.

```
switch(config-ospf-1)# default-metric 37
```

3. Optionally, set the cost of default-summary LSAs using the area command.

```
area <AREA-ID> default-metric <COST>
```

For example, setting the cost of default summary LSAs to 2.

```
switch (config-ospf-1)# area 1 default-metric 2
```

4. Optionally, use the `max-metric router-lsa` command to set the protocol to advertise a maximum metric so that other routers do not prefer this router as an intermediate hop in their shortest path first (SPF) calculations.

```
max-metric router-lsa [on-startup [<ADVERT-TIME>]]
```

For example, setting advertise max-metric router-lsa on startup.

```
switch(config-ospf-1)# max-metric router-lsa on-startup 3000
```

Influencing route choice by changing the administrative distance

The administrative distance value can be left in its default configuration setting (110) unless a change is needed to improve OSPF performance for a specific network configuration.

Prerequisites

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

Procedure

Reconfigure the administrative distance using the following command. For command details, see distance command.

```
distance <distance>
```

For example, use the following command to set administrative distance to 100.

```
switch(config-ospf-1)# distance 100
```

Configuring graceful restart of OSPF routing

OSPF routing can be gracefully restarted on the switch without losing packets that are in transit. OSPF neighbors are informed that the router is completing a graceful restart, which allows for maintenance on the switch without interrupting traffic in the network. There is no effect on the saved switch configuration.

Prerequisites

> **NOTE**
> Graceful restart is only applicable to the 5420 and 6200-VSF switches.

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

Manually create a router-id configuration to support graceful restart and management module failover. A dynamic router-id is chosen from the list of configured IPs at the time of the router-id election, and in the event of a graceful restart and management module failover, the router-id should be preserved before and after these events. Graceful restart would fail if there is a change in the router-id, causing traffic loss.

Procedure

Enable graceful restart of OSPF routing using the following command. For command details, see graceful-restart.

```
graceful-restart {restart-interval <seconds> | helper}
```

For example, the following command specifies 50 seconds as the maximum interval another router will wait for this router to gracefully restart.

```
switch(config-ospf-1)# graceful-restart restart-interval 50
```

Configuring OSPF interface settings

You can optionally adjust the following OSPF interface settings.

Prerequisites

You must be in the interface configuration context, as indicated by the `switch(config-if)#` prompt.

Procedure

1. Set the interface cost using the following command. For command details, see ip ospf cost.

   ```
   ip ospf cost <interface-cost>
   ```

   For example, the following command sets the cost associated with interface 1/1/1 to 100.

   ```
   switch(config)# interface 1/1/1
   switch(config-if)# ip ospf cost 100
   ```

2. Set the time interval between OSPF hello packets for the OSPF interface using the following command. For command details, see ip ospf hello-interval.

```
ip ospf hello-interval  <seconds>
```

3. Set the interval after which a neighbor is declared dead if no hello packet is received on the OSPF interface using the following command. For command details, see ip ospf dead-interval.

```
ip ospf dead-interval <seconds>
```

4. Set the time between retransmitting lost link state advertisements for the OSPF interface using the following command. For command details, see ip ospf retransmit-interval.

```
ip ospf retransmit-interval  <seconds>
```

5. Sets the transit delay in link state transmission for the OSPF interface using the following command. For command details, see ip ospf transit-delay.

```
ip ospf transit-delay <seconds>
```

6. Set the OSPF network type for the interface. For command details, see ip ospf network.

```
ip ospf network {broadcast|point-to-point}
```

7. Set the OSPF priority on the interface using the following command. For command details, see ip ospf priority.

```
ip ospf priority <number-value>
```

8. Set the OSPF interface as OSPF passive interface. The interface participates in OSPF but does not send or receive packets on that interface. For command details, see ip ospf passive.

```
ip ospf passive
```

Configuring OSPF interface authentication

Configure authentication on the interface. Only one method of authentication can be active on an interface at a time. If one method is already configured on an interface, configuring an alternative method on the same interface automatically overwrites the first method used.

Prerequisites

You must be in the interface configuration context, as indicated by the `switch(config-if)#` prompt.

Procedure

1. Set the authentication type that will be used for authentication with the neighbor router using the following command. For command details, see ip ospf authentication.
   ```
   ip ospf authentication {message-digest | simple-text | null | keychain}
   ```
   For example, the following command sets the authentication type to simple-text.
   ```
   switch(config-if)# ip ospf authentication simple-text
   ```
2. For simple-text authentication, set the password using the following command. For command details, see ip ospf authentication-key.
   ```
   ip ospf authentication-key <PASSWORD>
   ```
   For example:
   ```
   switch(config-if)# ip ospf authentication-key secure
   ```
3. For MD5 authentication, set the password using the following command. For command details, see ip ospf message-digest-key md5.
   ```
   ip ospf message-digest-key {Key ID} md5 {ciphertext | plaintext}  <KEY>
   ```
   For example:
   ```
   switch(config-if)# ip ospf message-digest-key 1 md5 ciphertext 100
   ```

Configuring all OSPF interfaces as passive

OSPF sends LSAs to all other routers in the same AS. To limit the flooding of LSAs throughout the AS, you can configure OSPF to be passive.

Prerequisites

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

Procedure

Configure all OSPF interfaces as passive using the following command. For command details, see passive-interface default.

```
passive-interface default
switch(config-ospf-1)# passive-interface default
```

Configuring SPF throttling timers

SPF calculation is throttled with default timer values (start-time 200ms, hold-time 1000ms, max-wait-time 5000ms). You can throttle SPF calculation by configuring non-default timers to improve performance of a specific network configuration.

Prerequisites

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

Procedure

Reconfigure SPF throttling timers using the following command. For command details, see timers throttle spf.

```
timers throttle spf start-time <milliseconds> hold-time <milliseconds> max-
wait-time <milliseconds>
```

For example, use the following command to set start-time to 500ms, hold-time to 3000ms and max-wait-time to 9000ms:

```
switch(config-ospf-1)# timers throttle spf start-time 500 hold-time
3000 max-wait-time 9000
```

Clearing OSPF statistics on a switch

Clear OSPF event statics using the following command. For command details, see clear ip ospf statistics .

```
clear ip ospf [<PROCESS-ID>] statistics [interface [<INTERFACE-NAME>]] [all-
vrfs | vrf <VRF-NAME>]
```

For example, the following command clears OSPF event statistics from interface 1/1/1.

```
switch(config-router)# clear ip ospf statistics interface 1/1/1
```

Configuring OSPFv2 over GRE IPv4 Tunnels

OSPFv2 can be configured on GRE IPv4 tunnel interfaces to establish OSPF adjacencies through tunneled connections. This feature is supported on the 6300, 6400, 8320, 8325, 9300, and 10000 Switch series.

Prerequisites

You must be in the global configuration context, as indicated by the `switch(config)#` prompt, to create and configure GRE tunnel interfaces.

You must be in the tunnel interface configuration context, indicated by the `switch(config-gre-if)#` prompt, to enable or disable OSPFv2 on the GRE tunnel interface.

> **NOTE**
>
> Configure the tunnel interface IP MTU to 1476 bytes using **ip mtu 1476** when running OSPFv2 over GRE. This accommodates the 24-byte GRE encapsulation overhead (20-byte outer IP header plus 4-byte GRE header), ensuring packets fit within the default Layer-2 MTU and avoid fragmentation.

Procedure

Configure the physical underlay interface.

```
switch(config)# interface 1/1/1
switch(config-gre-if)# no shutdown
switch(config-gre-if)# ip address 20.1.1.1/24
```

Create and configure the GRE IPv4 tunnel interface.

```
switch(config)# interface tunnel 1 mode gre ip
switch(config-gre-if)# source ip 20.1.1.1
switch(config-gre-if)# destination ip 20.1.1.2
switch(config-gre-if)# ip address 30.1.1.1/24
switch(config-gre-if)# ip ospf 1 area 0
switch(config-gre-if)# no shutdown
```

Configure the OSPFv2 process.

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 0.0.0.0
```

Enable OSPFv2 on the GRE tunnel interface and associate it with an OSPF process and area. If the OSPFv2 process or area does not exist, the system prompts you to create them.

- Use A.B.C.D to specify an OSPFv2 area ID in IPv4 address.
- Use 0-4294967295 to specify an OSPFv2 area ID.

```
switch(config)# interface tunnel 1
switch(config-gre-if)# ip ospf 1 area 0
switch(config-gre-if)# ip ospf 1 area 0.0.0.0
```

Disable OSPFv2 on the GRE tunnel interface.

```
switch(config-gre-if)# no ip ospf 1 area 0
switch(config-gre-if)# no ip ospf 1 area 0.0.0.0
```

In this configuration:

- A GRE tunnel interface is created using `mode gre ip`.
- The tunnel source is the IP address of the physical interface (20.1.1.1).
- The tunnel destination is the remote endpoint (20.1.1.2).
- The tunnel interface has its own IP address (30.1.1.1/24) for OSPFv2 communication.
- The tunnel interface is assigned to OSPFv2 process 1, area 0 using `ip ospf 1 area 0`.
- The physical interface (1/1/1) provides underlay connectivity with IP 20.1.1.1/24.

Use the `show ip ospf neighbors` command to view OSPF neighbor adjacencies established over GRE tunnel interfaces.

```
(config-gre-if)# show ip ospf neighbors
VRF : default                          Process : 1
===================================================


Total Number of Neighbors : 1


Neighbor ID      Priority  State       Nbr Address       Interface
------------------------------------------------------------------
20.1.1.2         n/a       FULL        30.1.1.2           tunnel1
```

This output shows:

- **Neighbor ID:** Router ID of the OSPFv2 neighbor (20.1.1.2).
- **Priority:** Not applicable (n/a) for point-to-point interfaces like GRE tunnels.
- **State:** FULL indicates the adjacency is fully established.
- **Nbr Address:** IP address of the neighbor on the tunnel interface (30.1.1.2).
- **Interface:** The GRE tunnel interface name (tunnel1).

Use the `show interface tunnel` command to verify the tunnel interface status.

```
(config-gre-if)# show interface tunnel 1

Interface tunnel1 is up
 Admin state is up
 Description:
 Tunnel type: gre ipv4
 Tunnel source: ipv4 address 20.1.1.1
 Tunnel destination: ipv4 address 20.1.1.2
 Tunnel ttl: 64
 Tunnel ip address: 30.1.1.1/24
```

Additional Configurations

All standard OSPFv2 interface configuration commands are supported on GRE tunnel interfaces, including:

- ip ospf hello-interval
- ip ospf dead-interval
- ip ospf retransmit-interval
- ip ospf transmit-delay
- ip ospf cost
- ip ospf network
- ip ospf priority (not applicable to point-to-point)
- ip ospf passive
- ip ospf shutdown

> **NOTE**
> The tunnel interface must have an IPv4 address and reachable underlay connectivity between the tunnel source and destination for OSPFv2 adjacencies to form. OSPFv2 packets are encapsulated in GRE and then in IPv4, using the source and destination IPv4 addresses configured on the tunnel.

An example of the OSPFv2 information in the show running-config command

When OSPFv2 is configured, the output of the `show running-config` command includes OSPFv2 information.

For example:

```
switch# show running-config
Current configuration:
!
!Version AOS-CX 10.0X.XXXX
!
lldp enable
timezone set utc
vrf red
mgmt
led base-loc_fdc on
```

```
led base-loc on
led base-hlth_fdc fast_blink
led base-pwr_fdc on
!
!
!
!
!
!
aaa authentication login default local
aaa authorization commands default none
!
!
!
!
router ospf 1 vrf vrf_default
    area 0.0.0.0
    area 0.0.0.1
    area 0.0.0.1 virtual-link 55.4.4.4
        hello-interval 15
        dead-interval 60
router ospf 1 vrf red
    area 0.0.0.4
    area 0.0.0.5
    area 0.0.0.5 range 10.1.0.0/16 type inter-area
vlan 1
    no shutdown
interface lag 44
    no shutdown
    ip address 33.1.1.1/24
    ip ospf 1 area 0.0.0.0
    ip ospf hello-interval 22
interface 1/1/1
    no shutdown
    ip address 33.44.1.1/24
    ip ospf 1 area 0.0.0.0
    ip ospf passive
    ip ospf dead-interval 88
    ip ospf transit-delay 44
    ip ospf priority 88
    ip ospf cost 8
    ip ospf network point-to-point
    ip ospf authentication simple-text
```

```
    ip ospf authentication-key kkk
    ip ospf shutdown
interface 1/1/2
interface loopback 2
    ip address 55.55.55.55/32
    ip ospf 1 area 0.0.0.0
switch# show running-config
Current configuration:
!
!Version AOS-CX 10.0X.XXXX
!
lldp enable
timezone set utc
mgmt
led base-loc_fdc on
led base-loc on
led base-hlth_fdc fast_blink
led base-pwr_fdc on
!
!
!
!
!
!
aaa authentication login default local
aaa authorization commands default none
!
!
!
!
router ospf 1 vrf vrf_default
    area 0.0.0.0
vlan 1
    no shutdown
interface lag 44
    no shutdown
    ip address 33.1.1.1/24
    ip ospf 1 area 0.0.0.0
    ip ospf hello-interval 22
interface 1/1/1
    no shutdown
    ip address 33.44.1.1/24
    ip ospf 1 area 0.0.0.0
```

```
    ip ospf passive
    ip ospf dead-interval 88
    ip ospf transit-delay 44
    ip ospf priority 88
    ip ospf cost 8
    ip ospf network point-to-point
    ip ospf authentication simple-text
    ip ospf authentication-key kkk
    ip ospf shutdown
interface 1/1/2
interface loopback 2
    ip address 55.55.55.55/32
    ip ospf 1 area 0.0.0.0
```

OSPFv2 commands

Select a command from the list in the left navigation menu.

Subtopics

active-backbone

Syntax

```
active-backbone stub-default-route
no active-backbone stub-default-route
```

Description

This command enables the router to send a default route to stub areas if there is an active loopback link in the backbone area. The configuration is not required if backbone area has neighbors or passive interfaces configured. By default active backbone detection is enabled.

Examples

```
switch(config)# router ospf 1
switch(config-ospf-1)# active-backbone stub-default-route
switch(config)# no active-backbone stub-default-route
```

> **NOTE**
>
> For more information on features that use this command, refer to the Multicast Guide for your switch model.

Command History

Release	Modification
10.10.1000	Command Introduced

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospf-<PROCESS-ID>` `config-ospfv3-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`area (ospf)`

Syntax

```
area <AREA-ID>
no area <AREA-ID>
```

Description

Creates a normal area, with **<AREA-ID>** set if not present. If the area is already present and it is not a normal area, then this command changes the area type to normal.

The **no** form of this command deletes the area with the **<AREA-ID>** specified. Area can be of any type (nssa, nssa no-summary, stub, stub no-summary, and default normal area).

Parameter	Description
<AREA-ID>	Specifies the area ID in one of the following formats. • OSPF area identifier in IPv4 format (**x.x.x.x**), where **x** is a d ecimal number from 0 to 255. • OSPF area identifier in decimal format. Range: 0 to 429496 7295.

Examples

Creating a normal area:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 1
Switch(config-ospf-1)# show running-config current-context router ospf 1
    router-id 1.1.1.1
    area 0.0.0.1
```

Deleting an area:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no area 1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-ospf-*<PROCESS-ID>*	Administrators or local user group members with execution righ ts for this command.

`area default-metric`

Syntax

```
area <AERA-ID> default-metric <COST>
no area <AREA-ID> default-metric
```

Description

Sets the cost of the default route announced to NSSA or stub areas.

The **no** form of this command resets the cost of the default route announced to NSSA or stub areas, to the default value of 1.

Parameter	Description
<AREA-ID>	Specifies area ID in one of the following formats. • OSPF area identifier in IPv4 format (**x.x.x.x**), where **x** is a d ecimal number from 0 to 255. • OSPF area identifier in decimal format. Range: 0 to 429496 7295.
`default-metric <COST>`	Sets the cost of default - summary LSAs announced to NSSA o r stub areas, to the specified value. Default cost: 1. Range: 0 to 16777215.

Examples

Setting cost for default LSA summary:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 1 default-metric 2
```

Setting cost for default LSA summary to default:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no area 1 default-metric
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospf-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`area nssa`

Syntax

```
area <AREA-ID> nssa [no-summary]
no area <AREA-ID> nssa [no-summary]
```

Description

Creates the NSSA area (Not So Stubby Area) with **<AREA-ID>** if not present. If area is present and not NSSA area, this command changes the area type to NSSA area. If **no-summary** is used, area type will be NSSA No-Summary.

The **no** form of this command unsets the area type as NSSA. That is, the configured area will be changed to default normal area. The **no area <AREA-ID> nssa no-summary** command enables sending inter-area routes into NSSA, but will not unset the area as NSSA.

Parameter	Description
`<AREA-ID>`	Specifies the area ID in one of the following formats. • OSPF area identifier in IPv4 format (**x.x.x.x**), where **x** is a d ecimal number from 0 to 255. • OSPF area identifier in decimal format. Range: 0 to 429496 7295.
`nssa [no-summary]`	Specifies Not So Stubby Area (NSSA) area type. If area is pres ent and not NSSA area, parameter changes the area type to N SSA area. If **no - summary** is specified, area type will be NSSA No - Summary, which means do not inject inter - area routes in to NSSA.

Examples

Creating an NSSA area:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 1 nssa
switch(config-ospf-1)# area 1 nssa no-summary
```

Unsetting the area as NSSA

```
switch(config)# router ospf 1
switch(config-ospf-1)# no area 1 nssa
switch(config-ospf-1)# no area 1 nssa no-summary
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospf-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`clear ip ospf neighbors`

Syntax

```
clear ip ospf [<PROCESS-ID>] neighbor [<NEIGHBOR>] [interface [<INTERFACE-
NAME>]]
[all-vrfs | vrf <VRF-NAME>]
```

Description

Resets the neighbor and clears the OSPF neighbor information.

Parameter	Description
<PROCESS-ID>	Specifies the OSPFv2 process ID to clear the statistics for the p articular OSPFv2 process. Range: 1 to 65535.
<NEIGHBOR>	Specifies the router ID of a neighbor.
<INTERFACE-NAME>	Specifies the OSPFv2 statistics to clear for the specified interfa ce.
`all-vrfs`	Select to clear the OSPFv2 statistics for all VRFs.
`vrf` *<VRF-NAME>*	Specifies the name of a VRF.

Example

Clearing the OSPFv2 neighbor information:

```
switch# clear ip ospf 1 neighbor
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ip ospf 1 neighbor 1.1.1.2
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ip ospf 1 neighbor interface 1/1/1
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ip ospf 1 neighbor 1.1.1.5 vrf red
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ip ospf neighbor
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ip ospf neighbor 1.1.1.4
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ip ospf neighbor interface 1/1/1
Performing clear ospf neighbor may result in traffic disruption.
```

```
Do you want to continue (y/n)? y
switch# clear ip ospf neighbor 1.1.1.5 vrf red
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ip ospf 1 neighbor 1.1.1.2
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ip ospf 1 neighbor interface 1/1/1
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
```

Command History

Release	Modification
10.14	Supported process ID range expanded from 1 - 63 to 1 - 65535.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`clear ip ospf statistics`

Syntax

```
clear ip ospf [<PROCESS-ID>] statistics [interface [<INTERFACE-NAME>]] [all-
vrfs | vrf <VRF-NAME>]
```

Description

Clear the OSPF event statistics.

Parameter	Description
<PROCESS-ID>	OSPF process ID. Clear the statistics for the particular OSPF p rocess. Range: 1 to 65535.
<INTERFACE-NAME>	Clear the OSPF statistics for the specified interface.
`all-vrfs`	Optionally select to clear the OSPF statistics for all VRFs.
`vrf` *<VRF-NAME>*	Optionally select to clear the OSPF statistics for a particular VR F. If the VRF is not specified, information for the default VRF is cleared.

Examples

Clearing the OSPF event statistics:

```
switch# clear ip ospf statistics
switch# clear ip ospf statistics interface 1/1/1
```

Command History

Release	Modification
10.14	Supported process ID range expanded from 1 - 63 to 1 - 65535.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

default-information originate

Syntax

```
default-information originate [metric <METRIC-VALUE>]
no default-information originate [metric <METRIC-VALUE>]
```

Description

Configures OSPF to advertise the default route (0.0.0.0/0) to its neighbors if it is present in the routing table. Optionally, the metric value can be set for default route ::/0. The default value is 1.

The **no** form of this command disables advertisement of the default route.

Parameter	Description
metric <METRIC-VALUE>	Specifies the OSPF metric value for the default route. Optional. Default: 1.

Examples

Setting advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# default-information originate
```

Disabling advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no default-information originate
```

Setting advertisement of the default route and specifying an optional metric value of 20:

```
switch(config)# router ospf 1
switch(config-ospfv3-1)# default-information originate
switch(config-ospfv3-1)# default-information originate metric 20
```

Disabling advertisement of the default route and setting metric to the default value:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no default-information originate metric
```

Command History

Release	Modification
10.09	Added parameter: **metric** <**METRIC - VALUE**>
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospf-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`default-information originate always`

Syntax

```
default-information originate always [metric <METRIC-VALUE>]
no default-information originate always [metric <METRIC-VALUE>]
```

Description

Configures OSPF to advertise the default route (0.0.0.0/0) to its neighbors, regardless if it is present in the routing table or not. Optionally, metric can be set for default route 0.0.0.0/0. The default value is 1.

The **no** form of this command disables advertisement of the default route.

Parameter	Description
metric <METRIC-VALUE>	Specifies the OSPF metric value for the default route. Default: 1.

Examples

Setting advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# default-information originate always
```

Disabling advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no default-information originate always
```

Setting advertisement of the default route with metric set to 20:

```
switch(config)# router ospf 1
switch(config-ospf-1)# default-information originate always metric 20
```

Disabling advertisement of the default route and setting the metric to the default value:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no default-information originate always metric
```

Command History

Release	Modification
10.09	Added parameter: metric <**METRIC - VALUE**>
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospf-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`default-metric`

Syntax

```
default-metric <METRIC-VALUE>
no default-metric
```

Description

Sets the default metric for redistributed routes in the OSPF.

The **no** form of this command sets the default metric to be used for redistributed routes into OSPF to the default of 25.

Parameter	Description
<METRIC-VALUE>	Specifies the default metric value to use for redistributed routes . Default: 25. Range: 0 - 1677214.

Examples

Setting default metric for redistributed routes:

```
switch(config)# router ospf 1
switch(config-ospf-1)# default-metric 37
```

Setting default metric for redistributed routes to the default:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no default-metric
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-ospf-*<PROCESS-ID>*	Administrators or local user group members with execution righ ts for this command.

`disable`

Syntax

```
disable
```

Description

Disables the OSPF process.

> **NOTE**
> This command does not remove the OSPF configurations.

Examples

Disabling OSPF process:

```
switch(config)# router ospf 1
switch(config-ospf-1)# disable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospf-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`distance`

Syntax

```
distance [<DISTANCE-VAL> | intra-area [<DISTANCE-VAL>] | inter-area
[<DISTANCE-VAL>] | external [<DISTANCE-VAL>]]
no distance [intra-area | inter-area | external]
```

Description

Defines an Administrative Distance (AD) for OSPF. Administrative Distance is used as a criteria to select the best route when multiple routes are present from different routing protocols.

The **no** form of this command sets the OSPF administrative distance to the default of 110. Optionally, administrative distance can be set to default for the specific OSPF route type: intra-area, inter-area, or external type-5 and type-7 routes.

Parameter	Description
`<DISTANCE-VAL>`	Specifies the OSPF administrative distance. Range: 1 to 255. De fault: 110.
`intra-area`	Specifies the OSPF distance for intra - area routes.
`inter-area`	Specifies the OSPF distance for inter - area routes.
`external`	Specifies the OSPF distance for external type 5 and type 7 rou tes.

Usage

Within a given OSPF process, intra-area routes are always given precedence even when distances are configured for inter-area or external type routes.

Examples

Setting OSPF administrative distance:

```
switch(config)# router ospf 1
switch(config-ospf-1)# distance 100
switch(config-ospf-1)# distance intra-area 24 external 55 inter-area 66
switch(config-ospf-1)# distance intra-area 24 external 55
switch(config-ospf-1)# distance external 55
switch(config-ospf-1)#exit
switch(config)# router ospf 2
switch(config-ospf-2)# distance 200
switch(config-ospf-2)# distance external 60
```

```
switch(config-ospf-2)# distance intra-area 24 inter-area 66
```

Setting OSPF administrative distance to the default:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no distance
switch(config-ospf-1)# no distance external
switch(config-ospf-1)# no distance inter-area
switch(config-ospf-1)# no distance intra-area
switch(config-ospf-1)# no distance 100
switch(config-ospf-1)# no distance 220
switch(config)# router ospf 2 vrf blue
switch(config-ospf-2)# no distance 200
switch(config-ospf-2)# no distance external 60
switch(config-ospf-2)# no distance intra-area 24 inter-area 66
```

Command History

Release	Modification
10.14	Added capability to have individual admin distance for multiple OSPF processes in a VRF.
10.09	Added parameters: **intra - area**, **inter - area**, **external**
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospf-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

distribute-list prefix

Syntax

```
distribute-list prefix <prefix-list-name> {in | out}
no distribute-list prefix <prefix-list-name> {in | out}
```

Description

This command uses an existing prefix list to filter routes that are being installed in the routing table or redistributed to another routing protocol.

The **distribute-list prefix** command filters routes in the inbound or the outbound direction. When this command is issued with the **in** parameter, it filters routes from being installed in the routing table, it does not filter LSAs. When this command is issued with the **out** parameter, it filters only the desired redistributed routes from other protocols.

> **NOTE**
> This command requires that your prefix list is already defined using the ip prefix commands. Route-maps are not supported with the distribute-list feature.

Parameter	Description
`prefix <prefix-list-name>`	Specify the name of an existing prefix.
`{in | out}`	Select one of the following parameters to set the filter direction: • **in**: Filter incoming routes into the routing table • **out**: Filter outgoing routing updates

Examples

The following commands enable the filtering of OSPFv2 routes in an IPv4 network, so routes are no longer installed in the routing table or redistributed from another routing protocol.

```
switch(config)# router ospfv2 1
switch(config-ospfv2-1)# distribute-list prefix listA in
switch(config-ospfv2-1)# distribute-list prefix listB out
```

The following command disables the filtering of OSPFv2 routes in an IPv4 network, so routes can be installed in the routing table or redistributed from another routing protocol.

```
switch# configure terminal
switch(config)# router ospfv2 1
switch(config-ospfv2-1)# no distribute-list prefix listA in
switch(config-ospfv2-1)# no distribute-list prefix listB out
```

Command History

Release	Modification
10.13	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv2-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

enable

Syntax

```
enable
```

Description

Enables the OSPF process, if disabled. By default the OSPF process is enabled.

Examples

Enabling OSPF process:

```
switch(config)# router ospf 1
switch(config-ospf-1)# enable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospf-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

graceful-restart

Syntax

```
graceful-restart
   restart-interval <INTERVAL>
   helper [strict-lsa-check]
   ignore-lost-interface
   no...
```

Description

Configures graceful restart parameters for OSPF.

The **no** form of this command sets the restart interval to the default interval of 120 seconds or disables the helper mode, depending on the parameter specified.

Parameter	Description
`restart-interval <INTERVAL>`	Specifies the time another router waits for this router to gracefu lly restart and selects the maximum time to wait in seconds. Ra nge: 5 to 1800. Default: 120.
`helper`	Specifies that the router will participate in the graceful restart o f a neighbor router.
`strict-lsa-check`	(Optional). Use with the **helper** parameter to enable strict Link state Advertisement (LSA) checking when acting as a restart he lper for a restarting peer. **NOTE** OSPF neighbors must disable strict LSA checki ng. If the local node has fewer OSPF interfaces after restarting, then the neighbors that were a

Parameter	Description
	djacent on those interfaces will clear up their a djacencies to the restarting node and will send out link state updates to advertise the dropped adjacency. If strict LSA checking is enabled, th e restarting router's neighbors will exit helper m ode when they receive the updated LSAs and th e graceful restart will still fail.
`ignore-lost-interface`	Enable the restarting router to ignore lost OSPF interfaces duri ng a graceful restart process. This setting should be enabled on a high availability system to ensure a graceful restart completes successfully, even if OSPF - enabled links fail due to High Avail ability events like a switchover or failover. **NOTE** Enabling this setting means that the hitless restart procedures do not strictly follow those defined in RFC 3623, Graceful OSPF Restart.
`no`	Negate any parameter or return the setting to its default.

Examples

Enabling OSPF graceful restart:

```
switch(config)# router ospf 1
switch(config-ospf-1)# graceful-restart restart-interval 40
switch(config-ospf-1)# graceful-restart helper strict-lsa-check
```

Enabling the switch to ignore lost OSPF interfaces during a graceful restart process:

```
switch(config)# router ospf 1
switch (config-ospf-1)# graceful-restart ignore-lost-interface
```

Setting the restart interval to default, and disabling helper mode:

```
switch(config)# router ospf 1
switch(config-ospfv3-1)# no graceful-restart restart-interval
switch(config-ospfv3-1)# no graceful-restart helper
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospf-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`ip ospf area`

Syntax

```
ip ospf <PROCESS-ID> area <AREA-ID>
no ip ospf <PROCESS-ID> area <AREA-ID>
```

Description

Runs the OSPF protocol on the interface with the configured IPv4 address for the area specified. The interfaces which have an IP address configured in this network or in a subset of this network, will participate in the OSPF protocol.

To move an interface to a new area, unmap the existing area and then associate a new area with the interface.

The **no** form of this command disables OSPF on the interface and removes the interface from the area. Interfaces which have an IP address configured on the network or in a subset of the network, stop participating in the OSPF protocol.

Parameter	Description
`<PROCESS-ID>`	Specifies the OSPF process Id. Range: 1 to 65535.

Parameter	Description
<AREA-ID>	Specifies the OSPF area ID in one of the following formats. • Area identifier in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255. • Area identifier in decimal format. Range: 0 to 4294967295.

Examples

Setting OSPF network for the area:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf 1 area 1
switch(config-if-vlan)# ip ospf 1 area 0.0.0.1
```

Disabling OSPF network for the area:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf 1 area 1
switch(config-if-vlan)# no ip ospf 1 area 0.0.0.1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ip ospf authentication`

Syntax

```
ip ospf authentication {message-digest | simple-text | null | keychain |
hmac-sha-1 | hmac-sha-256 | hmac-sha-384 | hmac-sha-512}
no ip ospf authentication
```

Description

Sets the authentication type that will be used for authentication with the neighbor router.

The **no** form of this command deletes the authentication type used for a particular authentication with the neighbor router and sets to null authentication.

Parameter	Description
`message-digest`	Sets authentication type as message - digest.
`simple-text`	Sets authentication type as simple - text.
`null`	Sets authentication type as null.
`keychain`	Sets the authentication type to use the key chain.
`hmac-sha-1`	Sets the authentication type to SHA - 1.
`hmac-sha-256`	Sets the authentication type to SHA - 256.
`hmac-sha-384`	Sets the authentication type to SHA - 384.
`hmac-sha-512`	Sets the authentication type to SHA - 512.

Examples

Setting OSPF authentication type on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf authentication simple-text
```

Deleting OSPF authentication type on the interface and sets it to null:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf authentication
```

Setting OSPF authentication type to SHA-384 on the interface:

```
switch(config)# interface vlan 5
switch(config-if-vlan)# ip ospf authentication hmac-sha-384
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ip ospf authentication-key`

Syntax

```
ip ospf authentication-key [{ciphertext | plaintext} <PASSWORD>]
no ip ospf authentication-key
```

Description

Sets the authentication password used for simple-text authentication. If the password is given in ciphertext it will be decrypted and applied to the protocol.

The **no** form of this command deletes the authentication password used for simple-text authentication.

Parameter	Description
`{ciphertext	plaintext}`	Selects the password format.
	Specifies the password.

Parameter	Description
<PASSWORD>	

NOTE

When the password is not provided on the command line, plaintext password prompting occurs upon pressing Enter. The entered password characters are masked with asterisks.

Examples

Setting the OSPF simple-text authentication password in plaintext format:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf authentication-key plaintext F82#450b
```

Setting the OSPF simple-text authentication password with a prompted plaintext password:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf authentication-key
Enter the authentication key: ********
Re-Enter the authentication key: ********
```

Setting the OSPF simple-text authentication password in ciphertext format:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf authentication-key ciphertext AQBaZ...ecopg=
```

Deleting the OSPF simple-text authentication password:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf authentication-key
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ip ospf cost`

Syntax

```
ip ospf cost <INTERFACE-COST>
no ip ospf cost
```

Description

Sets the cost (metric) associated with a particular interface. The interface cost is used as a parameter to calculate the best routes.

The **no** form of this command sets the cost (metric) associated with a particular interface to the default cost 1.

Parameter	Description
<INTERFACE-COST>	Specifies the interface cost value. Range: 1 to 65535. Default: 1.

Examples

Setting OSPF interface cost

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf cost 100
```

Setting the OSPF interface cost to default

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf cost
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ip ospf dead-interval`

Syntax

```
ip ospf dead-interval <INTERVAL>
no ip ospf dead-interval
```

Description

Sets the interval after which a neighbor is declared dead if no hello packet is received on the OSPF interface.

The **no** form of this command sets the interval after which a neighbor is declared dead, to the default for the OSPF interface. The default value is 40 seconds (generally 4 times the hello packet interval).

Parameter	Description
<INTERVAL>	Specifies the time interval for the dead interval, in seconds. Ran ge: 1 to 65535. Default: 40.

Examples

Setting OSPF dead interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf dead-interval 30
```

Setting OSPF dead interval to default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf dead-interval
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ip ospf hello-interval`

Syntax

```
ip ospf hello-interval <INTERVAL>
no ip ospf hello-interval
```

Description

Sets the time interval between OSPF hello packets for the OSPF interface.

The **no** form of this command sets the time interval OSPF hello packets to the default of 10 seconds for the OSPF interface.

Parameter	Description
<INTERVAL>	Specifies the time interval for the hello interval, in seconds. Ran ge: 1 to 65535. Default: 10.

Examples

Setting OSPF hello interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf hello-interval 30
```

Setting OSPF hello interval to the default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf hello-interval
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ip ospf keychain`

Syntax

```
ip ospf keychain <KEYCHAIN-NAME>
no ip ospf keychain
```

Description

Sets the key chain for md5 authentication. A key chain configures rotating keys for packet authenticating, reducing the risk of keys being compromised.

The **no** form of this command deletes the key chain used for md5 authentication.

Parameter	Description
<KEYCHAIN-NAME>	Name of key chain to be used for md5 authentication.

Examples

Setting OSPFv2 key chain authentication:

```
switch(config)# interface 1/1/1
switch(config-if)# ip ospf keychain ospf_keys
```

Deleting OSPFv2 key chain authentication:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip ospf keychain
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ip ospf message-digest-key md5`

Syntax

```
ip ospf message-digest-key <KEY-ID> md5 [{ciphertext | plaintext} <KEY>]
no ip ospf message-digest-key <KEY-ID>
```

Description

Sets the md5 message digest authentication key. If the md5 key is given in ciphertext, it will be decrypted and applied to the protocol.

The **no** form of this command deletes the md5 authentication key.

Parameter	Description
<KEY-ID>	Specifies the md5 key ID. Range: 1 to 255.
`{ciphertext	plaintext}`	Selects the md5 key format.
<KEY>	Specifies the md5 authentication key.

> **NOTE**
> When the authentication key is not provided on the command line, plaintext key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

Examples

Setting the md5 key in plaintext format:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf message-digest-key 1 md5 plaintext F82#450b
```

Setting the md5 key with a prompted plaintext key:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf message-digest-key 1 md5
Enter the MD5 authentication key: ********
Re-Enter the MD5 authentication key: ********
```

Setting the md5 key in ciphertext format:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf message-digest-key 1 md5 ciphertext
AQt6e...7qEa4=
```

Deleting the md5 key:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf message-digest-key 1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-if config-if-vlan	Administrators or local user group members with execution righ ts for this command.

ip ospf network

Syntax

```
ip ospf network {broadcast | point-to-point | point-to-multipoint[dynamic]}
no ip ospf network {broadcast | point-to-point | point-to-
multipoint[dynamic]}
```

Description

Configures the OSPFv2 network type for the interface.
The **no** form of this command sets the network type for the interface to the system default which is broadcast network.

> **NOTE**
> For an IP unnumbered interface, the network type should be configured as a **point-to-point**.

Parameter	Description
`broadcast`	Specifies the OSPFv2 network type as a broadcast multi - acc ess network.
`point-to-point`	Specifies the OSPFv2 network type as a point - to - point netw ork.
`point-to-multipoint`	Specifies the OSPFv2 network type as a point-to-multipoint net work. Supports non-broadcast mode (static neighbor configura tion) and broadcast mode (dynamic neighbor discovery with **dy namic** option).

NOTE

It is recommended to configure the same network type on all OSPFv2 peers to ensure proper route learning and adjacency formation.

Usage

When configuring a **point-to-multipoint dynamic**, consider the following best-practice guidelines to ensure optimal stability and performance:

- Configure the **hello-interval** to 30 seconds and the **dead-interval** to 120 seconds.
- Use multiple OSPFv2 areas rather than placing all interfaces within a single area.
- Observe scaling and performance limits, for OSPFv2 the number of neighbors per area should not exceed 128.

Examples

Setting OSPFv2 network type for the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf network broadcast
switch(config-if-vlan)# ip ospf network point-to-point
switch(config-if-vlan)# ip ospf network point-to-multipoint
switch(config-if-vlan)# ip ospf network point-to-multipoint dynamic
```

Disabling OSPFv2 network type for the interface to system default of broadcast network:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf network
switch(config-if-vlan)# no ip ospf network point-to-multipoint
switch(config-if-vlan)# no ip ospf network point-to-multipoint dynamic
```

Command History

Release	Modification
10.18	Added a new parameter for **point-to-multipoint**.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ip ospf passive`

Syntax

```
ip ospf passive
no ip ospf passive
```

Description

Configures the interface as an OSPF passive interface. With this setting, the interface participates in OSPF but does not send or receive packets on that interface.

The **no** form of this command resets the interface as active. With this setting, the interface starts sending and receiving OSPF packets.

Examples

Setting the interface as OSPF passive interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf passive
```

Setting the interface as OSPF active interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf passive
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ip ospf priority`

Syntax

```
ip ospf priority <PRIORITY-VALUE>
no ip ospf priority
```

Description

Sets the OSPF priority for the interface. The larger the numeric value of the priority, the higher the chances for it to become the designated router. Setting a priority of zero makes the router ineligible to become a designated router or back up designated router.

The **no** form of this command sets the OSPF priority for the interface to the default of 1.

Parameter	Description
<PRIORITY-VALUE>	Specifies the OSPF priority value. Range: 0 to 255. Default: 1.

Examples

Setting OSPF priority for the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf priority 50
```

Disabling OSPF priority for the interface to default:

```
switch(config)# interface vlan1
switch(config-if-vlan)# no ip ospf priority
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ip ospf retransmit-interval`

Syntax

```
ip ospf retransmit-interval <INTERVAL>
no ip ospf retransmit-interval
```

Description

Sets the time between retransmitting lost link state advertisements for the OSPF interface.

The **no** form of this command sets the time between retransmitting lost link state advertisements to the default of 5 seconds for the OSPF interface.

Parameter	Description
	Specifies the retransmit interval, in seconds. Range: 1 to 3600. Default: 5.

Parameter	Description
<INTERVAL>	

Examples

Setting OSPF retransmit interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf retransmit-interval 30
```

Setting OSPF retransmit interval to the default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf retransmit-interval
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ip ospf sha-key sha`

Syntax

```
ip ospf sha-key <KEY-ID> sha [{ciphertext | plaintext} <KEY>]
no ip ospf sha-key <KEY-ID>
```

Description

Sets the SHA (secure hash authentication) key for the selected interface. If the SHA key is given in ciphertext, it will be decrypted and applied to the protocol. This command accepts a key of up to 64 characters irrespective of the SHA version configured on the interface. OSPF will internally pad zeros to the key to obtain a 64-byte key. For all types of SHA, key length is adjusted to 64 bytes.

The **no** form of this command deletes the SHA authentication key.

Parameter	Description
`<KEY-ID>`	Specifies the SHA key ID. Range: 1 to 255.
`{ciphertext | plaintext}`	Selects the SHA key format.
`<KEY>`	Specifies the SHA authentication key.

> **NOTE**
> When the authentication key is not provided on the command line, plaintext key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

Examples

Setting the SHA authentication key in plaintext format:

```
switch(config)# interface 1/1/1
switch(config-if)# ip ospf sha-key 1 sha plaintext F82#450b
```

Setting the SHA authentication key in prompted plaintext format:

```
switch(config)# interface 1/1/1
switch(config-if)# ip ospf sha-key 1 sha
Enter the SHA authentication key: ********
Re-Enter the SHA authentication key: ********
```

Setting the SHA authentication key in ciphertext format:

```
switch(config)# interface 1/1/1
switch(config-if)# ip ospf sha-key 1 sha ciphertext AQapu...C2K47A=
```

Deleting the SHA authentication key:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip ospf sha-key 1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ip ospf shutdown`

Syntax

```
ip ospf shutdown
no ip ospf shutdown
```

Description

Disables OSPF on the interface. The interface state changes to Down. It does not remove the interface from the OSPF area. To remove the interface, use the command **no ip ospf area**.

The **no** form of this command re-enables OSPF on the interface

Examples

Disabling OSPF on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf shutdown
```

Re-enabling OSPF on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf shutdown
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ip ospf transit-delay`

Syntax

```
ip ospf transit-delay <DELAY>
no ip ospf transit-delay
```

Description

Sets the time delay in link state transmission for the OSPF interface.

The **no** form of this command sets the delay in link state transmission to the default of 1 second for the OSPF interface.

Parameter	Description
<DELAY>	Specifies the transit delay in seconds. Range: 1 to 3600. Default : 1.

Examples

Setting OSPF transit delay on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf transit-delay 30
```

Setting OSPF transit delay to the default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf transit-delay
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`keychain`

Syntax

```
keychain <KEYCHAIN-NAME>
no keychain
```

Description

Sets the key chain for md5 authentication. A key chain configures rotating keys for packet authenticating, reducing the risk of keys being compromised.

The **no** form of this command deletes the key chain used for md5 authentication.

Parameter	Description
	Name of key chain to be used for md5 authentication.

Parameter	Description
<KEYCHAIN-NAME>	

Examples

Setting OSPF virtual link key chain authentication:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink)# keychain ospf_keys
```

Deleting OSPF virtual link key chain authentication:

```
switch# configure terminal
switch(config)# router ospf 1
switch(config-ospf-1)# area 100 virtual-link  100.0.1.1
switch(config-router-vlink)# no keychain
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-router-vlink`	Administrators or local user group members with execution righ ts for this command.

max-metric router-lsa

Syntax

```
max-metric router-lsa [include-stub | on-startup [<ADVERT-TIME>]]
no max-metric router-lsa [include-stub | on-startup]
```

Description

Sets the protocol to advertise a maximum metric so that other routers do not prefer this router as an intermediate hop in their shortest path first (SPF) calculations. If the on-startup parameter is used, the router is configured to advertise a maximum metric at startup for the time mentioned in seconds or for a default value of 600 seconds.

The **no** form of this command advertises the normal cost metrics instead of advertising the maximized cost metric. This setting causes the router to be considered in traffic forwarding.

Parameter	Description
`include-stub`	Configures the stub Router - LSA to advertise a maximum metr ic for all links, including stub links.
`on-startup <ADVERT-TIME>`	Specifies the time in seconds to advertise self as stub - router o n startup. If no time is specified, the default time of 600 second s is used. Range: 5 to 86400. Default: 600.

Examples

Setting to maximize the cost metrics for Router LSA:

```
switch(config)# router ospf 1
switch(config-ospf-1)# max-metric router-lsa
switch(config-ospf-1)# max-metric router-lsa on-startup
switch(config-ospf-1)# max-metric router-lsa on-startup 3000
```

Setting to advertise the normal cost metrics instead of advertising the maximized cost metric:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no max-metric router-lsa
switch(config-ospf-1)# no max-metric router-lsa on-startup
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospf-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

Platforms	Command context	Authority

neighbor

Syntax

```
neighbor <neighbor-ip>
no neighbor <neighbor-ip>
```

Description

This command configures a static OSPFv2 neighbor for point-to-multipoint networks. Static neighbors are required for NBMA (Non-Broadcast Multiple Access) networks, where automatic neighbor discovery through multicast hello packets is not possible. The neighbor IP address should be the IP address of the interface on the neighboring router within the same network segment.

The **no** form of this command removes a previously configured static OSPFv2 neighbor from the point-to-multipoint network configuration.

Parameter	Description
`neighbor <neighbor-ip>`	Static neighbor IP address for point-to-multipoint networks.

Example

Configuring the static neighbor configuration:

```
switch# configure terminal
switch(config)# router ospf 1
switch(config-ospf-1)# neighbor 192.168.1.2
switch(config-ospf-1)# neighbor 192.168.1.3
```

Removing the static neighbor configuration from the interface:

```
switch# configure terminal
switch(config)# router ospf 1
switch(config-ospf-1)# no neighbor 192.168.1.2
```

> **NOTE**
>
> For more information on features that use this command, refer to the IP Routing Guide for your switch model.

Command History

Release	Modification
10.18	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config-ospf-1`	Administrators or local user group members with execution righ ts for this command.

`passive-interface default`

Syntax

```
passive-interface default
no passive-interface
```

Description

Configures all OSPF interfaces as passive.

The **no** form of this command sets all OSPF interfaces as active.

Examples

Setting OSPF-enabled interfaces as passive:

```
switch(config)# router ospf 1
switch(config-ospf-1)# passive-interface default
```

Setting OSPF-enabled interfaces as active:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no passive-interface
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-ospf- *<PROCESS-ID>*	Administrators or local user group members with execution righ ts for this command.

redistribute

Syntax

```
redistribute {bgp | connected | local loopback | static | rip | ospf
<PROCESS-ID>}[route-map <ROUTE-MAP-NAME>]
no redistribute {bgp | connected | local loopback | static | rip | ospf
<PROCESS-ID>}[route-map <ROUTE-MAP-NAME>]
```

Description

Redistributes routes originating from other protocols, or from another OSPFv2 process, to the current OSPFv2 process.

If a route map is specified, then only the routes that pass the match clause specified in the route map are redistributed to OSPFv2. Configuration is not allowed if the referenced route map has not yet been configured.

If you try to redistribute routes from an OSPFv2 process which is not created, you are prompted to allow the OSPFv2 process to be auto-created before proceeding with redistribution. If you confirm at the prompt, the OSPFv2 process is created with defaults and redistribution configuration applied. If you deny at the prompt, redistribution configuration is skipped.

If command **route-redistribute active-routes-only** has been issued, only the routes from other protocols which are selected for forwarding are considered for redistribution into OSPFv2.

The **no** form of this command disables redistribution of routes to the current OSPFv2 process.

Parameter	Description
`bgp`	Specifies redistributing BGP (Border Gateway Protocol) routes.
`connected`	Specifies redistributing connected (directly attached subnet or host).
`local loopback`	Specifies redistributing local routes of the loopback interface.
`static`	Specifies redistributing static routes.
`rip`	Specifies redistributing RIP routes.
`ospf` *<PROCESS-ID>*	Specifies redistributing routes from the specified OSPFv2 proc ess ID. Range: 1 to 65535.
`route-map` *<ROUTE-MAP-NAME>*	Specifies redistribution filtering by route map. To create a route map, use command **route - map**.

Examples

Redistributing routes to OSPFv2:

```
switch(config)# router ospf 1
switch(config-ospf-1)# redistribute connected
switch(config-ospf-1)# redistribute connected route-map connected_routes
switch(config-ospf-1)# redistribute local loopback
switch(config-ospf-1)# redistribute local loopback route-map local_routes
switch(config-ospf-1)# redistribute static
switch(config-ospf-1)# redistribute static route-map static_networks
switch(config-ospf-1)# redistribute rip
switch(config-ospf-1)# redistribute rip route-map rip-routes
switch(config-ospf-1)# redistribute ospf 2
```

Disabling redistributing routes to OSPFv2:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no redistribute connected
switch(config-ospf-1)# no redistribute connected route-map connected_routes
switch(config-ospf-1)# no redistribute local loopback
switch(config-ospf-1)# no redistribute local loopback route-map local_routes
switch(config-ospf-1)# no redistribute static
```

```
switch(config-ospf-1)# no redistribute static route-map static_networks
switch(config-ospf-1)# no redistribute rip
switch(config-ospf-1)# no redistribute rip route-map rip-routes
switch(config-ospf-1)# no redistribute ospf 2
```

Command History

Release	Modification
10.14	Supported process ID range expanded from 1 - 63 to 1 - 65535.
10.08	Added **route - map** support for supported redistribute source - protocols. Upd ated information and examples.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospf-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`reference-bandwidth`

Syntax

```
reference-bandwidth <BANDWIDTH>
no reference-bandwidth
```

Description

Sets the reference bandwidth for OSPFv2. If the OSPFv2 interface cost is not explicitly set, then the cost of all the OSPFv2 interfaces is recalculated based on the reference bandwidth and link speed of the interface.

For VLAN interfaces the link speed value is taken as 1 Gbps (if the OSPFv2 interface cost is not explicitly set).

The **no** form of this command sets the reference bandwidth for OSPF to the default of 100000 Mbps.

Parameter	Description
<BANDWIDTH>	Specifies the reference bandwidth used to calculate the cost of an interface in Mbps. Range: 1 to 4000000. Default: 100000.

Examples

Setting the reference bandwidth:

```
switch(config)# router ospf 1
switch(config-ospf-1)# reference-bandwidth 40000
```

Setting the reference bandwidth to the default value:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no reference-bandwidth
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-ospf-*<PROCESS-ID>*	Administrators or local user group members with execution righ ts for this command.

rfc1583-compatibility

Syntax

```
rfc1583-compatibility
no rfc1583-compatibility
```

Description

Enables OSPF compatibility with RFC1583 (backward compatibility). If RFC1583 compatibility is enabled, then the route cost calculation follows a different method.

The **no** form of this command disables OSPF compatibility with RFC1583 (backward compatibility). By default the RFC1583 compatibility is disabled.

Examples

Enabling OSPF RFC1583 compatibility:

```
switch(config)# router ospf 1
switch(config-ospf-1)# rfc1583-compatibility
```

Disabling OSPF RFC1583 compatibility:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no rfc1583-compatibility
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospf-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`router ospf`

Syntax

```
router ospf <PROCESS-ID> [vrf <VRF-NAME>]
no router ospf <PROCESS-ID> [vrf <VRF-NAME>]
```

Description

Creates an OSPF process (if not created already) on a VRF, and switches to the OSPF router instance context. Up to eight OSPF processes are supported per VRF.

The **no** form of this command removes the OSPF instance.

Parameter	Description
`<PROCESS-ID>`	Specifies an OSPF process ID. Range: 1 to 65535.
`vrf <VRF-NAME>`	Specifies a VRF name for the OSPF process. Default: default.

Examples

```
switch(config)# router ospf 1
switch(config-ospf-1)#
switch(config)# router ospf 1 vrf vrf_red
switch(config)# no router ospf 1
```

Command History

Release	Modification
10.14	Supported process ID range expanded from 1 - 63 to 1 - 65535.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`router-id`

Syntax

```
router-id <ROUTER-ADDR>
no router-id
```

Description

Sets an ID for the router in an IPv4 address format.

The **no** form of this command unconfigures the router-id for the instance and sets the router-id to the default as follows: the router-id is selected dynamically as equal to the highest loopback address on the router, or the highest active interface if there are no loopback addresses. If no IP address is configured on any interfaces on the router, OSPF will not form an adjacency.

Parameter	Description
<ROUTER-ADDR>	Specifies the Router ID in IPv4 format (**x.x.x.x**), where **x** is a de cimal number from 0 to 255.

Examples

Setting router-id in the OSPF context:

```
switch(config)# router ospf 1
switch(config-ospf-1)# router-id 1.1.1.1
```

Unconfiguring router-id:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no router-id
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospf-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`sha-key sha`

Syntax

```
sha-key <KEY-ID> sha [{ciphertext | plaintext} <KEY>]
no sha-key <KEY-ID>
```

Description

Sets the SHA (secure hash authentication) key for the selected virtual link. If the SHA key is given in ciphertext, it will be decrypted and applied to the protocol. This command accepts a key of up to 64 characters irrespective of the SHA version configured on virtual link. OSPF will internally pad zeros to the key to obtain a 64-byte key. For all types of SHA, key length is adjusted to 64 bytes.

The **no** form of this command deletes the virtual link SHA authentication key.

Parameter	Description
`<KEY-ID>`	Specifies the virtual link SHA key ID. Range: 1 to 255.
`{ciphertext | plaintext}`	Selects the virtual link SHA key format.
	Specifies the virtual link SHA authentication key.

Parameter	Description
<KEY>	

NOTE

When the authentication key is not provided on the command line, plaintext key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

Examples

Setting virtual link SHA authentication key in plaintext format:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink)# sha-key 1 sha plaintext F82#450b
```

Setting the virtual link SHA authentication key in prompted plaintext format:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink)# sha-key 1 sha
Enter the SHA authentication key: ********
Re-Enter the SHA authentication key: ********
```

Setting the virtual link SHA authentication key in ciphertext format:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink)# sha-key 1 sha ciphertext AQapu...C2K47A=
```

Deleting the virtual link SHA authentication key:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink)# no sha-key 1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-router-vlink`	Administrators or local user group members with execution righ ts for this command.

`show ip ospf`

Syntax

```
show ip ospf [<PROCESS-ID>] [all-vrfs | vrf <VRF-NAME>]
```

Description

Displays general OSPF, area, state, and configuration information.

Parameter	Description
`<PROCESS-ID>`	Enter an OSPF process ID to display general OSPF information for a particular OSPF process. Range: 1 to 65535.
`all-vrfs`	Optionally select to display general OSPF information for all VR Fs.
`vrf <VRF-NAME>`	Specifies the name of a VRF. Default: default.

Examples

Showing general OSPF configurations:

```
switch# show ip ospf 200
VRF : Default                             Process : 200
-------------------------------------------------------------------
Router ID               : 1.1.1.1             OSPFv2                 : Enabled
BFD                     : Disabled            SPF Start Interval     : 5000 ms
SPF Hold Interval       : 1000 ms             SPF Max Wait Interval : 1000 ms
LSA Start Interval      : 5000 ms             LSA Hold Interval      : 1000 ms
LSA Max Wait Interval   : 1000 ms             LSA Arrival Interval   : 1000 ms
```

```
External LSAs           : 4                    Checksum Sum           : 133302
ECMP                    : 4                    Reference Bandwidth    : 100000
Mbps
Area Border             : Yes                  AS Border              : No
GR Status               : Disabled             GR Interval            : 120 sec
GR State                : Inactive             GR Exit Status         : None
GR Helper               : Enabled              GR Strict LSA Check    : Enabled
GR Ignore Lost I/F      : Disabled             Internal Process ID    : 1
Summary address:
  prefix 10.1.1.0/24, advertise, tag 10
Area      Total    Active
-------------------------
Normal    2        1
Stub      2        1
NSSA      0        0
Area  : 0.0.0.1
----------------------------------
Area Type              : Normal     Status                 : Active
Total Interfaces       : 100        Active Interfaces      : 10
Passive Interfaces     : 5          Loopback Interfaces    : 85
SPF Calculation Count : 1500
Area ranges:
  ip-prefix 10.1.1.1/24, inter-area, advertise
Number of LSAs         : 5000       Checksum Sum           : 99122
```

Command History

Release	Modification
10.14	Supported process ID range expanded from 1 - 63 to 1 - 65535.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manage r (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

show ip ospf border-routers

Syntax

```
show ip ospf [<PROCESS-ID>]
     border-routers [all-vrfs | vrf <VRF-NAME>]
```

Description

Displays the OSPF routing table entries for Area Border Router (ABR) and Autonomous System Border Router (ASBR).

Parameter	Description
`<PROCESS-ID>`	Enter an OSPF process ID to display general OSPF information for a particular OSPF process. Range: 1 to 65535.
`all-vrfs`	Optionally select to display general OSPF information for all VR Fs.
`vrf <VRF-NAME>`	Specifies the name of a VRF. Default: default.

Examples

Showing OSPF border routers information:

```
switch# show ip ospf border-routers
VRF : default                           Process : 1
Internal Routing Table
-----------------------------------------------------
Codes: i - Intra-area route, I - Inter-area route
  Router-ID     Cost Type Area      SPF   Nexthop    Interface
i 40.40.40.40   10   ABR  0.0.0.0   71    192.0.2.1  1/1/1
i 60.60.60.60   20   ABR  0.0.0.0   71    192.0.2.1  1/1/1
i 40.40.40.40   10   ABR  0.0.0.1   71    192.0.2.1  1/1/2
i 60.60.60.60   20   ABR  0.0.0.1   71    192.0.2.1  1/1/2
```

Command History

Release	Modification
10.14	Supported process ID range expanded from 1 - 63 to 1 - 65535.

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show ip ospf interface`

Syntax

```
show ip ospf [<PROCESS-ID>] interface [<interface-name>] [brief]
[all-vrfs | vrf <VRF-NAME>]
```

Description

Displays general OSPF, area, state, and configuration information.

> **NOTE**
> For an IP unnumbered enabled OSPF interface, the output of the command displays the IP address borrowed from the donor interface.

Parameter	Description
`<PROCESS-ID>`	Enter an OSPF process ID to display general OSPF information for a particular OSPF process. Range: 1 to 65535.
`<interface-name>]`	Specify the name of an OSPF interface.
`brief`	Include this parameter to display a brief overview of the following OSPF configuration information.

Parameter	Description
	• Interface: OSPF interface name. • Area: OSPF area ID. • Cost: The metric OSPF uses to judge a path's feasibility, calc ulated as (reference bandwidth / interface bandwidth). • State: Indicates if the interface is a designated router (**Dr**) o r a backup designated router (**Backup - dr**). • Status: Indicates if the interface is **up** or **down**. • Flags: P - Passive A - Active.
`all-vrfs`	Optionally select to display general OSPF information for all VR Fs.
`vrf <VRF-NAME>`	Specifies the name of a VRF. Default: default.

Examples

Showing general OSPF configuration settings for the default VRF:

```
switch (config-if)# show ip ospf 1
VRF : default                          Process  : 1
----------------------------------------------------
RouterID                : 20.0.0.1          OSPFv2                  :
Enabled
SPF Start Interval      : 200   ms
SPF Hold Interval       : 1000  ms          SPF Max Wait Interval   : 5000
ms
LSA Start Time          : 5000  ms          LSA Hold Time           : 0
ms
LSA Max Wait Time       : 0     ms          LSA Arrival             : 1000
ms
External LSAs           : 0                 Checksum Sum            : 0
ECMP                    : 4                 Reference Bandwidth     :
100000 Mbps
Area Border             : false             AS Border               : false
GR Status               : Enabled           GR Interval             : 120
sec
GR State                : inactive          GR Exit Status          : none
GR Helper               : Enabled           GR Strict LSA Check     :
Enabled
GR Ignore Lost I/F      : Disabled
Summary address:
Area       Total     Active
```

```
------------------------------
Normal      1          1
Stub        0          0
NSSA        0          0
Area  : 0.0.0.0
----------------
Area Type               : Normal         Status                : Active
Total Interfaces        : 1              Active Interfaces     : 1
Passive Interfaces      : 0              Loopback Interfaces   : 0
SPF Calculation Count   : 4
Area ranges       :
Number of LSAs          : 3              Checksum Sum          : 82420
```

Showing OSPF configuration settings for all interfaces:

```
switch(config)# show ip ospf interface
Codes: DR - Designated router BDR - Backup designated router
Interface 1/1/1 is up, line protocol is up
--------------------------------------------
  VRF                  : default               Process              : 1
  IP Address           : 10.10.10.1/24         Area                 : 0.0.0.0
  Status               : Up                    Network Type         :
Broadcast
  Hello Interval       : 10    sec             Dead Interval        : 40
sec
  Transit Delay        : 1     sec             Retransmit Interval : 5
sec
  Link Speed           : 1000 Mbps
  Cost Configured      : 1                     Cost Calculated      : 1
  State/Type           : BDR                   Router Priority      : 1
  DR                   : 10.10.10.2            BDR                  :
10.10.10.1
  Link LSAs            : 0                     Checksum Sum         : 0
  Authentication       : Md5                   Passive              : Yes
Interface 1/1/2 is up, line protocol is up
--------------------------------------------
  VRF                  : default               Process              : 1
  IP Address           : 10.10.10.1/24         Area                 : 0.0.0.0
  Status               : Up                    Network Type         :
Broadcast
  Hello Interval       : 10    sec             Dead Interval        : 40
sec
  Transit Delay        : 1     sec             Retransmit Interval : 5
sec
  Link Speed           : 1000 Mbps
```

```
  Cost Configured       : 1                      Cost Calculated      : 1
  State/Type            : BDR                    Router Priority      : 1
  DR                    : 20.10.10.2             BDR                  :
20.10.10.1
  Link LSAs             : 0                      Checksum Sum         : 0
  Authentication        : Simple                 Passive              : No
```

Displaying brief OSPF information

```
switch(config-if)# show ip ospf interface brief
VRF : default   Process : 1
==============================
Total Number of Interfaces: 2
Flags: P - Passive  A - Active
Interface      Area             IP Address/Mask     Cost  State    Status
Flags
-----------------------------------------------------------------------------
-------------
1/1/1          0.0.0.0          10.10.10.1/24       40    DR       Up        P
1/1/2          255.255.255.255  200.200.200.123/24  4     Waiting  Up        A
```

Command History

Release	Modification
10.14.1000	, The output of the command displays the borrowed IP address for an IP unnum bered enabled OSPF interface.
10.14	Supported process ID range expanded from 1 - 63 to 1 - 65535.
10.09	Output of the **show ip ospf interface** command includes flags to indicate wheth er the interface is in passive or active mode.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manage r (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show ip ospf lsdb`

Syntax

```
show ip ospf [<process-id>] lsdb
   adv-router {<ROUTER-ID> | self}
   area <AREA-ID>
   lsid <link-state-id>
   all-vrfs |{vrf <VRF-NAME>}
asbr-summary
database-summary
external
lsid <LINK-STATE-ID>
network
nssa-external
router
summary
```

Description

Shows the OSPF link state database summary for different OSPF LSAs (Link State Advertisement). Use the parameters to get information for a particular LSA.

Parameter	Description
`<PROCESS-ID>`	Enter an OSPF process ID to display general OSPF information for a particular OSPF process. Range: 1 to 65535.
`adv-router {<ROUTER-ID>| self}`	Select to display link states for a particular advertising router. Specify either a Router ID of the advertising router or specify **self** to show self - originated link states.
`area <AREA-ID>`	Select to display information filtered for the specified area in one of the following formats. • OSPF area identifier in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255. • OSPF area identifier in decimal format. Value: 0 to 4294967295.
`lsid <LINK-STATE-ID>`	Select to display information filtered by link state identifier specified in IPv4 address format (A.B.C.D).

Parameter	Description
`all-vrfs|{vrf <vrf-name>`	Select **all - vrfs** to display general OSPF information for all VRFs, or use the **vrf** <VRF - NAME> option to display information for a specific VRF. Optionally select one of the following parameters to filter the link state database information.
`asbr-summary`	Show ASBR summary link states (LSA type 4).
`database-summary`	Select to display the count of each type of LSA and each area in the database. **NOTE** The **database - summary** parameter does not support the **area <area - id>**, **lsid <link - state - id>** or **adv - router {<router - id>|self}** parameters.
`external`	Show external link states (LSA type 5). **NOTE** The **external** parameter does not support the area **area <area - id>** parameter.
`network`	Show network LSAs (LSA type 2).
`nssa-external`	Show NSSA external link states (LSA type 7).
`router`	Show router LSAs (LSA type 1).
`summary`	Show network - summary link states (LSA type 3).

Examples

Showing OSPF link state database (LSDB) general information:

```
switch# show ip ospf lsdb
OSPF Router with ID (50.50.50.50) (Process ID 1 VRF default)
============================================================
Router Link State Advertisements (Area 0.0.0.0)
----------------------------------------------------------------------------
-
ADV Router       Age      Seq#         Checksum    LSID          Link Count
Bits
```

```
--------------------------------------------------------------------------------
-
40.40.40.40     930     0x80000004  0x2ea1      0            3
None
50.50.50.50     935     0x80000002  0x8b52      0            1            E
60.60.60.60     943     0x800003c5  0x9854      0            2
None
Network Link State Advertisements (Area 0.0.0.0)
----------------------------------------------------------------------------
ADV Router      Age     Seq#        Checksum    LSID        Router Count
----------------------------------------------------------------------------
60.60.60.60     944     0x80000001  0x7179      1360007168  2
50.50.50.50     935     0x80000001  0x516a      19          1
Inter Area Prefix Link State Advertisements (Area 0.0.0.0)
----------------------------------------------------------------------
ADV Router      Age     Seq#        Checksum    LSID        Prefix
----------------------------------------------------------------------
40.40.40.40     929     0x80000001  0x2498      131072      FEC0:3344::/32
50.50.50.50     928     0x80000001  0x5b2f      65536       111::/64
Inter Area Router Link State Advertisements (Area 0.0.0.0)
--------------------------------------------------------------------------------
-----
ADV Router      Age     Seq#        Checksum    LSID        Destination
Router ID
--------------------------------------------------------------------------------
-----
40.40.40.40     929     0x80000001  0x2498      1           33.33.33.33
AS External Link State Advertisements (Area 0.0.0.0)
----------------------------------------------------------------------
ADV Router      Age     Seq#        Checksum    LSID        Prefix
----------------------------------------------------------------------
40.40.40.40     264     0x80000001  0x24cc4     1           10::/64
40.40.40.40     675     0x80000001  0x5b00f     2           11::/64
NSSA External Link State Advertisements (Area 0.0.0.0)
----------------------------------------------------------------------
ADV Router      Age     Seq#        Checksum    LSID        Prefix
----------------------------------------------------------------------
3.3.3.3         264     0x80000001  0x24ac2     1           200::/64
Link-local Link State Advertisements (Area 0.0.0.0)
-------------------------------------------------------------------------
ADV Router      Age     Seq#        Checksum    LSID        Interface
-------------------------------------------------------------------------
50.50.50.50     264     0x80000001  0x653c4     19          1/1/1
```

```
Intra Area Prefix Link State Advertisements (Area 0.0.0.0)
----------------------------------------------------------------------------
-------------------
ADV Router      Age     Seq#        Checksum    LSID        Referenced LS
Type  Referenced LSID
----------------------------------------------------------------------------
-------------------
50.50.50.50     263     0x80000001  0x1da34     1
0x2001              0
50.50.50.50     264     0x80000001  0x2a45d     1
0x2002              19
```

Showing ASBR summary link states:

```
switch# show ip ospf lsdb asbr-summary
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
=========================================================
ASBR Summary Link State Advertisements (Area 0.0.0.0)
-----------------------------------------------------
LSID            ADV Router      Age     Seq#            Checksum
------------------------------------------------------------------
209.165.201.3   60.60.60.60     944     0x80000001      0x7179
192.0.2.1       50.50.50.50     935     0x80000001      0x516a
```

Showing external link states:

```
switch# show ip ospf lsdb external
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
=========================================================
AS External Link State Advertisements
--------------------------------------
LSID            ADV Router      Age     Seq#            Checksum
------------------------------------------------------------------
209.165.201.3   60.60.60.60     944     0x80000001      0x7179
192.0.2.1       50.50.50.50     935     0x80000001      0x516a
```

Showing database summary:

```
switch# show ip ospf lsdb database-summary
OSPF Router with ID (10.1.1.1) (Process ID 1 VRF default)
===========================================================
Area 0.0.0.0 database summary
------------------------------
LSA Type             Count
---------------------------
Router               2
Network              1
Inter-area Summary   1
```

```
ASBR Summary          0
NSSA External         0
Subtotal              4
Process 1 database summary
---------------------------
LSA Type              Count
---------------------------
Router                2
Network               1
Inter-area Summary    1
ASBR Summary          0
NSSA External         0
AS External           0
Total                 4
```

Showing router LSAs:

```
switch# show ip ospf lsdb router
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
========================================================
Router Link State Advertisements (Area 0.0.0.0)
-----------------------------------------------
LSID            ADV Router      Age         Seq#       Checksum Link Count
--------------------------------------------------------------------------
1.1.1.2         1.1.1.2         15          0x80000004 0xf526    1
2.2.2.1         2.2.2.1         14          0x80000005 0x6c5e    2
2.2.2.2         2.2.2.2         104         0x80000004 0xf51a    1
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
========================================================
Router Link State Advertisements (Area 0.0.0.0)
-----------------------------------------------
LSID            ADV Router      Age         Seq#       Checksum Link Count
--------------------------------------------------------------------------
1.1.1.2         1.1.1.2         15          0x80000004 0xf526    1
2.2.2.1         2.2.2.1         14          0x80000005 0x6c5e    2
2.2.2.2         2.2.2.2         104         0x80000004 0xf51a    1
```

Showing network LSAs:

```
switch# show ip ospf lsdb network
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
========================================================
Network Link State Advertisements (Area 0.0.0.0)
------------------------------------------------
LSID            ADV Router      Age         Seq#       Checksum
----------------------------------------------------------------
```

```
1.1.1.2         1.1.1.2         141         0x80000001 0xc55e
2.2.2.2         2.2.2.2         230         0x80000001 0xa179
```

Showing network-summary link states:

```
switch# show ip ospf lsdb summary
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
========================================================
Inter-area Summary Link State Advertisements (Area 0.0.0.0)
-----------------------------------------------------------
LSID            ADV Router      Age         Seq#       Checksum
----------------------------------------------------------------
1.1.1.0         2.2.2.1         133         0x80000002 0xa089
Inter-area Summary Link State Advertisements (Area 0.0.0.1)
-----------------------------------------------------------
LSID            ADV Router      Age         Seq#       Checksum
----------------------------------------------------------------
2.2.2.0         2.2.2.1         133         0x80000002 0x7caa
```

Showing NSSA external link states:

```
switch(config-ospf-1)# show ip ospf lsdb nssa-external
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
========================================================
NSSA External Link State Advertisements (Area 0.0.0.1)
------------------------------------------------------
LSID            ADV Router      Age         Seq#       Checksum
----------------------------------------------------------------
8.8.8.0         1.1.1.2         162         0x80000003 0xc7b2
```

Command History

Release	Modification
10.14	Supported process ID range expanded from 1 - 63 to 1 - 65535.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

show ip ospf neighbors

Syntax

```
show ip ospf [<PROCESS-ID>] neighbors [<NEIGHBOR-ID>]
     [interface <INTERFACE-NAME>] [detail | summary]
     [all-vrfs | vrf <VRF-NAME>]
```

Description

Displays information about OSPF neighbors.

Parameter	Description
`<PROCESS-ID>`	Enter an OSPF process ID to display OSPF neighbor informatio n for the particular OSPF process. Range: 1 to 65535.
`neighbors <NEIGHBOR-ID>`	Select to display information about a particular neighbor, specifi ed in IPv4 format (A.B.C.D).
`interface <INTERFACE-NAME>`	Select to display neighbor information only for the specified in terface.
`detail`	Select to display detailed information for all the neighbors.
`summary`	Select to display summary information for the neighbors.
`all-vrfs`	Select to display neighbor information for all VRFs.
`vrf <VRF-NAME>`	Specify the name of a VRF. Default: default.

Examples

Showing OSPF neighbors information for the default VRF:

```
switch# show ip ospf neighbors
OSPF Process ID 1 VRF default
==============================
Total Number of Neighbors: 1
Neighbor ID      Priority  State                 Nbr Address        Interface
```

```
-------------------------------------------------------------------------
2.2.2.2         1          FULL/DR            10.1.1.2            1/1/1
```

NOTE

The **Nbr Address** displays the borrowed IP address for IP unnumbered interfaces.

Showing OSPF neighbors information for a specific neighbor:

```
switch# show ip ospf neighbors 2.2.2.2
switch# show ip ospf neighbors 2.2.2.2
VRF : default                         Process : 1
----------------------------------------------------------------
Router-Id      : 2.2.2.2              Area Id           : 0.0.0.0
Interface      : 1/1/1                Address           : 10.10.10.2
State          : FULL                 Neighbor Priority : 1
Dead Timer Due : 00:00:36             Options           : 0x42
```

Showing OSPF neighbors information for a specific neighbor and interface:

```
switch# show ip ospf neighbors 2.2.2.2 interface 1/1/1
VRF : default                         Process : 1
-----------------------------------------------
Router-Id      : 2.2.2.2              Area Id           : 0.0.0.0
Interface      : 1/1/1                Address           : 10.10.10.2
State          : FULL                 Neighbor Priority : 1
Dead Timer Due : 00:00:36             Options           : 0x42
```

Showing detail information for OSPF neighbors:

```
switch# show ip ospf neighbors detail
VRF : default                         Process : 1
----------------------------------------------
Router-Id      : 2.2.2.2              Area Id           : 0.0.0.0
Interface      : 1/1/1                Address           : 10.10.10.2
State          : FULL                 Neighbor Priority : 1
DR             : 10.10.10.2           BDR               : 10.10.10.1
Dead Timer Due : 00:00:38             Options           : 0x42
Retransmission Queue Length  : 0
Time Since Last State Change : 00h:11m:37s
```

Showing summary information for OSPF neighbors in the **default** VRF:

```
switch# show ip ospf neighbors summary
OSPF Process ID 1 VRF default, Neighbor Summary
================================================
Interface  Down Attempt Init TwoWay ExStart Exchange Loading Full  Total
-------------------------------------------------------------------------
1/1/1       0    0       0    0      0       0        0       1     1
```

```
1/1/2        0     0       0     0      0       0        0       0     0
Total        0     0       0     0      0       0        0       1     1
```

Command History

Release	Modification
10.14.1000	The output parameter, **Nbr Address** displays the borrowed IP address for IP un numbered interfaces.
10.14	Supported process ID range expanded from 1 - 63 to 1 - 65535.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show ip ospf routes`

Syntax

```
show ip ospf [<PROCESS-ID>] routes
     [<IPV4-ADDR>/<MASK>] protected [all-vrfs | vrf <VRF-NAME|
```

Description

Displays OSPF routing information.

Parameter	Description
<IPV4-ADDR>	Specify an IP address in IPv4 format (**x.x.x.x**), where **x** is a deci mal number from 0 to 255.

Parameter	Description
<MASK>	Specifies the number of bits in the address mask in CIDR format (**x**), where **x** is a decimal number from 0 to 32.
<PROCESS-ID>	Enter an OSPF process ID to display OSPF neighbor informatio n for the particular OSPF process. Range: 1 to 65535.
`all-vrfs`	Select to display neighbor information for all VRFs.
`vrf` *<VRF-NAME>*	Specify the name of a VRF. Default: default.
`protected`	Selects a protected route. Specifying an IPv4 address will select protected route for that prefix.

Examples

Showing OSPF routing table information:

```
switch# show ip ospf routes
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2
OSPF Process ID 1 VRF default, Routing Table
---------------------------------------------
Total Number of Routes : 2
10.1.1.0/24        (i) area: 0.0.0.0
     directly attached to interface 1/1/1, cost 1 distance 110
20.1.1.0/24        (I)
     via 10.1.1.2 interface 1/1/1, cost 2 distance 110
```

Showing OSPF routing table information for a specific subnet:

```
switch# show ip ospf routes 10.1.1.0/2
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2
OSPF Process ID 1 VRF default, Routing Table for prefixes 10.1.1.0/24
----------------------------------------------------------------------
Total Number of Routes : 1
10.1.1.0/24        (i) area: 0.0.0.0
     directly attached to interface 1/1/1, cost 1 distance 110
```

Showing OSPF routing table information for a specific protected subnet on the default VRF:

```
switch# show ip ospf routes 211.0.0.0/24 protected vrf default
Codes: i - Intra-area route, I - Inter-area route
```

```
       E1 - External type-1, E2 - External type-2
OSPF Process ID 1, VRF default, Protected Routes for prefixes 211.0.0.0/24
---------------------------------------------------------------------------
211.0.0.0/24       (i) area: 0.0.0.0
    via 110.110.110.1, interface vlan20, cost 122
    bkp 130.130.130.1, interface vlan22, cost 132
Total Number of Routes : 1
```

Showing OSPF routing table information for a specific protected subnet on all VRFs

```
switch# show ip ospf routes 211.0.0.0/24 protected all-vrfs
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2
OSPF Process ID 1, VRF default, Protected Routes for prefixes 211.0.0.0/24
---------------------------------------------------------------------------
211.0.0.0/24       (i) area: 0.0.0.0
    via 110.110.110.1, interface vlan20, cost 122
    bkp 130.130.130.1, interface vlan22, cost 132
Total Number of Routes : 1
```

Showing OSPF routing table information for a specific protected subnet:

```
switch# show ip ospf routes 211.0.0.0/24 protected
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2
OSPF Process ID 1, VRF default, Protected Routes for prefixes 211.0.0.0/24
---------------------------------------------------------------------------
211.0.0.0/24       (i) area: 0.0.0.0
    via 110.110.110.1, interface vlan20, cost 122
    bkp 130.130.130.1, interface vlan22, cost 132
Total Number of Routes : 1
```

Showing OSPF routing table information for all protected routes:

```
switch# show ip ospf routes 211.0.0.0/24 protected
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2
OSPF Process ID 1, VRF default, Protected Routes
---------------------------------------------------------------------------
120.120.120.0/24       (i) area: 0.0.0.0
    via 110.110.110.1, interface vlan20, cost 120
    bkp 130.130.130.1, interface vlan22, cost 230
140.140.140.0/24       (i) area: 0.0.0.0
    via 110.110.110.1, interface vlan20, cost 130
    bkp 130.130.130.1, interface vlan22, cost 220
211.0.0.0/24       (i) area: 0.0.0.0
    via 110.110.110.1, interface vlan20, cost 122
    bkp 130.130.130.1, interface vlan22, cost 132
```

```
Total Number of Routes : 3
OSPFs Process ID 1, VRF red, Protected Routes
---------------------------------------------
Total Number of Protected Routes: 0
```

Showing OSPF routing table information for protected routes on the default VRF:

```
switch# show ip ospf routes protected vrf default
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2
OSPF Process ID 1, VRF default, Protected Routes
---------------------------------------------------------------------------
140.140.140.0/24       (i) area: 0.0.0.0
    via 130.130.130.1, interface vlan22, cost 130
    bkp 110.110.110.1, interface vlan20, cost 220
211.0.0.0/24      (i) area: 0.0.0.0
    via 110.110.110.1, interface vlan20, cost 122
    bkp 130.130.130.1, interface vlan22, cost 132
120.120.120.0/24       (i) area: 0.0.0.0
    via 110.110.110.1, interface vlan20, cost 120
    bkp 130.130.130.1, interface vlan22, cost 230
Total Number of Routes : 3
```

NOTE

For more information on features that use this command, refer to the Multicast Guide for your switch model.

Command History

Release	Modification
10.15	Introduces the **protected** parameter.
10.14	Supported process ID range expanded from 1 - 63 to 1 - 65535.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show ip ospf statistics`

Syntax

```
show ip ospf [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]
```

Description

Displays OSPF statistics.

Parameter	Description
`<PROCESS-ID>`	Enter an OSPF process ID to display OSPF neighbor informatio n for the particular OSPF process. Range: 1 to 65535.
`all-vrfs`	Select to display OSPF statistics information for all VRFs.
`vrf <VRF-NAME>`	Specify the name of a VRF. Default: default.

Examples

Showing OSPF statistics:

```
switch# show ip ospf statistics
OSPF Process ID 1 VRF default, Statistics (cleared 1h 16m 24s ago)
-------------------------------------------------------------
Unknown Interface Drops          : 0
Unknown Virtual Interface Drops : 0
Bad Instance ID Drops            : 0
Bad IP Header Length Drops       : 0
Wrong OSPF Version Drops         : 0
Bad Source IP Drops              : 0
Resource Failure Drops           : 0
Bad Header Length Drops          : 0
Total Drops                      : 0
```

Command History

Release	Modification
10.14	Supported process ID range expanded from 1 - 63 to 1 - 65535.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ip ospf statistics interface

Syntax

```
show ip ospf [<PROCESS-ID>] statistics interface [<INTERFACE-NAME>]
     [all-vrfs | vrf <VRF-NAME>]
```

Description

Displays OSPF statistics for the OSPF-enabled interfaces.

Parameter	Description
`<PROCESS-ID>`	Enter an OSPF process ID to display OSPF - enabled interface statistics information on the specified OSPF process. Range: 1 to 65535.
`<INTERFACE-NAME>`	Select to display information only for the specified interface.
`all-vrfs`	Select to display OSPF - enabled interface statistics information for all VRFs.

Parameter	Description
`vrf <VRF-NAME>`	Specify the name of a VRF. Default: default.

Examples

Showing OSPF-enabled interfaces information:

```
switch# show ip ospf statistics interface 1/1/1
OSPF Process ID 1 VRF default, interface 1/1/1 statistics (cleared 0h 30m
28s ago)
============================================================================
=
Tx Hello Packets      : 101             Rx Hello Packets       : 99
Tx Hello Bytes        : 101             Rx Hello Bytes         : 99
Tx DD Packets         : 101             Rx DD Packets          : 99
Tx DD Bytes           : 101             Rx DD Bytes            : 99
Tx LS Request Packets : 101             Rx LS Requests Packets : 99
Tx LS Request Bytes   : 101             Rx LS Request Bytes    : 99
Tx LS Update Packets  : 101             Rx LS Update Packets   : 99
Tx LS Update Bytes    : 101             Rx LS Update Bytes     : 99
Tx LS Ack Packets     : 101             Rx LS Ack Packets      : 99
Tx LS Ack Bytes       : 101             Rx LS Ack Bytes        : 99
Total Number of State Changes : 8
Number of LSAs                : 29
LSA Checksum Sum              : 2345
Total Transmit Failures       : 29
Total OSPF Packets Discarded  : 999
Reason                              Packets Dropped
-----------------------------------------------------
Invalid type                        19
Invalid length                      9
Invalid checksum                    0
Invalid version                     23
Bad or unknown source               67
Area mismatch                       1
Self-originated                     19
Duplicate router ID                 9
Interface standby                   0
Total Hello packets dropped         60
  Network Mask mismatch             10
  Hello interval mismatch           10
  Dead interval mismatch            10
  Options mismatch                  10
```

```
  MTU mismatch                          10
  Neighbor ignored                      10
Authentication errors                   12
  Type mismatch                         6
  Authentication failures               6
Wrong protocol                          0
Resource failures                       0
Bad LSA length                          0
Others                                  0
Total LSAs Ignored : 176
Bad Type           : 10
Bad Length         : 56
Invalid Data       : 55
Invalid Checksum   : 55
```

Command History

Release	Modification
10.14	Supported process ID range expanded from 1 - 63 to 1 - 65535.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`summary-address`

Syntax

```
summary-address  <IPV4-ADDR>/<MASK> [no-advertise | tag <TAG-VALUE>]
no summary-address <prefix/length> [no-advertise | tag <tag-value>
```

Description

Summarizes the external routes with the matching address and mask. When advertising this route, its metric is set to the lowest cost path from among the routes that were summarized.

The **no** form of this command disables route summarization.

> **NOTE**
> This command only works for an ASBR (Autonomous System Boundary Router).

Parameter	Description
<IPV4-ADDR>	Specifies an IP address in IPv4 format (**x.x.x.x**), where **x** is a de cimal number from 0 to 255.
<MASK>	Specifies the number of bits in the address mask in CIDR format (**x**), where **x** is a decimal number from 0 to 32.
`no-advertise`	Do not advertise the aggregate route. Suppress routes that m atch the specified prefix/mask pair.
`tag` *<TAG-VALUE>*	Specify the tag for the aggregate route. The summary prefix wi ll be advertised along with the tag value in External LSAs. Rang e: 0 to 4294967295

Examples

Setting OSPF route summarization:

```
switch(config)# router ospf 1
switch(config-ospf-1)# summary-address 10.1.0.0/16
```

Disabling route summarization:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no summary-address 10.1.0.0/16
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospf-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`timers lsa-arrival`

Syntax

```
timers lsa-arrival <DELAY>
no timers lsa-arrival
```

Description

Configures the minimum delay between receiving the same LSA from a peer. The same LSA is an LSA that contains the same LSA ID number, LSA type, and advertising router ID. If an instance of the same LSA arrives sooner before the delay expires, the LSA is dropped. Generally, the LSA arrival timer should be set to a value less than or equal to the start-time value for the command **timers throttle lsa start** on the neighbor.

The **no** form of this command sets the LSA timers to default values.

Parameter	Description
<DELAY>	Specifies the delay in milliseconds. Range: 0 to 600000. Default : 1000.

Examples

Setting the LSA arrival timer:

```
switch(config)# router ospf 1
switch(config-ospf-1)# timers lsa-arrival 10
```

Setting the LSA arrival timer to default:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no timers lsa-arrival
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospf-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

timers throttle lsa

Syntax

```
timers throttle lsa start-time <START-TIME> hold-time <HOLD-TIME> max-wait-
time <WAIT-TIME>
no timers throttle lsa
```

Description

Configures the timers for LSA generation.

The **no** form of this command sets the LSA timers to default values.

Parameter	Description
`start-time <START-TIME>`	Specifies the initial wait time in milliseconds after which LSAs a re generated. When set to 0, the LSAs are generated without a ny delay. Range: 0 to 600000. Default: 5000.
`hold-time <HOLD-TIME>`	Specifies the amount of time, in milliseconds, between regene ration of an LSA. The hold time doubles each time the same L SA must be regenerated, until **max - wait - time** is reached. W hen set to 0, LSA regeneration time is not increased. Range: 0 to 600000. Default: 0.
`max-wait-time <WAIT-TIME>`	Specifies the maximum wait time, in milliseconds, for regenerati on of the same LSA. When set to 0, LSA regeneration time is n ot increased. Range: 0 to 600000. Default: 0.

Examples

Setting the LSA timers:

```
switch(config)# router ospf 1
switch(config-ospf-1)# timers throttle lsa start-time 100 hold-time 1000
max-wait-time 10000
```

Setting LSA timers to default values:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no timers throttle lsa
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospf-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

timers throttle spf

Syntax

```
timers throttle spf start-time <START-TIME> hold-time <HOLD-TINME>
max-wait-time <WAIT-TIME>
no timers throttle spf
```

Description

Configures timers for SPF calculation. There are three timers:

- **start-time** Is the initial delay before an SPF calculation is started. Default is 200 milliseconds.
- **hold-time** Is the progressive backoff time to wait before next scheduled SPF calculation. Default is 1000 milliseconds. If a route change event occurs during this period, the value doubles until it reaches the max-wait-time.
- **max-wait-time** Is the maximum time to wait before the next scheduled SPF calculation. Default is 5000 milliseconds. This is used to limit the SPF hold timer and also defines the time to be considered for which the OSPF LSDB has to be stable, after which the SPF throttle mechanism is reset.

The **no** form of this command sets all the configured non-default timers to default value.

Parameter	Description
`<START-TIME>`	Time in milliseconds to set timer for initial SPF delay. Default: 200.
`<HOLD-TINME>`	Time in milliseconds to set the minimum hold time between two consecutive SPF calculations. Default: 1000.
`<WAIT-TIME>`	Time in milliseconds to set the maximum wait time between two consecutive SPF calculations. Default: 5000.

Examples

Setting non-default timer values for SPF throttling:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# timers throttling spf start-time 500 hold-time
3000 max-wait-time 9000
Switch(config-ospfv3-1)# show running-config current-context
router ospfv3 1
        area 0.0.0.0
        area 0.0.0.1
        area 0.0.0.2 nssa no-summary
        area 0.0.0.3 stub
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# timers throttling spf start-time 500 hold-time
3000 max-wait-time 9000
Switch(config-ospfv3-1)# show running-config current-context
router ospfv3 1
        area 0.0.0.0
```

Setting default timer values for SPF throttling after configuring non-default values:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no timers throttling spf
```

NOTE

For more information on features that use this command, refer to the Multicast Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

trap-enable

Syntax

```
trap-enable
no trap-enable
```

Description

Enables the notification of the events to be sent as traps to the SNMP management stations for OSPF.

The **no** form of this command disables the notification of the events to be sent as traps to the SNMP management stations for OSPF.

Examples

Enabling sending notification of events as traps:

```
switch(config)# router ospf 1
switch(config-ospf-1)# trap-enable
```

Disabling sending notification of events as traps:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no trap-enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospf-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

Open Shortest Path First version 3 (OSPFv3)

> **NOTE**
> OSPFv2 is not supported on the 6000 and 6100 Switch Series.

> **NOTE**
> OSPFv2 is not supported on the 4100i Switch Series.

OSPFv3 is the IPv6 implementation of Open Shortest Path First protocol (OSPFv2 is the IPv4 implementation of this protocol). OSPFv3 is a routing protocol which is described in RFC5340 entitled OSPF for IPv6. It is a link-state based IGP (Interior Gateway Protocol) routing protocol. It is widely used with medium to large-sized enterprise networks.

Subtopics

Overview

The characteristics of OSPFv3 are:

- Provides a loop-free topology using SPF algorithm.
- Allows hierarchical routing using area 0 (backbone area) as the top of the hierarchy.
- Supports load balancing with equal cost routes for same destination.

- OSPFv3 is a classless protocol and allows for a hierarchical design with VLSM (Variable Length Subnet Masking) and route summarization.
- Scales enterprise size network easily with area concept.
- Provides fast convergence with triggered, incremental updates through Link State Advertisements (LSAs).
- Supports OSPFv3 operation over GRE tunnel interfaces (platform-specific).

Some OSPFv3 configuration is done in the configuration context, others in the OSPFv3 router context, or in the interface context. OSPFv3 can be configured on routed ports, VLAN interfaces, LAG interfaces, GRE tunnel interfaces, and loopback interfaces. All such configurations work in the mentioned interfaces context. OSPFv3 mandates the associated interface to be routing interface.

The switch supports congestion control (prioritizing hello packets, inactivity timer reset, and adjacency throttling).

Supported features

- OSPFv3 neighbor adjacency, Hello protocol, multiple areas, Inter-area routing
- AS external routes, stub areas, totally stubby areas, NSSA, ASBR
- Designated Router/Backup Designated Router
- Point-to-point interfaces/broadcast interfaces
- Point-to-multipoint network types in both non-broadcast mode (static neighbor configuration using IPv6 link-local addresses) and broadcast mode (dynamic neighbor discovery using the **dynamic** option)
- Equal-cost multipath
- Null authentication, Simple password authentication, MD5 authentication
- Graceful Restart - un-planned, restart interval, Graceful Restart Helper
- Stub router advertisement
- SPF throttling
- LSA Throttling
- SPF throttling
- Graceful Restart - un-planned, restart interval, Graceful Restart Helper
- Stub router advertisement
- Configuration of interface parameters such as priority, cost, hello-interval, dead-interval, retransmit-interval, transit-delay, etc.

- Configuration of virtual link parameters such as hello-interval, dead-interval, retransmit-interval, transit-delay, etc.
- Route redistribution
- Passive interfaces
- Multi VRF support with each VRF having up to eight OSPF process instances
- Congestion control (prioritizing hello packets, inactivity timer reset, and adjacency throttling)

> **NOTE**
> An OSPFv3 session will not come up in 6in6 tunnel.

How OSPFv3 protocol works

OSPFv3 protocol

The protocol uses Link State Advertisements (LSAs) transmitted by each router to update neighboring routers regarding its interfaces and the routes available through those interfaces. Each routing switch in an area also maintains a link-state database (LSDB) that describes the area topology. (All routers in a given OSPF area have identical LSDBs.) The routing switches used to connect areas to each other flood summary link LSAs and external link LSAs to neighboring OSPF areas to update them regarding available routes. Through this means, each OSPF router determines the shortest path between itself and a desired destination router in the same OSPF domain (Autonomous System (AS)).

OSPFv3 concepts

OSPFv3 is a link-state routing protocol applied to IPv6 routers grouped into OSPFv3 areas identified by the IPv6 routing configuration on each routing switch. Each OSPFv3 area includes one or more networks. OSPFv3 routers use hello packets and LSAs to maintain OSPFv3 operation across networks within an area and between areas within an OSPFv3 domain.

OSPFv3 uses hello packets to initiate and preserve relationships between neighboring routers on the same interface.

Subtopics

OSPFv3 Link-state advertisement (LSA) types
OSPFv3 area types
OSPFv3 router types

OSPFv3 Link-state advertisement (LSA) types

OSPFv3 uses LSAs transmitted by each router to update neighboring routers regarding its interfaces and the routes available through those interfaces.

Each routing switch in an area also maintains an LSDB that describes the area topology. (All routers in a given OSPFv3 area have identical LSDBs, and each router uses the LSDB to build its own shortest-path tree.)

The routing switches used to connect areas to each other flood inter-area-prefix-LSAs, inter-area-router-LSAs, and AS-external-LSAs to backbone area to update it regarding available routes. Each OSPFv3 router determines the shortest path between itself and a desired destination router in the same OSPFv3 domain (AS). Routed traffic in an OSPFv3 AS is classified as one of the following:

- Intra-area traffic
- Inter-area traffic
- External traffic

The routing switches support the LSAs listed in the following table.

Link - state type	Description	Use	Flood scope
0x2001	Router - LSA	Describes the state of eac h active interface on a ro uter for Area a given area . (Excludes loopback inter faces and interfaces that have not achieved full adj acency.)	Area
0x2002	Network - LSA	Describes the OSPFv3 ro uters in a given network.	Area
0x2003	Inter - area - prefix - L SA	Describes the route to a p refix in another OSPFv3 a rea of Area the same AS. (Excludes prefixes for link - local addresses.) Propagated through back bone area to other areas.	Area
0x2004	Inter - area - router - L SA	Describes the route to an ASBR in another OSPFv3 normal area (including th e backbone area) of the s ame AS. Propagated through back bone area to other areas.	Area

Link - state type	Description	Use	Flood scope
		(Excludes any ASBR in th e same area as the router sending the LSA.)	
0x4005	AS - external - LSA	Describes the route to a destination prefix in anot her AS (external route). (Excludes prefixes for link - local addresses.) Origi nated by ASBR in norma l or backbone areas of an AS and propagates throu gh backbone area to othe r normal areas. Does not flood over virtu al links and is not summa rized in virtual links. For i njection into an NSSA, an NSSA ABR generates a t ype - 7 - default - LSA a dvertising the default rou te (::/0).	AS
0x2007	NSSA - LSA	Describes the route to a destination in another A S (external route). Origin ated by ASBR in NSSA. A BR translates type - 7 L SAs to AS - external - L SAs for injection into the backbone area.	NSSA
0x0008	Link - LSA	For other routers on the s ame VLAN interface (exc ept virtual links), descri bes the router link - loca l address and any other I Pv6 prefixes reachable o n the VLAN. Link LSAs a re not flooded over virtua l links.	Link - local
0x2009	Intra - area - prefix - L SA	Generated on transit links within an area by the DR operating on those links. Also, every OSPFv3 route r generates this LSA to re	Area

Link - state type	Description	Use	Flood scope
		fer to stub and loopback prefixes on the router.	

OSPFv3 area types

OSPFv3 is built upon a hierarchy of network areas. All areas for a given OSPFv3 domain reside in the same AS. An AS is defined as a number of contiguous networks that share an interior gateway routing protocol.

An AS can be divided into multiple areas, including the backbone (area 0). Because each area represents a collection of contiguous networks and hosts, the topology of a given area is not known by the internal routers in any other area. Areas define the boundaries to which router-LSAs and network-LSAs are broadcast. This functionality limits the amount of LSA flooding that occurs within the AS. It also helps to control the size of the link-state databases (LSDBs) maintained in OSPFv3 routers.

An area is represented in OSPFv3 by either a 32-bit dotted-decimal address or a number. Area types include: Backbone, Normal, Not-so-stubby (NSSA), and Stub.

Backbone area

Every AS must have one (and only one) backbone area (identified as area 0 or 0.0.0.0.) The ABRs of all other areas in the same AS connect to the backbone area, either physically through an ABR or through a configured, virtual link. The backbone is a special type of area and serves as a transit area for carrying the inter-area-prefix-LSAs, AS-external-LSAs, and routed traffic between non-backbone areas, as well as the router-LSAs, network-LSAs, and routed traffic internal to the area. ASBRs are allowed in backbone areas.

Normal area

A normal area allows inter-area-prefix-LSAs and AS-external-LSAs to and from the backbone area. A normal area connects to the AS backbone area through one or more ABRs (physically or through a virtual link) and allows router-LSAs and network LSAs within this area. ASBRs are allowed in normal areas.

Stub area

Stub area connects to the AS backbone through one or more ABRs. It does not allow an internal ASBR, and does not allow AS-external-LSAs. A stub area supports these actions:

- Advertise the area summary routes to the backbone area.
- Advertise inter-area routes from other areas.
- Use the inter-area-prefix-LSA default route to advertise routes to an ASBR and to other areas.

You can configure the stub area ABR to do the following:

- Suppress advertising on the area's summarized internal routes into the backbone area.

- Suppress LSA traffic from other areas in the AS by replacing inter-area-prefix-LSAs and the default external route from the backbone area with the default summary route (::/0.). This area is called totally stubby area.

Virtual links are not allowed for stub areas.

Not-so-stubby (NSSA) area

An NSSA area connects to the backbone area through one or more ABRs. NSSAs are used where an ASBR exists in an area where you want to:

- Block injection of external routes from other areas of the AS.
- Advertise type-7-LSA external routes (learned from the ASBR) to the backbone area as AS-external-LSAs.

NSSAs also support the following:

- Advertise inter-area-prefix-LSAs from the backbone area into the NSSA. (If no-summary is enabled, the NSSA ABR suppresses these LSAs from the backbone and, instead, injects the default route into the NSSA.)
- Advertise NSSA inter-area-prefix-LSAs to the backbone area.

Virtual links are not allowed for NSSAs.

OSPFv3 router types

Internal routers

Internal OSPFv3 routers belong to only one area. Internal routers flood router-LSAs to all routers in the same area and maintain identical LSDBs.

Area border routers (ABRs)

Area border routers have membership in multiple areas. ABRs are used to connect the various areas in an AS to the backbone area for that AS. Multiple ABRs can be used to connect a given area to the backbone, and a given ABR can belong to multiple areas other than the backbone.

An ABR maintains a separate LSDB for each area to which it belongs. (All routers within the same area have identical LSDBs.) The ABR is responsible for flooding inter-area-prefix-LSAs and inter-area router LSAs between its border areas.

You can reduce summary LSA flooding by configuring area ranges. An area range enables you to assign an aggregate address to a range of IPv6 addresses. This aggregate address is advertised instead of all the individual addresses it represents. You can assign up to eight ranges in an OSPFv3 area.

Autonomous system boundary router (ASBR)

Autonomous system boundary routers run one or more interior gateway protocols and serve as a gateway to other autonomous systems operating with interior gateway protocols. The ASBR imports and translates different protocol routes into OSPFv3 through redistribution. ASBRs can be used in backbone areas, normal areas, and NSSAs, but not in stub areas.

Designated routers (DRs)

In an OSPFv3 network having two or more routers, one router is elected to serve as the designated router (DR) and another router to act as the backup designated router (BDR). All other routers in the same network segment forward their routing information to the DR and BDR, and the DR forwards this information to all routers in the network. This functionality minimizes the amount of repetitive information that is forwarded on the network by eliminating the need for each individual router in the area to forward its routing information to all other routers in the network. If the area includes multiple networks, each network elects its own DR and BDR.

In an OSPFv3 network with no DR and no BDR, the neighboring router with the highest priority is elected the DR and the router with the next highest priority is elected the BDR. If the DR goes off-line, the BDR automatically becomes the DR, and the router with the next highest priority then becomes the new BDR. If multiple routing switches on the same OSPFv3 network are declaring themselves DRs, both priority and router ID are used to select the DR and BDR.

Priority is configurable using the `ipv6 ospfv3 priority` command at the interface level. You can use this command to help bias one router as the DR. If two neighbors share the same priority, the router with the highest router ID is designated the DR. The router with the next highest router ID is designated the BDR. Routers retain DR/BDR states throughout the period for which the adjacency is up and running.

OSPFv3 configuration task list

Tasks at a glance

- Configuring OSPFv3 on the routing switch
- Setting OSPFv3 network for the area
- Configuring external route redistribution and control
- Influencing route choice by changing the administrative distance
- Configuring graceful restart
- Configuring OSPFv3 interface settings
-
- Viewing OSPFv3 information
- Clearing OSPFv3 statistics on a switch

- Configuring OSPFv3 on the routing switch
- Assigning the routing switch to an OSPF area
- Setting OSPFv3 network for the area
- Configuring external route redistribution and control
- Configuring area ranges on an ABR to reduce advertisements to the backbone
- Influencing route choice by changing the administrative distance
- Configuring graceful restart
- Configuring OSPFv3 virtual link settings
- Configuring OSPFv3 interface settings
- Configuring BFD for OSPFv3
-
- Viewing OSPFv3 information
- Clearing OSPFv3 statistics on a switch

Configuring OSPFv3 on the routing switch

Prerequisites

- You must be in the global configuration context, as indicated by the `switch(config)#` prompt to create the OSPFv3 instance and enter the OSPFv3 configuration context.
- To configure a router ID, and other OSPFv3 configuration you must be in the OSPFv3 router configuration context, as indicated by the `switch(config-ospfv3-1 )#` prompt.

Create the OSPFv3 instance and enter the OSPFv3 configuration context. From this context, you can proceed with other OSPFv3 configuration.

Procedure

1. Create the OSPFv3 instance and enter the OSPFv3 configuration context using the following command. For command details, see router ospfv3.
   router ospfv3 <PROCESS-ID> [vrf <VRF-NAME>]

   For example, the following command creates OSPF instance 1.

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)#
```

2. Configure a global router ID using the following command. For command details, see router-id.
   router-id <ROUTER-ADDRESSW>

   For example, the following command sets the router ID to 1.1.1.1.

   ```
   switch(config-ospfv3-1)# router-id 1.1.1.1
   ```

3. Optionally, if the OSPFv3 process was disabled (it is enabled by default), enable the OSPFv3 process using the following command.
   enable

   For command details, see enable. (Refer to disable for disabling the OSPFv3 process).

Creating an OSPFv3 area

Prerequisites

You must be in the OSPFv3 router configuration context.

Create a Normal, Stub, or Not So Stubby (NSSA) area.

Procedure

Create an OSPFv3 area for the routing switch using one of the following commands:

- Create a Normal area using the following command: `area <area-id>` . For command details, see area.
- Create a Not So Stubby Area using the command: `area <area-id> nssa`.

For example, the following command creates a normal area with an area identifier of 10.1.1.1. Area identifier could alternatively be entered in decimal format such as 1.

```
switch(config-ospfv3-1)# area 1.1.1.1
```

Section	Owner - Group
Important Information	NTL/Dev
Products Supported (Define New HW Platforms)	PLM
Compatibility/interoperability	NTL/Dev

Transceiver Support	PLM
Enhancements and new features	PLM
Upgrade	NTL
Fixes (Defects)	
Define CFDs to include	PMO
Define IFDs to include	NTL
Issues & Workarounds	
Define IFDs to include	NTL

Setting OSPFv3 network for the area

Prerequisites

Routing must be enabled on the interface where you are planning to enable OSPFv3.

You must be in the interface configuration context, as indicated by the `switch(config-if)#` prompt.

After you define an OSPFv3 area, you can assign one or more networks to it. The OSPFv3 protocol will run on the interface with the configured link-local ipv6 address. The interfaces that have an IPv6 address configured in this network or in a subset of this network will participate in the OSPFv3 protocol.

Procedure

1. Set an OSPFv3 network for the area using the following command. For command details, see ipv6 ospfv3 area.
   ipv6 ospfv3 <PROCESS-ID> area <AREA-ID>

   For example, use the following command to assign interface 1/1/1 to OSPFv3 area 1. The area can alternatively be entered in IO address format.

   ```
   switch(config)# interface 1/1/1
   switch(config-if)# ipv6 ospfv3 1 area 1
   ```

2. Optionally, you can disable OSPFv3 on the interface using the following command. For command details, see ipv6 ospfv3 area.
   ipv6 ospfv3 shutdown

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ospfv3 shutdown
```

Configuring external route redistribution and control

About this task

Configuring route redistribution for OSPFv3 establishes the routing switch as an ASBR for importing and translating different protocol routes from other IGP domains into an OSPFv3 domain. When you configure redistribution for OSPFv3, you can specify that routes external to the OSPFv3 domain are imported as OSPFv3 routes.

Procedure

1. Enable route redistribution using the `redistribute` command.

   ```
      redistribute {bgp | connected | local loopback | static | ripng |
   ospf <PROCESS-ID>}
                   [route-map <ROUTE-MAP-NAME>]
   ```

   For example, setting redistribution of connected routes as OSPFv3 routes:

   ```
   switch(config-ospfv3-1)# redistribute connected
   ```

   If a route map is specified, then only the routes that pass the match clause specified in the route map are redistributed to OSPFv3. For example, setting redistribution of local loopback routes, that only match the routes specified by the route map:

   ```
   switch(config-ospf-1)# redistribute local loopback route-map local_routes
   ```

2. Optionally, modify the default metric for redistribution using the `default-metric` command.

   ```
      default-metric <METRIC-VALUE>
   ```

   For example, setting the default metric for redistribution to 37.

   ```
   switch(config-ospfv3-1)# default-metric 37
   ```

3. Optionally, set the cost of default-summary LSAs using the area command.

   ```
      area <AREA-ID> default-metric <COST>
   ```

   For example, setting the cost of default summary LSAs to 2.

```
switch (config-ospfv3-1)# area 1 default-metric 2
```

4. Optionally, use the `max-metric router-lsa` command to set the protocol to advertise a maximum metric so that other routers do not prefer this router as an intermediate hop in their shortest path first (SPF) calculations.

   ```
   max-metric router-lsa [on-startup [<ADVERT-TIME>]]
   ```

   For example, setting advertise max-metric router-lsa on startup.

   ```
   switch(config-ospfv3-1)# max-metric router-lsa on-startup 3000
   ```

Influencing route choice by changing the administrative distance

Prerequisites

You must be in the OSPFv3 router configuration context, as indicated by the `switch(config-ospfv3-1)#` prompt.

The administrative distance is used to influence the selection of routes learned by different protocols.

Procedure

Reconfigure the administrative distance using the following command. For command details, see distance.

```
distance <distance>
```

For example, use the following command to set administrative distance to 100.

```
switch(config-ospfv3-1)# distance 100
```

Configuring graceful restart

Prerequisites

You must be in the OSPFv3 router configuration context, as indicated by the `switch(config-ospfv3-1)#` prompt.

OSPFv3 routing can be gracefully restarted on switches without losing packets that are in transit. There is no effect on the saved switch configuration.

Manually create a router-id configuration to support graceful restart and management module failover. A dynamic router-id is chosen from the list of configured IPs at the time of the router-id election, and in the event of a graceful restart and management module failover, the router-id should be preserved before and after these events. Graceful restart would fail if there is a change in the router-id, causing traffic loss.

Procedure

Configure graceful restart of using the following command. For command details, see graceful-restart.

```
graceful-restart [restart-interval  <seconds> | helper]
```

For example, the following command specifies 50 seconds as the maximum interval another router will wait for this router to gracefully restart.

```
switch(config-ospfv3-1)# graceful-restart restart-interval 50
```

Configuring OSPFv3 interface settings

Prerequisites

You must be in the interface configuration context, as indicated by the `switch(config-if)#` prompt.

You can optionally adjust the following OSPFv3 interface settings.

1. Set the interface cost using the following command. For command details, see ipv6 ospfv3 cost.

   ```
   ipv6 ospfv3 cost <interface-cost>
   ```

   For example, the following command sets the cost associated with interface 1/1/1 to 100.

   ```
   switch(config)# interface 1/1/1
   switch(config-if)# ipv6 ospfv3 cost 100
   ```

2. Set the time interval between OSPFv3 hello packets for the OSPFv3 interface using the following command. For command details, see ipv6 ospfv3 hello-interval.

   ```
   ipv6 ospfv3 hello-interval  <seconds>
   ```

3. Set the interval after which a neighbor is declared dead if no hello packet is received on the OSPFv3 interface using the following command. For command details, see ipv6 ospfv3 dead-interval.
   ipv6 ospfv3 dead-interval <seconds>

4. Set the time between re-transmitting lost link state advertisements for the OSPFv3 interface using the following command. For command details, see ipv6 ospfv3 retransmit-interval.
   ipv6 ospfv3 retransmit-interval <seconds>

5. Sets the time delay in Link state transmission for the OSPFv3 interface using the following command. For command details, see ipv6 ospfv3 transit-delay.
   ipv6 ospfv3 transit-delay <seconds>

6. Set the OSPFv3 network type for the interface. For command details, see ipv6 ospfv3 network.
   ipv6 ospfv3 network {broadcast|point-to-point}
7. Set the OSPFv3 priority on the interface using the following command. For command details, see ipv6 ospfv3 priority.
   ipv6 ospfv3 priority <number-value>
8. Set the OSPFv3 interface as OSPFv3 passive interface. With this setting the interface participates in OSPFv3 but does not send or receive packets on that interface. For command details, see ipv6 ospfv3 passive.
   ipv6 ospfv3 passive

Configuring all OSPFv3 interfaces as passive

Prerequisites

You must be in the OSPFv3 router configuration context, as indicated by the `switch(config-ospfv3-1)#` prompt.

OSPFv3 sends LSAs to all other routers in the same AS. To limit the flooding of LSAs throughout the AS, you can configure OSPFv3 to be passive.

Procedure

Configure all OSPFv3 interfaces as passive using the following command. For command details, see passive-interface default.

```
passive-interface default
switch(config-ospfv3-1)# passive-interface default
```

Configuring SPF throttling timers

SPF calculation is throttled with default timer values (start-time 200ms, hold-time 1000ms, max-wait-time 5000ms). You can throttle SPF calculation by configuring non-default timers to improve performance of a specific network configuration.

Prerequisites

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

Procedure

Reconfigure SPF throttling timers using the following command. For command details, see timers throttle spf.

```
timers throttle spf start-time <milliseconds> hold-time <milliseconds> max-
wait-time <milliseconds>
```

For example, use the following command to set start-time to 500ms, hold-time to 3000ms and max-wait-time to 9000ms:

```
switch(config-ospf-1)# timers throttle spf start-time 500 hold-time
3000 max-wait-time 9000
```

Viewing OSPFv3 information

Prerequisites

Use these show commands from the Manager context, as indicated by the `switch#` prompt.

Procedure

To view OSPFv3 information, use the following commands. For command details and examples, click the links.

- To view general OSPFv3, area, state and configuration information use: show ipv6 ospfv3.
- To view information about OSPFv3 interfaces use: show ipv6 ospfv3 interface.
- To view information about OSPFv3 neighbors use: show ipv6 ospfv3 neighbors .
- To view OSPFv3 routing table information use: show ipv6 ospfv3 routes.
- To view OSPFv3 statistics for the OSPFv3 interfaces use show ipv6 ospfv3 statistics interface.

Clearing OSPFv3 statistics on a switch

Clear OSPFv3 event statics using the following command. For command details, see clear ipv6 ospfv3 statistics

```
clear ipv6 ospfv3 [<PROCESS-ID>] statistics [interface [<INTERFACE-NAME>]]
[all-vrfs | vrf <VRF-NAME>]
```

For example, the following command clears OSPFv3 event statistics from interface 1/1/1.

```
switch# clear ipv6 ospfv3 statistics interface 1/1/1
```

Configuring OSPFv3 over GRE Tunnels

OSPFv3 can be configured on GRE tunnel interfaces to establish OSPFv3 adjacencies through tunneled connections. This feature is supported on the 6300, 6400, 8320, 8325, 9300, and 10000 Switch series.

Prerequisites

You must be in the global configuration context, as indicated by the `switch(config)#` prompt, to create and configure GRE tunnel interfaces.

You must be in the tunnel interface configuration context, indicated by the `switch(config-gre-if)#` prompt, to enable or disable OSPFv3 on the GRE tunnel interface.

> **NOTE**
>
> Configure the tunnel interface IP MTU to 1476 bytes using **ip mtu 1476** when running OSPFv3 over GRE. This accommodates the 24-byte GRE encapsulation overhead (20-byte outer IP header plus 4-byte GRE header), ensuring packets fit within the default Layer-2 MTU and avoid fragmentation.

Procedure

Configure the physical underlay interface.

```
switch(config)# interface 1/1/8
switch(config-gre-if)# no shutdown
switch(config-gre-if)# ipv4 address 20.1.1.1/24
```

Create and configure the GRE tunnel interface.

```
switch(config)# interface tunnel 1 mode gre ip
switch(config-gre-if)# source ipv4 20.1.1.1
switch(config-gre-if)# destination ipv4 20.1.1.2
switch(config-gre-if)# ipv6 address 1001::1/64
switch(config-gre-if)# no shutdown
```

Configure the OSPFv3 process.

```
switch(config)# router ospfv3 1
switch(config-ospv3-1)# router-id 1.1.1.1
switch(config-ospv3-1)# area 0.0.0.0
```

Enable OSPFv3 on the GRE tunnel interface and associate it with an OSPFv3 process and area. If the OSPFv3 process or area does not exist, the system prompts you to create them.

- Use A.B.C.D to specify an OSPFv3 area ID in IPv6 address.
- Use 0-4294967295 to specify an OSPFv3 area ID.

```
switch(config)# interface tunnel 1
switch(config-gre-if)# ipv6 ospfv3 1 area 0
switch(config-gre-if)# ipv6 ospfv3 1 area 0.0.0.0
```

Disable OSPFv3 on the GRE tunnel interface.

```
switch(config-gre-if)# no ipv6 ospfv3 1 area 0
switch(config-gre-if)# no ipv6 ospfv3 1 area 0.0.0.0
```

In this configuration:

- OSPFv3 process 1 is configured with router-id 1.1.1.1 and area 0.0.0.0.
- The physical interface (1/1/8) provides underlay connectivity with IPv4 address 20.1.1.1/24.
- A GRE tunnel interface (tunnel 1) is created with `mode gre ipv4`.
- The tunnel source is the IPv4 address of the physical interface (20.1.1.1).
- The tunnel destination is the remote endpoint IPv4 address (20.1.1.2).
- The tunnel interface has an IPv6 address (1001::1/64) for OSPFv3 communication.
- The tunnel interface is assigned to OSPFv3 process 1, area 0.0.0.0 using `ipv6 ospfv3 1 area 0.0.0.0`.

Use the `show ipv6 ospfv3 neighbors` command to view OSPFv3 neighbor adjacencies established over GRE tunnel interfaces.

```
(config-gre-if)# show ipv6 ospfv3 neighbors
VRF : default                          Process : 1
===================================================


Total Number of Neighbors : 1


Neighbor ID      Priority  State         Nbr Address         Interface
----------------------------------------------------------------------
2.2.2.2          0         FULL          fe80::2             tunnel1
```

This output shows:

- **Neighbor ID:** Router ID of the OSPFv3 neighbor (2.2.2.2).
- **Priority:** 0 for point-to-point interfaces like GRE tunnels.
- **State:** FULL indicates the adjacency is fully established.
- **Nbr Address:** IPv6 link-local address of the neighbor on the tunnel interface (fe80::2).
- **Interface:** The GRE tunnel interface name (tunnel1).

Use the `show interface tunnel` command to verify the tunnel interface status.

```
(config-gre-if)# show interface tunnel 1

 Interface tunnel1 is up
 Admin state is up
 Description:
 Tunnel type: gre ipv4
 Tunnel source: IPv4 address 20.1.1.1
 Tunnel destination: IPv4 address 20.1.1.2
 Tunnel ttl: 64
 IPv6 address: 1001::1/64
```

Additional Configurations

All standard OSPFv3 interface configuration commands are supported on GRE tunnel interfaces, including:

- ipv6 ospfv3 hello-interval
- ipv6 ospfv3 dead-interval
- ipv6 ospfv3 retransmit-interval
- ipv6 ospfv3 transmit-delay
- ipv6 ospfv3 cost
- ipv6 ospfv3 network
- ipv6 ospfv3 priority (not applicable for point-to-point)
- ipv6 ospfv3 passive
- ipv6 ospfv3 shutdown

> **NOTE**
> The tunnel interface must have an IPv4 address and reachable underlay connectivity between the tunnel source and destination for OSPFv3 adjacencies to form. OSPFv3 packets are encapsulated in GRE and then in IPv4, using the source and destination IPv4 addresses configured on the tunnel.

OSPFv3 commands

Select a command from the list in the left navigation menu.

Subtopics

[timers lsa-arrival](#)
[timers throttle lsa](#)
[timers throttle spf](#)
[transit-delay](#)
[trap-enable](#)

active-backbone

Syntax

```
active-backbone stub-default-route
no active-backbone stub-default-route
```

Description

This command enables the router to send a default route to stub areas if there is an active loopback link in the backbone area. The configuration is not required if backbone area has neighbors or passive interfaces configured. By default active backbone detection is enabled.

Examples

```
switch(config)# router ospf 1
switch(config-ospf-1)# active-backbone stub-default-route
switch(config)# no active-backbone stub-default-route
```

NOTE

For more information on features that use this command, refer to the Multicast Guide for your switch model.

Command History

Release	Modification
10.10.1000	Command Introduced

Command Information

Platforms	Command context	Authority
5420	`config-ospf-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

Platforms	Command context	Authority
6200	`config-ospfv3-<PROCESS-ID>`	

`area`

Syntax

```
area <AREA-ID>
no area <AREA-ID>
```

Description

Creates a normal area with **<AREA-ID>** set if not present. If area is present and is not the normal area, this command changes the area type to normal area.

The **no** form of this command deletes the area with the **<AREA-ID>** specified. The area can be of any type (stub, stub no-summary, and default normal area).

Parameter	Description
`<AREA-ID>`	Specifies the area ID is one of the following formats. • OSPF area identifier in IPv4 format (**x.x.x.x**), where **x** is a d ecimal number from 0 to 255. • OSPF area identifier in decimal format. Range: 0 to 429496 7295.

Examples

Creating a normal area:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 1
switch(config-ospfv3-1)# area 1.1.1.1
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 1
```

Deleting an area:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no area 1
switch(config-ospfv3-1)# no area 1.1.1.1
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no area 1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`area authentication ipsec`

Syntax

```
area <AREA-ID> authentication ipsec spi <SPI-INDEX>
            <AUTH-TYPE> [<KEY-TYPE>
            <AUTH-KEY>]
no area <AREA-ID> authentication
```

Description

Configures IPsec AH authentication for the specified area. OSPFv3 interfaces which have IPsec configured at the interface context will not use area level IPsec.

The **no** form of this command removes IPsec AH authentication for the specified area.

> **NOTE**
> IPsec is not supported for 6in6 tunnel interfaces.

Parameter	Description
<AREA-ID>	Specifies the area ID is one of the following formats: • OSPF area identifier in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255. • OSPF area identifier in decimal format. Range: 0 to 4294967295.
spi *<SPI-INDEX>*	Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. Range: 256 to 4294967295.
<AUTH-TYPE>	Specifies the authentication type: **md5** or **sha1**.
<KEY-TYPE>	Specifies the key type to use: **plaintext** (unencrypted), **hex-string** (encrypted) or **ciphertext** (encrypted).
<AUTH-KEY>	Specifies the authentication key.

NOTE

When the authentication key is not provided on the command line, plaintext key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

Examples

Setting area 0 to use IPsec authentication with a provided plaintext authentication key:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 0 authentication ipsec spi 256 sha1 plaintext
F82#450
```

Setting area 5 to use IPsec authentication with a prompted plaintext authentication key:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 5 authentication ipsec spi 256 sha1
Enter the IPsec authentication key: ********
Re-Enter the IPsec authentication key: ********
```

Removing IPsec authentication from area 1:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no area 1 authentication
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

area encryption ipsec

Syntax

```
area <AREA-ID> encryption ipsec spi <SPI-INDEX>
            <AUTH-TYPE>
    [<KEY-TYPE>
            <AUTH-KEY>
            <ENCR-TYPE> [<KEY-TYPE>
            <ENCR-KEY>]]
no area <AREA-ID> encryption
```

Description

Configures IPsec ESP with the authentication and encryption algorithm types and keys for the specified area. OSPFv3 interfaces with IPsec configured at the interface context will not use area level IPsec ESP configuration.

The **no** form of this command removes IPsec ESP from the specified area.

> **NOTE**
> IPsec is not supported for 6in6 tunnel interfaces.

Parameter	Description
<AREA-ID>	Specifies the area ID is one of the following formats. • OSPF area identifier in IPv4 format (**x.x.x.x**), where **x** is a d ecimal number from 0 to 255. • OSPF area identifier in decimal format. Range: 0 to 429496 7295.
`spi` *<SPI-INDEX>*	Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables t he receiving OSPF process to select and use the Security Associ ation (SA) from the SA table. The SPI must be unique on the sw itch. Range: 256 to 4294967295.
<AUTH-TYPE>	Specifies the authentication type: **md5** or **sha1**.
<KEY-TYPE>	Specifies the key type to use: **plaintext** (unencrypted), **hex - st ring** (encrypted) or **ciphertext** (encrypted).
<AUTH-KEY>	Specifies the authentication key.
<ENCR-TYPE>	Specifies the encryption type: **des**, **3des**, **aes**, or **null**. **NOTE**

Parameter	Description
	Encryption type **aes** is considered to be AES128, AES192 or AES256 based on key length.
<ENCR-KEY>	Specifies the encryption key.

NOTE

When the authentication key is not provided on the command line, plaintext authentication key prompting occurs upon pressing Enter, followed by encryption type prompting, and finally plaintext encryption key prompting. The entered key characters are masked with asterisks.

NOTE

When the authentication key and encryption type are provided on the command line but the encryption key is not provided, plaintext encryption key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

Examples

Setting area 0 to use IPSec ESP with provided authentication and encryption keys:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 0 encryption ipsec spi 256 md5 plaintext
F824eva
                                                        des plaintext F82#450b
```

Setting area 5 to use IPSec ESP with prompted authentication and encryption keys:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 5 encryption ipsec spi 256 md5
Enter the IPsec authentication key: ********
Re-Enter the IPsec authentication key: ********
Enter the IPsec encryption type (3des/aes/des/null)? des
Enter the IPsec encryption key: ********
Re-Enter the IPsec encryption key: ********
```

Setting area 2 to use IPsec ESP with provided authentication password and encryption type but a prompted encryption key:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)#  area 2 encryption ipsec spi 256 md5 plaintext
F82# des
Enter the IPsec encryption key: ********
Re-Enter the IPsec encryption key: ********
```

Setting area 0 to use IPSec ESP with provided plaintext authentication key and null encryption:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 0 encryption ipsec spi 256 md5 plaintext axtw
null
```

Removing IPSec ESP from area 0:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no area 0 encryption
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`clear ipv6 ospfv3 neighbors`

Syntax

```
clear ipv6 ospfv3 [<PROCESS-ID>] neighbor [<NEIGHBOR>] [interface
[<INTERFACE-NAME>]]
[all-vrfs | vrf <VRF-NAME>]
```

Description

Resets the neighbor and clears the OSPF neighbor information.

Parameter	Description
<PROCESS-ID>	Specifies the OSPFv3 process ID to clear the statistics for the p articular OSPFv3 process. Range: 1 to 65535.
<NEIGHBOR>	Specifies the router ID of a neighbor.
<INTERFACE-NAME>	Specifies the OSPFv3 statistics to clear for the specified interfa ce.
`all-vrfs`	Select to clear the OSPFv3 statistics for all VRFs.
`vrf` *<VRF-NAME>*	Specifies the name of a VRF.

Example

Clearing the OSPFv3 neighbor information:

```
switch# clear ipv6 ospfv3 1 neighbor 3.3.3.3
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ipv6 ospfv3 1 neighbor interface 1/1/1
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`clear ipv6 ospfv3 statistics`

Syntax

```
clear ipv6 ospfv3 [<PROCESS-ID>] statistics [interface [<INTERFACE-NAME>]]
[all-vrfs | vrf <VRF-NAME>]
```

Description

Clears the OSPFv3 event statistics.

Parameter	Description
`<PROCESS-ID>`	Specifies the OSPFv3 process ID to clear the statistics for the particular OSPFv3 process. Range: 1 to 65535.
`<INTERFACE-NAME>`	Specifies the OSPFv3 statistics to clear for the specified interface.
`all-vrfs`	Select to clear the OSPFv3 statistics for all VRFs.
`vrf <VRF-NAME>`	Specifies the name of a VRF.

Example

Clearing the OSPFv3 event statistics:

```
switch# clear ipv6 ospfv3 statistics
switch# clear ipv6 ospfv3 statistics interface 1/1/1
switch# clear ipv6 ospfv3 statistics interface 1/1/1 vrf vrf_red
```

```
switch# clear ipv6 ospfv3 statistics
switch# clear ipv6 ospfv3 statistics interface 1/1/1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`default-information originate`

Syntax

```
default-information originate [metric <METRIC-VALUE>]
no default-information originate [metric <METRIC-VALUE>]
```

Description

Configures OSPFv3 to advertise the default route (::/0) to its neighbors if it is present in the routing table. Optionally, the metric value can be set for default route ::/0. The default value is 1.

The **no** form of this command disables advertisement of the default route.

Parameter	Description
metric <METRIC-VALUE>	Specifies the OSPF metric value for the default route. Optional. Default: 1.

Examples

Setting advertisement of the default route:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# default-information originate
```

Disabling advertisement of the default route:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no default-information originate
```

Setting advertisement of the default route and specifying an optional metric value of 20:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# default-information originate
switch(config-ospfv3-1)# default-information originate metric 20
```

Disabling advertisement of the default route and setting metric to the default value:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no default-information originate metric
```

NOTE

For more information on features that use this command, refer to the Multicast Guide for your switch model.

Command History

Release	Modification
10.09	Added parameter: **metric <METRIC - VALUE>**
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospf-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`default-metric`

Syntax

```
default-metric <METRIC-VALUE>
no default-metric
```

Description

Sets the default metric for redistributed routes in the OSPFv3.

The **no** form of this command sets the default metric to be used for redistributed routes into OSPFv3 to the default of 25.

Parameter	Description
<METRIC-VALUE>	Specifies the default metric value to use for redistributed routes . Range: 1 to 1677214. Default: 25.

Examples

Setting default metric for redistributed routes:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# default-metric 36
```

Setting default metric for redistributed routes to the default:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no default-metric
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`disable`

Syntax

```
disable
```

Description

Disables the OSPFv3 process. By default OSPFv3 process is enabled.

This command does not remove the OSPFv3 configurations.

Example

Disabling OSPFv3 process:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

Platforms	Command context	Authority

`distance`

Syntax

```
distance [<DISTANCE-VAL> | intra-area [<DISTANCE-VAL>] | inter-area
[<DISTANCE-VAL>] | external [<DISTANCE-VAL>]]
no distance [<DISTANCE-VAL> | intra-area [<DISTANCE-VAL>] | inter-area
[<DISTANCE-VAL>] | external [<DISTANCE-VAL>]]
```

Description

Defines an administrative distance for OSPFv3. Administrative distance is used as a criteria to select the best route when the same route is learned by multiple routing protocols.

The **no** form of this command sets the OSPFv3 administrative distance to the default value of 110. Optionally, administrative distance can be set to default for the specific OSPF route type: intra-area, inter-area, or external type-5 and type-7 routes.

Parameter	Description
`<DISTANCE-VAL>`	Specifies the OSPFv3 administrative distance. Range: 1 to 255. Default: 110.
`intra-area`	Specifies the OSPFv3 distance for intra - area routes.
`inter-area`	Specifies the OSPFv3 distance for inter - area routes.
`external`	Specifies the OSPFv3 distance for external type 5 and type 7 ro utes.

Usage

Within a given OSPF process, intra-area routes are always given precedence even when distances are configured for inter-area or external type routes.

Examples

Setting OSPFv3 administrative distance:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# distance 100
switch(config-ospfv3-1)# distance intra-area 24 external 55 inter-area 66
switch(config-ospfv3-1)# distance intra-area 24 external 55
switch(config-ospfv3-1)# distance external 55
switch(config-ospfv3-1)#exit
switch(config)# router ospfv3 2
switch(config-ospfv3-2)# distance 200
switch(config-ospfv3-2)# distance external 60
switch(config-ospfv3-2)# distance intra-area 24 inter-area 66
```

Setting OSPFv3 administrative distance to the default:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no distance
switch(config-ospfv3-1)# no distance external
switch(config-ospfv3-1)# no distance intra-area
switch(config-ospfv3-1)# no distance inter-area
switch(config-ospfv3-1) # no distance 1
switch(config-ospfv3-1)#exit
switch(config)# router ospfv3 2 vrf blue
switch(config-ospfv3-2)# no distance 200
switch(config-ospfv3-2)# no distance external 60
switch(config-ospfv3-2)# no distance intra-area 24 inter-area 66
```

Command History

Release	Modification
10.14	Added capability to have individual admin distance for multiple OSPF processes in a VRF.
10.09	Added parameters: **intra - area**, **inter - area**, **external**
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

Platforms	Command context	Authority

distribute-list prefix

Syntax

```
distribute-list prefix <prefix-list-name> {in | out}
no distribute-list prefix <prefix-list-name> {in | out}
```

Description

This command uses an existing prefix list to filter routes that are being installed in the routing table or redistributed to another routing protocol.

The **distribute-list prefix** command filters routes in the inbound or the outbound direction. When this command is issued with the **in** parameter, it filters routes from being installed in the routing table, it does not filter LSAs. When this command is issued with the **out** parameter, it filters only the desired redistributed routes from other protocols.

> **NOTE**
> This command requires that your prefix list is already defined using the ipv6 prefix commands. Route-maps are not supported with the distribute-list feature.

Parameter	Description
`prefix <prefix-list-name>`	Specify the name of an existing prefix.
`{in | out}`	Select one of the following parameters to set the filter direction: • **in**: Filter incoming routes into the routing table • **out**: Filter outgoing routing updates

Examples

The following commands enable the filtering of OSPFv3 routes in an IPv6 network, so routes are no longer installed in the routing table or redistributed from another routing protocol.

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# distribute-list prefix listA in
switch(config-ospfv3-1)# distribute-list prefix listB out
```

The following command disables the filtering of OSPFv3 routes in an IPv6 network, so routes can be installed in the routing table or redistributed from another routing protocol.

```
switch# configure terminal
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no distribute-list prefix listA in
switch(config-ospfv3-1)# no distribute-list prefix listB out
```

Command History

Release	Modification
10.13	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-<PROCESS-ID>`	Administrators or local user group members with execution rights for this command.

enable

Syntax

```
enable
```

Description

Enables OSPFv3 process when disabled. By default OSPFv3 process is enabled.

Example

Enabling OSPFv3 process:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# enable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`default-information originate`

Syntax

```
default-information originate [metric <METRIC-VALUE>]
no default-information originate [metric <METRIC-VALUE>]
```

Description

Configures OSPFv3 to advertise the default route (::/0) to its neighbors if it is present in the routing table. Optionally, the metric value can be set for default route ::/0. The default value is 1.

The **no** form of this command disables advertisement of the default route.

Parameter	Description
metric <METRIC-VALUE>	Specifies the OSPF metric value for the default route. Optional. Default: 1.

Examples

Setting advertisement of the default route:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# default-information originate
```

Disabling advertisement of the default route:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no default-information originate
```

Setting advertisement of the default route and specifying an optional metric value of 20:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# default-information originate
switch(config-ospfv3-1)# default-information originate metric 20
```

Disabling advertisement of the default route and setting metric to the default value:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no default-information originate metric
```

NOTE

For more information on features that use this command, refer to the Multicast Guide for your switch model.

Command History

Release	Modification
10.09	Added parameter: **metric <METRIC - VALUE>**
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospf-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`default-information originate always`

Syntax

```
default-information originate always [metric <METRIC-VALUE>]
no default-information originate always [metric <METRIC-VALUE>]
```

Description

Configures OSPFv3 to advertise the default route (::/0) to its neighbors, regardless if it is present in the routing table or not. Optionally, metric can be set for default route ::/0. The default value is 1.

The **no** form of this command disables advertisement of the default route.

Parameter	Description
metric <METRIC-VALUE>	Specifies the OSPFv3 metric value for the default route. Default: 1.

Examples

Setting advertisement of the default route:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# default-information originate always
```

Disabling advertisement of the default route:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no default-information originate always
```

Setting advertisement of the default route with metric set to 20:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# default-information originate always metric 20
```

Disabling advertisement of the default route and setting the metric to the default value:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no default-information originate always metric
```

Command History

Release	Modification
10.09	Added parameter: **metric <METRIC - VALUE>**
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospf-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`graceful-restart`

Syntax

```
graceful-restart
   restart-interval <INTERVAL>
   helper [strict-lsa-check]
   ignore-lost-interface
   no...
```

Description

Configures graceful restart for OSPFv3. By default graceful restart is enabled on the OSPFv3 router.

The **no** form of this command sets the restart interval to the default of 120 seconds or disables helper mode depending on the specified parameters.

Parameter	Description
`restart-interval <INTERVAL>`	Specifies the time another router waits for this router to gracefu lly restart and selects the maximum time to wait in seconds. Ra nge: 5 to 1800. Default: 120.

Parameter	Description
`helper`	Specifies that the router will participate in the graceful restart o f a neighbor router.
`strict-lsa-check`	(Optional). Use with the **helper** parameter to enable strict Link state Advertisement (LSA) checking when acting as a restart he lper for a restarting peer. **NOTE** OSPF neighbors must disable strict LSA checki ng. If the local node has fewer OSPF interfaces after restarting, then the neighbors that were a djacent on those interfaces will clear up their a djacencies to the restarting node and will send out link state updates to advertise the dropped adjacency. If strict LSA checking is enabled, th e restarting router's neighbors will exit helper m ode when they receive the updated LSAs and th e graceful restart will still fail.
`ignore-lost-interface`	Enable the restarting router to ignore lost OSPF interfaces duri ng a graceful restart process. This setting should be enabled on a high availability system to ensure a graceful restart completes successfully, even if OSPF - enabled links fail due to High Avail ability events like a switchover or failover. **NOTE** Enabling this setting means that the hitless restart procedures do not strictly follow those defined in RFC 3623, Graceful OSPF Restart.
`no`	Negate any parameter or return the setting to its default..

Examples

Enabling OSPF graceful restart:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# graceful-restart restart-interval 40
switch(config-ospfv3-1)# graceful-restart helper strict-lsa-check
```

Enabling the switch to ignore lost OSPF interfaces during a graceful restart process:

```
switch(config)# router ospfv3 1
switch (config-ospfv3-1)# graceful-restart ignore-lost-interface
```

Setting the restart interval to default, and disabling helper mode:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no graceful-restart restart-interval
switch(config-ospfv3-1)# no graceful-restart helper
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

ipv6 ospfv3 area

Syntax

```
ipv6 ospfv3 <PROCESS-ID> area <AREA-ID>
no ipv6 ospfv3 <PROCESS-ID> area <area-id>
```

Description

Runs the OSPFv3 protocol on the interface for the area specified.

To move an interface to a new area, unmap the existing area and then associate a new area with the interface.

The **no** form of this command disables OSPF on the interface and removes the interface from the area. Interfaces which have an IP address configured on the network or in a subset of the network, stop participating in the OSPF protocol

Parameter	Description
<AREA-ID>	Specifies the area ID is one of the following formats. • Area ID in IPv4 format (**x.x.x.x**), where **x** is a decimal num ber from 0 to 255. • Area ID as a decimal value. Range: 0 - 4294967295.
<PROCESS-ID>	Specifies the OSPFv3 process ID. Range: 1 to 65535.

Examples

Setting OSPFv3 network for the area:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 1 area 1
switch(config-if-vlan)# ipv6 ospfv3 1 area 0.0.0.1
```

Disabling OSPFv3 network for the area:

```
switch(config)# interface 1/1/1
switch(config-if-vlan)# no ipv6 ospfv3 1 area 1
switch(config-if-vlan)# no ipv6 ospfv3 1 area 0.0.0.1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

ipv6 ospfv3 authentication ipsec

Syntax

```
ipv6 ospfv3 authentication ipsec spi <SPI-INDEX>
            <AUTH-TYPE> [<KEY-TYPE>
            <AUTH-KEY>]
no ipv6 ospfv3 authentication
```

Description

Configures IPSec AH authentication. OSPFv3 interfaces that have IPsec configured at the interface context will not use area level IPsec.

The **no** form of this command removes IPsec AH authentication for the specified area.

Parameter	Description
`spi <SPI-INDEX>`	Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables t he receiving OSPF process to select and use the Security Associ ation (SA) from the SA table. The SPI must be unique on the sw itch. Range: 256 to 4294967295.
`<AUTH-TYPE>`	Specifies the authentication type: **md5** or **sha1**.
`<KEY-TYPE>`	Specifies the key type to use: **plaintext** (unencrypted), **hex - st ring** (encrypted) or **ciphertext** (encrypted).
`<AUTH-KEY>`	Specifies the authentication key.

> **NOTE**
> When the authentication key is not provided on the command line, plaintext key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

Examples

Setting interface VLAN **1** to use IPsec authentication with a provided plaintext authentication key:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 authentication ipsec spi 256 md5
plaintext F82#
```

Setting interface VLAN **4** to use IPsec authentication with a prompted plaintext authentication key:

```
switch(config)# interface vlan 4
switch(config-if-vlan)# ipv6 ospfv3 authentication ipsec spi 256 md5
Enter the IPsec authentication key: ********
Re-Enter the IPsec authentication key: ********
```

Removing IPsec authentication from interface VLAN **1**:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 authentication
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ipv6 ospfv3 authentication null`

Syntax

```
ipv6 ospfv3 authentication null
```

Description

Configures null authentication on an interface which disables IPsec authentication.

Examples

Disabling IPsec on interface VLAN **1**:

```
switch(config)# interface van 1
switch(config-if-vlan)# ipv6 ospfv3 authentication null
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-if config-if-vlan	Administrators or local user group members with execution righ ts for this command.

ipv6 ospfv3 cost

Syntax

```
ipv6 ospfv3 cost <INTERFACE-COST>
no ipv6 ospfv3 cost
```

Description

Sets the cost (metric) associated with a particular interface. The interface cost is used as a parameter to calculate the best routes.

The **no** form of this command sets the cost (metric) associated with a particular interface to the default of 1.

Parameter	Description
	Specifies the interface cost value. Range: 1 to 65535. Default: 1.

Parameter	Description
<INTERFACE-COST>	

Examples

Setting OSPFv3 interface cost:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 cost 100
```

Setting the OSPFv3 interface cost to default:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 cost
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	config-if config-if-vlan	Administrators or local user group members with execution righ ts for this command.

ipv6 ospfv3 dead-interval

Syntax

```
ipv6 ospfv3 dead-interval <INTERVAL>
no ipv6 ospfv3 dead-interval
```

Description

Sets the interval after a neighbor is declared dead when no hello packet is received on the OSPFv3 interface.

The **no** form of this command sets the interval after which a neighbor is declared dead, to the default for the OSPFv3 interface. The default value is 40 seconds (generally four times the hello packet interval).

Parameter	Description
<INTERVAL>	Specifies the time interval for the dead interval, in seconds. Range: 1 to 65535. Default: 40.

Examples

Setting OSPFv3 dead interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 dead-interval 30
```

Setting OSPFv3 dead interval to default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 dead-interval
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if-vlan`	Administrators or local user group members with execution rights for this command.

ipv6 ospfv3 encryption ipsec

Syntax

```
ipv6 ospfv3 encryption ipsec spi <SPI-INDEX>
            <AUTH-TYPE>
    [<KEY-TYPE>
            <AUTH-KEY>
            <ENCR-TYPE> [<KEY-TYPE>
            <ENCR-KEY>]]
no ipv6 ospfv3 encryption
```

Description

Configures IPsec ESP authentication. OSPFv3 interfaces that have IPsec configured at the interface context will not use area level IPsec ESP.

The **no** form of this command removes IPsec ESP for the specified area.

Parameter	Description
`spi <SPI-INDEX>`	Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables t he receiving OSPF process to select and use the Security Associ ation (SA) from the SA table. The SPI must be unique on the sw itch. Range: 256 to 4294967295.
`<AUTH-TYPE>`	Specifies the authentication type: **md5** or **sha1**.
`<KEY-TYPE>`	Specifies the key type to use: **plaintext** (unencrypted), **hex - st ring** (encrypted) or **ciphertext** (encrypted).
`<AUTH-KEY>`	Specifies the authentication key.
`<ENCR-TYPE>`	Specifies the encryption type: **des**, **3des**, **aes**, or **null**. **NOTE** **aes** is considered to be AES128, AES192, or AE S256 based on key length.
	Specifies the encryption key.

Parameter	Description
<ENCR-KEY>	

NOTE

When the authentication key is not provided on the command line, plaintext authentication key prompting occurs upon pressing Enter, followed by encryption type prompting, and finally plaintext encryption key prompting. The entered key characters are masked with asterisks.

NOTE

When the authentication key and encryption type are provided on the command line but the encryption key is not provided, plaintext encryption key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

Examples

Setting interface VLAN **1** to use IPsec ESP with provided authentication and encryption keys:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 encryption ipsec spi 256 sha1 plaintext
F82
                                des plaintext F82#450b
```

Setting interface VLAN **3** to use IPsec ESP with prompted authentication and encryption keys:

```
switch(config)# interface vlan 3
switch(config-if-vlan)# ipv6 ospfv3 encryption ipsec spi 256 sha1
Enter the IPsec authentication key: ********
Re-Enter the IPsec authentication key: ********
Enter the IPsec encryption type (3des/aes/des/null)? des
Enter the IPsec encryption key: ********
Re-Enter the IPsec encryption key: ********
```

Setting interface VLAN **4** to use IPsec ESP with provided authentication password and encryption type but a prompted encryption key:

```
switch(config)# interface vlan 4
switch(config-if-vlan)# ipv6 ospfv3 encryption ipsec spi 256 sha1 plaintext
F82 des
Enter the IPsec encryption key: ********
```

```
Re-Enter the IPsec encryption key: ********
```

Setting interface VLAN **1** to use IPSec ESP with provided plaintext authentication key and null encryption:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 encryption ipsec spi 256 sha1 plaintext
F82 null
```

Removing IPsec from interface VLAN **1**:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 encryption
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ipv6 ospfv3 encryption null`

Syntax

```
ipv6 ospfv3 encryption null
```

Description

Configures NULL ESP on an interface which disables IPsec ESP.

Examples

Disable IPsec ESP on interface VLAN 1:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 encryption null
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ipv6 ospfv3 hello-interval`

Syntax

```
ipv6 ospfv3 hello-interval <INTERVAL>
no ipv6 ospfv3 hello-interval
```

Description

Sets the time interval between OSPFv3 hello packets for the OSPFv3 interface.

The **no** form of this command sets the time interval between OSPFv3 hello packets to the default for the OSPFv3 interface of 10 seconds.

Parameter	Description
<INTERVAL>	Specifies the time interval between hello packets, in seconds. R ange: 1 to 65535. Default: 10.

Examples

Setting OSPFv3 hello interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 hello-interval 30
```

Setting OSPFv3 hello interval to default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 hello-interval
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ipv6 ospfv3 network`

Syntax

```
ipv6 ospfv3 network {broadcast | point-to-point | point-to-
multipoint[dynamic]}
no ipv6 ospfv3 network {broadcast | point-to-point | point-to-
multipoint[dynamic]}
```

Description

Configures the network type for the interface. By default, the network type is broadcast.

The **no** form of this command sets the network type for the interface to the system default of broadcast network.

Parameter	Description
`broadcast`	Specifies the OSPFv3 network type as a broadcast multiaccess network.
`point-to-point`	Specifies the OSPFv3 network type as a point - to - point netw ork.
`point-to-multipoint`	Specifies the OSPFv3 network type as a point-to-multipoint net work. Supports non-broadcast mode (static neighbor configura tion) and broadcast mode (dynamic neighbor discovery with **dy namic** option).

NOTE

It is recommended to configure the same network type on all OSPFv3 peers to ensure proper route learning and adjacency formation.

Examples

Setting OSPFv3 network type for the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 network broadcast
switch(config-if-vlan)# ipv6 ospfv3 network point-to-point
switch(config-if-vlan)# ipv6 ospfv3 network point-to-multipoint
switch(config-if-vlan)# ipv6 ospfv3 network point-to-multipoint dynamic
```

Disabling OSPFv3 network type for the interface to system default of broadcast network:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 network
switch(config-if-vlan)# no ipv6 ospfv3 network broadcast
switch(config-if-vlan)# no ipv6 ospfv3 network point-to-point
switch(config-if-vlan)# no ipv6 ospfv3 network point-to-multipoint
switch(config-if-vlan)# no ipv6 ospfv3 network point-to-multipoint dynamic
```

Command History

Release	Modification
10.18	Added a new parameter for **point-to-multipoint**.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ipv6 ospfv3 passive`

Syntax

```
ipv6 ospfv3 passive
no ipv6 ospfv3 passive
```

Description

Configures the interface as an OSPFv3 passive interface. With this setting, the interface participates in the OSPF, but does not send or receive OSPF packets on that interface.

The **no** form of this command resets the interface as active. With this setting, the interface starts sending and receiving OSPF packets.

Examples

Setting the interface as OSPFv3 passive interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 passive
```

Setting the interface as OSPFv3 active interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 passive
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ipv6 ospfv3 priority`

Syntax

```
ipv6 ospfv3 priority <number-value>
no ipv6 ospfv3 priority
```

Description

Sets the OSPFv3 priority for the interface. The larger the numeric value of the priority, the higher the chance it will become the designated router. Setting a priority of 0 makes the router ineligible to become a designated router or back up designated router.

The **no** form of this command sets the OSPFv3 priority for the interface to the default of 1.

Parameter	Description
<number-value>	Specifies the OSPFv3 priority value. Default: 1. Range: 0 to 255.

Examples

Setting the OSPFv3 priority for the interface:

```
switch(config)# interface vlan /1
switch(config-if-vlan)# ipv6 ospfv3 priority 50
```

Setting the OSPFv3 priority for the interface to the default of 1:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 priority
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ipv6 ospfv3 shutdown`

Syntax

```
ipv6 ospfv3 shutdown
no ipv6 ospfv3 shutdown
```

Description

Disables OSPFv3 on the interface. The interface state changes to Down. It does not remove the interface from the OSPF area. To remove the interface, use the command **no ip ospf area**.

The **no** form of this command re-enables OSPFv3 on the interface.

Examples

Disabling OSPFv3 on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 shutdown
```

Re-enabling OSPFv3 on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 shutdown
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`ipv6 ospfv3 transit-delay`

Syntax

```
ipv6 ospfv3 transit-delay <DELAY>
no ipv6 ospfv3 transit-delay
```

Description

Sets the time delay in Link state transmission for the OSPFv3 interface.

The **no** form of this command sets the transit delay in Link state transmission to the default of 1 second.

Parameter	Description
<DELAY>	Specifies the time delay for the transit delay, in seconds. Range: 1 to 3600. Default: 1.

Examples

Setting OSPFv3 transit delay on the interface

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 transit-delay 30
```

Setting OSPFv3 transit delay to default on the interface

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 transit-delay
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-if` `config-if-vlan`	Administrators or local user group members with execution righ ts for this command.

`maximum-paths`

Syntax

```
maximum-paths <MAXIMUM>
no maximum-paths
```

Description

Sets the maximum number of ECMP routes that OSPFv3 can support.

The **no** form of this command sets the maximum number of ECMP routes that OSPFv3 can support to the default value of 4.

Parameter	Description
<MAXIMUM>	Specifies the maximum number of ECMP routes. Range: 1 to 32. Default: 4.

Examples

Setting maximum number of parallel routes:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# maximum-paths 32
```

Setting maximum number of parallel paths to the default value of 4:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no maximum-paths
```

Command History

Release	Modification
10.10	Increased upper limit of range of **<MAXIMUM>** parameter to 32.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`max-metric router-lsa`

Syntax

```
max-metric router-lsa [include-stub |on-startup <INTERVAL>]
no max-metric router-lsa [include-stub | on-startup <INTERVAL>]
```

Description

Sets the protocol to advertise a maximum metric so that other routers do not prefer the router as an intermediate hop in their shortest path first (SPF) calculations. If the `on-startup` parameter is used, the Router is configured to advertise a maximum metric at startup.

To disable advertisement of the maximum metric, use the **no** form of the command.

The **no** form of this command advertises the normal cost metrics instead of advertising the maximized cost metric. This setting causes the router to be considered in traffic forwarding.

Parameter	Description
`include-stub`	Configures the stub Router - LSA to advertise a maximum metr ic for all links, including stub links.
`on-startup <INTERVAL>`	Automatically advertises the stub Router - LSA (or maximize th e router - LSA cost metric) for a specified time interval upon O SPFv3 startup. Specifies the time in seconds. Range: 5 to 86400. Default: 600.

Examples

Setting to maximize the cost metrics for Router LSA:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# max-metric router-lsa
switch(config-ospfv3-1)# max-metric router-lsa on-startup
```

Setting to advertise the normal cost metrics instead of advertising the maximized cost metric:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no max-metric router-lsa
switch(config-ospfv3-1)# no max-metric router-lsa on-startup
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-` *`<PROCESS-ID>`*	Administrators or local user group members with execution righ ts for this command.

`passive-interface default`

Syntax

```
passive-interface default
no passive-interface default
```

Description

Sets all OSPFv3 interfaces as passive.

The **no** form of this command sets all the OSPFv3 interfaces as active.

Examples

Setting OSPFv3-enabled interfaces as passive:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# passive-interface default
```

Setting OSPFv3-enabled interfaces as active:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no passive-interface default
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`redistribute`

Syntax

```
redistribute {bgp | connected | local loopback | static | ripng | ospf
<PROCESS-ID>}
             [route-map <ROUTE-MAP-NAME>]
```

```
no redistribute {bgp | connected | local loopback | static | ripng | ospf
<PROCESS-ID>}
                [route-map <ROUTE-MAP-NAME>]
```

Description

Redistributes routes originating from other protocols, or from another OSPFv3 process, to the current OSPFv3 process.

If a route map is specified, then only the routes that pass the match clause specified in the route map are redistributed to OSPFv3. Configuration is not allowed if the referenced route map has not yet been configured.

If you try to redistribute routes from an OSPFv3 process which is not created, you are prompted to allow the OSPFv3 process to be auto-created before proceeding with redistribution. If you confirm at the prompt, the OSPFv3 process is created with defaults and redistribution configuration applied. If you deny at the prompt, redistribution configuration is skipped.

If command **route-redistribute active-routes-only** has been issued, only the routes from other protocols which are selected for forwarding are considered for redistribution into OSPFv3.

The **no** form of this command disables redistribution of routes to the current OSPFv3 process.

Parameter	Description
`bgp`	Specifies redistributing BGP (Border Gateway Protocol) routes.
`connected`	Specifies redistributing connected (directly attached subnet or host).
`local loopback`	Specifies redistributing local routes of the loopback interface.
`static`	Specifies redistributing static routes.
`ripng`	Specifies redistributing RIPng routes.
`ospf <PROCESS-ID>`	Specifies redistributing routes from the specified OSPFv3 proc ess ID. Range: 1 to 65535.
`route-map <ROUTE-MAP-NAME>`	Specifies redistribution filtering by route map. To create a route map, use command **route - map**.

Examples

Redistributing routes to OSPFv3:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# redistribute bgp
switch(config-ospfv3-1)# redistribute bgp route-map BGP_routes

switch(config-ospfv3-1)# redistribute connected
switch(config-ospfv3-1)# redistribute connected route-map connected_routes
switch(config-ospfv3-1)# redistribute local loopback
switch(config-ospfv3-1)# redistribute local loopback route-map local_routes
switch(config-ospfv3-1)# redistribute static
switch(config-ospfv3-1)# redistribute static route-map static_networks
switch(config-ospfv3-1)# redistribute ripng
switch(config-ospfv3-1)# redistribute ripng route-map rip-routes
switch(config-ospfv3-1)# redistribute ospf 2
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# redistribute connected
switch(config-ospfv3-1)# redistribute connected route-map connected_routes
switch(config-ospfv3-1)# redistribute local loopback
switch(config-ospfv3-1)# redistribute local loopback route-map local_routes
switch(config-ospfv3-1)# redistribute static
switch(config-ospfv3-1)# redistribute static route-map static_networks
switch(config-ospfv3-1)# redistribute ripng
switch(config-ospfv3-1)# redistribute ripng route-map rip-routes
switch(config-ospfv3-1)# redistribute ospf 2
```

Disabling redistributing routes to OSPFv3:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no redistribute bgp
switch(config-ospfv3-1)# no redistribute bgp route-map BGP_routes

switch(config-ospfv3-1)# no redistribute connected
switch(config-ospfv3-1)# no redistribute connected route-map
connected_routes
switch(config-ospfv3-1)# no redistribute local loopback
switch(config-ospfv3-1)# no redistribute local loopback route-map
local_routes
switch(config-ospfv3-1)# no redistribute static
switch(config-ospfv3-1)# no redistribute static route-map static_networks
switch(config-ospfv3-1)# no redistribute ripng
switch(config-ospfv3-1)# no redistribute ripng route-map rip-routes
switch(config-ospfv3-1)# no redistribute ospf 2
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no redistribute connected
switch(config-ospfv3-1)# no redistribute connected route-map
```

```
connected_routes
switch(config-ospfv3-1)# no redistribute local loopback
switch(config-ospfv3-1)# no redistribute local loopback route-map
local_routes
switch(config-ospfv3-1)# no redistribute static
switch(config-ospfv3-1)# no redistribute static route-map static_networks
switch(config-ospfv3-1)#
switch(config-ospfv3-1)# no redistribute ripng
switch(config-ospfv3-1)# no redistribute ripng route-map rip-routes
switch(config-ospfv3-1)# no redistribute ospf 2
```

Command History

Release	Modification
10.08	Added **route - map** support for supported redistribute source - protocols. Updated information and examples.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-<PROCESS-ID>`	Administrators or local user group members with execution rights for this command.

reference-bandwidth

Syntax

```
reference-bandwidth <BANDWIDTH>
no reference-bandwidth
```

Description

Sets the reference bandwidth for OSPFv3. If the OSPFv3 interface cost is not explicitly set, then the cost of all the OSPFv3 interfaces is recalculated based on the reference bandwidth and link speed of the interface.

For VLAN interfaces the calculated link speed value is 1 Gbps (if the OSPFv3 interface cost is not explicitly set).

The **no** form of this command sets the reference bandwidth for OSPF to the default of 100000 Mbps.

Parameter	Description
<BANDWIDTH>	Specifies the reference bandwidth used to calculate the cost of an interface in Mbps. Range: 1 to 4000000. Default: 100000.

Examples

Setting the reference bandwidth:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# reference-bandwidth 40000
```

Setting the reference bandwidth to the default value:

```
switch(config)# routerv3 ospf 1
switch(config-ospfv3-1)# no reference-bandwidth
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	config-ospfv3-*<PROCESS-ID>*	Administrators or local user group members with execution righ ts for this command.

`retransmit-interval`

Syntax

```
retransmit-interval <INTERVAL>
no retransmit-interval
```

Description

Sets the time between retransmitting lost link state advertisements for virtual links.

The **no** form of this command sets the time between retransmitting lost link state advertisements to the default of 5 seconds for virtual links.

Parameter	Description
<INTERVAL>	Specifies the time interval for the retransmit interval, in seconds . Range: 1 to 3600. Default: 5.

Examples

Setting OSPFv3 virtual links retransmit interval:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# retransmit-interval 30
```

Setting OSPFv3 virtual links retransmit interval to default:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# no retransmit-interval
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-router-vlink`	Administrators or local user group members with execution righ ts for this command.

`router-id`

Syntax

```
router-id <ROUTER-ADDRESS>
no router-id
```

Description

Sets an ID for the router in an IPv4 address format.

The **no** form of this command unconfigures the router-id for the instance and sets the router-id to the default. The router-id is changed to the dynamically selected router-id. The default router-id 0.0.0.0 updates to the routing stack that triggers to auto-elect a router-id based on the highest IP address of loopback interface, or the highest IP address of interfaces.

Parameter	Description
<ROUTER-ADDRESS>	Specifies the router address in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255.

Examples

Setting router-id in the OSPFv3 context:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1) # router-id 1.1.1.1
```

Unconfiguring router-id:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no router-id
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`router ospfv3`

Syntax

```
router ospfv3 <PROCESS-ID> [vrf <VRF-NAME>]
no router ospfv3 <PROCESS-ID> [vrf <VRF-NAME>]
```

Description

Creates the OSPFv3 process (if not created already) and enters the router OSPFv3 instance context. Optionally if specified, you can specify a named VRF, or the default VRF if the <vrf-name> is not specified. Only one OSPFv3 process is allowed per VRF.

The **no** form of this command removes the OSPFv3 instance. If a VRF is specified, it removes the OSPF instance from the named VRF, or the default VRF if the <var-name> is not specified.

Parameter	Description
`<PROCESS-ID>`	Specifies an OSPFv3 process ID. Length: 1 to 65535.
`vrf <VRF-NAME>`	Specifies the name of a VRF. Default: default.

Examples

Entering the router OSPFv3 instance:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)#
```

Setting the router OSPFv3 VRF instance:

```
switch(config)# router ospfv3 1 vrf vrf_red
```

Removing the router OSPFv3 instance:

```
switch#(config)# no router ospfv3 1
```

Command History

Release	Modification
10.14	Supported process ID range expanded from 1 - 63 to 1 - 65535.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`show ipv6 ospfv3`

Syntax

```
show ipv6 ospfv3 [<PROCESS-ID>] [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows OSPFv3 information including area, state, and configuration information.

Parameter	Description
<PROCESS-ID>	Specifies an OSPFv3 process ID optionally to show OSPFv3 info rmation for a particular OSPFv3 process. Range: 1 to 65535.
all-vrfs	Select to show OSPFv3 information for all VRFs.
vrf *<VRF-NAME>*	Specifies the name of a VRF. Default: default.

Example

Showing general OSPFv3 configurations:

```
switch# show ipv6 ospfv3 200
VRF : default                                       Process : 200
---------------------------------------------------------------------
Router ID                : 1.1.1.1             OSPFv3                 : Enabled
BFD                      : Disabled            SPF Start Interval     : 200 ms
SPF Hold Interval        : 1000 ms             SPF Max Wait Interval  : 5000 ms
LSA Start Interval       : 5000 ms             LSA Hold Interval      : 1000 ms
LSA Max Wait Interval    : 1000 ms             LSA Arrival Interval   : 1000 ms
External LSAs            : 0                   Checksum Sum           : 0
ECMP                     : 4                   Reference Bandwidth    : 100000
Mbps
Area Border              : Yes                 AS Border              : No
GR Status                : Enabled             GR Interval            : 120 sec
GR State                 : Inactive            GR Exit Status         : None
GR Helper                : Enabled             GR Strict LSA Check    : Enabled
GR Ignore Lost I/F       : Disabled            Internal Process ID    : 1
Summary address:
  prefix fd00::1/64, advertise, tag 10
Area      Total Active
----------------------
Normal      2      2
Stub        0      0
Area   : 0.0.0.0
----------------------------------
Area Type              : Normal       Status                : Active
Total Interfaces       : 1            Active Interfaces     : 1
Passive Interfaces     : 0            Loopback Interfaces   : 0
SPF calculation count  : 4
```

```
Area ranges:
  fd00::1/64, inter-area, no-advertise
AH Authentication      : SHA1, SPI 256
Number of LSAs         : 5            Checksum Sum         : 99122
Area  : 0.0.0.1
----------------------------------
Area Type              : Normal       Status               : Active
Total Interfaces       : 1            Active Interfaces    : 1
Passive Interfaces     : 0            Loopback Interfaces  : 0
SPF Calculation Count : 4
Area ranges:
  fd00::1/64, inter-area, no-advertise
ESP Authentication     : SHA1
Encryption             : 3DES, SPI 256
Number of LSAs         : 5            Checksum Sum         : 99122
```

Command History

Release	Modification
10.14	Supported process ID range expanded from 1 - 63 to 1 - 65535.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show ipv6 ospfv3 border-routers`

Syntax

```
show ipv6 ospfv3 [<PROCESS-ID>] border-routers [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows the OSPFv3 routing table entries for Area Border Router (ABR) and Autonomous System Border Router (ASBR).

Parameter	Description
<PROCESS-ID>	Specifies an OSPFv3 process ID to show the OSPFv3 routing ta ble entries for ABR and ASBR for the particular OSPFv3 proces s. Range: 1 to 65535.
`all-vrfs`	Select to show OSPFv3 border router information for all VRFs.
`vrf` *<VRF-NAME>*	Specifies the name of a VRF. Default: default.

Example

Showing OSPFv3 border routers information for VRF **vrf_red**:

```
switch# show ipv6 ospfv3 border-routers vrf vrf_red
VRF : vrf_red       Process ID : 1
             Internal Routing Table
--------------------------------------------------------
Codes: i - Intra-area route, I - Inter-area route
   Router-ID Cost  Type    Area    SPF  Nexthop
Interface
i  1.1.1.1   1     ASBR    0.0.0.0 9    fe80::7272:cfff:fe9a:a15d   1/1/2
i  3.3.3.3   1     ASBR    1.1.1.1 9    fe80::7272:cfff:fe1f:d80   **
tunnel1
```

Command History

Release	Modification
10.14	Supported process ID range expanded from 1 - 63 to 1 - 65535.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ipv6 ospfv3 interface

Syntax

```
show ipv6 ospfv3 [<PROCESS-ID>] interface [<INTERFACE-NAME>] [brief]
[all-vrfs | vrf <VRF-NAME>]
```

Description

Shows information about OSPFv3 enabled interfaces.

Parameter	Description
`<PROCESS-ID>`	Specifies an OSPFv3 process ID optionally to show the OSPFv3 enabled interfaces for the particular OSPFv3 process. Range: 1 to 65535.
`<INTERFACE-NAME>`	Selects to show information only for the specified OSPFv3 - enabled interface.
`brief`	Include this parameter to display a brief overview of the following OSPF configuration information. • Interface: OSPF interface name. • Area: OSPF area ID. • Cost: The metric OSPF uses to judge a path's feasibility, calculated as (reference bandwidth / interface bandwidth). • State: Indicates if the interface is a designated router (**Dr**) or a backup designated router (**Backup - dr**). • Status: Indicates if the interface is **up** or **down**. • Flags: P - Passive A - Active.

Parameter	Description
`all-vrfs`	Select to show OSPF - enabled interface information for all VR Fs.
`vrf <VRF-NAME>`	Specifies the name of a VRF. Default: default.

Examples

Showing OSPFv3 information for all interfaces in default VRF:

```
switch# show ipv6 ospfv3 interface
Codes: DR - Designated router  BDR - Backup Designated router
Interface 1/3/10 is Up, Line Protocol is Up
--------------------------------------------
VRF                     : default
Process                 : 1
IPv6 address            : fe80::9020:c203:280a:e800
Area                    : 0.0.0.0
Status                  : Up                            Network
Type            : Broadcast
Hello Interval          : 10    sec                     Dead
Interval          : 40    sec
Transit Delay           : 1     sec                     Retransmit
Interval  : 5     sec
BFD                     : Disabled                      Link
Speed             : 1000 Mbps
Cost Configured         : NA                            Cost
Calculated        : 100
State/Type              : DR                            Router
Priority        : 2
DR                      : 1.1.1.1
BDR                     : 2.2.2.2
Link LSAs               : 2                             Checksum
Sum          : 39245
Authentication          : no
Passive                 : No
Codes: DR - Designated router  BDR - Backup Designated router
Interface 1/3/11 is Up, Line Protocol is Up
--------------------------------------------
VRF                     : default
Process                 : 1
IPv6 address            : fe80::9020:c203:2c0a:e800
Area                    : 0.0.0.1
```

```
Status                 : Up                                   Network
Type          : Broadcast
Hello Interval         : 10     sec                           Dead
Interval         : 40     sec
Transit Delay          : 1      sec                           Retransmit
Interval   : 5      sec
BFD                    : Disabled                             Link
Speed            : 1000 Mbps
Cost Configured        : NA                                   Cost
Calculated       : 100
State/Type             : BDR                                  Router
Priority       : 1
DR                     : 3.3.3.3
BDR                    : 1.1.1.1
Link LSAs              : 2                                    Checksum
Sum          : 83119
Authentication         : no
Passive                : No
```

Showing overview information for OSPFv3 enabled interfaces for all VRFs in brief:

```
switch# show ipv6 ospfv3 interface brief all-vrfs
VRF : default                          Process : 1
===================================================
Total Number of Interfaces: 2
Flags: P - Passive  A - Active
Interface     Area                Cost      State              Status
Flags
----------------------------------------------------------------------------
-----
1/3/10        0.0.0.0             100       DR                 Up          A
1/3/11        0.0.0.1             100       BDR                Up          A
```

Showing overview information for OSPFv3 enabled interfaces for all VRFs:

```
6200(config)# show ipv6 ospfv3 interface br all-vrfs
VRF : default                          Process : 1
===================================================
Total Number of Interfaces: 1
Flags: P - Passive  A - Active
Interface     Area                Cost      State              Status
Flags
----------------------------------------------------------------------------
-----
1/1/1         0.0.0.0             100       BDR                Up          A
```

Command History

Release	Modification
10.14	Supported process ID range expanded from 1 - 63 to 1 - 65535.
10.09	Output of the **show ipv6 ospfv3 interface** command includes flags to indicate whether the interface is in passive or active mode.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show ipv6 ospfv3 lsdb`

Syntax

```
show ipv6 ospfv3 [<process-id>] lsdb
   adv-router {<ROUTER-ID>|self}
   area <AREA-ID>
   lsid <link-state-id>
   all-vrfs|vrf <VRF-NAME>}
as-external
asbr-summary
database-summary
inter-area-prefix
inter-area-router
intra-area-prefix
link
network
nssa-external
router
summary
```

Description

Shows the OSPFv3 link state database summary for different OSPF LSAs (Link State Advertisement). Use the parameters to get information for a particular LSA.

Parameter	Description
<PROCESS-ID>	Enter an OSPFv3 process ID to display general OSPF information for a particular OSPF process. Range: 1 to 65535.
adv-router {*<ROUTER-ID>*|self}	Select to display link states for a particular advertising router. Specify either a Router ID of the advertising router or specify **self** to show self - originated link states.
area *<AREA-ID>*	Select to display information filtered for the specified area in one of the following formats. • OSPFv3 area identifier in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255. • OSPFv3 area identifier in decimal format. Value: 0 to 4294967295.
lsid *<LINK-STATE-ID>*	Select to display information filtered by link state identifier specified in IPv4 address format (A.B.C.D).
all-vrfs|{vrf <vrf-name>	Select **all - vrfs** to display general OSPF information for all VRFs, or use the **vrf** <VRF - NAME> option to display information for a specific VRF. Optionally select one of the following parameters to filter the link state database information.
as-external	Show external link states (LSA type 5)
database-summary	Select to display the count of each type of LSA and each area in the database. **NOTE** The **database - summary** parameter does not support the **area <area - id>**, **lsid <link - state - id>** or **adv - router {<router - id>|self}** parameters.

Parameter	Description
`inter-area-prefix`	Show inter - area prefix link states (LSA type 3)
`inter-area-router`	Show inter - area router link states (LSA type 4)
`intra-area-prefix`	Show intra - area prefix link states (LSA type 9
`link`	Show link states (LSA type 8)
`network`	Show network LSAs (LSA type 2).
`nssa-external`	Show NSSA external link states (LSA type 7).
`router`	Show router LSAs (LSA type 1).

Examples

Showing OSPFv3 link state database (LSDB) general information:

```
switch# show ipv6 ospfv3 lsdb
OSPF Router with ID (50.50.50.50) (Process ID 1 VRF default)
============================================================
Router Link State Advertisements (Area 0.0.0.0)
-----------------------------------------------------------------------------
-
ADV Router       Age      Seq#         Checksum     LSID          Link Count
Bits
-----------------------------------------------------------------------------
-
40.40.40.40      930      0x80000004   0x2ea1       0             3
None
50.50.50.50      935      0x80000002   0x8b52       0             1            E
60.60.60.60      943      0x800003c5   0x9854       0             2
None
Network Link State Advertisements (Area 0.0.0.0)
-------------------------------------------------------------------------
ADV Router       Age      Seq#         Checksum     LSID         Router Count
-------------------------------------------------------------------------
60.60.60.60      944      0x80000001   0x7179       1360007168   2
50.50.50.50      935      0x80000001   0x516a       19           1
```

```
Inter Area Prefix Link State Advertisements (Area 0.0.0.0)
------------------------------------------------------------------
ADV Router       Age      Seq#         Checksum    LSID         Prefix
------------------------------------------------------------------
40.40.40.40      929      0x80000001   0x2498      131072       FEC0:3344::/32
50.50.50.50      928      0x80000001   0x5b2f      65536        111::/64
Inter Area Router Link State Advertisements (Area 0.0.0.0)
----------------------------------------------------------------------------
-----
ADV Router       Age      Seq#         Checksum    LSID         Destination
Router ID
----------------------------------------------------------------------------
-----
40.40.40.40      929      0x80000001   0x2498      1            33.33.33.33
AS External Link State Advertisements (Area 0.0.0.0)
------------------------------------------------------------------
ADV Router       Age      Seq#         Checksum    LSID         Prefix
------------------------------------------------------------------
40.40.40.40      264      0x80000001   0x24cc4     1            10::/64
40.40.40.40      675      0x80000001   0x5b00f     2            11::/64
NSSA External Link State Advertisements (Area 0.0.0.0)
------------------------------------------------------------------
ADV Router       Age      Seq#         Checksum    LSID         Prefix
------------------------------------------------------------------
3.3.3.3          264      0x80000001   0x24ac2     1            200::/64
Link-local Link State Advertisements (Area 0.0.0.0)
---------------------------------------------------------------------
ADV Router       Age      Seq#         Checksum    LSID         Interface
---------------------------------------------------------------------
50.50.50.50      264      0x80000001   0x653c4     19           1/1/1
Intra Area Prefix Link State Advertisements (Area 0.0.0.0)
----------------------------------------------------------------------------
-------------------
ADV Router       Age      Seq#         Checksum    LSID         Referenced LS
Type  Referenced LSID
----------------------------------------------------------------------------
-------------------
50.50.50.50      263      0x80000001   0x1da34     1
0x2001              0
50.50.50.50      264      0x80000001   0x2a45d     1
0x2002              19
```
Showing AS external link states:

```
switch# show ipv6 ospfv3 lsdb as-external
OSPF Router with ID (60.60.60.60) (Process ID 1 VRF default)
========================================================
AS External Link State Advertisements (Area 0.0.0.0)
------------------------------------------------------------------
ADV Router      Age     Seq#        Checksum    LSID        Prefix
------------------------------------------------------------------
40.40.40.40     264     0x80000001  0x24cc4     1           10::/64
40.40.40.40     675     0x80000001  0x5b00f     2           11::/64
```

Showing the LSDB database summary:

```
switch# show ipv6 ospfv3 lsdb database-summary
OSPF Router with ID (10.1.1.1) (Process ID 1 VRF default)
==========================================================
Area 0.0.0.0 database summary
--------------------------
LSA Type              Count
--------------------------
Router                2
Network               1
Inter Area Prefix     1
Inter Area Router     0
NSSA External         0
Link                  3
Intra Area Prefix     3
--------------------------
Total                 10
Process 1 database summary
--------------------------
LSA Type              Count
--------------------------
Router                2
Network               1
Inter Area Prefix     1
Inter Area Router     0
AS External           2
NSSA External         0
Link                  3
Intra Area Prefix     3
--------------------------
Total                 12
```

Showing inter-area prefix LSAs:

```
switch# show ipv6 ospfv3 lsdb inter-area-prefix
OSPF Router with ID (6.6.6.6) (Process ID 1 VRF default)
```

```
=========================================================
Inter Area Prefix Link State Advertisements (Area 0.0.0.0)
-------------------------------------------------------------------
ADV Router       Age      Seq#         Checksum     LSID         Prefix
-------------------------------------------------------------------
40.40.40.40      929      0x80000001   0x2498       131072       FEC0:3344::/32
50.50.50.50      928      0x80000001   0x5b2f       65536        111::/64
```

Showing network LSAs:

```
switch# show ipv6 ospfv3 lsdb network
OSPF Router with ID (50.50.50.50) (Process ID 1 VRF default)
=========================================================
Network Link State Advertisements (Area 0.0.0.0)
-------------------------------------------------------------------------
ADV Router       Age      Seq#         Checksum     LSID         Router Count
-------------------------------------------------------------------------
60.60.60.60      944      0x80000001   0x7179       1360007168   2
50.50.50.50      935      0x80000001   0x516a       19           1
```

Showing NSSA external link states:

```
switch# show ipv6 ospfv3 lsdb nssa-external
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
=========================================================
NSSA External Link State Advertisements (Area 0.0.0.0)
-------------------------------------------------------------------
ADV Router       Age      Seq#         Checksum     LSID         Prefix
-------------------------------------------------------------------
3.3.3.3          264      0x80000001   0x24ac2      1            200::/64
```

Command History

Release	Modification
10.14	Supported process ID range expanded from 1 - 63 to 1 - 65535.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show ipv6 ospfv3 neighbors`

Syntax

```
show ipv6 ospfv3 [<PROCESS-ID>] neighbors [<NEIGHBOR-ID>]
     [interface <INTERFACE-NAME>] [detail | summary]
     [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows information about OSPFv3 neighbors.

Parameter	Description
`<PROCESS-ID>`	Specifies an OSPFv3 process ID to show OSPFv3 neighbor infor mation for the particular OSPFv3 process. Range: 1 to 65535.
`neighbors <NEIGHBOR-ID>`	Shows information about a particular neighbor, specified in IPv 4 format (A.B.C.D).
`interface <INTERFACE-NAME>`	Shows neighbor information only for the specified interface.
`detail`	Shows detailed information for the neighbors.
`summary`	Shows summary information for the neighbors.
`all-vrfs`	Shows neighbor information for all VRFs.
`vrf <VRF-NAME>`	Specifies the name of a VRF. Default: default.

Examples

Showing OSPFv3 neighbors information:

Showing OSPFv3 neighbors information for a specific neighbor:

```
switch# show ipv6 ospfv3 neighbors 3.3.3.3
VRF : default                          Process : 1
--------------------------------------------------------------
Router-Id          : 3.3.3.3           Area              : 0.0.0.0
Interface          : 1/1/1             Address           :
```

```
fe80::7272:cfff:fe79:7510
State                : FULL              Neighbor Priority : 1
Dead Timer Due       : 00:00:36          Options           : 0x13
Time since last state change : 00h:14m:45s
```

Showing detail OSPFv3 neighbors information for a specific neighbor:

```
switch# show ipv6 ospfv3 neighbors 2.2.2.2 detail
VRF : default                            Process : 1
--------------------------------------------------------------
Router-Id            : 3.3.3.3           Area              : 0.0.0.0
Interface            : 1/1/1             Address           :
fe80::7272:cfff:fe79:7510
State                : FULL              Neighbor Priority : 1
DR                   : 3.3.3.3           BDR               : 1.1.1.3
Dead Timer Due       : 00:00:36          Options           : 0x13
Retransmission Queue Length  : 0
Time Since Last State Change : 00h:14m:45s
```

Showing OSPFv3 neighbors information for interface **1/1/1**:

```
switch# show ipv6 ospfv3 neighbors 3.3.3.3 interface 1/1/1
VRF : default                            Process : 1
--------------------------------------------------------------
Router-Id            : 3.3.3.3           Area              : 0.0.0.0
Interface            : 1/1/1             Address           :
fe80::7272:cfff:fe79:7510
State                : FULL              Neighbor Priority : 1
Dead Timer Due       : 00:00:36          Options           : 0x13
Time Since Last State Change : 00h:14m:45s
```

Command History

Release	Modification
10.14	Supported process ID range expanded from 1 - 63 to 1 - 65535.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show ipv6 ospfv3 routes`

Syntax

```
show ipv6 ospfv3 [<PROCESS-ID>] routes [<PREFIX/LENGTH>]
     [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows the OSPFv3 routing table information.

Parameter	Description
<PROCESS-ID>	Specifies an OSPFv3 process ID that shows information from th e OSPFv3 routing table for the particular OSPFv3 process. Ran ge: 1 to 65535.
<PREFIX/LENGTH>	Specifies the IPv6 destination prefix showing information about a particular destination prefix. For example, 2010:bd9::/32.
`all-vrfs`	Select to show OSPFv3 routing table information for all VRFs.
`vrf` *<VRF-NAME>*	Specifies the name of a VRF. Default: default.

Examples

Showing OSPFv3 routing table information:

```
switch# show ipv6 ospfv3 routes
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2
OSPFv3 Process ID 1 VRF default, Routing Table
-----------------------------------------------
Total Number of OSPFv3 Routes : 2
111::/64           (i) area:0.0.0.0
     directly attached to interface 1/1/1, cost 1 distance 110
fd00::/64          (i) area:0.0.0.1
     directly attached to interface vlan10, cost 1 distance 110
```

Showing OSPFv3 routing table information for VRF red:

```
switch# show ipv6 ospfv3 routes vrf red
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2
OSPFv3 Process ID 2 VRF red, Routing Table
-------------------------------------------
Total Number of OSPFv3 Routes : 1
222::/64           (i) area:0.0.0.1
     directly attached to interface 1/1/2, cost 1 distance 110
```

Showing OSPFv3 routing table information for destination prefix fd00::/64:

```
switch# show ipv6 ospfv3 1 routes fd00::/64
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2
OSPFv3 Process ID 1 VRF default, Routing Table for prefixes fd00::/64
----------------------------------------------------------------------
Total Number of OSPFv3 Routes : 1
fd00::/64          (i) area:0.0.0.1
     directly attached to interface vlan10, cost 1 distance 110
```

Command History

Release	Modification
10.14	Supported process ID range expanded from 1 - 63 to 1 - 65535.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show ipv6 ospfv3 statistics`

Syntax

```
show ipv6 ospfv3 [<PROCESS-ID>] statistics
     [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows OSPFv3 statistics.

Parameter	Description
<PROCESS-ID>	Specifies an OSPFv3 process ID that shows information on the OSPFv3 SPF statistics for the particular OSPFv3 process. Range: 1 to 65535.
`all-vrfs`	Select to show OSPFv3 SPF statistics information for all VRFs.
`vrf` *<VRF-NAME>*	Specifies the name of a VRF. Default: default.

Examples

Showing OSPFv3 statistics information:

```
switch# show ipv6 ospfv3 statistics
OSPFv3 Process ID 1 VRF default, Statistics (cleared 3h 2m 21s ago)
---------------------------------------------------------------------
Unknown Interface Drops          : 0
Unknown Virtual Interface Drops : 0
Bad IPv6 Header Length Drops     : 0
Wrong OSPFv3 Version Drops       : 0
Bad Source IPv6 Drops            : 0
Resource Failure Drops           : 0
Bad Header Length Drops          : 0
Total Drops                      : 0
```

Showing OSPFv3 statistics information for VRF red:

```
switch# show ipv6 ospfv3 2 statistics vrf red
OSPFv3 Process ID 2 VRF red, Statistics (cleared 3h 2m 30s ago)
-----------------------------------------------------------------
Unknown Interface Drops          : 0
Unknown Virtual Interface Drops : 0
Bad IPv6 Header Length Drops     : 0
Wrong OSPFv3 Version Drops       : 0
Bad Source IPv6 Drops            : 0
Resource Failure Drops           : 0
Bad Header Length Drops          : 0
Total Drops                      : 0
```

Command History

Release	Modification
10.14	Supported process ID range expanded from 1 - 63 to 1 - 65535.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show ipv6 ospfv3 statistics interface`

Syntax

```
show ipv6 ospfv3 [<PROCESS-ID>] statistics interface [<INTERFACE-NAME>]
     [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows the OSPFv3 statistics for the OSPFv3-enabled interfaces.

Parameter	Description
`<PROCESS-ID>`	Specifies an OSPFv3 process ID to show OSPF - enabled inter face statistics information on the specified OSPFv3 process. Ra nge: 1 to 65535.
`<INTERFACE-NAME>`	Selects to show information only for the specified interface.
`all-vrfs`	Select to show OSPF - enabled interface statistics information for all VRFs.

Parameter	Description
`vrf <VRF-NAME>`	Specifies the name of a VRF. Default: default.

Example

Showing OSPFv3-enabled interfaces information for interface 1/1/1:

```
switch# show ipv6 ospfv3 statistics interface all-vrfs
OSPFv3 Process ID 1 VRF default, Interface vlan2000 Statistics  (cleared 0h
2m 52s ago)
============================================================================
==========
Tx Hello packets      : 0                    Rx Hello packets      : 0
Tx Hello bytes        : 0                    Rx Hello bytes        : 0
Tx DD packets         : 0                    Rx DD packets         : 0
Tx DD bytes           : 0                    Rx DD bytes           : 0
Tx LS request packets : 0                    Rx LS request packets : 0
Tx LS request bytes   : 0                    Rx LS request bytes   : 0
Tx LS update packets  : 0                    Rx LS update packets  : 0
Tx LS update bytes    : 0                    Rx LS update bytes    : 0
Tx LS ack packets     : 0                    Rx LS ack packets     : 0
Tx LS ack bytes       : 0                    Rx LS ack bytes       : 0
Total IPsec packets processed : 0
Total IPsec bytes processed   : 0
Total Number of State Changes : 0
Number of LSAs                : 0
LSA Checksum Sum              : 0
Total OSPFv3 Packets Discarded: 0
----------------------------------
Reason                           Packets Dropped
--------------------------------------------------
Invalid Type                     0
Invalid Length                   0
Invalid Version                  0
Bad or Unknown Source            0
Area Mismatch                    0
Self-originated                  0
Duplicate Router ID              0
Interface Standby                0
Total Hello Packets Dropped      0
  Hello Interval Mismatch        0
  Dead Interval Mismatch         0
  Options Mismatch               0
```

```
  MTU Mismatch                  0
  Neighbor Ignored              0
Resource Failures               0
Bad LSA Length                  0
Others                          0
IPsec Authentication Errors     0
IPsec ESP Errors                0
Total LSAs Ignored : 0
-----------------------
Bad Type            : 0
Bad Length          : 0
Invalid Data        : 0
Invalid Checksum    : 0
```

Command History

Release	Modification
10.14	Supported process ID range expanded from 1 - 63 to 1 - 65535.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manage r (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`summary-address`

Syntax

```
summary-address  <IPV6-ADDR>/<MASK> [no-advertise | tag <TAG-VALUE>]
no summary-address <prefix/length> [no-advertise | tag <tag-value>
```

Description

Summarizes the external routes with the matching address and mask. When advertising this route, its metric is set to the lowest cost path from among the routes that were summarized.

The **no** form of this command disables route summarization.

> **NOTE**
> This command only works for an ASBR (Autonomous System Boundary Router).

Parameter	Description
<IPV6-ADDR>	Specifies an IP address in IPv6 format (**xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx**), where **x** is a hexadecimal number from 0 to F.
<MASK>	Specifies the number of bits in the address mask in CIDR format (**x**), where **x** is a decimal number from 0 to 128.
`no-advertise`	Do not advertise the aggregate route. Suppress routes that m atch the specified prefix/mask pair.
`tag` *<TAG-VALUE>*	Specify the tag for the aggregate route. The summary prefix wi ll be advertised along with the tag value in External LSAs. Rang e: 0 to 4294967295

Examples

Setting OSPF route summarization:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# summary-address 2001:DB8::1/32
```

Disabling route summarization:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no summary-address 2001:DB8::1/32
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`timers lsa-arrival`

Syntax

```
timers lsa-arrival <DELAY>
no timers lsa-arrival
```

Description

Configures the minimum delay between receiving the same LSA from a peer. The same LSA is an LSA that contains the same LSA ID number, LSA type, and advertising router ID. If an instance of the same LSA arrives sooner before the delay expires, the LSA is dropped. Generally, the LSA arrival timer should be set to a value less than or equal to the start-time value for the command **timers throttle lsa start** on the neighbor.

The **no** form of this command sets the LSA timers to default values.

Parameter	Description
<DELAY>	Specifies the delay in milliseconds. Range: 0 to 600000. Default : 1000.

Examples

Setting the LSA arrival timer:

```
switch(config)# router ospf 1
switch(config-ospfv3-1)# timers lsa-arrival 10
```

Setting the LSA arrival timer to default:

```
switch(config)# router ospf 1
switch(config-ospfv3-1)# no timers lsa-arrival
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

timers throttle lsa

Syntax

```
timers throttle lsa start-time <START-TIME> hold-time <HOLD-TIME> max-wait-
time <WAIT-TIME>
no timers throttle lsa
```

Description

Configures the timers for LSA generation.

The **no** form of this command sets the LSA timers to default values.

Parameter	Description
`start-time <START-TIME>`	Specifies the initial wait time in milliseconds after which LSAs a re generated. When set to 0, the LSAs are generated without a ny delay. Range: 0 to 600000. Default: 5000.
`hold-time <HOLD-TIME>`	Specifies the amount of time, in milliseconds, between regene ration of an LSA. The hold time doubles each time the same L SA must be regenerated, until **max - wait - time** is reached. W

Parameter	Description
	hen set to 0, LSA regeneration time is not increased. Range: 0 to 600000. Default: 0.
`max-wait-time <WAIT-TIME>`	Specifies the maximum wait time, in milliseconds, for regenerati on of the same LSA. When set to 0, LSA regeneration time is n ot increased. Range: 0 to 600000. Default: 0.

Examples

Setting the LSA timers:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# timers throttle lsa start-time 100 hold-time 1000
max-wait-time 10000
```

Setting LSA timers to default values:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no timers throttle lsa
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`timers throttle spf`

Syntax

```
timers throttle spf start-time <START-TIME> hold-time <HOLD-TINME>
max-wait-time <WAIT-TIME>
no timers throttle spf
```

Description

Configures timers for SPF calculation. There are three timers:

- **start-time** Is the initial delay before an SPF calculation is started. Default is 200 milliseconds.
- **hold-time** Is the progressive backoff time to wait before next scheduled SPF calculation. Default is 1000 milliseconds. If a route change event occurs during this period, the value doubles until it reaches the max-wait-time.
- **max-wait-time** Is the maximum time to wait before the next scheduled SPF calculation. Default is 5000 milliseconds. This is used to limit the SPF hold timer and also defines the time to be considered for which the OSPF LSDB has to be stable, after which the SPF throttle mechanism is reset.

The **no** form of this command sets all the configured non-default timers to default value.

Parameter	Description
`<START-TIME>`	Time in milliseconds to set timer for initial SPF delay. Default: 2 00.
`<HOLD-TINME>`	Time in milliseconds to set the minimum hold time between two consecutive SPF calculations. Default: 1000.
`<WAIT-TIME>`	Time in milliseconds to set the maximum wait time between tw o consecutive SPF calculations. Default: 5000.

Examples

Setting non-default timer values for SPF throttling:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# timers throttling spf start-time 500 hold-time
3000 max-wait-time 9000
Switch(config-ospfv3-1)# show running-config current-context
router ospfv3 1
```

```
        area 0.0.0.0
        area 0.0.0.1
        area 0.0.0.2 nssa no-summary
        area 0.0.0.3 stub
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# timers throttling spf start-time 500 hold-time
3000 max-wait-time 9000
Switch(config-ospfv3-1)# show running-config current-context
router ospfv3 1
        area 0.0.0.0
```

Setting default timer values for SPF throttling after configuring non-default values:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no timers throttling spf
```

NOTE

For more information on features that use this command, refer to the Multicast Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`transit-delay`

Syntax

```
transit-delay <DELAY>
no transit-delay
```

Description

Sets the time delay in Link state transmission for virtual links.

The **no** form of this command sets the delay in Link state transmission to the default of 1 second for virtual links.

Parameter	Description
`<DELAY>`	Specifies the time delay for the transit delay, in seconds. Range: 1 to 3600. Default: 1.

Examples

Setting OSPFv3 virtual links transit delay:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# transit-delay 30
```

Setting OSPFv3 virtual links transit delay to default:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# no transit-delay
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420	`config-router-vlink`	Administrators or local user group members with execution righ ts for this command.

Platforms	Command context	Authority
6200		

`trap-enable`

Syntax

```
trap-enable
no trap-enable
```

Description

Enables the notification of the events to be sent as traps to the SNMP management stations for OSPFv3.

The **no** form of this command disables the notification of the events to be sent as traps to the SNMP management stations for OSPFv3.

Examples

Enabling sending notification of events as traps:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# trap-enable
```

Disabling sending notification of events as traps:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no trap-enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

Platforms	Command context	Authority

Open Shortest Path First version 3 Address Family (OSPFv3 AF)

OSPFv3 AF extends the existing OSPFv3 protocol by adding support for multiple unicast address families as described in RFC5838. As an extension of OSPFv3, OSPFv3 AF retains all the functionalities, LSA types, and concepts of the original OSPFv3 protocol.

Subtopics

- **Overview**
- **Limitations**
- **How OSPFv3 AF protocol works**
- **OSPFv3 AF concepts**
- **OSPFv3 AF configuration task list**
- **Configuring OSPFv3 AF on the routing switch**
- **Assigning the routing switch to an OSPFv3 AF area**
- **Setting OSPFv3 AF network for the area**
- **Configuring external route redistribution and control**
- **Configuring area ranges on an ABR to reduce advertisements to the backbone**
- **Influencing route choice by changing the administrative distance**
- **Configuring graceful restart**
- **Configuring OSPFv3 AF virtual link settings**
- **Configuring OSPFv3 AF interface settings**
- **Configuring BFD for OSPFv3 AF**
- **Configuring all OSPFv3 AF interfaces as passive**
- **Configuring SPF throttling timers**
- **Viewing OSPFv3 AF information**
- **Clearing OSPFv3 AF statistics on a switch**
- **Configuring OSPFv3 AF over GRE Tunnels**
- **OSPFv3 AF commands**

Overview

The characteristics of OSPFv3 AF are:

- Provides a loop-free topology using SPF algorithm.

- Allows hierarchical routing using area 0 (backbone area) as the top of the hierarchy.
- Supports load balancing with equal cost routes for same destination.
- OSPFv3 AF is a classless protocol and allows for a hierarchical design with VLSM (Variable Length Subnet Masking) and route summarization.
- Scales enterprise size network easily with area concept.
- Provides fast convergence with triggered, incremental updates through Link State Advertisements (LSAs).
- Supports OSPFv3 AF operation over GRE IPv4 and IPv6 tunnel interfaces (platform-specific).

OSPFv3 AF configurations are supported in global router context, address-family context, and interface context. Some OSPFv3 AF router configurations are supported only in address-family context, while others are supported in both AF router and global router context. OSPFv3 AF can be configured on routed ports, VLAN interfaces, LAG interfaces, GRE tunnel interfaces, and loopback interfaces. All such configurations work in the mentioned interfaces context. OSPFv3 AF mandates the associated interface to be routing interface.

The switch supports congestion control (prioritizing hello packets, inactivity timer reset, and adjacency throttling).

Supported features

- OSPFv3 neighbor adjacency, Hello protocol, multiple areas, Inter-area routing
- AS external routes, stub areas, totally stubby areas, NSSA, ABR, ASBR
- Designated Router/Backup Designated Router
- Point-to-point interfaces/broadcast interfaces
- Virtual Links (Supported only by IPv6 AF)
- Bidirectional Forwarding Detection (BFD) - refer to the High Availability Guide for additional information
- Equal-cost multipath
- Null authentication, Simple password authentication, MD5 authentication
- Area range aggregation - Type-3/Type-7
- External route aggregation - Type-5 LSAs
- Graceful Restart - un-planned, restart interval, Graceful Restart Helper
- Stub router advertisement
- SPF throttling
- LSA Throttling

- Configuration of interface parameters such as priority, cost, hello-interval, dead-interval, retransmit-interval, transit-delay, etc.
- Configuration of virtual link parameters such as hello-interval, dead-interval, retransmit-interval, transit-delay, etc.
- Route redistribution
- Passive interfaces
- Multi VRF support with each VRF having up to eight OSPF process instances
- Congestion control (prioritizing hello packets, inactivity timer reset, and adjacency throttling)

> **NOTE**
> An OSPFv3 AF session will not come up in 6in6 tunnel.

Limitations

- The following configurations are not supported in router context:
  - area
  - bfd all-interfaces
  - default-metric
  - passive-interface-default
  - redistribute
  - maximum-paths
  - traps
  - default-information originate
  - summary-address
  - distribute-list
- Only one instance per address-family can be configured on the interface.
- Multicast address-families are not supported.
- OSPFv3 AF processes cannot establish neighborships with peers that do not support OSPFv3 AF or OSPFv3 Instance-IDs.

- OSPFv3 AF area-level IPsec authentication is not supported.
- Instance ID mismatch counter is not supported per interface. To identify a mismatch in instance IDs, the user must review the configuration of that particular interface. To help identify debug logs for mismatched instance IDs, the user must enable debug ospfv3-af and monitor debug logs for the specific interface.
- The Options field in the show ospfv3-af neighbor detail output does not display the address-family information.
- Simultaneous configurations of OSPFv3-AF with OSPFv2 or OSPFv3 on the same interface are not supported.

How OSPFv3 AF protocol works

OSPFv3 AF protocol

The protocol uses Link State Advertisements (LSAs) transmitted by each router to update neighboring routers regarding its interfaces and the routes available through those interfaces. Each routing switch in an area also maintains a link-state database (LSDB) that describes the area topology. (All routers in a given OSPF area have identical LSDBs). The routing switches used to connect areas to each other flood summary link LSAs and external link LSAs to neighboring OSPF areas to update them regarding available routes. Through this means, each OSPF router determines the shortest path between itself and a desired destination router in the same OSPF domain (Autonomous System (AS)). OSPFv3 AF supports unicast address-families by mapping each AF to a separate Instance ID and OSPFv3 AF instance.

The following ranges are defined for different AFs. The first value of each range is the default value for the corresponding AF.

Instance ID	Description
0 - 31	IPv6 unicast AF
64 - 95	IPv4 unicast AF

OSPFv3 AF concepts

OSPFv3 Address Family is a link-state routing protocol applied to routers grouped into OSPFv3 AF areas identified by the IPv4 and IPv6 routing configuration on each routing switch. Each OSPFv3 AF area includes one or more networks. OSPFv3 AF routers use hello packets and LSAs to maintain OSPFv3 AF operation across networks within an area and between areas within an OSPFv3 AF domain.

OSPFv3 AF uses hello packets to establish and maintain connections between neighboring routers on the same interface. Because neighborship and LSA exchanges are conducted using IPv6 link-local addresses, OSPFv3 AF requires an IPv6 address, even when operating with the IPv4 unicast address family.

Subtopics

OSPFv3 AF Link-state advertisement (LSA) types

OSPFv3 AF uses LSAs transmitted by each router to update neighboring routers regarding its interfaces and the routes available through those interfaces.

Each routing switch in an area also maintains an LSDB that describes the area topology. (All routers in a given OSPFv3 AF area have identical LSDBs, and each router uses the LSDB to build its own shortest-path tree.)

The routing switches used to connect areas to each other flood inter-area-prefix-LSAs, inter-area-router-LSAs, and AS-external-LSAs to backbone area to update it regarding available routes. Each OSPFv3 AF router determines the shortest path between itself and a desired destination router in the same OSPFv3 AF domain (AS). Routed traffic in an OSPFv3 AF AS is classified as one of the following:

- Intra-area traffic
- Inter-area traffic
- External traffic

The routing switches support the LSAs listed in the following table.

Link - state type	Description	Use	Flood scope
0x2001	Router - LSA	Describes the state of each active interface on a router for Area a given area. (Excludes loopback interfaces and interfaces that have not achieved full adjacency.)	Area
0x2002	Network - LSA	Describes the OSPFv3 AF routers in a given network.	Area

Link - state type	Description	Use	Flood scope
0x2003	Inter - area - prefix - L SA	Describes the route to a prefix in another OSPFv3 AF area of Area the same AS. (Excludes prefixes for link - local addresses.) Propagated through back bone area to other areas.	Area
0x2004	Inter - area - router - L SA	Describes the route to an ASBR in another OSPFv3 AF normal area (including the backbone area) of the same AS. Propagated through back bone area to other areas. (Excludes any ASBR in th e same area as the router sending the LSA.)	Area
0x4005	AS - external - LSA	Describes the route to a destination prefix in anot her AS (external route). (Excludes prefixes for link - local addresses.) Origi nated by ASBR in norma l or backbone areas of an AS and propagates throu gh backbone area to othe r normal areas. Does not flood over virtu al links and is not summa rized in virtual links. For i njection into an NSSA, an NSSA ABR generates a t ype - 7 - default - LSA a dvertising the default rou te (::/0).	AS
0x2007	NSSA - LSA	Describes the route to a destination in another A S (external route). Origin ated by ASBR in NSSA. A BR translates type - 7 L SAs to AS - external - L SAs for injection into the backbone area.	NSSA

Link - state type	Description	Use	Flood scope
0x0008	Link - LSA	For other routers on the s ame VLAN interface (exc ept virtual links), descri bes the router link - loca l address and any other I Pv6 prefixes reachable o n the VLAN. Link LSAs a re not flooded over virtua l links.	Link - local
0x2009	Intra - area - prefix - L SA	Generated on transit links within an area by the DR operating on those links. Also, every OSPFv3 AF ro uter generates this LSA to refer to stub and loopb ack prefixes on the route r.	Area

OSPFv3 AF area types

OSPFv3 AF is built upon a hierarchy of network areas. All areas for a given OSPFv3 AF domain reside in the same AS. An AS is defined as a number of contiguous networks that share an interior gateway routing protocol.

An AS can be divided into multiple areas, including the backbone (area 0). Because each area represents a collection of contiguous networks and hosts, the topology of a given area is not known by the internal routers in any other area. Areas define the boundaries to which router-LSAs and network-LSAs are broadcast. This functionality limits the amount of LSA flooding that occurs within the AS. It also helps to control the size of the link-state databases (LSDBs) maintained in OSPFv3 AF routers.

An area is represented in OSPFv3 AF by either a 32-bit dotted-decimal address or a number. Area types include: Backbone, Normal, Not-so-stubby (NSSA), and Stub.

Backbone area

Every AS must have one (and only one) backbone area (identified as area 0 or 0.0.0.0.) The ABRs of all other areas in the same AS connect to the backbone area, either physically through an ABR or through a configured, virtual link. The backbone is a special type of area and serves as a transit area for carrying the inter-area-prefix-LSAs, AS-external-LSAs, and routed traffic between non-backbone areas, as well as the router-LSAs, network-LSAs, and routed traffic internal to the area. ASBRs are allowed in backbone areas.

Normal area

A normal area allows inter-area-prefix-LSAs and AS-external-LSAs to and from the backbone area. A normal area connects to the AS backbone area through one or more ABRs (physically or through a virtual link) and allows router-LSAs and network LSAs within this area. ASBRs are allowed in normal areas.

Stub area

Stub area connects to the AS backbone through one or more ABRs. It does not allow an internal ASBR, and does not allow AS-external-LSAs. A stub area supports these actions:

- Advertise the area summary routes to the backbone area.
- Advertise inter-area routes from other areas.
- Use the inter-area-prefix-LSA default route to advertise routes to an ASBR and to other areas.

You can configure the stub area ABR to do the following:

- Suppress advertising on the area's summarized internal routes into the backbone area.
- Suppress LSA traffic from other areas in the AS by replacing inter-area-prefix-LSAs and the default external route from the backbone area with the default summary route (::/0.). This area is called totally stubby area.

Virtual links are not allowed for stub areas.

Not-so-stubby (NSSA) area

An NSSA area connects to the backbone area through one or more ABRs. NSSAs are used where an ASBR exists in an area where you want to:

- Block injection of external routes from other areas of the AS.
- Advertise type-7-LSA external routes (learned from the ASBR) to the backbone area as AS-external-LSAs.

NSSAs also support the following:

- Advertise inter-area-prefix-LSAs from the backbone area into the NSSA. (If no-summary is enabled, the NSSA ABR suppresses these LSAs from the backbone and, instead, injects the default route into the NSSA.)
- Advertise NSSA inter-area-prefix-LSAs to the backbone area.

Virtual links are not allowed for NSSAs.

OSPFv3 AF router types

Internal routers

Internal OSPFv3 AF routers belong to only one area. Internal routers flood router-LSAs to all routers in the same area and maintain identical LSDBs.

Area border routers (ABRs)

Area border routers have membership in multiple areas. ABRs are used to connect the various areas in an AS to the backbone area for that AS. Multiple ABRs can be used to connect a given area to the backbone, and a given ABR can belong to multiple areas other than the backbone.

An ABR maintains a separate LSDB for each area to which it belongs. (All routers within the same area have identical LSDBs.) The ABR is responsible for flooding inter-area-prefix-LSAs and inter-area router LSAs between its border areas.

You can reduce summary LSA flooding by configuring area ranges. An area range enables you to assign an aggregate address to a range of IPv6 addresses. This aggregate address is advertised instead of all the individual addresses it represents. You can assign up to eight ranges in an OSPFv3 AF area.

Autonomous system boundary router (ASBR)

Autonomous system boundary routers run one or more interior gateway protocols and serve as a gateway to other autonomous systems operating with interior gateway protocols. The ASBR imports and translates different protocol routes into OSPFv3 AF through redistribution. ASBRs can be used in backbone areas, normal areas, and NSSAs, but not in stub areas.

Designated routers (DRs)

In an OSPFv3 AF network having two or more routers, one router is elected to serve as the designated router (DR) and another router to act as the backup designated router (BDR). All other routers in the same network segment forward their routing information to the DR and BDR, and the DR forwards this information to all routers in the network. This functionality minimizes the amount of repetitive information that is forwarded on the network by eliminating the need for each individual router in the area to forward its routing information to all other routers in the network. If the area includes multiple networks, each network elects its own DR and BDR.

In an OSPFv3 AF network with no DR and no BDR, the neighboring router with the highest priority is elected the DR and the router with the next highest priority is elected the BDR. If the DR goes off-line, the BDR automatically becomes the DR, and the router with the next highest priority then becomes the new BDR. If multiple routing switches on the same OSPFv3 AF network are declaring themselves DRs, both priority and router ID are used to select the DR and BDR.

Priority is configurable using the `ospfv3-af ipv4 priority` or `ospfv3-af ipv6 priority` command at the interface level. You can use this command to help bias one router as the DR. If two neighbors share the same priority, the router with the highest router ID is designated the DR. The router with the next highest router ID is designated the BDR. Routers retain DR/BDR states throughout the period for which the adjacency is up and running.

OSPFv3 AF configuration task list

Tasks at a glance

- Configuring OSPFv3 AF on the routing switch
- Assigning the routing switch to an OSPFv3 AF area
- Setting OSPFv3 AF network for the area
- Configuring external route redistribution and control
- Configuring area ranges on an ABR to reduce advertisements to the backbone
- Influencing route choice by changing the administrative distance
- Configuring graceful restart
- Configuring OSPFv3 AF virtual link settings
- Configuring OSPFv3 AF interface settings
- Configuring BFD for OSPFv3 AF
-
- Viewing OSPFv3 AF information
- Clearing OSPFv3 AF statistics on a switch

Configuring OSPFv3 AF on the routing switch

Prerequisites

- You must be in the global configuration context, as indicated by the `switch(config)#` prompt to create the OSPFv3 AF instance and enter the OSPFv3 AF configuration context.
- To configure a router ID, and other OSPFv3 AF configuration you must be in the OSPFv3 AF router configuration context, as indicated by the `switch(config-ospfv3-af-1)#` prompt.

Create the OSPFv3 AF instance and enter the OSPFv3 AF configuration context. From this context, you can proceed with other OSPFv3 AF configuration.

Procedure

1. Create the OSPFv3 AF instance and enter the OSPFv3 AF configuration context using the following command. For command details, see router ospfv3-af.
   router ospfv3-af <PROCESS-ID> [vrf <VRF-NAME>]

   For example, the following command creates OSPF instance 1.

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)#
```

For example, the following commands enter the IPv4, or IPv6 AF context respectively.

```
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)#
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)#
```

2. Configure a global router ID using the following command. For command details, see router-id.
   router-id <ROUTER-ADDRESS>

   For example, the following command sets the router ID to 1.1.1.1.

```
switch(config-ospfv3-af-1)# router-id 1.1.1.1
```

3. Optionally, if the OSPFv3 AF process was disabled (it is enabled by default), enable the OSPFv3 AF process using the following command.
   enable

   For command details, see enable. (Refer to disable for disabling the OSPFv3 AF process).

Assigning the routing switch to an OSPFv3 AF area

Create a Normal, Stub, or Not So Stubby (NSSA) area.

Prerequisites

You must be in the router configuration context, as indicated by the `switch(config-router)#` prompt.

Procedure

Create an OSPFv3 AF area for the routing switch using one of the following commands:

- Create a normal area using the following command: `area <AREA-ID>`. For command details, see area.
- Create a stub area using the following command: `area <AREA-ID> stub`. For command details, see area stub.
- Create a Not-So-Stubby-Area using the command: `area <AREA-ID> nssa`. For command details, see area nssa.

For example, the following commands create either an IPv4 or IPv6 normal area with an area identifier of 1.1.1.1. Area identifier could alternatively be entered in decimal format such as 1.

```
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# area 1
switch(config-ospfv3-ipv4-uc)# area 1.1.1.1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1
switch(config-ospfv3-ipv6-uc)# area 1.1.1.1
```

Setting OSPFv3 AF network for the area

Prerequisites

Routing must be enabled on the interface where you are planning to enable OSPFv3 AF.

You must be in the interface configuration context, as indicated by the `switch(config-if)#` prompt.

After you define an OSPFv3 AF area, you can assign one or more networks to it. The OSPFv3 AF protocol will run on the interface with the configured link-local ipv6 address. The interfaces that have an IPv6 address configured in this network or in a subset of this network will participate in the OSPFv3 AF protocol.

Procedure

1. Set an OSPFv3 AF network for the area using any of the following commands. For command details, see ospfv3-af ipv4 area or ospfv3-af ipv6 area.
   ospfv3-af <PROCESS-ID> ipv4 area <AREA-ID>

   ospfv3-af <PROCESS-ID> ipv6 area <AREA-ID>

   For example, use the following command to assign interface 1/1/1 to OSPFv3 AF area 1. The area can alternatively be entered in IO address format.

   ```
   switch(config)# interface 1/1/1
   switch(config-if)# ospfv3-af 1 ipv4 area 1
   switch(config)# interface 1/1/1
   switch(config-if)# ospfv3-af 1 ipv6 area 1
   ```

2. Optionally, you can disable OSPFv3 AF on the interface using any of the following commands. For command details, see ospfv3-af ipv4 shutdown or ospfv3-af ipv6 shutdown.
   ospfv3-af ipv4 shutdown

   ospfv3-af ipv6 shutdown

   ```
   switch(config)# interface 1
   switch(config-if)# ospfv3-af ipv4 shutdown
   switch(config)# interface 1
   ```

```
switch(config-if)# ospfv3-af ipv6 shutdown
```

Configuring external route redistribution and control

About this task

Configuring route redistribution for OSPFv3 AF establishes the routing switch as an ASBR for importing and translating different protocol routes from other IGP domains into an OSPFv3 AF domain. When you configure redistribution for OSPFv3 AF, you can specify that routes external to the OSPFv3 AF domain are imported as OSPFv3 AF routes.

Procedure

1. Enable route redistribution using the `redistribute` command.

   ```
      redistribute {bgp | connected | host-routes | local loopback | static
   | ripng | ospfv3-af <PROCESS-ID>}
                       [route-map <ROUTE-MAP-NAME>]
   ```

   For example, setting redistribution of connected routes as OSPFv3 AF routes:

   ```
   switch(config-ospfv3-ipv4-uc)# redistribute connected
   ```

   If a route map is specified, then only the routes that pass the match clause specified in the route map are redistributed to OSPFv3 AF. For example, setting redistribution of local loopback routes, that only match the routes specified by the route map:

   ```
   switch(config-ospfv3-ipv4-uc)# redistribute local loopback route-map
   local_routes
   ```

2. Optionally, modify the default metric for redistribution using the `default-metric` command.

   ```
      default-metric <METRIC-VALUE>
   ```

   For example, setting the default metric for redistribution to 37.

   ```
   switch(config-ospfv3-ipv4-uc)# default-metric 37
   ```

3. Optionally, set the cost of default-summary LSAs using the area command.

   ```
      area <AREA-ID> default-metric <COST>
   ```

   For example, setting the cost of default summary LSAs to 2.

```
switch(config-ospfv3-ipv4-uc)# area 1 default-metric 2
```

4. Optionally, use the `max-metric router-lsa` command to set the protocol to advertise a maximum metric so that other routers do not prefer this router as an intermediate hop in their shortest path first (SPF) calculations.

   ```
   max-metric router-lsa [include-stub] | [on-startup <INTERVAL>]
   ```

   For example, setting advertise max-metric router-lsa on startup.

   ```
   switch(config-ospfv3-af-1)# max-metric router-lsa on-startup 3000
   ```

5. Optionally set the maximum number of ECMP routes that OSPFv3 can support using the `maximum-paths` command.

   ```
   maximum-paths <MAXIMUM>
   ```

   For example, setting the maximum number of ECMP routes to 8.

   ```
   switch(config-ospfv3-ipv4-uc)# maximum-paths 8
   ```

Configuring area ranges on an ABR to reduce advertisements to the backbone

You can configure area ranges to reduce inter-area advertisements by summarizing a range of IP addresses into a single route advertisement. This action prevents an ABR from advertising specific networks or subnets to the backbone area.

Prerequisites

You must be in the address-family configuration context, as indicated by the `switch(config-ospfv3-ipv4-uc)#` prompt.

Procedure

Summarize inter-area or NSSA paths using the following command. For command details, see area range.

```
area <AREA-ID> range <IP-PREFIX> type {inter-area | nssa} [no-advertise]
```

For example, use any of the following commands to summarize routes matching the area range fd00::64 using inter-area as the type of address aggregation.

```
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# area 1 range fd00::/64 type inter-area
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1 range fd00::/64 type inter-area
```

In another example, use the following command to specify DoNotAdvertise status for routes matching the area range 10.1.0.0./16. Use nssa as the type of address aggregation.

```
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# area 1 range 10.1.0.0/16 type inter-area no
advertise
```

Influencing route choice by changing the administrative distance

Prerequisites

You must be in the OSPFv3 AF router configuration context, as indicated by the `switch(config-ospfv3-af-1)#` prompt.

The administrative distance is used to influence the selection of routes learned by different protocols.

Procedure

Reconfigure the administrative distance using the following command. For command details, see distance.

```
distance [<DISTANCE-VAL> | intra-area [<DISTANCE-VAL>] | inter-area
[<DISTANCE-VAL>] | external [<DISTANCE-VAL>]]
```

For example, use the following command to set administrative distance to 100.

```
switch(config-ospfv3-ipv4-uc)# distance 100
```

Configuring graceful restart

Prerequisites

You must be in the OSPFv3 address-family configuration context, as indicated by the `switch(config-ospfv3-ipv4-uc)#` prompt.

OSPFv3 AF routing can be gracefully restarted on switches without losing packets that are in transit. There is no effect on the saved switch configuration.

Manually create a router-id configuration to support graceful restart and management module failover. A dynamic router-id is chosen from the list of configured IPs at the time of the router-id election, and in the event of a graceful restart and management module failover, the router-id should be preserved before and after these events. Graceful restart would fail if there is a change in the router-id, causing traffic loss.

Procedure

Configure graceful restart of using the following command. For command details, see graceful-restart.

```
graceful-restart {restart-interval <INTERVAL> | helper [strict-lsa-check]}
```

For example, the following command specifies 50 seconds as the maximum interval another router will wait for this router to gracefully restart.

```
switch(config-ospfv3-ipv4-uc)# graceful-restart restart-interval 50
```

Configuring OSPFv3 AF virtual link settings

About this task

You must be in the router vlink configuration context, as indicated by the `switch(config-router-vlink)#` prompt.

The OSPFv3 AF interface parameters for this process are automatically set to their default values for virtual links. You can optionally adjust the following OSPFv3 AF virtual link settings.

Procedure

1. Set the time interval between OSPF hello packets for the OSPFv3 AF virtual links using the following command. For command details, see hello-interval.

   hello-interval <seconds>

   For example:

   ```
   switch(config)# router ospfv3-af 1
   switch(config)# address-family ipv6 unicast
   switch(config-ospfv3-ipv4-uc)# area 100 virtual-link 100.0.1.1
   switch(config-router-vlink)# hello-interval 30
   ```

2. Set the interval after which a neighbor is declared dead if no hello packet is received on the OSPFv3 AF virtual links. Use the following command. For command details, see dead-interval.

   dead-interval <seconds>

3. Set the time between retransmitting lost link state advertisements for the OSPFv3 AF virtual links using the following command. For command details, see retransmit-interval.

   retransmit-interval <seconds>

4. Sets the transit delay in Link state transmission for the OSPFv3 AF virtual links using the following command. For command details, see transit-delay.

   transit-delay <seconds>

Configuring OSPFv3 AF interface settings

Prerequisites

You must be in the interface configuration context, as indicated by the `switch(config-if)#` prompt.

You can optionally adjust the following OSPFv3 AF interface settings.

1. Set the interface cost using any of the following commands. For command details, see ospfv3-af ipv4 cost or ospfv3-af ipv6 cost.

   ```
   ospfv3-af ipv4 cost <INTERFACE-COST>
   ospfv3-af ipv6 cost <INTERFACE-COST>
   ```

   For example, the following command sets the cost associated with interface 1/1/1 to 100.

   ```
   switch(config)# interface 1/1/1
   switch(config-if)# ospfv3-af ipv4 cost 100
   switch(config)# interface 1/1/1
   switch(config-if)# ospfv3-af ipv6 cost 100
   ```

2. Set the time interval between OSPFv3 AF hello packets for the OSPFv3 AF interface using the following command. For command details, see ospfv3-af ipv6 hello-interval.

   ```
   ospfv3-af ipv6 hello-interval <INTERVAL>
   ```

3. Set the interval after which a neighbor is declared dead if no hello packet is received on the OSPFv3 AF interface using any of the following command. For command details, see ospfv3-af ipv4 dead-interval or ospfv3-af ipv6 dead-interval.
   ospfv3-af ipv4 dead-interval <INTERVAL>

   ospfv3-af ipv6 dead-interval <INTERVAL>

4. Set the time between re-transmitting lost link state advertisements for the OSPFv3 AF interface using any of the following command. For command details, see ospfv3-af ipv4 retransmit-interval or ospfv3-af ipv6 retransmit-interval.
   ospfv3-af ipv4 retransmit-interval <INTERVAL>

   ospfv3-af ipv6 retransmit-interval <INTERVAL>

5. Sets the time delay in Link state transmission for the OSPFv3 AF interface using any of the following command. For command details, seeospfv3-af ipv4 transit-delay or ospfv3-af ipv6 transit-delay.
   ospfv3-af ipv4 transit-delay <INTERVAL>

   ospfv3-af ipv6 transit-delay <INTERVAL>
6. Set the OSPFv3 AF network type for the interface. For command details, see ospfv3-af ipv4 network or ospfv3-af ipv6 network.
   ospfv3-af ipv4 network {broadcast|point-to-point}

   ospfv3-af ipv6 network {broadcast|point-to-point}
7. Set the OSPFv3 AF priority on the interface using any of the following command. For command details, see ospfv3-af ipv4 priority or ospfv3-af ipv6 priority.
   ospfv3-af ipv4 priority <number-value>

   ospfv3-af ipv6 priority <number-value>
8. Set the OSPFv3 AF interface as OSPFv3 AF passive interface. With this setting the interface participates in OSPFv3 AF but does not send or receive packets on that interface. For command details, see ospfv3-af ipv4 passive orospfv3-af ipv6 passive.
   ospfv3-af ipv4 passive ospfv3-af ipv6 passive

Configuring BFD for OSPFv3 AF

About this task

Enabling BFD Support for OSPFv3 AF over IPv6 enables OSPFv3 AF to register with Bidirectional Forwarding Detection (BFD) to receive forwarding path detection failure messages. You can either configure BFD for OSPFv3 AF globally on all interfaces or configure it selectively on one or more interfaces. BFD creates a session in asynchronous mode as soon as the switch reaches the 2-Way state with a neighbor. Once the session is established BFD will commence sending echos for path failure detection.

OSPFv3 AF uses IPv6 for neighborship in both IPv4 and IPv6 unicast address-families. Therefore, only a single BFD session is created to monitor the IPv6 link-local address, even when both address-families are enabled on the same interface.

There are two methods for enabling BFD for OSPFv3 AF:

Procedure

1. Enable BFD for all interfaces enabled for OSPFv3 AF by using the BFD all-interfaces command in address-family configuration mode
2. Enable BFD for a subset of interfaces that have OSPFv3 AF enabled by using the `ospfv3-af bfd` command in interface configuration mode

Results

> **NOTE**
> OSPFv3 AF needs to be enabled prior to enabling BFD on one or more interfaces.

Subtopics

[Examples](#examples)

Examples

Enable BFD on all OSPFv3 AF Interfaces

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# bfd all-interfaces
```

Disable BFD on all OSPFv3 AF Interfaces

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# no bfd all-interfaces
```

Enable BFD on a single OSPFv3 AF interface

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af ipv4 bfd
switch(config-if)# ospfv3-af ipv6 bfd
```

Disable BFD on a single OSPFv3 AF interface

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af ipv4 bfd disable
switch(config-if)# ospfv3-af ipv6 bfd disable
```

Set BFD as default on an OSPFv3 AF interface

```
switch(config)# interface 1/1/1
switch(config-if)# no ospfv3-af ipv4 bfd
switch(config-if)# no ospfv3-af ipv6 bfd
```

Configuring all OSPFv3 AF interfaces as passive

Prerequisites

You must be in the OSPFv3 address-family configuration context, as indicated by the `switch(config-ospfv3-ipv4-uc)#` prompt.

OSPFv3 AF sends LSAs to all other routers in the same AS. To limit the flooding of LSAs throughout the AS, you can configure OSPFv3 AF to be passive.

Procedure

Configure all OSPFv3 AF interfaces as passive using the following command. For command details, see passive-interface default.

```
passive-interface default
switch(config-ospfv3-ipv4-uc)# passive-interface default
switch(config-ospfv3-ipv6-uc)# passive-interface default
```

Configuring SPF throttling timers

SPF calculation is throttled with default timer values (start-time 200ms, hold-time 1000ms, max-wait-time 5000ms). You can throttle SPF calculation by configuring non-default timers to improve performance of a specific network configuration.

Prerequisites

SPF throttling timers can be configured on any of the following contexts:

- Address-family configuration context, as indicated by the `switch(config-ospfv3-ipv4-uc)#` or `switch(config-ospfv3-ipv6-uc)#` prompt.
- Router configuration context, as indicated by the `switch(config-router)#` prompt.

Procedure

Reconfigure SPF throttling timers using the following command. For command details, see timers throttle spf.

```
timers throttle spf start-time <milliseconds> hold-time <milliseconds> max-wait-time <milliseconds>
```

For example, use the following command to set start-time to 500ms, hold-time to 3000ms and max-wait-time to 9000ms:

```
switch(config-ospfv3-ipv4-uc)# timers throttle spf start-time 500 hold-time 3000 max-wait-time 9000
```

Viewing OSPFv3 AF information

Prerequisites

Use these show commands from the Manager context, as indicated by the `switch#` prompt.

Procedure

To view OSPFv3 AF information, use the following commands. For command details and examples, click the links.

- To view general OSPFv3 AF, area, state and configuration information use: show ospfv3-af ipv4 or show ospfv3-af ipv6.
- To view information about OSPFv3 AF interfaces use: show ospfv3-af ipv4 interface or show ospfv3-af ipv6 interface.
- To view information about OSPFv3 AF neighbors use: show ospfv3-af ipv4 neighbors or show ospfv3-af ipv6 neighbors .
- To view OSPFv3 AF routing table information use: show ospfv3-af ipv4 routes or show ospfv3-af ipv6 routes.
- To view OSPFv3 AF statistics for the OSPFv3 AF interfaces use show ospfv3-af ipv4 statistics interface or show ospfv3-af ipv6 statistics interface.

Clearing OSPFv3 AF statistics on a switch

Clear OSPFv3 AF event statics using the following command. For command details, see clear ospfv3-af ipv4 statistics or clear ospfv3-af ipv6 statistics .

```
clear ospfv3-af [<PROCESS-ID>] <address-family> statistics [interface
[interface <INTERFACE-NAME> | vlan <VLAN-ID> | lag <LAG-ID>] | loopback
<LOOPBACK-ID> | tunnel <TUNNEL-ID>] [all-vrfs | vrf <VRF-NAME>]
```

For example, the following command clears OSPFv3 AF event statistics from interface 1/1/1.

```
switch# clear ospfv3-af ipv4 statistics interface 1/1/1
switch# clear ospfv3-af ipv6 statistics interface 1/1/1
```

Configuring OSPFv3 AF over GRE Tunnels

OSPFv3 Address Family (AF) can be configured on GRE IPv4 tunnel interfaces to establish OSPFv3 AF adjacencies through tunneled connections. Both IPv4 and IPv6 address families are supported over GRE tunnels. This feature is supported on the 5420, 6300, 6400, 8320, 8325, 8360, 9300, and 10000 Switch series.

Prerequisites

You must be in the global configuration context, as indicated by the `switch(config)#` prompt, to create and configure GRE tunnel interfaces.

You must be in the tunnel interface configuration context, indicated by the `switch(config-gre-if)#` prompt, to enable or disable OSPFv3 AF on the GRE tunnel interface.

> **NOTE**
>
> Configure the tunnel interface IP MTU to 1476 bytes using **ip mtu 1476** when running OSPFv3 AF over GRE. This accommodates the 24-byte GRE encapsulation overhead (20-byte outer IP header plus 4-byte GRE header), ensuring packets fit within the default Layer-2 MTU and avoid fragmentation

Procedure

Configure the physical underlay interface with an IPv4 address, regardless of whether you are using OSPFv3 AF for IPv4 or IPv6.

```
switch(config)# interface 1/1/4
switch(config-if)# no shutdown
switch(config-if)# ip address 20.1.1.1/24
switch(config-if)# ip address 30.1.1.1/24
```

Create and configure the GRE tunnel interface.

For and IPv4 address:

```
switch(config)# interface tunnel 1 mode gre ip
switch(config-gre-if)# source ip 20.1.1.1
switch(config-gre-if)# destination ip 20.1.1.2
switch(config-gre-if)# ip address 30.1.1.1/24
switch(config-gre-if)# ospfv3-af 1 ipv4 area 0.0.0.0 instance 64
switch(config-gre-if)# no shutdown
```

For an IPv6 address:

```
switch(config)# interface tunnel 1 mode gre ip
switch(config-gre-if)# source ip 30.1.1.1
switch(config-gre-if)# destination ip 30.1.1.2
switch(config-gre-if)# ipv6 address 3001::1/64
switch(config-gre-if)# ospfv3-af 1 ipv6 area 0.0.0.0 instance 0
switch(config-gre-if)# no shutdown
```

Configure the OSPFv3 AF process for an IPv4 address.

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# router-id 1.1.1.1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-af-1)# area 0.0.0.0
switch(config-ospfv3-af-1)# exit-address-family
```

Configure the OSPFv3 AF process for an IPv6 address.

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# router-id 1.1.1.1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-af-1)# area 0.0.0.0
switch(config-ospfv3-af-1)# exit-address-family
```

Enable OSPFv3 AF on the GRE tunnel interface. If the OSPFv3 AF process or area does not exist, the system prompts you to create them.

- Use A.B.C.D to specify an OSPFv3 AF area ID in the specific IP address.
- Use 0-4294967295 to specify an OSPFv3 AF area ID.

For an IPv4 address:

```
switch(config)# interface tunnel 1
switch(config-gre-if)# ospfv3-af 1 ipv4 area 0.0.0.0
```

For an IPv6 address:

```
switch(config)# interface tunnel 1
switch(config-gre-if)# ospfv3-af 1 ipv6 area 0.0.0.0
```

In this configuration:

- OSPFv3 process 1 is configured with router-id 1.1.1.1.
- IPv4 and IPv6 unicast address family are enabled with area 0.0.0.0.
- The physical interface (1/1/4) provides underlay connectivity with IPv4 address 20.1.1.1/24 or 30.1.1.1/24.
- A GRE tunnel interface (tunnel 1) is created with `mode gre ipv4`.
- The tunnel source is the IPv4 address of the physical interface (20.1.1.1 or 30.1.1.1).
- The tunnel destination is the remote endpoint IPv4 address (20.1.1.2 or 30.1.1.2).
- The tunnel interface has an IPv4 address (30.1.1.1/24).
- The tunnel interface has an IPv6 address (3001::1/64).

Disable OSPFv3 AF on the IPv4 or IPv6 GRE tunnel interface

```
switch(config-gre-if)# no ospfv3-af 1 ipv4 area 0
switch(config-gre-if)# no ospfv3-af 1 ipv4 area 0.0.0.0
switch(config-gre-if)# no ospfv3-af 1 ipv6 area 0
switch(config-gre-if)# no ospfv3-af 1 ipv6 area 0.0.0.0
```

Use the `show ospfv3-af 1 ipv4 neighbors` or the `show ospfv3-af 1 ipv6 neighbors` command to view OSPFv3 AF neighbor adjacencies established over GRE tunnel interfaces.

```
(config-gre-if)# show ospfv3-af 1 ipv4 neighbors
Address-family : IPv4 Unicast
------------------------------

VRF : default                          Process : 1
===================================================

Total Number of Neighbors: 1

Neighbor ID      Priority  State            Interface
-------------------------------------------------------
2.2.2.2          n/a       FULL             tunnel1
  Neighbor address fe80::b86a:97a0:122:1c42


(config-gre-if)# show ospfv3-af 1 ipv6 neighbors
Address-family : IPv6 Unicast
------------------------------

VRF : default                          Process : 1
===================================================

Total Number of Neighbors: 1

Neighbor ID      Priority  State            Interface
-------------------------------------------------------
2.2.2.2          n/a       FULL             tunnel1
  Neighbor address fe80::b86a:97a0:122:1c42
```

This output shows:

- **Address-family:** Indicates whether IPv4 Unicast or IPv6 Unicast address family.
- **Neighbor ID:** Router ID of the OSPFv3 AF neighbor.
- **Priority:** n/a for point-to-point interfaces like GRE tunnels.
- **State:** FULL indicates the adjacency is fully established.

- **Neighbor address:** IPv4 or IPv6 link-local address of the neighbor on the tunnel interface.
- **Interface:** The GRE tunnel interface name (tunnel1).

Use the `show interface tunnel` command to verify the tunnel interface status.

For OSPFv3 AF using IPv4 addresses:

```
(config)# show interface tunnel 1

 Interface tunnel1 is up
 Admin state is up
 Description:
 Tunnel type: gre ipv4
 Tunnel source: IPv4 address 20.1.1.1
 Tunnel destination: IPv4 address 20.1.1.2
 Tunnel ttl: 64
 Tunnel IP address: 30.1.1/24
 IPv6 address: 30::1/64
```

For OSPFv3 AF using IPv6 addresses:

```
(config)# show interface tunnel 1

 Interface tunnel1 is up
 Admin state is up
 Description:
 Tunnel type: gre ipv4
 Tunnel source: IPv4 address 30.1.1.1
 Tunnel destination: IPv4 address 30.1.1.2
 Tunnel ttl: 64
 IPv6 address: 30::1/64
```

Additional Configurations

All standard OSPFv3 AF interface configuration commands are supported on GRE tunnel interfaces, including:

- ospfv3-af ipv4 hello-interval
- ospfv3-af ipv4 dead-interval
- ospfv3-af ipv4 retransmit-interval
- ospfv3-af ipv4 transmit-delay
- ospfv3-af ipv4 cost
- ospfv3-af ipv4 priority (not applicable for point-to-point)

- ospfv3-af ipv4 shutdown
- ospfv3-af ipv6 hello-interval
- ospfv3-af ipv6 dead-interval
- ospfv3-af ipv6 retransmit-interval
- ospfv3-af ipv6 transmit-delay
- ospfv3-af ipv6 cost
- ospfv3-af ipv6 network
- ospfv3-af ipv6 priority (not applicable for point-to-point)
- ospfv3-af ipv6 shutdown

> **NOTE**
> For OSPFv3 AF to operate over a GRE tunnel, the tunnel interface must be configured with the appropriate IPv4 address and have reachable underlay connectivity between the tunnel source and destination. OSPFv3 AF packets are encapsulated in GRE and then in the IP address, using the source and destination address configured on the tunnel. Neighbor adjacencies over GRE are established using IPv6 link-local addresses. Both IPv4 and IPv6 address families can be configured simultaneously on the same GRE tunnel interface.

OSPFv3 AF commands

Select a command from the list in the left navigation menu.

Subtopics

`active-backbone stub-default-rout`

Syntax

```
active-backbone stub-default-route
```

Description

This command enables the router to send a default route to stub areas if there is an active loopback link in the backbone area. The configuration is not required if backbone area has neighbors or passive interfaces configured. By default, active backbone detection is disabled.

Parameter	Description
stub-default-route	Advertise default route to stub if loopback in backbone area.

Examples

Enabling the router to send an IPv4 default route to stub areas:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# active-backbone stub-default-route
```

Enabling the router to send an IPv6 default route to stub areas:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# active-backbone stub-default-route
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	config-ospfv3-ipv4-uc<*PROCESS-ID*> config-ospfv3-ipv6-uc<*PROCESS-ID*>	Administrators or local user group members with execution righ ts for this command.

`address-family unicast`

Syntax

```
address-family { <ipv4> | <ipv6> } unicast
```

Description

This command specifies the IPv4/IPv6, address-family and enters address family configuration mode.

Examples

Specifying an IPv4 address-family:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
```

Specifying an IPv6 address-family:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-af-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`area default-metric`

Syntax

```
area <AREA-ID> default-metric <METRIC>
no area <AREA-ID> default-metric
```

Description

Sets the cost of default-summary LSAs announced to the stub/nssa areas.

The **no** form of this command resets the cost of the default-summary LSAs announced to stub/nssa areas to the default of 1.

Parameter	Description
<AREA-ID>	Specifies the area ID as one of the following formats. • OSPF area identifier in IPv4 format (**x.x.x.x**), where **x** is a d ecimal number from 0 to 255. • OSPF area identifier in decimal format. Range: 0 to 429496 7295.
`default-metric` *<METRIC>*	Specifies the default metric of default - summary LSAs announ ced to the stub/nssa areas, to the specified value. Default: 1. Ra nge: 0 to 16777215.

Examples

Setting cost for default IPv4 LSA summary:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# area 1 default-metric 2
switch(config-ospfv3-ipv4-uc)# area 1.1.1.1 default-metric 2
```

Setting cost for default IPv4 LSA summary to default:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# no area 1 default-metric
switch(config-ospfv3-ipv4-uc)# no area 0.0.0.1 default-metric
switch(config-ospfv3-ipv4-uc)# no area 0.0.0.1 default-metric 2
```

Setting cost for default IPv6 LSA summary:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1 default-metric 2
```

```
switch(config-ospfv3-ipv6-uc)# area 1.1.1.1 default-metric 2
```

Setting cost for default IPv6 LSA summary to default:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# no area 1 default-metric
switch(config-ospfv3-ipv6-uc)# no area 0.0.0.1 default-metric
switch(config-ospfv3-ipv6-uc)# no area 0.0.0.1 default-metric 2
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-ipv4-uc<PROCESS-ID>` `config-ospfv3-ipv6-uc<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

area nssa (OSPFv3-AF)

Syntax

```
area <AREA-ID> nssa [no-summary]
no area <AREA-ID> nssa [no-summary]
```

Description

This command sets the area type as NSSA. If area is present and not NSSA area, changes the area type to NSSA area. If **no-summary** is configured the area type will be NSSA No-Summary.

Parameter	Description
`<AREA-ID>`	Specifies the area ID is one of the following formats. • OSPF area identifier in IPv4 format (**x.x.x.x**), where **x** is a d ecimal number from 0 to 255. • OSPF area identifier in decimal format. Range: 0 to 429496 7295. Required
`nssa`	Configure OSPFv3 AF area as NSSA. Required
`no-summary`	Do not inject inter - area routes into NSSA. Optional
no	Unsets the area type as NSSA. This means that the configured area becomes an area type default. The **no area area - id nssa no - summary** command will stop sending Area Border Router (ABR) summary link advertisements into the NSSA area, but wil l not unset the area as NSSA. Optional

Examples

Creating an NSSA area for OSPFv3 AF:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# area 1 nssa
switch(config-ospfv3-ipv4-uc)# area 1 nssa no-summary
```

Clearing the NSSA area for OSPFv3 AF:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# no area 1 nssa
switch(config-ospfv3-ipv6-uc)# no area 1 nssa no-summary
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-ipv4-uc`*<PROCESS-ID>* `config-ospfv3-ipv6-uc`*<PROCESS-ID>*	Administrators or local user group members with execution righ ts for this command.

`area`

Syntax

```
area <AREA-ID>
no area <AREA-ID>
```

Description

Creates a normal area with **<AREA-ID>** set if not present. If area is present and is not the normal area, this command changes the area type to normal area.

The **no** form of this command deletes the area with the **<AREA-ID>** specified. The area can be of any type (stub, stub no-summary, and default normal area).

Parameter	Description
<AREA-ID>	Specifies the area ID is one of the following formats. • OSPFv3 AF area identifier in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255.

Parameter	Description
	• OSPFv3 AF area identifier in decimal format. Range: 0 to 4294967295.

Examples

Creating an IPv4 normal area:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# area 1
switch(config-ospfv3-ipv4-uc)# area 1.1.1.1
```

Creating an IPv6 normal area:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1
switch(config-ospfv3-ipv6-uc)# area 1.1.1.1
```

Deleting an IPv4 area:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# no area 1
switch(config-ospfv3-ipv4-uc)# no area 0.0.0.1
```

Deleting an IPv6 area:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# no area 1
switch(config-ospfv3-ipv6-uc)# no area 1.1.1.1
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-ipv4-uc<PROCESS-ID>` `config-ospfv3-ipv6-uc<PROCESS-ID>`	Administrators or local user group members with execution rights for this command.

area range

Syntax

```
area <AREA-ID> range <IP-PREFIX> type {inter-area | nssa} [no-advertise]
no area <AREA-ID> range <IP-PREFIX> type {inter-area | nssa} [no-advertise]
```

Description

Summarizes the routes with the matching address or masks for OSPFv3 AF. This command only works for border routers.

The **no** form of this command unsets the route summarization for the configured IP prefix address on the ABR. When using the **no** form of the command with the **no-advertise** option, enables advertising this range to other areas.

Parameter	Description
`<AREA-ID>`	Specifies the area identifier as one of the following formats. • OSPFv3 AF area identifier in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255. • OSPFv3 AF area identifier in decimal format. Range: 0 to 4294967295.
`range <IP-PREFIX>`	Specifies summarizing routes matching the area range prefix/mask in IPv4 or IPv6 format.

Parameter	Description
`type {inter-area | nssa}`	Specifies the type this address aggregation applies to as either inter - area range prefix or NSSA range prefix.
`no-advertise`	Specifies the address range status as **DoNotAdvertise** (do not advertise this range to other areas).

Examples

Summarizing inter-area or NSSA paths on OSPFv3 AF for IPv4:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# area 1 range 10.1.0.0/16 type inter-area
switch(config-ospfv3-ipv4-uc)# area 1 range 10.1.0.0/16 type inter-area no
advertise
switch(config-ospfv3-ipv4-uc)# area 1 range 10.1.0.0/16 type nssa
switch(config-ospfv3-ipv4-uc)# area 1 range 10.1.0.0/16 type nssa no-
advertise
```

Unsetting summarization on OSPFv3 AF for IPv4:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# no area 1 range 10.1.0.0/16 type inter-area
switch(config-ospfv3-ipv4-uc)# no area 1 range 10.1.0.0/16 type inter-area
no advertise
switch(config-ospfv3-ipv4-uc)# no area 1 range 10.1.0.0/16 type nssa
switch(config-ospfv3-ipv4-uc)# no area 1 range 10.1.0.0/16 type nssa no-
advertise
```

Summarizing inter-area or NSSA paths on OSPFv3 AF for IPv6:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1 range fd00::1/64 type inter-area
switch(config-ospfv3-ipv6-uc)# area 1 range fd00::1/64 type inter-area no
advertise
switch(config-ospfv3-ipv6-uc)# area 1 range fd00::1/64 type nssa
switch(config-ospfv3-ipv6-uc)# area 1 range fd00::1/64 type nssa no-
advertise
```

Unsetting summarization on OSPFv3 AF for IPv6:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
```

```
switch(config-ospfv3-ipv6-uc)# no area 1 range fd00::1/64 type inter-area
switch(config-ospfv3-ipv6-uc)# no area 1 range fd00::1/64 type inter-area
no advertise
switch(config-ospfv3-ipv6-uc)# no area 1 range fd00::1/64 type nssa
switch(config-ospfv3-ipv6-uc)# no area 1 range fd00::1/64 type nssa no-
advertise
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-ipv4-uc<PROCESS-ID>` `config-ospfv3-ipv6-uc<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

area stub

Syntax

```
area <AREA-ID> stub [no-summary]
no area <AREA-ID> stub [no-summary]
```

Description

Creates the stub area with **<AREA-ID>** if not present. If area is present and is not the stub area, this command changes the area type to stub area. If **no-summary** is used, area type will be totally stubby area.

The **no** form of this command unsets the area type as stub. The configured area will be changed to the default normal area. The **no area <AREA-ID> stub no-summary** command will start sending Area Border Router (ABR) summary link advertisements into the stub area, but will not unset the stub area.

> **NOTE**
>
> ABR does not inject the default route in a Totally Stubby Area with loopback in Area 0.0.0.0. As a workaround, configure a passive interface or active neighbors in the backbone area.

Parameter	Description
<AREA-ID>	Specifies the area ID as one of the following formats. • OSPF area identifier in IPv4 format (**x.x.x.x**), where **x** is a d ecimal number from 0 to 255. • OSPF area identifier in decimal format. Range: 0 to 429496 7295.
`stub [no-summary]`	Specifies stub area type. If area is present and not stub area, thi s parameter changes the area type to stub area. If **no - summa ry** is specified, area type will be totally stubby area, which mean s do not inject inter - area routes into stub.

Examples

Creating an IPv4 stub area:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# area 1 stub
switch(config-ospfv3-ipv4-uc)# area 1 stub no-summary
```

Unsetting the IPv4 stub area type:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc) # no area 1 stub
switch(config-ospfv3-ipv4-uc) # no area 1 sub no-summary
```

Creating an IPv6 stub area:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1 stub
```

```
switch(config-ospfv3-ipv6-uc)# area 1 stub no-summary
```

Unsetting the IPv6 stub area type:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc) # no area 1 stub
switch(config-ospfv3-ipv6-uc) # no area 1 sub no-summary
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	config-ospfv3-ipv4-uc<*PROCESS-ID*> config-ospfv3-ipv6-uc<*PROCESS-ID*>	Administrators or local user group members with execution righ ts for this command.

area virtual-link

Syntax

```
area <AREA-ID> virtual-link <ROUTER-ID>
no area <AREA-ID> virtual-link <ROUTER-ID>
```

Description

Creates an OSPF virtual link with remote ABR (if not created already) and enters the vlink context. The configuration is supported only in ipv6-unicast address-family.

The **no** form of this command deletes an OSPF virtual link with the specified router ID of the remote ABR. If no **<ROUTER-ID>** is specified, the **no** form of the command sets the virtual link to the default settings.

Parameter	Description
`<AREA-ID>`	Specifies the area ID is one of the following formats. • OSPF area identifier in IPv4 format (**x.x.x.x**), where **x** is a d ecimal number from 0 to 255. • OSPF area identifier in decimal format. Range: 0 to 429496 7295.
`virtual-link <ROUTER-ID>`	Configures a virtual link with the specified router ID of the rem ote ABR.

Examples

Configuring OSPFv3 AF virtual links:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 100 virtual-link 100.0.1.1
```

Deleting OSPFv3 AF virtual links:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# no area 100 virtual-link 100.0.1.1
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-ipv4-uc<PROCESS-ID>` `config-ospfv3-ipv6-uc<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

Platforms	Command context	Authority

authentication ipsec spi (OSPFv3-AF)

Syntax

```
authentication ipsec spi <SPI-INDEX>
            <AUTH-TYPE> [<KEY-TYPE>
            <AUTH-KEY>]
no authentication
```

Description

Configures IPsec AH authentication for the selected Vlink.

The **no** form of this command removes IPsec AH authentication for the selected Vlink.

Parameter	Description
`spi <SPI-INDEX>`	Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables t he receiving OSPF process to select and use the Security Associ ation (SA) from the SA table. The SPI must be unique on the sw itch. Range: 256 to 4294967295.
`<AUTH-TYPE>`	Specifies the authentication type: **md5** or **sha1**.
`<KEY-TYPE>`	Specifies the key type to use: **plaintext** (unencrypted), **hex - st ring** (encrypted) or **ciphertext** (encrypted).
	Specifies the authentication key.

Parameter	Description
<AUTH-KEY>	

NOTE

When the authentication key is not provided on the command line, plaintext key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

Examples

Setting area 1 to use IPsec AH authentication for Vlink with provided plaintext key:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1
switch(config-ospfv3-ipv6-uc)# area 1 virtual-link 3.3.3.3
switch(config-router-vlink6)# authentication ipsec spi 256 sha1 plaintext
abcd
```

Setting area 1 to use IPsec AH authentication for Vlink with prompted plaintext key:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1
switch(config-ospfv3-ipv6-uc)# area 1 virtual-link 3.3.3.3
switch(config-router-vlink6)# authentication ipsec spi 256 sha1
Enter the IPsec authentication key: **********
Re-Enter the IPsec authentication key: **********
```

Removing IPsec AH authentication for Vlink on area 1:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1 virtual-link 3.3.3.3
switch(config-router-vlink6)# no authentication
switch(config-router-vlink6)# no authentication ipsec spi 1234 md5
ciphertext
                              qwertyuioplkjhgfdsazxcvbnmkjhgfdsa
switch(config-router-vlink6)# no authentication ipsec spi 1234 sha1
ciphertext
                              qwertyuioplkjhgfdsazxcvbnmkjhgfdsa
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-router-vlink`	Administrators or local user group members with execution righ ts for this command.

`clear ospfv3-af ipv4 neighbor`

Syntax

```
clear ospfv3-af [<PROCESS-ID>] ipv4 neighbor [<NEIGHBOR-ID>] [interface
<INTERFACE-NAME> | vlan <VLAN-ID> | lag <LAG-ID>] | loopback <LOOPBACK-ID>
| tunnel <TUNNEL-ID>] [all-vrfs | vrf <VRF-NAME>]
```

Description

Resets the neighbor and clears the OSPFv3 AF neighbor information.

Parameter	Description
`<PROCESS-ID>`	Specifies the OSPFv3 AF process ID to clear the statistics for th e particular OSPFv3 AF process. Range: 1 to 65535.
`neighbor <NEIGHBOR-ID>`	Specifies the router ID of a neighbor.
`interface <INTERFACE-NAME>`	Specifies the OSPFv3 AF statistics to clear for the specified inte rface.
`vlan <VLAN-ID>`	Specifies the vlan interface name.

Parameter	Description
`lag <LAG-ID>`	Specifies the lag interface name.
`loopback <LOOPBACK-ID>`	Specifies the loopback interface name.
`tunnel <TUNNEL-ID>`	Specifies the tunnel interface name.
`all-vrfs`	Select to clear the OSPFv3 AF statistics for all VRFs.
`vrf <VRF-NAME>`	Specifies the name of a VRF.

Example

Clearing the OSPFv3 AF neighbor information:

```
switch# clear ospfv3-af 1 ipv4 neighbor
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af 1 ipv4 neighbor 3.3.3.3
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af 1 ipv4 neighbor interface 1/1/1
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af 1 ipv4 neighbor 3.3.3.3 vrf red
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv4 neighbor
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv4 neighbor 5.5.5.5
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv4 neighbor interface 1/1/1
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv4 neighbor 5.5.5.5 vrf red
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`clear ospfv3-af ipv4 statistics`

Syntax

```
clear ospfv3-af [<PROCESS-ID>] ipv4 statistics [interface [interface
<INTERFACE-NAME> | vlan <VLAN-ID> | lag <LAG-ID>] | loopback <LOOPBACK-ID>
| tunnel <TUNNEL-ID>] [all-vrfs | vrf <VRF-NAME>]
```

Description

Clears the OSPFv3 AF event statistics.

Parameter	Description
`<PROCESS-ID>`	Specifies the OSPFv3 AF process ID to clear the statistics for the particular OSPFv3 AF process. Range: 1 to 65535.
`interface <INTERFACE-NAME>`	Specifies the OSPFv3 AF statistics to clear for the specified interface.
`vlan <VLAN-ID>`	Specifies the vlan interface name.
`lag <LAG-ID>`	Specifies the lag interface name.

Parameter	Description
`loopback <LOOPBACK-ID>`	Specifies the loopback interface name.
`tunnel <TUNNEL-ID>`	Specifies the tunnel interface name.
`all-vrfs`	Select to clear the OSPFv3 AF statistics for all VRFs.
`vrf <VRF-NAME>`	Specifies the name of a VRF.

Example

Clearing the OSPFv3 AF event statistics:

```
switch# clear ospfv3-af ipv4 statistics
switch# clear ospfv3-af ipv4 statistics interface 1/1/1
switch# clear ospfv3-af ipv4 statistics interface 1/1/1 vrf vrf_red
switch# clear ospfv3-af ipv4 statistics interface tunnel1
switch# clear ospfv3-af ipv4 statistics interface tunnel1 vrf vrf_red
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`clear ospfv3-af ipv6 neighbor`

Syntax

```
clear ospfv3-af [<PROCESS-ID>] ipv6 neighbor [<NEIGHBOR-ID>] [interface
<INTERFACE-NAME> | vlan <VLAN-ID> | lag <LAG-ID>] | loopback <LOOPBACK-ID>
| tunnel <TUNNEL-ID>] [all-vrfs | vrf <VRF-NAME>]
```

Description

Resets the neighbor and clears the OSPFv3 AF neighbor information.

Parameter	Description
`<PROCESS-ID>`	Specifies the OSPFv3 AF process ID to clear the statistics for th e particular OSPFv3 AF process. Range: 1 to 65535.
`neighbor <NEIGHBOR-ID>`	Specifies the router ID of a neighbor.
`interface <INTERFACE-NAME>`	Specifies the OSPFv3 AF statistics to clear for the specified inte rface.
`vlan <VLAN-ID>`	Specifies the vlan interface name.
`lag <LAG-ID>`	Specifies the lag interface name.
`loopback <LOOPBACK-ID>`	Specifies the loopback interface name.
`tunnel <TUNNEL-ID>`	Specifies the tunnel interface name.
`all-vrfs`	Select to clear the OSPFv3 AF statistics for all VRFs.
`vrf <VRF-NAME>`	Specifies the name of a VRF.

Example

Clearing the OSPFv3 AF neighbor information:

```
switch# clear ospfv3-af 1 ipv6 neighbor
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af 1 ipv6 neighbor 3.3.3.3
Performing clear ospf neighbor may result in traffic disruption.
```

```
Do you want to continue (y/n)? y
switch# clear ospfv3-af 1 ipv6 neighbor interface 1/1/1
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af 1 ipv6 neighbor 3.3.3.3 vrf red
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv6 neighbor
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv6 neighbor 5.5.5.5
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv6 neighbor interface 1/1/1
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
switch# clear ospfv3-af ipv6 neighbor 5.5.5.5 vrf red
Performing clear ospf neighbor may result in traffic disruption.
Do you want to continue (y/n)? y
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`clear ospfv3-af ipv6 statistics`

Syntax

```
clear ospfv3-af [<PROCESS-ID>] ipv6 statistics [interface [interface
<INTERFACE-NAME> | vlan <VLAN-ID> | lag <LAG-ID>] | loopback <LOOPBACK-ID>
| tunnel <TUNNEL-ID>] [all-vrfs | vrf <VRF-NAME>]
```

Description

Clears the OSPFv3 AF event statistics.

Parameter	Description
`<PROCESS-ID>`	Specifies the OSPFv3 AF process ID to clear the statistics for th e particular OSPFv3 AF process. Range: 1 to 65535.
`interface <INTERFACE-NAME>`	Specifies the OSPFv3 AF statistics to clear for the specified inte rface.
`vlan <VLAN-ID>`	Specifies the vlan interface name.
`lag <LAG-ID>`	Specifies the lag interface name.
`loopback <LOOPBACK-ID>`	Specifies the loopback interface name.
`tunnel <TUNNEL-ID>`	Specifies the tunnel interface name.
`all-vrfs`	Select to clear the OSPFv3 AF statistics for all VRFs.
`vrf <VRF-NAME>`	Specifies the name of a VRF.

Example

Clearing the OSPFv3 AF event statistics:

```
switch# clear ospfv3-af ipv6 statistics
switch# clear ospfv3-af ipv6 statistics interface 1/1/1
switch# clear ospfv3-af ipv6 statistics interface 1/1/1 vrf vrf_red
switch# clear ospfv3-af ipv6 statistics interface tunnel1
switch# clear ospfv3-af ipv6 statistics interface tunnel1 vrf vrf_red
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`dead-interval`

Syntax

```
dead-interval <INTERVAL>
no dead-interval
```

Description

Sets the interval after which a neighbor is declared dead if no hello packet comes in for virtual links.

The **no** form of this command sets the dead interval to default for virtual links. The default value is 40 seconds (generally four times the hello packet interval).

Parameter	Description
`<INTERVAL>`	Specifies the time interval for the dead interval, in seconds. Range: 1 to 65535. Default: 40.

Examples

Setting OSPFv3 AF virtual links dead interval:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 100 virtual-link 100.0.1.1
```

```
switch(config-router-vlink6)# dead-interval 30
```

Setting OSPFv3 AF virtual links dead interval to default:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# no dead-interval
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-router-vlink`	Administrators or local user group members with execution righ ts for this command.

debug ospfv3-af

Syntax

```
debug ospfv3-af <SUBMODULE-NAME> [instance <PROCESS-ID>] [ip <IP-ADDRESS>]
[port <IF-NAME>] [vrf <VRF-NAME>] [severity {alert | crit | debug | emer |
err | info | notice | warning}]
no debug ospfv3-af <SUBMODULE-NAME> [instance <PROCESS-ID>] [ip <IP-
ADDRESS>] [port <IF-NAME>] [vrf <VRF-NAME>] [severity {alert | crit | debug
| emer | err | info | notice | warning}]
```

Description

Enables problem logs to the debug logs. If <SEVERITY> is omitted, all severities are logged.

The **no** form of this command disables problem logs to the debug logs.

Parameter	Description
<SUBMODULE-NAME>	Enables debug logging for a specific submodule. For a list of s upported submodules, enter the debug <MODULE - NAME> co mmand followed by a space and a question mark (?).
instance *<PROCESS-ID>*	Displays debug logs for the specified OSPFv3 AF process ID. Ra nge: 1 to 65535.
ip *<IP-ADDRESS>*	Displays debug logs for the specified IP Address.
port *<INTERFACE-NAME>*	Displays debug logs for the OSPFv3 AF specified interfaces.
vrf *<VRF-NAME>*	Displays debug logs for the specified VRF name.
severity {alert|crit|debug|emer| err|info|notice|warning}	Enables the debug logs for OSPv3 AF based on the required s everity level for the destination. Optional.

Examples

Enables debug logging of event submodules for port 1/1/1 and debug severity:

```
switch# debug ospfv3-af ipv4 event port 1/1/1 severity debug
```

Enables all debug logging for VRF red:

```
switch# debug ospfv3-af ipv4 all vrf vrf_red
```

Disables all OSPFv3 AF debug logging:

```
switch# no debug ospfv3-af ipv4 all
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`default-information originate always`

Syntax

```
default-information originate always [metric <METRIC-VALUE>]
no default-information originate always [metric <METRIC-VALUE>]
```

Description

Configures OSPFv3 AF to advertise the default route (::/0) to its neighbors, regardless if it is present in the routing table or not. Optionally, metric can be set for default route ::/0. The default value is 1.

The **no** form of this command disables advertisement of the default route.

Parameter	Description
metric <METRIC-VALUE	Specifies the OSPF metric value for the default route. Default: 1.

Examples

Setting advertisement of the default route:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# default-information originate always
```

Disabling advertisement of the default route:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# no default-information originate always
```

Setting advertisement of the default route with metric set to 20:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# default-information originate always metric
20
```

Disabling advertisement of the default route and setting the metric to the default value:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# no default-information originate always
metric
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-ipv4-uc<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`default-information originate`

Syntax

```
default-information originate [metric <METRIC-VALUE>]
no default-information originate [metric <METRIC-VALUE>]
```

Description

Configures OSPFv3 AF to advertise the default route (::/0) to its neighbors if it is present in the routing table. Optionally, the metric value can be set for default route ::/0. The default value is 1.

The **no** form of this command disables advertisement of the default route.

Parameter	Description
metric <METRIC-VALUE>	Specifies the OSPF metric value for the default route. Optional. Default: 1.

Examples

Setting advertisement of the default route and specifying an optional metric value of 20: :

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# default-information originate
switch(config-ospfv3-ipv4-uc)# default-information originate metric 20
```

Disabling advertisement of the default route:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# no default-information originate metric
switch(config-ospfv3-ipv4-uc)# no default-information originate
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-ipv4-uc<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`default-metric`

Syntax

```
default-metric <METRIC-VALUE>
no default-metric
```

Description

Sets the default metric for redistributed routes in the OSPFv3 AF.

The **no** form of this command sets the default metric to be used for redistributed routes into OSPFv3 AF to the default of 25.

Parameter	Description
<METRIC-VALUE>	Specifies the default metric value to use for redistributed routes . Range: 1 to 1677214. Default: 25.

Examples

Setting default metric for redistributed routes:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# default-metric 36
```

Setting default metric for redistributed routes to the default:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1) # address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc) # no default-metric
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-ipv4-uc<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

diag ospfv3 lsdb dump

Syntax

```
diag ospfv3 lsdb dump
```

Description

This command dumps the OSPFv3 AF protocol LSA information.

Examples

```
switch# diag ospfv3 lsdb dump
ospfv3PmAreaLsdbTable row:
ospfv3PmAreaLsdbApplIndex:   537265152
ospfv3PmAreaLsdbAreaId:      0x00000000
ospfv3PmAreaLsdbType:        ospfv3RouterLsa
ospfv3PmAreaLsdbRouterId:    0x01010103
ospfv3PmAreaLsdbLsid:        0x00000000
ospfv3PmAreaLsdbSequence:    -2147483646
ospfv3PmAreaLsdbAge:         872
ospfv3PmAreaLsdbChecksum:    12468
ospfv3PmAreaLsdbAdvertisement:  0x00 0x01 0x20 0x01 0x00 0x00 0x00 0x00
                0x01 0x01 0x01 0x03 0x80 0x00 0x00 0x02
                0x30 0xB4 0x00 0x28 0x01 0x00 0x00 0x13
                0x02 0x00 0x00 0x01 0x10 0x01 0x01 0x01
                0x10 0x01 0x01 0x01 0x03 0x03 0x03 0x03
ospfv3PmAreaLsdbTypeKnown:   true
ospfv3PmAreaLsdbTable row:
ospfv3PmAreaLsdbApplIndex:   537265152
ospfv3PmAreaLsdbAreaId:      0x00000000
ospfv3PmAreaLsdbType:        ospfv3RouterLsa
```

```
ospfv3PmAreaLsdbRouterId:   0x03030303
ospfv3PmAreaLsdbLsid:       0x00000000
ospfv3PmAreaLsdbSequence:   -2147483646
ospfv3PmAreaLsdbAge:        873
ospfv3PmAreaLsdbChecksum:   64738
ospfv3PmAreaLsdbAdvertisement:  0x00 0x01 0x20 0x01 0x00 0x00 0x00 0x00
                0x03 0x03 0x03 0x03 0x80 0x00 0x00 0x02
                0xFC 0xE2 0x00 0x28 0x00 0x00 0x00 0x13
                0x02 0x00 0x00 0x01 0x10 0x01 0x01 0x01
                0x10 0x01 0x01 0x01 0x03 0x03 0x03 0x03
ospfv3PmAreaLsdbTypeKnown:  true
ospfv3PmAreaLsdbTable row:
ospfv3PmAreaLsdbApplIndex:  537265152
ospfv3PmAreaLsdbAreaId:     0x00000000
ospfv3PmAreaLsdbType:       ospfv3NetworkLsa
ospfv3PmAreaLsdbRouterId:   0x03030303
ospfv3PmAreaLsdbLsid:       0x10010101
ospfv3PmAreaLsdbSequence:   -2147483647
ospfv3PmAreaLsdbAge:        878
ospfv3PmAreaLsdbChecksum:   45127
ospfv3PmAreaLsdbAdvertisement:  0x00 0x01 0x20 0x02 0x10 0x01 0x01 0x01
                0x03 0x03 0x03 0x03 0x80 0x00 0x00 0x01
                0xB0 0x47 0x00 0x20 0x00 0x00 0x00 0x13
                0x01 0x01 0x01 0x03 0x03 0x03 0x03 0x03
ospfv3PmAreaLsdbTypeKnown:  true
ospfv3PmAreaLsdbTable row:
ospfv3PmAreaLsdbApplIndex:  537265152
ospfv3PmAreaLsdbAreaId:     0x00000000
ospfv3PmAreaLsdbType:       ospfv3InterAreaPrfxLsa
ospfv3PmAreaLsdbRouterId:   0x01010103
ospfv3PmAreaLsdbLsid:       0x00000001
ospfv3PmAreaLsdbSequence:   -2147483647
ospfv3PmAreaLsdbAge:        913
ospfv3PmAreaLsdbChecksum:   13755
ospfv3PmAreaLsdbAdvertisement:  0x00 0x24 0x20 0x03 0x00 0x00 0x00 0x01
                0x01 0x01 0x01 0x03 0x80 0x00 0x00 0x01
                0x35 0xBB 0x00 0x24 0x00 0x00 0x00 0x01
                0x40 0x00 0x00 0x00 0xFD 0x00 0x00 0x00
                0x00 0x00 0x00 0x00
ospfv3PmAreaLsdbTypeKnown:  true
ospfv3PmAreaLsdbTable row:
ospfv3PmAreaLsdbApplIndex:  537265152
ospfv3PmAreaLsdbAreaId:     0x00000000
```

```
ospfv3PmAreaLsdbType:        ospfv3IntraAreaPrfxLsa
ospfv3PmAreaLsdbRouterId:    0x03030303
ospfv3PmAreaLsdbLsid:        0x00010000
ospfv3PmAreaLsdbSequence:    -2147483645
ospfv3PmAreaLsdbAge:         873
ospfv3PmAreaLsdbChecksum:    2173
ospfv3PmAreaLsdbAdvertisement:  0x00 0x01 0x20 0x09 0x00 0x01 0x00 0x00
                0x03 0x03 0x03 0x03 0x80 0x00 0x00 0x03
                0x08 0x7D 0x00 0x2C 0x00 0x01 0x20 0x02
                0x10 0x01 0x01 0x01 0x03 0x03 0x03 0x03
                0x40 0x00 0x00 0x00 0x01 0x11 0x00 0x00
                0x00 0x00 0x00 0x00
ospfv3PmAreaLsdbTypeKnown:  true
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Administrators or local user group members with execution rights for this command.

disable

Syntax

```
disable
```

Description

Disables the OSPFv3 AF process. By default OSPFv3 AF process is enabled.

This command does not remove the OSPFv3 AF configurations.

Example

Disabling OSPFv3 AF process:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1) # disable
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`distance`

Syntax

```
distance [<DISTANCE-VAL> | intra-area [<DISTANCE-VAL>] | inter-area
[<DISTANCE-VAL>] | external [<DISTANCE-VAL>]]
no distance [<DISTANCE-VAL> | intra-area [<DISTANCE-VAL>] | inter-area
[<DISTANCE-VAL>] | external [<DISTANCE-VAL>]]
```

Description

Defines an administrative distance for OSPFv3 AF. Administrative distance is used as a criteria to select the best route when the same route is learned by multiple routing protocols.

The **no** form of this command sets the OSPFv3 AF administrative distance to the default value of 110. Optionally, administrative distance can be set to default for the specific OSPF route type: intra-area, inter-area, or external type-5 and type-7 routes.

Parameter	Description
<DISTANCE-VAL>	Specifies the OSPFv3 AF administrative distance. Range: 1 to 255. Default: 110.
`intra-area`	Specifies the OSPFv3 AF distance for intra - area routes.
`inter-area`	Specifies the OSPFv3 AF distance for inter - area routes.
`external`	Specifies the OSPFv3 AF distance for external type 5 and type 7 routes.

Usage

Within a given OSPF process, intra-area routes are always given precedence even when distances are configured for inter-area or external type routes.

Examples

Setting OSPFv3 AF administrative distance for address-family ipv6-unicast:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# distance 100
switch(config-ospfv3-ipv6-uc)# distance intra-area 24 external 55 inter-
area 66
switch(config-ospfv3-ipv6-uc)# distance intra-area 24 external 55
switch(config-ospfv3-ipv6-uc)# distance external 55
switch(config-ospfv3-ipv6-uc)#exit
```

Setting OSPFv3 AF administrative distance for address-family ipv4-unicast to default:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# no distance
switch(config-ospfv3-ipv4-uc)# no distance external
switch(config-ospfv3-ipv4-uc)# no distance intra-area
switch(config-ospfv3-ipv4-uc)# no distance inter-area
switch(config-ospfv3-ipv4-uc)# no distance 1
switch(config-ospfv3-ipv4-uc)#exit
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-ipv4-uc<PROCESS-ID>` `config-ospfv3-ipv6-uc<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`distribute-list prefix`

Syntax

```
distribute-list prefix <PREFIX-LIST-NAME> {in | out}
no distribute-list prefix <PREFIX-LIST-NAME> {in | out}
```

Description

This command uses an existing prefix list to filter routes that are being installed in the routing table or redistributed to another routing protocol.

The **distribute-list prefix** command filters routes in the inbound or the outbound direction. When this command is issued with the **in** parameter, it filters routes from being installed in the routing table, it does not filter LSAs. When this command is issued with the **out** parameter, it filters only the desired redistributed routes from other protocols.

Parameter	Description
`prefix <PREFIX-LIST-NAME>`	Specify the name of an existing prefix.
`{in | out}`	Select one of the following parameters to set the filter direction:

Parameter	Description
	• **in**: Filter incoming routes into the routing table • **out**: Filter outgoing routing updates

Examples

Enabling the filtering of OSPFv3 AF routes for an IPv4 network, preventing routes from being installed in the routing table or redistributed from another routing protocol.

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# distribute-list prefix list1 in
switch(config-ospfv3-ipv4-uc)# distribute-list prefix list2 out
```

Disabling the filtering of OSPFv3 AF routes for an IPv4 network, allowing routes to be installed in the routing table or redistributed from another routing protocol.

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# no distribute-list prefix list1 in
switch(config-ospfv3-ipv4-uc)# no distribute-list prefix list2 out
```

Enabling the filtering of OSPFv3 AF routes for an IPv6 network, preventing routes from being installed in the routing table or redistributed from another routing protocol.

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# distribute-list prefix list1 in
switch(config-ospfv3-ipv6-uc)# distribute-list prefix list2 out
```

Disabling the filtering of OSPFv3 AF routes for an IPv6 network, allowing routes to be installed in the routing table or redistributed from another routing protocol.

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# no distribute-list prefix list1 in
switch(config-ospfv3-ipv6-uc)# no distribute-list prefix list2 out
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-ipv4-uc<PROCESS-ID>` `config-ospfv3-ipv6-uc<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`enable`

Syntax

```
enable
```

Description

Enables OSPFv3 AF process when disabled. By default OSPFv3 AF process is enabled.

Example

Enabling OSPFv3 AF process:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# enable
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-af-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`encryption ipsec`

Syntax

```
encryption ipsec spi <SPI-INDEX>
             <AUTH-TYPE> [<KEY-TYPE>
            <AUTH-KEY>
            {<ENCR-TYPE> [<KEY-TYPE>
             <ENCR-KEY>] | null}]
no encryption
```

Description

Configures IPSec ESP authentication and encryption for the selected Vlink.

The **no** form of this command removes IPSec ESP authentication and encryption for the selected Vlink.

Parameter	Description
`spi <SPI-INDEX>`	Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables t he receiving OSPFv3 AF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. Range: 256 to 4294967295.
`<AUTH-TYPE>`	Specifies the authentication type: **md5** or **sha1**.
`<KEY-TYPE>`	Specifies the key type to use: **plaintext** (unencrypted), **hex - st ring** (encrypted) or **ciphertext** (encrypted).

Parameter	Description
<AUTH-KEY>	Specifies the authentication key.
<ENCR-TYPE>	Specifies the encryption type: **des**, **3des**, **aes**, or **null**. **NOTE** Encryption type **aes** is considered to be AES128, AES192, or AES256 based on key length.
<ENCR-KEY>	Specifies the encryption key.

NOTE

When the authentication key is not provided on the command line, plaintext authentication key prompting occurs upon pressing Enter, followed by encryption type prompting, and finally plaintext encryption key prompting. The entered key characters are masked with asterisks.

NOTE

When the authentication key and encryption type are provided on the command line but the encryption key is not provided, plaintext encryption key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

Examples

Setting area 1 to use IPSec ESP authentication and encryption for Vlink with provided plaintext keys:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1
switch(config-ospfv3-ipv6-uc)# area 1 virtual-link 3.3.3.3
switch(config-router-vlink6)# encryption ipsec spi 256 md5 plaintext abcd
                                         des plaintext abcdefab
```

Setting area 1 to use IPSec md5 authentication and encryption for Vlink with a prompted plaintext authentication key and encryption type:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1
switch(config-ospfv3-ipv6-uc)# area 1 virtual-link 3.3.3.3
switch(config-router-vlink6)# encryption ipsec spi 256 md5
Enter the IPsec authentication key: ********
Re-Enter the IPsec authentication key: *******
Enter the IPsec encryption type (3des/aes/des/null)? des
Enter the IPsec encryption key: ********
Re-Enter the IPsec encryption key: ********
```

Setting area 1 to use IPSec sha1 authentication and encryption for Vlink with a provided plaintext authentication key and prompted encryption type:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1
switch(config-ospfv3-ipv6-uc)# area 1 virtual-link 3.3.3.3
switch(config-router-vlink6)# encryption ipsec spi 256 sha1 plaintext pass
des
  ciphertext  Configure ciphertext key.
  hex-string  Configure hex-string key. Key length must be 16 Hex digits.
  plaintext   Configure plaintext key. Key length must be 8 characters.
switch(config-router-vlink6)# encryption ipsec spi 256 sha1 plaintext pass
des
Enter the IPsec encryption key: ********
Re-Enter the IPsec encryption key: ********
```

Setting area 1 to use IPSec ESP authentication for Vlink with a provided plaintext authentication key and null encryption:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-1)# area 1
switch(config-ospfv3-1)# area 1 virtual-link 3.3.3.3
switch(config-router-vlink6)# encryption ipsec spi 256 md5 plaintext abcd
null
```

Removing IPSec ESP authentication and encryption for Vlink on area 1:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 1 virtual-link 3.3.3.3
switch(config-router-vlink6)# no encryption
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-router-vlink`	Administrators or local user group members with execution righ ts for this command.

`graceful-restart ignore-lost-interface`

Syntax

```
graceful-restart ignore-lost-interface
no graceful-restart ignore-lost-interface
```

Description

Enables the restarting router to ignore lost OSPFv3 AF interfaces during a graceful restart process. It is recommended to enable the ignore-lost-interface parameter on the VSF device configured with OSPF enabled on ROPs to ensure the graceful restart completes successfully, as a VSF switchover or failover can render a ROP non-operational, leading to a Router LSA change published to neighbors.

The **no** form of this commands disables the restarting router to ignore OSPFv3 AF interfaces during a graceful restart process.

Parameter	Description
`ignore-lost-interface`	Enable the restarting router to ignore lost OSPFv3 AF interfac es during a graceful restart process. This setting should be ena bled on a high availability system to ensure a graceful restart co mpletes successfully, even if OSPFv3 AF - enabled links fail due to High Availability events like a switchover or failover. **NOTE**

Parameter	Description
	Enabling this setting means that the hitless restart procedures do not strictly follow those defined in RFC 3623, Graceful OSPF Restart. **NOTE** OSPFv3 AF neighbors must disable strict LSA c hecking. If the local node has fewer OSPFv3 AF interfaces after restarting, then the neighbors th at were adjacent on those interfaces will clear u p their adjacencies to the restarting node and w ill send out link state updates to advertise the d ropped adjacency. If strict LSA checking is enab led, the restarting router's neighbors will exit he lper mode when they receive the updated LSAs and the graceful restart will still fail.

Examples

Enabling the restarting router to ignore lost OSPFv3 AF interfaces during a graceful restart process:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-ipv4-uc)# graceful-restart ignore-lost-interface
```

Disabling the restarting router:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-ipv4-uc)# no graceful-restart ignore-lost-interface
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-ipv4-uc<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

Platforms	Command context	Authority

graceful-restart

Syntax

```
graceful-restart {restart-interval <INTERVAL> | helper [strict-lsa-check]}
no graceful-restart {restart-interval <INTERVAL> | helper [strict-lsa-
check]}
```

Description

Configures graceful restart for OSPFv3 AF. By default graceful restart is enabled on the OSPFv3 AF router.

The **no** form of this command sets the restart interval to the default of 120 seconds or disables helper mode depending on the specified parameters.

Parameter	Description
`restart-interval <INTERVAL>`	Specifies the time another router waits for this router to gracefully restart and selects the maximum time to wait in seconds. Range: 5 to 1800. Default: 120.
`helper`	Specifies that the router will participate in the graceful restart of a neighbor router.
`strict-lsa-check`	(Optional). Use with the **helper** parameter to enable strict Link state Advertisement (LSA) checking when acting as a restart helper for a restarting peer. **NOTE** If strict LSA checking is enabled, the restarting router's neighbors will exit helper mode when they receive the updated LSAs and the graceful restart will still fail.

Examples

Enabling OSPFv3 AF graceful restart:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-ipv4-uc)# graceful-restart helper
switch(config-ospfv3-ipv4-uc)# graceful-restart helper strict-lsa-check
switch(config-ospfv3-ipv4-uc)# graceful-restart restart-interval 40
```

Setting the restart interval to default, and disabling helper mode:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-ipv4-uc)# no graceful-restart
switch(config-ospfv3-ipv4-uc)# no graceful-restart helper
switch(config-ospfv3-ipv4-uc)# no graceful-restart helper strict-lsa-check
switch(config-ospfv3-ipv4-uc)# no graceful-restart restart-interval
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-ipv4-uc<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

hello-interval

Syntax

```
hello-interval <INTERVAL>
no hello-interval
```

Description

Sets the time interval between OSPFv3 AF hello packets for virtual links.

The **no** form of this command sets the hello interval to the default value of 10 seconds for virtual links.

Parameter	Description
<INTERVAL>	Specifies the time interval for the hello interval, in seconds. Range: 1 to 65535. Default: 10.

Examples

Setting OSPFv3 AF virtual links hello interval:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# hello-interval 30
```

Setting OSPFv3 AF virtual links hello interval to default:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# no hello-interval
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-router-vlink`	Administrators or local user group members with execution rights for this command.

`max-metric router-lsa`

Syntax

```
max-metric router-lsa [include-stub] | [on-startup <INTERVAL>]
no max-metric router-lsa [include-stub] | [on-startup]
```

Description

Sets the protocol to advertise a maximum metric so that other routers do not prefer the router as an intermediate hop in their shortest path first (SPF) calculations. If the `on-startup` parameter is used, the Router is configured to advertise a maximum metric at start up.

To disable advertisement of the maximum metric, use the **no** form of the command.

The **no** form of this command advertises the normal cost metrics instead of advertising the maximized cost metric. This setting causes the router to be considered in traffic forwarding.

Parameter	Description
`include-stub`	Configures the stub Router - LSA to advertise a maximum metr ic for all links, including stub links.
`on-startup <INTERVAL>`	Automatically advertises the stub Router - LSA (or maximize th e router - LSA cost metric) for a specified time interval upon O SPFv3 AF startup. Specifies the time in seconds. Range: 5 to 86400. Default: 600.

Examples

Setting to maximize the cost metrics for Router LSA:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# max-metric router-lsa on-startup 100
```

Setting to advertise the normal cost metrics instead of advertising the maximized cost metric:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# no max-metric router-lsa
switch(config-ospfv3-ipv6-uc)# no max-metric router-lsa on-startup
```

Setting to advertise Router LSA for stub links:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# max-metric router-lsa
switch(config-ospfv3-ipv6-uc)# max-metric router-lsa include-stub on-
startup 100
```

Setting to advertise the normal cost metrics instead of advertising the maximized cost metric for stub links:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-af-1)# no max-metric router-lsa
```

```
switch(config-ospfv3-ipv6-uc)# no max-metric router-lsa include-stub on-startup
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-ipv4-uc<PROCESS-ID>` `config-ospfv3-ipv6-uc<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

maximum-paths

Syntax

```
maximum-paths <MAXIMUM>
no maximum-paths
```

Description

Sets the maximum number of ECMP routes that OSPFv3 AF can support.

The **no** form of this command sets the maximum number of ECMP routes that OSPFv3 AF can support to the default value of 4.

Parameter	Description
	Specifies the maximum number of ECMP routes. Range: 1 to 32. Default: 4.

Parameter	Description
<MAXIMUM>	

Examples

Setting maximum number of parallel routes:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)#address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# maximum-paths 3
```

Setting maximum number of parallel paths to the default value of 4:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)#address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc)# no maximum-paths
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-ipv4-uc<PROCESS-ID>` `config-ospfv3-ipv6-uc<PROCESS-ID>`	Administrators or local user group members with execution rights for this command.

`ospfv3-af authentication ipsec spi`

Syntax

```
ospfv3-af authentication ipsec spi <SPI-INDEX>
            <AUTH-TYPE> [<KEY-TYPE>
            <AUTH-KEY>]
no ospfv3-af authentication ipsec spi
```

Description

Configures IPSec AH authentication. OSPFv3 AF interfaces that have IPsec configured at the interface context will not use area level IPsec. The Security Parameters Index (SPI) is an identification tag carried in IPsec AH header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table.

The **no** form of this command removes IPsec AH authentication for the specified area.

> **NOTE**
>
> IPsec is not supported for 6in6 tunnel interfaces, and IPsec AH SPI must be unique on the router.

Parameter	Description
`spi <SPI-INDEX>`	Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables t he receiving OSPF process to select and use the Security Associ ation (SA) from the SA table. The SPI must be unique on the sw itch. Range: 256 to 4294967295.
`<AUTH-TYPE>`	Specifies the authentication type: **md5** or **sha1**.
`<KEY-TYPE>`	Specifies the key type to use: **plaintext** (unencrypted), **hex - st ring** (encrypted) or **ciphertext** (encrypted).
	Specifies the authentication key.

Parameter	Description
<AUTH-KEY>	

NOTE

When the authentication key is not provided on the command line, plaintext key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

Examples

Setting interface**1/1/1** to use IPsec authentication with a provided plaintext authentication key:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af authentication ipsec spi 256 md5 plaintext abcd
```

Setting interface**1/1/1** to use IPsec authentication with a prompted plaintext authentication key:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af authentication ipsec spi 256 md5
Enter the IPsec authentication key: ********
Re-Enter the IPsec authentication key: ********
```

Removing IPsec authentication from interface**1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no ospfv3-af authentication
switch(config-if)# no ospfv3-af authentication null
switch(config-if)# no ospfv3-af authentication ipsec spi 300 md5 ciphertext
qwertyuioplkjhgfdsxcvbnm
switch(config-if)# no ospfv3-af authentication ipsec spi 300 sha1
ciphertext qwertyuioplkjhgfdsxcvbnm
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ospfv3-af authentication null`

Syntax

```
ospfv3-af authentication null
```

Description

This command configures NULL authentication on an interface which disables IPsec AH authentication.

Examples

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af authentication null
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manage r (`#`)	Administrators or local user group members with execution righ ts for this command.

`ospfv3-af encryption null`

Syntax

```
ospfv3-af encryption null
```

Description

This command configures NULL ESP on an interface, which disables IPsec ESP.

Example

Configuring null ESP:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af encryption null
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`ospfv3-af encryption ipsec spi`

Syntax

```
ospfv3-af encryption ipsec spi <SPI-INDEX>
            <AUTH-TYPE>
    [<KEY-TYPE>
            <AUTH-KEY>
            <ENCR-TYPE> [<KEY-TYPE>
            <ENCR-KEY>]]
```

```
no ospfv3-af encryption {null| [ipsec spi <SPI-INDEX>
            <AUTH-TYPE>
    [<KEY-TYPE>
            <AUTH-KEY>
            <ENCR-TYPE> [<KEY-TYPE>
            <ENCR-KEY>]]}
```

Description

Configures IPsec ESP authentication. OSPFv3 interfaces that have IPsec configured at the interface context will not use area level IPsec ESP.

The **no** form of this command removes IPsec ESP for the specified area.

Parameter	Description
`spi <SPI-INDEX>`	Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables t he receiving OSPF process to select and use the Security Associ ation (SA) from the SA table. The SPI must be unique on the sw itch. Range: 256 to 4294967295.
`<AUTH-TYPE>`	Specifies the authentication type: **md5** or **sha1**.
`<KEY-TYPE>`	Specifies the key type to use: **plaintext** (unencrypted), **hex - st ring** (encrypted) or **ciphertext** (encrypted).
`<AUTH-KEY>`	Specifies the authentication key.
`<ENCR-TYPE>`	Specifies the encryption type: **des**, **3des**, **aes**, or **null**. **NOTE** **aes** is considered to be AES128, AES192, or AE S256 based on key length.
	Specifies the encryption key.

Parameter	Description
<ENCR-KEY>	

> **NOTE**
>
> When the authentication key is not provided on the command line, plaintext authentication key prompting occurs upon pressing Enter, followed by encryption type prompting, and finally plaintext encryption key prompting. The entered key characters are masked with asterisks.

> **NOTE**
>
> When the authentication key and encryption type are provided on the command line but the encryption key is not provided, plaintext encryption key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

Examples

Setting interface VLAN **1** to use IPsec ESP with provided authentication and encryption keys:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ospfv3-af encryption ipsec spi 256 sha1 plaintext
F82
                                des plaintext F82#450b
```

Setting interface VLAN **3** to use IPsec ESP with prompted authentication and encryption keys:

```
switch(config)# interface vlan 3
switch(config-if-vlan)# ospfv3-af encryption ipsec spi 256 sha1
Enter the IPsec authentication key: ********
Re-Enter the IPsec authentication key: ********
Enter the IPsec encryption type (3des/aes/des/null)? des
Enter the IPsec encryption key: ********
Re-Enter the IPsec encryption key: ********
```

Setting interface VLAN **4** to use IPsec ESP with provided authentication password and encryption type but a prompted encryption key:

```
switch(config)# interface vlan 4
switch(config-if-vlan)# ospfv3-af encryption ipsec spi 256 sha1 plaintext
F82 des
Enter the IPsec encryption key: ********
Re-Enter the IPsec encryption key: ********
```

Setting interface VLAN **1** to use IPSec ESP with provided plaintext authentication key and null encryption:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ospfv3-af encryption ipsec spi 256 sha1 plaintext
F82 null
```

Removing IPsec from interface VLAN **1**:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ospfv3-af encryption ipsec spi
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ospfv3-af encryption ipsec`

Syntax

```
ospfv3-af encryption ipsec spi <SPI-INDEX>
            <AUTH-TYPE> [<KEY-TYPE>
            <AUTH-KEY> {<ENCR-TYPE> [<KEY-TYPE>
            <ENCR-KEY>] | null}]
no ospfv3-af encryption {null | [ipsec spi <SPI-INDEX>
            <AUTH-TYPE> ciphertext <AUTH-KEY> [<ENCR-TYPE> ciphertext <ENCR-
KEY>] | null]}
```

Description

Configures IPsec ESP authentication. OSPFv3 AF interfaces that have IPsec configured at the interface context will not use area level IPsec ESP.

The **no** form of this command removes IPsec ESP for the specified area.

> **NOTE**
>
> IPsec is not supported in 6in6 tunnel interfaces.

Parameter	Description
`spi <SPI-INDEX>`	Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. Range: 256 to 4294967295.
`<AUTH-TYPE>`	Specifies the authentication type: **md5** or **sha1**.
`<KEY-TYPE>`	Specifies the key type to use: **plaintext** (unencrypted), **hex-string** (encrypted) or **ciphertext** (encrypted).
`<AUTH-KEY>`	Specifies the authentication key.
`<ENCR-TYPE>`	Specifies the encryption type: **des**, **3des**, **aes**, or **null**. **NOTE** **aes** is considered to be AES128, AES192, or AES256 based on key length.
	Specifies the encryption key.

Parameter	Description
<ENCR-KEY>	

> **NOTE**
>
> When the authentication key is not provided on the command line, **plaintext** authentication key prompting occurs upon pressing Enter, followed by encryption type prompting, and finally **plaintext** encryption key prompting. The entered key characters are masked with asterisks.

> **NOTE**
>
> When the authentication key and encryption type are provided on the command line but the encryption key is not provided, **plaintext** encryption key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

Examples

Setting interface **1/1/1** to use IPsec ESP with provided authentication and encryption keys:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af encryption ipsec spi 256 sha1 plaintext
abcdef
                        aes plaintext abcdefabcdefabcdefab
```

Setting interface **1/1/1** to use IPSec ESP with provided plaintext authentication key and null encryption:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af encryption ipsec spi 256 sha1 plaintext abcdef
null
```

Setting interface **1/1/1** to use IPsec ESP with prompted authentication and encryption keys:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af encryption ipsec spi 256 sha1
Enter the IPsec authentication key: **********
Re-Enter the IPsec authentication key: **********
Enter the IPsec encryption type (3des/aes/des/null)? des
Enter the IPsec encryption key: **********
Re-Enter the IPsec encryption key: **********
```

Setting interface **1/1/1** to use IPsec ESP with provided authentication password and encryption type but a prompted encryption key:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af encryption ipsec spi 256 sha1 plaintext abcdef
des
  ciphertext   Configure ciphertext key.
  hex-string   Configure hex-string key. Key length must be 16 Hex digits.
  plaintext    Configure plaintext key. Key length must be 8 characters.
Enter the IPsec encryption key: ********
Re-Enter the IPsec encryption key: ********
```

Removing IPsec from interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no ospfv3-af encryption
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ospfv3-af ipv4 retransmit-interval`

Syntax

```
ospfv3-af ipv4 retransmit-interval <INTERVAL>
no ospfv3-af ipv4 retransmit-interval
```

Description

Sets the time between retransmitting lost link state advertisements for the OSPFv3 AF interface.

The **no** form of this command sets the time between retransmitting lost link state advertisements to the default of 5 seconds for the OSPFv3 AF interface.

Parameter	Description
`<INTERVAL>`	Specifies the retransmit interval, in seconds. Range: 1 to 3600. Default: 5.

Examples

Setting OSPFv3 AF retransmit interval on the interface:

```
switch(config)# interface 1
switch(config-if)# ospfv3-af ipv4 retransmit-interval 30
```

Setting OSPFv3 AF retransmit interval to the default on the interface:

```
switch(config)# interface 1
switch(config-if)# no ospfv3-af ipv4 retransmit-interval
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ospfv3-af ipv4 area`

Syntax

```
ospfv3-af <process-id> ipv4 area <area-id>
no ospfv3-af <process-id> ipv4 area <area-id>
```

Description

This command will run the OSPFv3 AF protocol on the interface with the configured link-local IPv6 address for the area mentioned. For moving an interface to a new area, you need to unmap the existing area and then associate a new area to the interface. If the OSPFv3 AF process or area is not created, user confirmation is requested to create the OSPFv3 AF process and area.

The **no** form of this command disables OSPFv3 AF on the interface.

Parameter	Description
`<process-id>`	Specify the OSPFv3 AF Process ID. Range: 1 - 65535
`area-id`	Area identifier in IPv4 format (**x.x.x.x**), where **x** is a decimal nu mber from 0 to 255 or using a decimal value (0 - 4294967295).

Examples

Enabling OSPFv3 AF on an interface using IPv4 address:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af 1 ipv4 area 1
switch(config-if)# ospfv3-af 1 ipv4 area 0.0.0.1
```

Disabling OSPFv3 AF on an interface using IPv4 address:

```
switch(config)# interface 1/1/1
switch(config-if)# no ospfv3-af 1 ipv4 area 1
switch(config-if)# no ospfv3-af 1 ipv4 area 0.0.0.1
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ospfv3-af ipv4 cost`

Syntax

```
ospfv3-af ipv4 cost <INTERFACE-COST>
no ospfv3-af ipv4 cost
```

Description

Sets the cost (metric) associated with a particular interface. The interface cost is used as a parameter to calculate the best routes.

The **no** form of this command sets the cost (metric) associated with a particular interface to the default of 1. The default cost is calculated automatically depending on the bandwidth of the interface.

Parameter	Description
<INTERFACE-COST>	Specifies the interface cost value. Range: 1 to 65535. Default: 1.

Examples

Setting OSPFv3 AF interface cost:

```
switch(config)# interface 1
switch(config-if)# ospfv3-af ipv4 cost 100
```

Setting the OSPFv3 AF interface cost to default:

```
switch(config)# interface 1
switch(config-if)# no ospfv3-af ipv4 cost
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ospfv3-af ipv6 dead-interval`

Syntax

```
ospfv3-af ipv6 dead-interval <seconds>
no ospfv3-af ipv6 dead-interval
```

Description

Sets the interval after a neighbor is declared dead when no hello packet is received on the OSPFv3 AF interface. Dead interval must be greater than hello interval, otherwise, the OSPF protocol will have an undefined behavior.

The **no** form of this command sets the interval after which a neighbor is declared dead, to the default for the OSPFv3 AF interface. The default value is 40 seconds (generally four times the hello packet interval).

Parameter	Description
<seconds>	Specifies the time interval for the dead interval, in seconds. Ran ge: 1 to 65535. Default: 40.

Examples

Setting OSPFv3 AF dead interval on the interface:

```
switch(config)# interface 1
switch(config-if)# ospfv3-af ipv6 dead-interval 30
```

Setting OSPFv3 AF dead interval to default on the interface:

```
switch(config)# interface 1
switch(config-if)# no ospfv3-af ipv6 dead-interval
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ospfv3-af ipv4 hello-interval`

Syntax

```
ospfv3-af ipv4 hello-interval <INTERVAL>
no ospfv3-af ipv4 hello-interval
```

Description

Sets the time interval between OSPFv3 AF hello packets for the OSPFv3 AF interface.

The **no** form of this command sets the time interval between OSPFv3 AF hello packets to the default for the OSPFv3 AF interface of 10 seconds.

Parameter	Description
	Specifies the time interval between hello packets, in seconds. R ange: 1 to 65535. Default: 10.

Parameter	Description
<INTERVAL>	

Examples

Setting OSPFv3 AF hello interval on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af ipv4 hello-interval 30
```

Setting OSPFv3 AF hello interval to default on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ospfv3-af ipv4 hello-interval
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ospfv3-af ipv4 network`

Syntax

```
ospfv3-af ipv4 network {broadcast|point-to-point}
no ospfv3-af ipv4 network {broadcast|point-to-point}
```

Description

Configures the network type for the interface. By default the network type is broadcast network.

The **no** form of this command sets the network type for the interface to the system default as broadcast network for ethernet interface and point-to-point network for tunnel interface.

Parameter	Description
`broadcast`	Specifies the OSPFv3 AF network type as a broadcast multiacc ess network.
`point-to-point`	Specifies the OSPFv3 AF network type as a point - to - point n etwork.

Examples

Setting OSPFv3 AF network type for the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af ipv4 network broadcast
switch(config-if)# ospfv3-af ipv4 network point-to-point
```

Disabling OSPFv3 AF network type for the interface to system default of broadcast network:

```
switch(config)# interface 1/1/1
switch(config-if)# no ospfv3-af ipv4 network
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ospfv3-af ipv4 priority`

Syntax

```
ospfv3-af ipv4 priority <number-value>
no ospfv3-af ipv4 priority
```

Description

Sets the OSPFv3 AF priority for the interface. The larger the numeric value of the priority, the higher the chance it will become the designated router. Setting a priority of 0 makes the router ineligible to become a designated router or back up designated router.

The **no** form of this command sets the OSPFv3 AF priority for the interface to the default of 1.

Parameter	Description
<number-value>	Specifies the priority value. Default: 1. Range: 0 to 255.

Examples

Setting the OSPFv3 AF priority for the interface:

```
switch(config)# interface 1
switch(config-if)# ospfv3-af ipv4 priority 50
```

Setting the OSPFv3 AF priority for the interface to the default of 1:

```
switch(config)# interface 1
switch(config-if)# no ospfv3-af ipv4 priority
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

ospfv3-af ipv6 retransmit-interval

Syntax

```
ospfv3-af ipv6 retransmit-interval <INTERVAL>
no ospfv3-af ipv6 retransmit-interval
```

Description

Sets the time between retransmitting lost link state advertisements for the OSPFv3 AF interface.

The **no** form of this command sets the time between retransmitting lost link state advertisements to the default 5 seconds.

Parameter	Description
<INTERVAL>	Specifies the time interval for the retransmit interval, in seconds . Range: 1 to 3600. Default: 5

Examples

Setting OSPFv3 AF retransmit interval on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af ipv6 retransmit-interval 30
```

Setting OSPFv3 AF retransmit interval to the default on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ospfv3-af ipv6 retransmit-interval
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ospfv3-af ipv4 shutdown`

Syntax

```
ospfv3-af ipv4 shutdown
no ospfv3-af ipv4 shutdown
```

Description

Disables OSPFv3 AF on the interface. The interface state changes to Down. It does not remove the interface from the OSPF area. To remove the interface, use the command **no ospfv3-af ipv4 area**.

The **no** form of this command re-enables OSPFv3 AF on the interface.

Examples

Disabling OSPFv3 AF on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af ipv4 shutdown
```

Re-enabling OSPFv3 AF on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ospfv3-af ipv4 shutdown
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ospfv3-af ipv4 transit-delay`

Syntax

```
ospfv3-af ipv4 transit-delay <seconds>
no ospfv3-af ipv4 transit-delay <seconds>
```

Description

Sets the time delay in Link state transmission for the OSPFv3 AF interface.

The **no** form of this command sets the transit delay in Link state transmission to the default of 1 second.

Parameter	Description
<DELAY>	Specifies the time delay for the transit delay, in seconds. Range: 1 to 3600. Default: 1.
no	Sets the delay in Link state transmission to the default of 1 for virtual links.

Examples

Setting OSPFv3 AF transit delay on the interface

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af ipv4 transit-delay 30
```

Setting OSPFv3 AF transit delay to default on the interface

```
switch(config)# interface 1/1/1
switch(config-if)# no ospfv3-af ipv4 transit-delay
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ospfv3-af ipv6 area`

Syntax

```
ospfv3-af <PROCESS-ID> ipv6 area <AREA-ID>
no ospfv3-af <PROCESS-ID> ipv6 area <area-id>
```

Description

This command runs the OSPFv3 AF IPv6 on the interface with the configured link-local IPv6 address for the area specified. To move an interface to a new area, unmap the existing area and then associate a new area with the interface. If the OSPFv3 AF process or area is not created, user confirmation is requested to create the OSPFv3 AF process and area.

The **no** form of this command disables OSPFv3 AF on the interface and removes the interface from the area. Interfaces which have an IP address configured on the network or in a subset of the network, stop participating in the OSPF protocol

Parameter	Description
	Specifies the area ID is one of the following formats. • Area ID in IPv4 format (**x.x.x.x**), where **x** is a decimal num ber from 0 to 255.

Parameter	Description
<AREA-ID>	• Area ID as a decimal value. Range: 0 - 4294967295.
<PROCESS-ID>	Specifies the OSPFv3 AF process ID. Range: 1 to 65535.

Examples

Setting OSPFv3 AF network for the area:

```
switch(config)# interface 1
switch(config-if)# ospfv3-af 1 ipv6 area 1
switch(config-if)# ospfv3-af 1 ipv6 area 0.0.0.1
```

Disabling OSPFv3 AF network for the area:

```
switch(config)# interface 1
switch(config-if)# no ospfv3-af 1 ipv6 area 1
switch(config-if)# no ospfv3-af 1 ipv6 area 0.0.0.1
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ospfv3-af ipv6 cost`

Syntax

```
ospfv3-af ipv6 cost <INTERFACE-COST>
no ospfv3-af ipv6 cost
```

Description

Sets the cost (metric) associated with a particular interface. The interface cost is used as a parameter to calculate the best routes.

The **no** form of this command sets the cost (metric) associated with a particular interface to the default which is calculated automatically depending on the bandwidth of the interface.

Parameter	Description
<INTERFACE-COST>	Specifies the interface cost value. Range: 1 to 65535. Default: 1.

Examples

Setting OSPFv3 AF interface cost:

```
switch(config)# interface vlan 1
switch(config-if)# ospfv3-af ipv6 cost 20
```

Setting the OSPFv3 AF interface cost to default:

```
switch(config)# interface vlan 1
switch(config-if)# no ospfv3-af ipv6 cost
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ospfv3-af ipv4 dead-interval`

Syntax

```
ospfv3-af ipv4 dead-interval <seconds>
```

Description

This command sets the interval after which a neighbor is declared dead if no hello packet is received for the OSPFv3 AF interface. Dead interval must always be greater than hello-interval, otherwise the OSPFv3 AF protocol will have undefined behavior.

The **no** form of this command sets the interval to the default value of 40 seconds.

Parameter	Description
seconds	Specifies Dead interval value in second. Default is 40. Range: 1 - 65535

Examples

Configuring an interval of 30 to consider an unresponsive neighbor as down:

```
switch(config)# interface 1
switch(config-if)# ospfv3-af ipv4 dead-interval 30
```

Setting the dead interval to the default value:

```
switch(config)# interface 1
switch(config-if)# no ospfv3-af ipv4 dead-interval
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ospfv3-af ipv6 hello-interval`

Syntax

```
ospfv3-af ipv6 hello-interval <INTERVAL>
no ospfv3-af ipv6 hello-interval
```

Description

Sets the time interval between OSPFv3 AF hello packets for the OSPFv3 AF interface.

The **no** form of this command sets the time interval between OSPFv3 AF hello packets to the default for the OSPFv3 AF interface of 10 seconds.

Parameter	Description
<INTERVAL>	Specifies the time interval between hello packets, in seconds. R ange: 1 to 65535. Default: 10.

Examples

Setting OSPFv3 AF hello interval on the interface:

```
switch(config)# interface 1
switch(config-if)# ospfv3-af ipv6 hello-interval 30
```

Setting OSPFv3 AF hello interval to default on the interface:

```
switch(config)# interface 1
switch(config-if)# no ospfv3-af ipv6 hello-interval
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ospfv3-af ipv6 network`

Syntax

```
ospfv3-af ipv6 network {broadcast|point-to-point}
no ospfv3-af ipv6 network {broadcast|point-to-point}
```

Description

Configures the network type for the interface. By default the network type is broadcast network.

The **no** form of this command sets the network type for the interface to the system default as broadcast network for ethernet interface and point-to-point network for tunnel interface.

Parameter	Description
`broadcast`	Specifies the OSPFv3 AF network type as a broadcast multiacc ess network.
`point-to-point`	Specifies the OSPFv3 AF network type as a point - to - point n etwork.

Examples

Setting OSPFv3 AF network type for the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af ipv6 network broadcast
```

```
switch(config-if)# ospfv3-af ipv6 network point-to-point
```

Disabling OSPFv3 AF network type for the interface to system default of broadcast network:

```
switch(config)# interface 1/1/1
switch(config-if)# no ospfv3-af ipv6 network
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ospfv3-af ipv6 passive`

Syntax

```
ospfv3-af ipv6 passive
no ospfv3-af ipv6 passive
```

Description

Configures the interface as an OSPFv3 AF passive interface. With this setting, the interface participates in the OSPF, but does not send or receive OSPF packets on that interface.

The **no** form of this command resets the interface as active. With this setting, the interface starts sending and receiving OSPF packets.

Examples

Setting the interface as OSPFv3 AF passive interface:

```
switch(config)# interface vlan 1
switch(config-if)# ospfv3-af ipv6 passive
```

Setting the interface as OSPFv3 AF active interface:

```
switch(config)# interface vlan 1
switch(config-if)# no ospfv3-af ipv6 passive
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ospfv3-af ipv6 priority`

Syntax

```
ospfv3-af ipv6 priority <number-value>
no ospfv3-af ipv6 priority
```

Description

Sets the OSPFv3 AF priority for the interface. The larger the numeric value of the priority, the higher the chance it will become the designated router. Setting a priority of 0 makes the router ineligible to become a designated router or back up designated router.

The **no** form of this command sets the OSPFv3 AF priority for the interface to the default of 1.

Parameter	Description
<number-value>	Specifies the OSPFv3 AF priority value. Default: 1. Range: 0 to 2 55.

Examples

Setting the OSPFv3 AF priority for the interface:

```
switch(config)# interface vlan /1
switch(config-if)# ospfv3-af ipv6 priority 50
```

Setting the OSPFv3 AF priority for the interface to the default of 1:

```
switch(config)# interface vlan 1
switch(config-if)# no ospfv3-af ipv6 priority
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ospfv3-af ipv6 shutdown`

Syntax

```
ospfv3-af ipv6 shutdown
no ospfv3-af ipv6 shutdown
```

Description

Disables OSPFv3 AF on the interface. The interface state changes to Down. It does not remove the interface from the OSPF area. To remove the interface, use the command **no ospfv3-af ipv6 area**.

The **no** form of this command re-enables OSPFv3 AF on the interface.

Examples

Disabling OSPFv3 AF on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af ipv6 shutdown
```

Re-enabling OSPFv3 AF on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ospfv3-af ipv6 shutdown
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ospfv3-af ipv6 transit-delay`

Syntax

```
ospfv3-af ipv6 transit-delay <seconds>
no ospfv3-af ipv6 transit-delay <seconds>
```

Description

Sets the time delay in Link state transmission for the OSPFv3 AF virtual link.

The **no** form of this command sets the transit delay in Link state transmission to the default of 1 second.

Parameter	Description
	Specifies the time delay for the transit delay, in seconds. Range: 1 to 3600. Default: 1.

Parameter	Description
<DELAY>	
no	Sets the delay in Link state transmission to the default of 1 for virtual links.

Examples

Setting OSPFv3 AF transit delay on the interface

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# transit-delay 30
```

Setting OSPFv3 AF transit delay to default on the interface

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 100 virtual-link  100.0.1.1
switch(config-router-vlink6)# no transit-delay
switch(config-router-vlink6)# no transit-delay 30
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-router-vlink`	Administrators or local user group members with execution righ ts for this command.

`max-metric router-lsa`

Syntax

```
max-metric router-lsa [include-stub | on-startup <INTERVAL>]
no max-metric router-lsa [include-stub | on-startup <INTERVAL>]
```

Description

Sets the protocol to advertise a maximum metric so that other routers do not prefer the router as an intermediate hop in their shortest path first (SPF) calculations. If the `on-startup` parameter is used, the Router is configured to advertise a maximum metric at start up.

To disable advertisement of the maximum metric, use the **no** form of the command.

The **no** form of this command advertises the normal cost metrics instead of advertising the maximized cost metric. This setting causes the router to be considered in traffic forwarding.

Parameter	Description
`include-stub`	Configures the stub Router - LSA to advertise a maximum metr ic for all links, including stub links.
`on-startup <INTERVAL>`	Automatically advertises the stub Router - LSA (or maximize th e router - LSA cost metric) for a specified time interval upon O SPFv3 AF startup. Specifies the time in seconds. Range: 5 to 86400. Default: 600.

Examples

Setting to maximize the cost metrics for Router LSA:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# max-metric router-lsa on-startup 100
```

Setting to advertise the normal cost metrics instead of advertising the maximized cost metric:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# no max-metric router-lsa
switch(config-ospfv3-ipv6-uc)# no max-metric router-lsa on-startup
```

Setting to advertise Router LSA for stub links:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# max-metric router-lsa
switch(config-ospfv3-ipv6-uc)# max-metric router-lsa include-stub on-
startup 100
```

Setting to advertise the normal cost metrics instead of advertising the maximized cost metric for stub links:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-af-1)# no max-metric router-lsa
switch(config-ospfv3-ipv6-uc)# no max-metric router-lsa include-stub on-
startup
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-ipv4-uc<PROCESS-ID>` `config-ospfv3-ipv6-uc<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

ospfv3-af ipv4 passive

Syntax

```
ospfv3-af ipv4 passive
```

Description

This command configures the interface as an OSPFv3 AF passive interface. With this setting the interface participates in OSPF but does not receive or send OSPF packets on that interface.

Example

Configuring interface 1/1/1 as an OSPFv3 AF passive interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ospfv3-af ipv4 passive
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`passive-interface default`

Syntax

```
passive-interface default
no passive-interface default
```

Description

Sets all OSPFv3 AF interfaces as passive.

The **no** form of this command sets all the OSPFv3 AF interfaces as active.

Examples

Setting OSPFv3 AF enabled interfaces as passive:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-ipv4-uc)# passive-interface default
```

Setting OSPFv3 AF enabled interfaces as active:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-ipv4-uc)# no passive-interface default
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-ipv4-uc<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

redistribute

Syntax

```
redistribute {bgp | connected | host-routes | local loopback | static |
ripng |
   ospf <PROCESS-ID> | ospfv3-af <PROCESS-ID>} [route-map <ROUTE-MAP-NAME>]
no redistribute {bgp | connected | host-routes | local loopback | static |
ripng |
   ospf <PROCESS-ID> | ospfv3-af <PROCESS-ID>} [route-map <ROUTE-MAP-NAME>]
```

Description

This command redistributes routes originating from other protocols to OSPFv3 AF. This command allows to redistribute routes from other OSPFv3 AF instances. IPv4 address family allows redistribution from other OSPFv3 Ipv4 AFs. IPv6 address family allows redistribution from other OSPFv3 Ipv6 AFs. In addition, if a route-map is specified, then routes that pass the match clause specified in the route-map are only redistributed to OSPFv3 AF. Configuration is not allowed if the referenced route-map has not yet been configured.

Only BGP leaked routes will be given to OSPFv3 AF for redistribution. If you try to redistribute routes from an OSPF process which is not created, user confirmation is asked to create the process before proceeding with redistribution. If accepted, the process is created with defaults and redistribution configuration is applied. If denied, redistribution configuration is skipped.

If command **route-redistribute active-routes-only** has been issued, only the routes from other protocols which are selected for forwarding are considered for redistribution into OSPFv3 AF. With the change in redistribute behavior, additional explicit redistribution of connected routes may be needed if any of the deployment assumes redistribution of connected networks will happen via redistribution of dynamic protocol routes into OSPF.

The **no** form of this command disables redistribution of routes to the current OSPFv3 AF process.

Parameter	Description
`bgp`	Specifies redistributing BGP (Border Gateway Protocol) routes into OSPFv3 AF.
`connected`	Specifies redistributing connected (directly attached subnet or host).
`host-routes`	Redistribute dynamic, IVRL and static ARP/ND entries into OSPFv3 AF. ARP/ND entries learnt from EVPN will not be redistributed. The **host - routes** parameter is applicable to HPE Aruba Networking 8325, 10000, 8360, and 9300/9300S Switch series only.
`local loopback`	Specifies redistributing local routes of the loopback interface.
`static`	Specifies redistributing static routes into OSPFv3 AF.
`ripng`	Specifies redistributing RIPng routes into OSPFv3 AF.
`ospf` *<PROCESS-ID>*	Specifies redistributing routes from the specified OSPF process ID. Range: 1 to 65535.
`ospfv3-af` *<PROCESS-ID>*	Specifies redistributing routes from the specified OSPFv3 AF process ID. Range: 1 to 65535.
`route-map` *<ROUTE-MAP-NAME>*	Specifies redistribution filtering by route map. To create a route map, use command **route - map**.

Examples

Redistributing routes to OSPFv3 AF:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-ipv4-uc) # redistribute bgp
switch(config-ospfv3-ipv4-uc) # redistribute connected
switch(config-ospfv3-ipv4-uc) # redistribute host-routes
switch(config-ospfv3-ipv4-uc) # redistribute static
switch(config-ospfv3-ipv4-uc) # redistribute ripng
switch(config-ospfv3-ipv4-uc) # redistribute ospfv3-af 2
switch(config-ospfv3-ipv4-uc) # redistribute local loopback
```

```
switch(config-ospfv3-ipv4-uc) # redistribute static route-map
% Command incomplete.
switch(config-ospfv3-ipv4-uc) # redistribute static route-map
static_networks
switch(config-ospfv3-af-1)#  redistribute static route-map static_networks1
Route Map static_networks1 not found
switch(config-ospfv3-af-1)#  redistribute static route-map static_networks12
Redistribute with route-map config exists
switch(config-ospfv3-ipv4-uc) # redistribute ripng route-map
% Command incomplete.
switch(config-ospfv3-ipv4-uc) # redistribute ripng route-map ripng_networks
switch(config-ospfv3-af-1)#  redistribute ripng route-map ripng_networks1
Route Map ripng_networks1 not found
switch(config-ospfv3-af-1)#  redistribute ripng route-map ripng_networks12
Redistribute with route-map config exists
switch(config-ospfv3-ipv4-uc) # redistribute ospfv3-af 2 route-map
% Command incomplete.
switch(config-ospfv3-ipv4-uc) # redistribute ospfv3-af 2 route-map
tagged_routes
switch(config-ospfv3-ipv4-uc)#  redistribute ospfv3-af 2 route-map
tagged_routes1
Route Map tagged_routes1 not found
switch(config-ospfv3-ipv4-uc)#  redistribute ospfv3-af 2 route-map
tagged_routes12
Redistribute with route-map config exists
switch(config-ospfv3-ipv4-uc)# redistribute local loopback route-map
% Command incomplete.
switch(config-ospfv3-ipv4-uc)# redistribute local loopback route-map
local_routes
switch(config-ospfv3-af-1)# redistribute local loopback route-map
local_routes1
Route Map local_routes1 not found
switch(config-ospfv3-ipv4-uc)# redistribute local loopback route-map
local_routes12
Redistribute with route-map config exists
switch(config-ospfv3-ipv4-uc)# redistribute host-routes route-map
% Command incomplete.
switch(config-ospfv3-ipv4-uc)# redistribute host-routes route-map
host_routes
switch(config-ospfv3-ipv4-uc)# redistribute host-routes route-map
host_routes1
Route Map local_routes1 not found
switch(config-ospfv3-ipv4-uc)# redistribute host-routes route-map
```

```
host_routes12
Redistribute with route-map config exists
```

Disabling redistributing routes to OSPFv3 AF:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-ipv6-uc) # no redistribute bgp
switch(config-ospfv3-ipv6-uc) # no redistribute connected
switch(config-ospfv3-ipv6-uc) # no redistribute host-routes
switch(config-ospfv3-ipv6-uc) # no redistribute static
switch(config-ospfv3-ipv6-uc) # no redistribute ripng
switch(config-ospfv3-ipv6-uc) # no redistribute ospfv3-af 2
switch(config-ospfv3-ipv6-uc) # no redistribute local loopback
switch(config-ospfv3-ipv6-uc) # no redistribute static route-map
% Command incomplete.
switch(config-ospfv3-ipv6-uc) # no redistribute static route-map
static_networks
switch(config-ospfv3-ipv6-uc) # no redistribute ripng route-map
% Command incomplete.
switch(config-ospfv3-ipv6-uc) # no redistribute ripng route-map
ripng_networks
switch(config-ospfv3-ipv6-uc) # no redistribute ospfv3-af 2 route-map
% Command incomplete.
switch(config-ospfv3-ipv6-uc) # no redistribute ospfv3-af 2 route-map
tagged_routes
switch(config-ospfv3-ipv6-uc) # no redistribute local loopback route-map
% Command incomplete.
switch(config-ospfv3-ipv6-uc) # no redistribute local loopback route-map
local_routes
switch(config-ospfv3-ipv6-uc)# no redistribute host-routes route-map
% Command incomplete.
switch(config-ospfv3-ipv6-uc)# no redistribute host-routes route-map
host_routes
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-ipv4-uc<PROCESS-ID>` `config-ospfv3-ipv6-uc<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

reference-bandwidth

Syntax

```
reference-bandwidth <BANDWIDTH-VALUE>
no reference-bandwidth
```

Description

Sets the reference bandwidth for OSPFv3 AF. If the OSPFv3 AF interface cost is not explicitly set, then the cost of all the OSPFv3 AF interfaces is recalculated based on the reference bandwidth and link speed of the interface.

For VLAN interfaces the calculated link speed value is 1 Gbps (if the OSPFv3 AF interface cost is not explicitly set).

The **no** form of this command sets the reference bandwidth for OSPFv3 AF to the default of 100000 Mbps.

Parameter	Description
<BANDWIDTH-VALUE>	Specifies the reference bandwidth used to calculate the cost of an interface in Mbps. Range: 1 to 4000000. Default: 100000.

Examples

Setting the reference bandwidth:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-1)# reference-bandwidth 40000
```

Setting the reference bandwidth to the default value:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-1)# no reference-bandwidth
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`retransmit-interval`

Syntax

```
retransmit-interval <INTERVAL>
no retransmit-interval
```

Description

Sets the time between retransmitting lost link state advertisements for virtual links.

The **no** form of this command sets the time between retransmitting lost link state advertisements to the default of 5 seconds for virtual links.

Parameter	Description
	Specifies the time interval for the retransmit interval, in seconds . Range: 1 to 3600. Default: 5.

Parameter	Description
<INTERVAL>	

Examples

Setting OSPFv3 AF virtual links retransmit interval:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# retransmit-interval 30
```

Setting OSPFv3 AF virtual links retransmit interval to default:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# no retransmit-interval
switch(config-router-vlink6)# no retransmit-interval 30
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-router-vlink`	Administrators or local user group members with execution righ ts for this command.

`router-id`

Syntax

```
router-id <ROUTER-ADDRESS>
no router-id
```

Description

Sets an ID for the router in an IPv4 address format.

Parameter	Description
`<ROUTER-ADDRESS>`	Specifies the router address in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255. Required
`no`	Unconfigures the router - id for the instance. The Router - id is changed to the dynamically selected router - id. The default ro uter - id 0.0.0.0 is updated to the routing stack that triggers to auto - elect a router - id based on the highest ip - address of l oopback interface or highest ip - address of interfaces. The co nfiguration is supported globally as well as per address - famili es ipv4 - unicast, ipv6 - unicast. If address - family specific rou ter - id is configured, it takes precedence over global configurat ion.

Examples

Setting router-id in the OSPFv3 AF context:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1) # router-id 1.1.1.1
switch(config-ospfv3-af-1) # address-family ipv4 unicast
switch(config-ospfv3-ipv4-uc) # router-id 2.2.2.2
switch(config-ospfv3-af-1) # address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc) # router-id 3.3.3.3
```

Unconfiguring router-id:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-ipv4-uc)# no router-id
switch(config-ospfv3-ipv4-uc)# no router-id 1.1.1.1
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-ipv4-uc<PROCESS-ID>` `config-ospfv3-ipv6-uc<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`router ospfv3-af`

Syntax

```
router ospfv3-af <PROCESS-ID> [vrf <VRF-NAME>]
no router ospfv3-af <PROCESS-ID> [vrf <VRF-NAME>]
```

Description

Creates the OSPFv3 AF process (if not created already) and enters the router OSPFv3 AF instance context. Optionally if specified, you can specify a named VRF, or the default VRF if the <vrf-name> is not specified. Maximum of eight OSPFv3 processes are allowed per VRF.

The switch allows configuration of OSPFv2, OSPFv3, OSPFv3 IPv4 AF and OSPFv3 IPv6 AF. But at any point only 8 OSPF instances can run per address-family, 8 instances for IPv4 (OSPFv2 + OSPFv3 IPv4 AF) and 8 instances for IPv6 (OSPFv3 + OSPFv3 IPv6 AF).

The **no** form of this command removes the OSPFv3 AF instance. If a VRF is specified, it removes the OSPF instance from the named VRF, or the default VRF if the <var-name> is not specified.

Parameter	Description
`<PROCESS-ID>`	Specifies an OSPFv3 AF process ID. Length: 1 to 65535.
`vrf <VRF-NAME>`	Specifies the name of a VRF. Default: default.

Examples

Entering the router OSPFv3 AF instance:

```
switch(config)# router ospfv3-af 1
```

Setting the router OSPFv3 VRF instance:

```
switch(config)# router ospfv3-af 1 vrf vrf_red
```

Removing the router OSPFv3 AF instance:

```
switch#(config)# no router ospfv3-af 1
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`show ospfv3-af ipv4 statistics`

Syntax

```
show ospfv3-af [<process-id>] ipv4 statistics [all-vrfs | vrf <vrf-name>]
```

Description

This command displays the OSPFv3 AF shortest path first (SPF) calculation statistics.

Parameter	Description
<PROCESS-ID>	Specifies an OSPFv3 AF process ID. Range: 1 to 65535
`all-vrfs`	Display the OSPFv3 AF statistics information for all the VRFs.
`vrf <VRF-NAME>`	Specify the VRF name.

Example

```
switch# show ospfv3-af ipv4 statistics
Address-family : IPv4 Unicast
-----------------------------
OSPFv3 AF Process ID 1 VRF default, Statistics (cleared 3h 2m 21s ago)
---------------------------------------------------------------------
Unknown Interface Drops          : 0
Unknown Virtual Interface Drops : 0
Bad IPv6 Header Length Drops     : 0
Wrong OSPFv3 Version Drops       : 0
Bad Source IPv6 Drops            : 0
Resource Failure Drops           : 0
Bad Header Length Drops          : 0
Total Drops                      : 0
switch# show ospfv3-af 2 ipv4 statistics vrf red
Address-family : IPv4 Unicast
-----------------------------
OSPFv3 AF Process ID 2 VRF red, Statistics (cleared 3h 2m 30s ago)
-----------------------------------------------------------------
Unknown Interface Drops          : 0
Unknown Virtual Interface Drops : 0
Bad IPv6 Header Length Drops     : 0
Wrong OSPFv3 Version Drops       : 0
Bad Source IPv6 Drops            : 0
Resource Failure Drops           : 0
```

```
Bad Header Length Drops          : 0
Total Drops                      : 0
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show debug buffer module ospfv3-af`

Syntax

```
show debug buffer module ospfv3-af
   [severity (alert|crit|debug|emer|err|info|notice|warn)]
```

Description

Displays debug logs for OSPFv3 AF stored in the specified debug buffer.

Parameter	Description
<MODULE-NAME>	Filters debug logs displayed by the specified module name.
`severity (alert|crit|debug| emer| err|info|notice|warn)`	Displays debug logs for OSPFv3 AF with a specified severity l evel. Defaults to**debug**. Optional.
`alert`	Displays debug logs with a severity level of **alert (6)** and above.

Parameter	Description
`crit`	Displays debug logs with a severity level of **critical (5)** and a bove.
`debug`	Displays debug logs with all severities.
`emer`	Displays debug logs with a severity level of **emergency (7)** onl y.
`err`	Specifies storage of debug logs with severity level of **error (4)** a nd above.
`info`	Displays debug logs with a severity level of **info (1)** and above.
`notice`	Specifies storage of debug logs with severity level of **notice (2)** and above.
`warning`	Displays debug logs with a severity level of **warning (3)** and a bove.

Examples

```
switch(config)# show debug
----------------------------------------------------------------------------
-------------
module      sub_module      severity   vlan  port     ip        mac
instance       vrf
----------------------------------------------------------------------------
--------------
ospfv3-af   all             debug
switch(config)# show debug buffer module ospfv3-af
----------------------------------------------------------------------------
-----------------------------------------------------------
show debug buffer
----------------------------------------------------------------------------
-----------------------------------------------------------
2017-09-18:10:35:38.621611|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
OSPF 537265152 hitless restart state changed (None -> None),
reason: None.
2017-09-18:10:35:38.621626|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
(OSPFv3 process Id = 1, VRF Name = default)
2017-09-18:10:35:38.621836|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
0X01111000 Protocol Manager has initialized successfully.
2017-09-18:10:35:38.621846|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
Entity index              = OSPF 537265152
```

```
2017-09-18:10:35:38.621854|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
(OSPFv3 process Id = 1, VRF Name = default)
2017-09-18:10:35:38.622188|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
OSPF 537265152 hitless restart state changed (None -> None),
reason: None.
2017-09-18:10:35:38.622197|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
(OSPFv3 process Id = 1, VRF Name = default)
2017-09-18:10:35:38.622423|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
OSPF 537265152 created.
2017-09-18:10:35:38.622433|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
(OSPFv3 process Id = 0, VRF Name = default)
2017-09-18:10:35:38.622516|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|A
master join has been successfully registered.
2017-09-18:10:35:38.622524|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
Join interface type       = 2
2017-09-18:10:35:38.622536|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
Join interface ID         = 0X00000001
2017-09-18:10:35:38.622544|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
(OSPFv3 process Id = 1, VRF Name = default)
2017-09-18:10:35:38.622717|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
OSPF 537265152 i/f idx 0X10010101: OSPF has been told over
the I3 interface about a new address on an
2017-09-18:10:35:38.622728|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
interface, or about updated attributes of an existing address.
2017-09-18:10:35:38.622738|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
Address                          = 111::1
2017-09-18:10:35:38.622746|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
Scope ID                         = 268501249
2017-09-18:10:35:38.622754|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
Prefix length                    = 64
2017-09-18:10:35:38.622762|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
Remote router address            = ::
2017-09-18:10:35:38.622771|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
Address use flags                = 0X00
2017-09-18:10:35:38.622780|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
(OSPFv3 process Id = 1, VRF Name = default)
2017-09-18:10:35:38.625212|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
OSPF 537265152 activated successfully
2017-09-18:10:35:38.625225|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
Router ID              rtr ID 1.1.1.1
2017-09-18:10:35:38.625235|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
(OSPFv3 process Id = 1, VRF Name = default)
2017-09-18:10:35:38.625286|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
```

```
OSPF 537265152 was Down and is now Going Up.
2017-09-18:10:35:38.625295|hpe-routing|LOG_INFO|AMM|-|OSPFV3-AF|OSPFV3-AF|
(OSPFv3 process Id = 1, VRF Name = default)
2017-09-18:10:35:38.625309|hpe-routing|LOG_ERR|AMM|-|OSPFV3-AF|
OSPFv3_AF_EVENT|Failed to configure interface cost for stub link port 1/1/1.
2017-09-18:10:35:38.625312|hpe-routing|LOG_ERR|AMM|-|OSPFV3-AF|
OSPFv3_AF_EVENT|Could not send ospfv3VirtNbrStateChange Trap
2020-07-21:11:36:47.324342|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|OSPF 537265152 NM has sent a Hello packet.
2020-07-21:11:36:47.324717|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|Sending interface index =  i/f idx 0X51101000.
2020-07-21:11:36:47.324952|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|Router ID of target neighbor
(null string if sent per interface) = .
2020-07-21:11:36:47.325217|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|Destination IP address = IP addr FF02::5.
2020-07-21:11:36:47.325460|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|(End of packet, OSPFv3 process Id = 1, VRF Name = default)
2020-07-21:11:36:47.326710|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|OSPF message details:
2020-07-21:11:36:47.326946|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|Header: Src IP:FF02::5 Dst IP:FE80::800:901:4C3:C58B
Type: 1 Rtr Id:2.2.2.2 Area:0.0.0.1 Inst ID:0
2020-07-21:11:36:47.327166|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|Packet: Hello: I/f 0X51101000 Pri:1 Opt:19 Hello
Int:10 Dead Int:40 DR:2.2.2.2 BDR:1.1.1.1
2020-07-21:11:36:47.327350|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|(End of packet, OSPFv3 process Id = 1, VRF Name = default)
2020-08-12:11:59:50.183653|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|OSPF message details:
2020-08-12:11:59:50.183666|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|Header: Src IP:FE80::800:901:4C1:3751 Dst IP:FF02::5
Type: 4 Rtr Id:2.2.2.2 Area:0.0.0.1 Inst ID:0
2020-08-12:11:59:50.183676|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|Packet: LS Upd: Num LSAs 2
2020-08-12:11:59:50.183687|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|(End of packet, OSPFv3 process Id = 1, VRF Name = default)
2020-08-12:11:59:50.272301|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|OSPF message details:
2020-08-12:11:59:50.272313|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|Header: Src IP:FE80::800:901:4C1:3751 Dst IP:FF02::5
Type: 5 Rtr Id:2.2.2.2 Area:0.0.0.1 Inst ID:0
2020-08-12:11:59:50.272323|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
```

```
OSPFv3_AF_PACKET|Packet: LS Ack
2020-08-12:11:59:50.272332|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|(End of packet, OSPFv3 process Id = 1, VRF Name = default)
2020-08-12:11:59:45.072407|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|OSPF message details:
2020-08-12:11:59:45.072424|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|Header: Src IP:FE80::800:901:4C1:3751
Dst IP:FE80::800:901:40D:A274 Type: 2 Rtr Id:2.2.2.2 Area:0.0.0.1 Inst ID:0
2020-08-12:11:59:45.072443|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|Packet: DB Desc: If MTU:1500 Opt:19 I,M,MS
bits:0X01 Seq No:1020993
2020-08-12:11:59:45.072460|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|(End of packet, OSPFv3 process Id = 1, VRF Name = default)
2020-08-12:11:59:45.073077|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|OSPF message details:
2020-08-12:11:59:45.073096|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|Header: Src IP:FE80::800:901:4C1:3751
Dst IP:FE80::800:901:40D:A274 Type: 3 Rtr Id:1.1.1.1 Area:0.0.0.1 Inst ID:0
2020-08-12:11:59:45.073114|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|Packet: LS Req: First LS Type:8193 LS id:0.0.0.0
Adv Router:2.2.2.2
2020-08-12:11:59:45.073130|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|(End of packet, OSPFv3 process Id = 1, VRF Name = default)
2020-08-12:12:11:22.344623|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|OSPF message details:
2020-08-12:12:11:22.344635|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|Header: Src IP:FE80::800:901:4C1:3751
Dst IP:FE80::800:901:40D:A274 Type: 2 Rtr Id:2.2.2.2 Area:0.0.0.1 Inst ID:0
2020-08-12:12:11:22.344646|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|Packet: DB Desc: If MTU:1500 Opt:19 I,M,MS bits:0X07
Seq No:1728268
2020-08-12:12:11:22.344663|hpe-routing|LOG_DEBUG|AMM|-|OSPFV3-AF|
OSPFv3_AF_PACKET|(End of packet, OSPFv3 process Id = 1, VRF Name = default)
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show ospfv3-af ipv4`

Syntax

```
show ospfv3-af [<PROCESS-ID>] ipv4 [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows OSPFv3 AF information including area, state, and configuration information.

Parameter	Description
`<PROCESS-ID>`	Specifies an OSPFv3 AF process ID optionally to show OSPFv3 AF information for a particular OSPF process. Range: 1 to 65535.
`all-vrfs`	Select to show OSPFv3 AF information for all VRFs.
`vrf <VRF-NAME>`	Specifies the name of a VRF. The default VRF OSPFv3 AF process information is displayed if VRF is not mentioned. Default: default.

Example

Showing general OSPFv3 AF configurations:

```
switch# show ospfv3-af 200 ipv4
Address-family : IPv4 Unicast
-----------------------------
VRF : default                              Process : 20200
---------------------------------------------------------------------
Router ID                : 1.1.1.1          OSPFv3-AF              : Enabled
BFD                      : Disabled         SPF Start Interval     : 200 ms
SPF Hold Interval        : 1000 ms          SPF Max Wait Interval  : 5000 ms
```

```
LSA Start Interval      : 5000 ms          LSA Hold Interval       : 1000 ms
LSA Max Wait Interval   : 1000 ms          LSA Arrival Interval    : 1000 ms
External LSAs           : 0                Checksum Sum            : 0
ECMP                    : 4                Reference Bandwidth     : 100000
Mbps
Area Border             : Yes              AS Border               : No
GR Status               : Enabled          GR Interval             : 120 sec
GR State                : Inactive         GR Exit Status          : None
GR Helper               : Enabled          GR Strict LSA Check     : Enabled
GR Ignore Lost I/F      : Disabled         Internal Process ID     : 1
Summary address:
  prefix fd00::1/64, advertise, tag 10
Area      Total Active
----------------------
Normal      2      2
Stub        0      0
Area  : 0.0.0.0
----------------------------------
Area Type              : Normal       Status                 : Active
Total Interfaces       : 1            Active Interfaces      : 1
Passive Interfaces     : 0            Loopback Interfaces    : 0
SPF calculation count  : 4
Area ranges:
  fd00::1/64, inter-area, no-advertise
AH Authentication      : SHA1, SPI 256
Number of LSAs         : 5            Checksum Sum           : 99122
Area  : 0.0.0.1
----------------------------------
Area Type              : Normal       Status                 : Active
Total Interfaces       : 1            Active Interfaces      : 1
Passive Interfaces     : 0            Loopback Interfaces    : 0
SPF Calculation Count  : 4
Area ranges:
  fd00::1/64, inter-area, no-advertise
ESP Authentication     : SHA1
Encryption             : 3DES, SPI 256
Number of LSAs         : 5            Checksum Sum           : 99122
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show ospfv3-af ipv4 border-routers`

Syntax

```
show ospfv3-af [<PROCESS-ID>] ipv4 border-routers | [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows the OSPFv3 AF routing table entries for Area Border Router (ABR) and Autonomous System Border Router (ASBR).

Parameter	Description
`<PROCESS-ID>`	Specifies an OSPFv3 AF process ID to show the OSPFv3 AF routing table entries for ABR and ASBR for the particular OSPFv3 AF process. Range: 1 to 65535.
`all-vrfs`	Select to show OSPFv3 AF border router information for all VRFs.
`vrf <VRF-NAME>`	Specifies the name of a VRF. Default: default.

Example

Showing OSPFv3 AF border routers information for VRF **vrf_red**:

```
switch# show ospfv3-af ipv4 border-routers vrf vrf_red
Address-family : IPv4 Unicast
-----------------------------
VRF : vrf_red       Process ID : 1
             Internal Routing Table
--------------------------------------------------------
```

```
Codes: i - Intra-area route, I - Inter-area route
   Router-ID Cost  Type     Area      SPF    Nexthop       Interface
i  1.1.1.1   1     ASBR     0.0.0.0   9      10.10.10.1    1/1/2
i  3.3.3.3   1     ASBR     1.1.1.1   9      20.20.20.1    ** tunnel1
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show ospfv3-af ipv4 interface`

Syntax

```
show ospfv3-af [<PROCESS-ID>] ipv4 interface [<INTERFACE-NAME>] | vlan
<VLAN-ID> | lag <LAG-ID> | loopback <LOOPBACK-ID> | tunnel <TUNNEL-ID>]
   [brief] [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows information about OSPFv3 AF enabled IPv4 interfaces.

Parameter	Description
<PROCESS-ID>	Specifies an OSPFv3 AF process ID optionally to show the OSPFv3 AF enabled interfaces for the particular OSPFv3 AF process. Range: 1 to 65535.
	Selects to show information only for the specified OSPFv3 AF enabled interface.

Parameter	Description
<INTERFACE-NAME>	
`vlan <VLAN-ID>`	Specifies the vlan interface name.
`lag <LAG-ID>`	Specifies the lag interface name.
`loopback <LOOPBACK-ID>`	Specifies the loopback interface name.
`tunnel <TUNNEL-ID>`	Specifies the tunnel interface name.
`brief`	Include this parameter to display a brief overview of the followi ng OSPFv3 AF configuration information. • Interface: OSPFv3 AF interface name. • Area: OSPFv3 AF area ID. • Cost: The metric OSPFv3 AF uses to judge a path's feasibil ity, calculated as (reference bandwidth / interface bandwidt h). • State: Indicates if the interface is a designated router (**Dr**) o r a backup designated router (**Backup - dr**). • Status: Indicates if the interface is **up** or **down**. • Flags: P - Passive A - Active.
`all-vrfs`	Select to show OSPFv3 AF - enabled interface information for all VRFs.
`vrf <VRF-NAME>`	Specifies the name of a VRF. Default: default.

Examples

Showing OSPFv3 AF information for all interfaces in default VRF:

```
switch# show ospfv3-af ipv4 interface
Codes: DR - Designated router BDR - Backup Designated router
Address-family : IPv4 Unicast
-----------------------------
Interface 1/1/1 is up, Line Protocol is up, Instance Id 64
----------------------------------------------------------
 VRF                      : default
Process                  : 1
 IP Address               : 1.1.1.2
```

```
Area                    : 0.0.0.0
 Status                  : Up                              Network
Type          : Broadcast
 Hello Interval        : 10    sec                         Dead
Interval         : 40     sec
 Transit Delay         : 1     sec                         Retransmit
Interval   : 5      sec
 BFD                     : Disabled                        Link
Speed            : 1000 Mbps
 Cost Configured       : 1                                 Cost
Calculated       : 4
 State/Type              : BDR                             Router
Priority       : 1
 DR                      : 3.3.3.3
BDR                     : 1.1.1.3
 Link LSAs               : 2                               Checksum
Sum          : 26868
 AH Authentication     : SHA1, SPI 256 (Area)
Passive                 : Yes
Interface 1/1/2 is up, Line Protocol is up, Instance Id 64
----------------------------------------------------------
 VRF                     : default
Process                 : 1
 IP Address              : 10.10.10.2
Area                    : 0.0.0.0
 Status                  : Up                              Network
Type          : Broadcast
 Hello Interval        : 10    sec                         Dead
Interval         : 40     sec
 Transit Delay         : 1     sec                         Retransmit
Interval   : 5      sec
 Link Speed              : 1000 Mbps
 Cost Configured       : 1                                 Cost
Calculated       : 4
 State/Type              : BDR                             Router
Priority       : 1
 DR                      : 10.10.10.2
BDR                     : 10.10.10.1
 Link LSAs               : 0                               Checksum
Sum          : 0
 AH Authentication     : SHA1, SPI 256 (Area)
Passive                 : No
Interface tunnel1 is up, Line Protocol is up, Instance Id 64
```

```
-------------------------------------------------------------
 VRF                   : default
Process                : 1
 IPv6 Address          : 10.10.10.1
Area                   : 0.0.0.1
 Status                : Up                               Network
Type         : Broadcast
 Hello Interval        : 10    sec                        Dead
Interval        : 40     sec
 Transit Delay         : 1     sec                        Retransmit
Interval  : 5      sec
 BFD                   : Disabled                         Link
Speed           : 1000 Mbps
 Cost Configured       : 1                                Cost
Calculated      : 4
 State/Type            : DR                               Router
Priority      : 1
 DR                    : 10.10.10.2
BDR                    : 10.10.10.1
 Link LSAs             : 1                                Checksum
Sum         : 9168
 AH Authentication     : SHA1, SPI 256 (Area)
Passive                : Yes
Interface vlan10 is up, Line Protocol is up, Instance Id 64
------------------------------------------------------------
 VRF                   : default
Process                : 1
 IPv6 Address          : 10.10.10.3
Area                   : 0.0.0.1
 Status                : Up                               Network
Type         : Broadcast
 Hello Interval        : 10    sec                        Dead
Interval        : 40     sec
 Transit Delay         : 1     sec                        Retransmit
Interval  : 5      sec
 BFD                   : Disabled                         Link
Speed           : 1000 Mbps
 Cost Configured       : 1                                Cost
Calculated      : 4
 State/Type            : DR                               Router
Priority      : 1
 DR                    : 10.10.10.2
BDR                    : 10.10.10.1
```

```
 Link LSAs            : 1                               Checksum
Sum         : 9168
 AH Authentication    : MD5, SPI 257
Passive              : No
Interface vlan20 is up, Line Protocol is up, Instance Id 64
------------------------------------------------------------
 VRF                  : default
Process              : 1
 IPv6 Address         : 1.1.1.1
Area                 : 0.0.0.1
 Status               : Up                              Network
Type         : Broadcast
 Hello Interval       : 10    sec                       Dead
Interval        : 40     sec
 Transit Delay        : 1     sec                       Retransmit
Interval  : 5      sec
 Link Speed           : 1000 Mbps
 Cost Configured      : 1                               Cost
Calculated      : 4
 State/Type           : DR                              Router
Priority      : 1
 DR                   : 1.1.1.3
BDR                  : NA
 Link LSAs            : 1                               Checksum
Sum          : 9168
 AH Authentication    : MD5, SPI 257
Passive              : No
```

Showing overview information for OSPFv3 AF enabled interfaces for all VRFs in brief:

```
switch# show ospfv3-af ipv4 interface brief all-vrfs
Address-family : IPv4 Unicast
-----------------------------
OSPFv3 AF Process ID 1 VRF default
================================
Total Number of Interfaces: 2
Flags: P - Passive  A - Active
Interface     Area          Cost      State             Status    Flags
----------------------------------------------------------------------
1/1/1         0.0.0.0       1         BDR               Up        P
vlan10        0.0.0.1       1         DR                Up        A
OSPFv3 AF Process ID 2 VRF red
============================
Total Number of Interfaces: 1
Flags: P - Passive  A - Active
```

```
Interface      Area         Cost      State            Status     Flags
---------------------------------------------------------------------
1/1/2          0.0.0.1      1         BDR              Up         P
tunnel1        0.0.0.0      100       DR               Up         A
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show ospfv3-af ipv4 lsdb`

Syntax

```
show ospfv3-af [<PROCESS-ID>] ipv4 lsdb [all-vrfs | vrf <VRF-NAME>] [router
| network | inter-area-prefix | inter-area-router | as-external | nssa-
external | link | intra-area-prefix]
   [area <AREA-ID>] [lsid <LINK-STATE-ID>] [adv-router <ROUTER-ID> | self]
show ospfv3-af [<PROCESS-ID>] ipv4 lsdb [all-vrfs | vrf <VRF-NAME>]
database-summary
```

Description

Shows the OSPFv3 AF link state database summary for different OSPFv3 AF LSAs (Link State Advertisement). Use the parameters to get information for a particular LSA.

Parameter	Description
	Enter an OSPFv3 AF process ID to display general OSPF information for a particular OSPFv3 AF process. Range: 1 to 65535.

Parameter	Description
<PROCESS-ID>	
`all-vrfs|vrf <vrf-name>`	Select **all - vrfs** to display general OSPF information for all VR Fs, or use the **vrf** <VRF - NAME> option to display information for a specific VRF. Optionally select one of the following parameters to filter the lin k state database information.
`as-external`	Show external link states (LSA type 5)
`database-summary`	Select to display the count of each type of LSA and each area in the database. **NOTE** The **database - summary** parameter does not support the **area <area - id>**, **lsid <link - state - id>** or **adv - router {<router - id>|self}** para meters.
`inter-area-prefix`	Show inter - area prefix link states (LSA type 3)
`inter-area-router`	Show inter - area router link states (LSA type 4)
`intra-area-prefix`	Show intra - area prefix link states (LSA type 9
`link`	Show link states (LSA type 8)
`network`	Show network LSAs (LSA type 2).
`nssa-external`	Show NSSA external link states (LSA type 7).
`router`	Show router LSAs (LSA type 1).
`area` *<AREA-ID>*	Select to display information filtered for the specified area in on e of the following formats. • OSPFv3 AF area identifier in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255. • OSPFv3 AF area identifier in decimal format. Value: 0 to 42 94967295.

Parameter	Description
`lsid <LINK-STATE-ID>`	Select to display information filtered by link state identifier specified in IPv4 address format (A.B.C.D).
`adv-router {<ROUTER-ID>|self}`	Select to display link states for a particular advertising router. Specify either a Router ID of the advertising router or specify **self** to show self - originated link states.

Examples

Showing OSPFv3 AF link state database (LSDB) general information:

```
switch# show ospfv3-af ipv4 lsdb
Address-family : IPv4 Unicast
-----------------------------
OSPF Router with ID (50.50.50.50) (Process ID 1 VRF default)
============================================================
Router Link State Advertisements (Area 0.0.0.0)
-----------------------------------------------------------------------------
-
ADV Router       Age      Seq#         Checksum    LSID          Link Count
Bits
-----------------------------------------------------------------------------
-
40.40.40.40      930      0x80000004   0x2ea1      0             3
None
50.50.50.50      935      0x80000002   0x8b52      0             1           E
60.60.60.60      943      0x800003c5   0x9854      0             2
None
Network Link State Advertisements (Area 0.0.0.0)
-------------------------------------------------------------------------
ADV Router       Age      Seq#         Checksum    LSID         Router Count
-------------------------------------------------------------------------
60.60.60.60      944      0x80000001   0x7179      1360007168   2
50.50.50.50      935      0x80000001   0x516a      19           1
Inter Area Prefix Link State Advertisements (Area 0.0.0.0)
-------------------------------------------------------------------
ADV Router       Age      Seq#         Checksum    LSID         Prefix
-------------------------------------------------------------------
40.40.40.40      929      0x80000001   0x2498      131072       11.0.0.0/8
50.50.50.50      928      0x80000001   0x5b2f      65536        100.100.0.0/16
Inter Area Router Link State Advertisements (Area 0.0.0.0)
-----------------------------------------------------------------------------
-----
```

```
ADV Router       Age     Seq#        Checksum    LSID        Destination
Router ID
----------------------------------------------------------------------------
-----
40.40.40.40      929     0x80000001  0x2498      1           33.33.33.33
AS External Link State Advertisements (Area 0.0.0.0)
------------------------------------------------------------------
ADV Router       Age     Seq#        Checksum    LSID        Prefix
------------------------------------------------------------------
40.40.40.40      264     0x80000001  0x24cc4     1           10.10.0.0/16
40.40.40.40      675     0x80000001  0x5b00f     2           11.11.0.0/16
NSSA External Link State Advertisements (Area 0.0.0.0)
------------------------------------------------------------------
ADV Router       Age     Seq#        Checksum    LSID        Prefix
------------------------------------------------------------------
3.3.3.3          264     0x80000001  0x24ac2     1           200.200.0.0/16
Link-local Link State Advertisements (Area 0.0.0.0)
---------------------------------------------------------------------
ADV Router       Age     Seq#        Checksum    LSID        Interface
---------------------------------------------------------------------
50.50.50.50      264     0x80000001  0x653c4     19          1/1/1
Intra Area Prefix Link State Advertisements (Area 0.0.0.0)
----------------------------------------------------------------------------
-------------------
ADV Router       Age     Seq#        Checksum    LSID        Referenced LS
Type  Referenced LSID
----------------------------------------------------------------------------
-------------------
50.50.50.50      263     0x80000001  0x1da34     1
0x2001              0
50.50.50.50      264     0x80000001  0x2a45d     1
0x2002               19
```

Showing AS external link states:

```
switch# show ospfv3-af ipv4 lsdb as-external
Address-family : IPv4 Unicast
----------------------------
OSPF Router with ID (60.60.60.60) (Process ID 1 VRF default)
========================================================
AS External Link State Advertisements (Area 0.0.0.0)
------------------------------------------------------------------
ADV Router       Age     Seq#        Checksum    LSID        Prefix
------------------------------------------------------------------
```

```
40.40.40.40      264      0x80000001  0x24cc4     1             10.10.0.0/16
40.40.40.40      675      0x80000001  0x5b00f     2             11.11.0.0/16
```

Showing the LSDB database summary:

```
switch# show ospfv3-af ipv4 lsdb database-summary
Address-family : IPv4 Unicast
-----------------------------
OSPF Router with ID (10.1.1.1) (Process ID 1 VRF default)
===========================================================
Area 0.0.0.0 database summary
--------------------------
LSA Type             Count
--------------------------
Router               2
Network              1
Inter Area Prefix    1
Inter Area Router    0
NSSA External        0
Link                 3
Intra Area Prefix    3
--------------------------
Total                10
Process 1 database summary
--------------------------
LSA Type             Count
--------------------------
Router               2
Network              1
Inter Area Prefix    1
Inter Area Router    0
AS External          2
NSSA External        0
Link                 3
Intra Area Prefix    3
--------------------------
Total                12
```

Showing inter-area prefix LSAs:

```
switch# show ospfv3-af ipv4 lsdb inter-area-prefix
Address-family : IPv4 Unicast
-----------------------------
OSPF Router with ID (6.6.6.6) (Process ID 1 VRF default)
=========================================================
Inter Area Prefix Link State Advertisements (Area 0.0.0.0)
------------------------------------------------------------------
```

```
ADV Router       Age     Seq#         Checksum    LSID         Prefix
-----------------------------------------------------------------
40.40.40.40      929     0x80000001   0x2498      131072       1.0.0.0/8
50.50.50.50      928     0x80000001   0x5b2f      65536        11.11.0.0./16
```

Showing inter-area router LSAs:

```
switch# show ospfv3-af ipv4 lsdb inter-area-router
Address-family : IPv4 Unicast
-----------------------------
OSPF Router with ID (60.60.60.60) (Process ID 1 VRF default)
============================================================
Inter Area Prefix Link State Advertisements (Area 0.0.0.0)
----------------------------------------------------------------------------
-----
ADV Router       Age     Seq#         Checksum    LSID         Destination
Router ID
----------------------------------------------------------------------------
-----
40.40.40.40      929     0x80000001   0x2498      1            33.33.33.33
```

Showing intra-area prefix LSAs:

```
switch# show ospfv3-af ipv4 lsdb intra-area-prefix
Address-family : IPv4 Unicast
-----------------------------
OSPF Router with ID (0.0.0.0) (Process ID 1 VRF default)
========================================================
Intra Area Prefix Link State Advertisements (Area 0.0.0.0)
----------------------------------------------------------------------------
-----------------------
ADV Router       Age     Seq#         Checksum    LSID         Referenced LS
Type      Referenced LSID
----------------------------------------------------------------------------
-----------------------
50.50.50.50      263     0x80000001   0x1da34      1
0x2001                   0
50.50.50.50      264     0x80000001   0x2a45d      1
0x2002                   19
```

Showing link LSAs:

```
switch# show ospfv3-af ipv4 lsdb link
Address-family : IPv4 Unicast
-----------------------------
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
========================================================
Link-local Link State Advertisements (Area 0.0.0.0)
```

```
-----------------------------------------------------------------------
ADV Router      Age     Seq#        Checksum    LSID        Interface
-----------------------------------------------------------------------
50.50.50.50     264     0x80000001  0x653c4     19          1/1/1
Link-local Link State Advertisements (Area 0.0.0.0)
-----------------------------------------------------------------------
ADV Router      Age     Seq#        Checksum    LSID        Interface
-----------------------------------------------------------------------
50.50.50.50     264     0x80000001  0xad324     20          1/1/2
```

Showing network LSAs:

```
switch# show ospfv3-af ipv4 lsdb network
Address-family : IPv4 Unicast
-----------------------------
OSPF Router with ID (50.50.50.50) (Process ID 1 VRF default)
========================================================
Network Link State Advertisements (Area 0.0.0.0)
--------------------------------------------------------------------------
ADV Router      Age     Seq#        Checksum    LSID        Router Count
--------------------------------------------------------------------------
60.60.60.60     944     0x80000001  0x7179      1360007168  2
50.50.50.50     935     0x80000001  0x516a      19          1
```

Showing NSSA external link states:

```
switch# show ospfv3-af ipv4 lsdb nssa-external
Address-family : IPv4 Unicast
-----------------------------
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
========================================================
NSSA External Link State Advertisements (Area 0.0.0.0)
--------------------------------------------------------------------
ADV Router      Age     Seq#        Checksum    LSID        Prefix
--------------------------------------------------------------------
3.3.3.3         264     0x80000001  0x24ac2     1           200.200.0.0/16
```

Showing router LSAs:

```
switch# show ospfv3-af ipv4 lsdb router
Address-family : IPv4 Unicast
-----------------------------
OSPF Router with ID (50.50.50.50) (Process ID 1 VRF default)
========================================================
Router Link State Advertisements (Area 0.0.0.0)
----------------------------------------------------------------------------
-
ADV Router      Age     Seq#        Checksum    LSID        Link Count
```

```
Bits
--------------------------------------------------------------------------------
-
40.40.40.40     930     0x80000004   0x2ea1       0           3
None
50.50.50.50     935     0x80000002   0x8b52       0           1            E
60.60.60.60     943     0x800003c5   0x9854       0           2
None
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show ospfv3-af ipv4 neighbors`

Syntax

```
show ospfv3-af [<PROCESS-ID>] ipv4 neighbors [<NEIGHBOR-ID>]
   [interface <INTERFACE-NAME> | vlan <VLAN-ID> | lag <LAG-ID> | loopback
<LOOPBACK-ID> | tunnel <TUNNEL-ID>]
   [detail | summary] [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows information about OSPFv3 AF neighbors.

Parameter	Description
<PROCESS-ID>	Specifies an OSPFv3 AF process ID to show OSPFv3 AF neigh bor information for the particular OSPFv3 AF process. Range: 1 to 65535.
`neighbors` *<NEIGHBOR-ID>*	Shows information about a particular neighbor, specified in IPv 4 format (A.B.C.D).
`interface` *<INTERFACE-NAME>*	Shows neighbor information only for the specified interface.
`vlan` *<VLAN-ID>*	Specifies the vlan interface name.
`lag` *<LAG-ID>*	Specifies the lag interface name.
`loopback` *<LOOPBACK-ID>*	Specifies the loopback interface name.
`tunnel` *<TUNNEL-ID>*	Specifies the tunnel interface name.
`detail`	Shows detailed information for the neighbors.
`summary`	Shows summary information for the neighbors.
`all-vrfs`	Shows neighbor information for all VRFs.
`vrf` *<VRF-NAME>*	Specifies the name of a VRF. Default: default.

Examples

Showing OSPFv3 AF neighbor information for the specified interface:

```
switch# show ospfv3-af ipv4 neighbors 3.3.3.3
Address-family : IPv4 Unicast
-----------------------------
VRF : default                         Process : 1
--------------------------------------------------------------
Router-Id          : 3.3.3.3          Area              : 0.0.0.0
Interface          : 1/1/1            Address           : 1.1.1.1
State              : FULL             Neighbor Priority : 1
Dead Timer Due     : 00:00:36         Options           : 0x13
Time since last state change : 00h:14m:45s
```

Showing detail OSPFv3 AF neighbor information:

```
switch# show ospfv3-af ipv4 neighbors 3.3.3.3 detail
Address-family : IPv4 Unicast
-----------------------------
VRF : default                          Process : 1
--------------------------------------------------------------
Router-Id          : 3.3.3.3           Area              : 0.0.0.0
Interface          : 1/1/1             Address           : 1.1.1.1
State              : FULL              Neighbor Priority : 1
DR                 : 3.3.3.3           BDR               : 1.1.1.3
Dead Timer Due     : 00:00:36          Options           : 0x13
Retransmission Queue Length  : 0
Time Since Last State Change : 00h:14m:45s
```

Showing OSPFv3 AF neighbor information for interface **1/1/1**:

```
switch# show ospfv3-af ipv4 neighbors 3.3.3.3 interface 1/1/1
Address-family : IPv4 Unicast
-----------------------------
VRF : default                          Process : 1
--------------------------------------------------------------
Router-Id          : 3.3.3.3           Area              : 0.0.0.0
Interface          : 1/1/1             Address           : 1.1.1.1
State              : FULL              Neighbor Priority : 1
Dead Timer Due     : 00:00:36          Options           : 0x13
Time Since Last State Change : 00h:14m:45s
```

Showing OSPFv3 AF neighbor information for tunnel interface:

```
switch# show ospfv3-af ipv4 neighbors 3.3.3.3
Address-family : IPv4 Unicast
-----------------------------
VRF : default                          Process : 1
--------------------------------------------------------------
Router-Id          : 3.3.3.3           Area              : 0.0.0.0
Interface          : tunnel1           Address           : 1.1.1.1
State              : FULL              Neighbor Priority : 1
Dead Timer Due     : 00:00:36          Options           : 0x13
Time since last state change : 00h:14m:45s
```

Showing detail OSPFv3 AF neighbor information for tunnel interface:

```
switch# show ospfv3-af ipv4 neighbors 3.3.3.3 detail
Address-family : IPv4 Unicast
-----------------------------
VRF : default                          Process : 1
--------------------------------------------------------------
Router-Id          : 3.3.3.3           Area              : 0.0.0.0
```

```
Interface          : tunnel1            Address            : 1.1.1.1
State              : FULL               Neighbor Priority : 1
DR                 : 3.3.3.3            BDR                : 1.1.1.3
Dead Timer Due     : 00:00:36           Options            : 0x13
Retransmission Queue Length  : 0
Time Since Last State Change : 00h:14m:45s
```

Showing summary OSPFv3 AF neighbors information for a specific neighbor for all VRFs:

```
switch# show ospfv3-af ipv4 neighbors 3.3.3.3 summary all-vrfs
Address-family : IPv4 Unicast
-----------------------------
OSPFv3 AF Process ID 1 VRF default, Neighbor Summary
==================================================
Interface     Down Attempt Init TwoWay ExStart Exchange Loading Full Total
--------------------------------------------------------------------------
1/1/1         0    0       0    0      0       0        0       1    1
Total         0    0       0    0      0       0        0       1    1
OSPFv3 AF Process ID 1 VRF red, Neighbor Summary
==============================================
Interface     Down Attempt Init TwoWay ExStart Exchange Loading Full Total
--------------------------------------------------------------------------
1/1/2         0    0       0    0      0       0        0       1    1
Total         0    0       0    0      0       0        0       1    1
```

Showing OSPFv3 AF neighbors information for VRF red:

```
switch# show ospfv3-af ipv4 neighbors vrf red
Address-family : IPv4 Unicast
-----------------------------
OSPFv3 AF Process ID 2 VRF red
============================
Total Number of Neighbors: 1
Neighbor ID      Priority  State            Interface
-------------------------------------------------------
4.4.4.4          1         FULL/DR          1/1/2
  Neighbor address 1.1.1.1
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show ospfv3-af ipv4 routes`

Syntax

```
show ospfv3-af [<PROCESS-ID>] ipv4 routes [<PREFIX/LENGTH>]
   [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows the OSPFv3 AF routing table information.

Parameter	Description
<PROCESS-ID>	Specifies an OSPFv3 AF process ID that shows information from the OSPFv3 AF routing table for the particular OSPFv3 AF process. Range: 1 to 65535.
<PREFIX/LENGTH>	Specifies an IP address in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255.
`all-vrfs`	Select to show OSPFv3 AF routing table information for all VRFs.
`vrf` *<VRF-NAME>*	Specifies the name of a VRF. Default: default.

Examples

Showing OSPFv3 AF routing table information:

```
switch# show ospfv3-af ipv4 routes
Codes: i - Intra-area route, I - Inter-area route
```

```
       E1 - External type-1, E2 - External type-2
Address-family : IPv4 Unicast
-----------------------------
OSPFv3 AF Process ID 1 VRF default, Routing Table
-------------------------------------------------
Total Number of OSPFv3 AF Routes : 3
11.11.0.0/16       (i) area:0.0.0.0
     directly attached to interface 1/1/1, cost 1 distance 110
200.200.0.0/16     (i) area:0.0.0.0
     directly attached to interface tunnel1, cost 100 distance 110
1.1.1.1/132        (i) area:0.0.0.1
     directly attached to interface vlan10, cost 1 distance 110
```

Showing OSPFv3 AF routing table information for VRF red:

```
switch# show ospfv3-af ipv4 routes vrf red
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2
Address-family : IPv4 Unicast
-----------------------------
OSPFv3 AF Process ID 2 VRF red, Routing Table
---------------------------------------------
Total Number of OSPFv3 AF Routes : 1
2.2.2.1/24           (i) area:0.0.0.1
     directly attached to interface 1/1/2, cost 1 distance 110
```

Showing OSPFv3 AF routing table information for destination prefix 2.2.2.1/24:

```
switch# show ospfv3-af 1 ipv4 routes 2.2.2.1/24
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2
Address-family : IPv4 Unicast
-----------------------------
OSPFv3 AF Process ID 1 VRF default, Routing Table for prefixes 2.2.2.1/24
-------------------------------------------------------------------------
Total Number of OSPFv3 AF Routes : 1
2.2.2.1/24          (i) area:0.0.0.1
     directly attached to interface vlan10, cost 1 distance 110
```

Showing OSPFv3 AF routing table information for destination prefix 200.0.0.0/8:

```
switch# show ospfv3-af 1 ipv4 routes 200.0.0.0/8
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2
Address-family : IPv4 Unicast
-----------------------------
OSPFv3 AF Process ID 1 VRF default, Routing Table for prefixes 200.0.0.0/8
--------------------------------------------------------------------------
```

```
Total Number of OSPFv3 AF Routes : 1
200.0.0.0/8          (i) area:0.0.0.0
     directly attached to interface tunnel1, cost 100 distance 110
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ospfv3-af ipv4 statistics interface

Syntax

```
show ospfv3-af [<PROCESS-ID>] ipv4 statistics interface [<INTERFACE-NAME>]
| vlan <VLAN-ID> | lag <LAG-ID> | loopback <LOOPBACK-ID> | tunnel <TUNNEL-
ID>]
   [brief] [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows information about OSPFv3 AF enabled IPv4 interfaces.

Parameter	Description
<PROCESS-ID>	Specifies an OSPFv3 AF process ID optionally to show the OSPFv3 AF enabled interfaces for the particular OSPFv3 AF process. Range: 1 to 65535.
	Selects to show information only for the specified OSPFv3 AF enabled interface.

Parameter	Description
<INTERFACE-NAME>	
vlan *<VLAN-ID>*	Specifies the vlan interface name.
lag *<LAG-ID>*	Specifies the lag interface name.
loopback *<LOOPBACK-ID>*	Specifies the loopback interface name.
tunnel *<TUNNEL-ID>*	Specifies the tunnel interface name.
brief	Include this parameter to display a brief overview of the followi ng OSPFv3 AF configuration information. • Interface: OSPFv3 AF interface name. • Area: OSPFv3 AF area ID. • Cost: The metric OSPFv3 AF uses to judge a path's feasibil ity, calculated as (reference bandwidth / interface bandwidt h). • State: Indicates if the interface is a designated router (**Dr**) o r a backup designated router (**Backup - dr**). • Status: Indicates if the interface is **up** or **down**. • Flags: P - Passive A - Active.
all-vrfs	Select to show OSPFv3 AF - enabled interface information for all VRFs.
vrf *<VRF-NAME>*	Specifies the name of a VRF. Default: default.

Examples

Showing OSPFv3 AF information for all interfaces in default VRF:

```
switch# show ospfv3-af ipv4 interface
Codes: DR - Designated router BDR - Backup Designated router
Address-family : IPv4 Unicast
-----------------------------
Interface 1/1/1 is up, Line Protocol is up, Instance Id 64
----------------------------------------------------------
 VRF                      : default
Process                  : 1
 IP Address               : 1.1.1.2
```

```
Area                    : 0.0.0.0
 Status                  : Up                             Network
Type          : Broadcast
 Hello Interval        : 10    sec                        Dead
Interval         : 40     sec
 Transit Delay         : 1     sec                        Retransmit
Interval   : 5      sec
 BFD                   : Disabled                         Link
Speed            : 1000 Mbps
 Cost Configured       : 1                                Cost
Calculated       : 4
 State/Type            : BDR                              Router
Priority       : 1
 DR                    : 3.3.3.3
BDR                    : 1.1.1.3
 Link LSAs             : 2                                Checksum
Sum          : 26868
 AH Authentication     : SHA1, SPI 256 (Area)
Passive                : Yes
Interface 1/1/2 is up, Line Protocol is up, Instance Id 64
----------------------------------------------------------
 VRF                   : default
Process                : 1
 IP Address            : 10.10.10.2
Area                   : 0.0.0.0
 Status                : Up                               Network
Type          : Broadcast
 Hello Interval        : 10    sec                        Dead
Interval         : 40     sec
 Transit Delay         : 1     sec                        Retransmit
Interval   : 5      sec
 Link Speed            : 1000 Mbps
 Cost Configured       : 1                                Cost
Calculated       : 4
 State/Type            : BDR                              Router
Priority       : 1
 DR                    : 10.10.10.2
BDR                    : 10.10.10.1
 Link LSAs             : 0                                Checksum
Sum          : 0
 AH Authentication     : SHA1, SPI 256 (Area)
Passive                : No
Interface tunnel1 is up, Line Protocol is up, Instance Id 64
```

```
------------------------------------------------------------
 VRF                    : default
Process                : 1
 IPv6 Address          : 10.10.10.1
Area                   : 0.0.0.1
 Status                : Up                                 Network
Type           : Broadcast
 Hello Interval        : 10    sec                          Dead
Interval          : 40     sec
 Transit Delay         : 1     sec                          Retransmit
Interval   : 5       sec
 BFD                   : Disabled                           Link
Speed             : 1000 Mbps
 Cost Configured       : 1                                  Cost
Calculated        : 4
 State/Type            : DR                                 Router
Priority       : 1
 DR                    : 10.10.10.2
BDR                    : 10.10.10.1
 Link LSAs             : 1                                  Checksum
Sum           : 9168
 AH Authentication     : SHA1, SPI 256 (Area)
Passive                : Yes
Interface vlan10 is up, Line Protocol is up, Instance Id 64
-----------------------------------------------------------
 VRF                    : default
Process                : 1
 IPv6 Address          : 10.10.10.3
Area                   : 0.0.0.1
 Status                : Up                                 Network
Type           : Broadcast
 Hello Interval        : 10    sec                          Dead
Interval          : 40     sec
 Transit Delay         : 1     sec                          Retransmit
Interval   : 5       sec
 BFD                   : Disabled                           Link
Speed             : 1000 Mbps
 Cost Configured       : 1                                  Cost
Calculated        : 4
 State/Type            : DR                                 Router
Priority       : 1
 DR                    : 10.10.10.2
BDR                    : 10.10.10.1
```

```
 Link LSAs             : 1                              Checksum
Sum          : 9168
 AH Authentication     : MD5, SPI 257
Passive               : No
Interface vlan20 is up, Line Protocol is up, Instance Id 64
-----------------------------------------------------------
 VRF                   : default
Process               : 1
 IPv6 Address          : 1.1.1.1
Area                  : 0.0.0.1
 Status                : Up                             Network
Type          : Broadcast
 Hello Interval        : 10    sec                      Dead
Interval         : 40     sec
 Transit Delay         : 1     sec                      Retransmit
Interval   : 5      sec
 Link Speed            : 1000 Mbps
 Cost Configured       : 1                              Cost
Calculated       : 4
 State/Type            : DR                             Router
Priority       : 1
 DR                    : 1.1.1.3
BDR                   : NA
 Link LSAs             : 1                              Checksum
Sum          : 9168
 AH Authentication     : MD5, SPI 257
Passive               : No
```

Showing overview information for OSPFv3 AF enabled interfaces for all VRFs in brief:

```
switch# show ospfv3-af ipv4 interface
Address-family : IPv4 Unicast
-----------------------------
OSPFv3 AF Process ID 1 VRF default
================================
Total Number of Interfaces: 2
Flags: P - Passive  A - Active
Interface     Area          Cost      State            Status    Flags
----------------------------------------------------------------------
1/1/1         0.0.0.0       1         BDR              Up        P
vlan10        0.0.0.1       1         DR               Up        A
OSPFv3 AF Process ID 2 VRF red
============================
Total Number of Interfaces: 1
Flags: P - Passive  A - Active
```

```
Interface      Area           Cost      State             Status    Flags
--------------------------------------------------------------------------
1/1/2          0.0.0.1        1         BDR               Up        P
tunnel1        0.0.0.0        100       DR                Up        A
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show ospfv3-af ipv6`

Syntax

```
show ospfv3-af [<PROCESS-ID>] ipv6 [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows OSPFv3 AF information including area, state, and configuration information.

Parameter	Description
`<PROCESS-ID>`	Specifies an OSPFv3 AF process ID optionally to show OSPFv3 AF information for a particular OSPF process. Range: 1 to 65535.
`all-vrfs`	Select to show OSPFv3 AF information for all VRFs.
`vrf <VRF-NAME>`	Specifies the name of a VRF. The default VRF OSPFv3 AF process information is displayed if VRF is not mentioned. Default: default.

Example

Showing general OSPFv3 AF configurations:

```
switch# show ospfv3-af 200 ipv6
Address-family : IPv6 Unicast
-----------------------------
VRF : default                       Process : 20200
---------------------------------------------------------------------
Router ID               : 1.1.1.1          OSPFv3-AF               : Enabled
BFD                     : Disabled         SPF Start Interval      : 200 ms
SPF Hold Interval       : 1000 ms          SPF Max Wait Interval   : 5000 ms
LSA Start Interval      : 5000 ms          LSA Hold Interval       : 1000 ms
LSA Max Wait Interval   : 1000 ms          LSA Arrival Interval    : 1000 ms
External LSAs           : 0                Checksum Sum            : 0
ECMP                    : 4                Reference Bandwidth     : 100000
Mbps
Area Border             : Yes              AS Border               : No
GR Status               : Enabled          GR Interval             : 120 sec
GR State                : Inactive         GR Exit Status          : None
GR Helper               : Enabled          GR Strict LSA Check     : Enabled
GR Ignore Lost I/F      : Disabled         Internal Process ID     : 1
Summary address:
  prefix fd00::1/64, advertise, tag 10
Area      Total Active
----------------------
Normal      2      2
Stub        0      0
Area  : 0.0.0.0
----------------------------------
Area Type              : Normal       Status                 : Active
Total Interfaces       : 1            Active Interfaces      : 1
Passive Interfaces     : 0            Loopback Interfaces    : 0
SPF calculation count : 4
Area ranges:
  fd00::1/64, inter-area, no-advertise
AH Authentication      : SHA1, SPI 256
Number of LSAs         : 5            Checksum Sum           : 99122
Area  : 0.0.0.1
----------------------------------
Area Type              : Normal       Status                 : Active
Total Interfaces       : 1            Active Interfaces      : 1
Passive Interfaces     : 0            Loopback Interfaces    : 0
SPF Calculation Count : 4
Area ranges:
```

```
  fd00::1/64, inter-area, no-advertise
ESP Authentication     : SHA1
Encryption             : 3DES, SPI 256
Number of LSAs         : 5             Checksum Sum          : 99122
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show ospfv3-af ipv6 border-routers`

Syntax

```
show ospfv3-af [<PROCESS-ID>] ipv6 border-routers | [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows the OSPFv3 AF routing table entries for Area Border Router (ABR) and Autonomous System Border Router (ASBR).

Parameter	Description
`<PROCESS-ID>`	Specifies an OSPFv3 AF process ID to show the OSPFv3 AF routing table entries for ABR and ASBR for the particular OSPFv3 AF process. Range: 1 to 65535.
`all-vrfs`	Select to show OSPFv3 AF border router information for all VRFs.

Parameter	Description
`vrf <VRF-NAME>`	Specifies the name of a VRF. Default: default.

Example

Showing OSPFv3 AF border routers information for VRF **vrf_red**:

```
switch# show ospfv3-af ipv6 border-routers vrf vrf_red
Address-family : IPv6 Unicast
-----------------------------
VRF : vrf_red       Process ID : 1
             Internal Routing Table
--------------------------------------------------------
Codes: i - Intra-area route, I - Inter-area route
   Router-ID Cost  Type    Area    SPF  Nexthop
Interface
i  1.1.1.1   1     ASBR    0.0.0.0 9    fe80::7272:cfff:fe9a:a15d   1/1/2
i  3.3.3.3   1     ASBR    1.1.1.1 9    fe80::7272:cfff:fe1f:d80   **
tunnel1
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show ospfv3-af ipv6 interface`

Syntax

```
show ospfv3-af [<PROCESS-ID>] ipv6 interface [<INTERFACE-NAME>] | vlan
<vlan id> |
lag <lag id> | loopback <loopback id> | tunnel <tunnel id>] [brief]
[all-vrfs | vrf <VRF-NAME>]
```

Description

Shows information about OSPFv3 AF enabled interfaces.

Parameter	Description
`<PROCESS-ID>`	Specifies an OSPFv3 AF process ID optionally to show the OSPFv3 AF enabled interfaces for the particular OSPFv3 AF process. Range: 1 to 65535.
`<INTERFACE-NAME>`	Selects to show information only for the specified OSPFv3 AF enabled interface.
`vlan <VLAN-ID>`	Specifies the vlan interface name.
`lag <LAG-ID>`	Specifies the lag interface name.
`loopback <LOOPBACK-ID>`	Specifies the loopback interface name.
`tunnel <TUNNEL-ID>`	Specifies the tunnel interface name.
`brief`	Include this parameter to display a brief overview of the following OSPFv3 AF configuration information. • Interface: OSPFv3 AF interface name. • Area: OSPFv3 AF area ID. • Cost: The metric OSPFv3 AF uses to judge a path's feasibility, calculated as (reference bandwidth / interface bandwidth). • State: Indicates if the interface is a designated router (**Dr**) or a backup designated router (**Backup - dr**). • Status: Indicates if the interface is **up** or **down**. • Flags: P - Passive A - Active.

Parameter	Description
`all-vrfs`	Select to show OSPFv3 AF - enabled interface information for all VRFs.
`vrf <VRF-NAME>`	Specifies the name of a VRF. Default: default.

Examples

Showing OSPFv3 AF information for all interfaces in default VRF:

```
switch# show ospfv3-af ipv6 interface
Codes: DR - Designated router  BDR - Backup Designated router
Interface 1/1/1 is Up, Line Protocol is Up, Instance ID 10
--------------------------------------------
VRF                      : default
Process                  : 1
IPv6 address             : fe80::7272:cfff:fe70:67ae
Area                     : 0.0.0.0
Status                   : Up                          Network
Type             : Broadcast
Hello Interval           : 10    sec                   Dead
Interval          : 40    sec
Transit Delay            : 1     sec                   Retransmit
Interval    : 5     sec
BFD                      : Disabled                    Link
Speed             : 1000 Mbps
Cost Configured          : 1                           Cost
Calculated        : 4
State/Type               : BDR                         Router
Priority        : 1
DR                       : 3.3.3.3
BDR                      : 1.1.1.3
Link LSAs                : 2                           Checksum
Sum           : 26868
AH Authentication        : SHA1, SPI 256 (Area)
Passive                  : Yes
Codes: DR - Designated router  BDR - Backup Designated router
Interface 1/1/2 is Up, Line Protocol is Up, Instance ID 70
--------------------------------------------
VRF                      : default
Process                  : 1
IPv6 address             : fe80::7272:cfff:fe70:67ae
Area                     : 0.0.0.0
```

```
Status                  : Up                                 Network
Type         : Broadcast
Hello Interval          : 10    sec                          Dead
Interval         : 40    sec
Transit Delay           : 1     sec                          Retransmit
Interval   : 5     sec
BFD                     : Disabled                           Link
Speed            : 1000 Mbps
Cost Configured         : 1                                  Cost
Calculated       : 4
State/Type              : BDR                                Router
Priority       : 1
DR                      : 10.10.10.2
BDR                     : 10.10.10.1
Link LSAs               : 0                                  Checksum
Sum          : 0
AH Authentication       : SHA1, SPI 256 (Area)
Passive                 : No
Codes: DR - Designated router  BDR - Backup Designated router
Interface tunnel1 is Up, Line Protocol is Up, Instance ID 70
--------------------------------------------
VRF                     : default
Process                 : 1
IPv6 address            : fe80::7272:cfff:fe70:679ae
Area                    : 0.0.0.1
Status                  : Up                                 Network
Type         : Broadcast
Hello Interval          : 10    sec                          Dead
Interval         : 40    sec
Transit Delay           : 1     sec                          Retransmit
Interval   : 5     sec
BFD                     : Disabled                           Link
Speed            : 1000 Mbps
Cost Configured         : 1                                  Cost
Calculated       : 4
State/Type              : DR                                 Router
Priority       : 1
DR                      : 10.10.10.2
BDR                     : 10.10.10.1
Link LSAs               : 1                                  Checksum
Sum          : 9168
AH Authentication       : SHA1, SPI 256 (Area)
Passive                 : Yes
```

```
Codes: DR - Designated router  BDR - Backup Designated router
Interface vlan10 is Up, Line Protocol is Up, Instance ID 10
--------------------------------------------
VRF                     : default
Process                 : 1
IPv6 address            : fe80::7272:cfff:fe70:67ae
Area                    : 0.0.0.1
Status                  : Up                            Network
Type        : Broadcast
Hello Interval          : 10   sec                      Dead
Interval         : 40   sec
Transit Delay           : 1    sec                      Retransmit
Interval   : 5     sec
BFD                     : Disabled                      Link
Speed            : 1000 Mbps
Cost Configured         : 1                             Cost
Calculated       : 4
State/Type              : DR                            Router
Priority       : 1
DR                      : 10.10.10.2
BDR                     : 10.10.10.1
Link LSAs               : 1                             Checksum
Sum          : 9168
AH Authentication       : MD5, SPI 257
Passive                 : No
Codes: DR - Designated router  BDR - Backup Designated router
Interface vlan20 is Up, Line Protocol is Up, Instance ID 10
--------------------------------------------
VRF                     : default
Process                 : 1
IPv6 address            : fe80::7272:cfff:fe70:67ae
Area                    : 0.0.0.1
Status                  : Up                            Network
Type        : Broadcast
Hello Interval          : 10   sec                      Dead
Interval         : 40   sec
Transit Delay           : 1    sec                      Retransmit
Interval   : 5     sec
BFD                     : Disabled                      Link
Speed            : 1000 Mbps
Cost Configured         : 1                             Cost
Calculated       : 4
State/Type              : DR                            Router
```

```
Priority      : 1
DR                     : 1.1.1.3
BDR                    : NA
Link LSAs              : 1                       Checksum
Sum         : 9168
AH Authentication      : MD5, SPI 257
Passive                : No
```

Showing overview information for OSPFv3 AF enabled interfaces for all VRFs in brief:

```
switch# show ospfv3-af ipv6 interface brief all-vrfs
VRF : default                          Process : 1
===================================================
Total Number of Interfaces: 2
Flags: P - Passive  A - Active
Interface     Area                Cost      State                Status
Flags
-----------------------------------------------------------------------------
-----
1/1/1         0.0.0.0             1         BDR                   Up
P
vlan10        0.0.0.1             1         DR                    Up
A
VRF : red                              Process : 1
===================================================
Total Number of Interfaces: 1
Flags: P - Passive  A - Active
Interface     Area                Cost      State                Status
Flags
-----------------------------------------------------------------------------
-----
1/1/2         0.0.0.1         1      BDR            Up       P
tunnel1       0.0.0.0             100       DR                    Up
A
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ospfv3-af ipv6 lsdb

Syntax

```
show ospfv3-af [<process-id>] ipv6 lsdb
   adv-router {<ROUTER-ID>|self}
   area <AREA-ID>
   lsid <link-state-id>
   all-vrfs|vrf <VRF-NAME>}
as-external
asbr-summary
database-summary
inter-area-prefix
inter-area-router
intra-area-prefix
link
network
nssa-external
router
summary
```

Description

Shows the OSPFv3 AF link state database summary for different OSPF LSAs (Link State Advertisement). Use the parameters to get information for a particular LSA.

Parameter	Description
<PROCESS-ID>	Enter an OSPFv3 AF process ID to display general OSPF information for a particular OSPF process. Range: 1 to 65535.

Parameter	Description
`adv-router {<ROUTER-ID>|self}`	Select to display link states for a particular advertising router. Specify either a Router ID of the advertising router or specify **self** to show self - originated link states.
`area <AREA-ID>`	Select to display information filtered for the specified area in one of the following formats. • OSPFv3 AF area identifier in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255. • OSPFv3 AF area identifier in decimal format. Value: 0 to 4294967295.
`lsid <LINK-STATE-ID>`	Select to display information filtered by link state identifier specified in IPv4 address format (A.B.C.D).
`all-vrfs|{vrf <vrf-name>`	Select **all - vrfs** to display general OSPF information for all VRFs, or use the **vrf** <VRF - NAME> option to display information for a specific VRF. Optionally select one of the following parameters to filter the link state database information.
`as-external`	Show external link states (LSA type 5)
`database-summary`	Select to display the count of each type of LSA and each area in the database. **NOTE** The **database - summary** parameter does not support the **area <area - id>**, **lsid <link - state - id>** or **adv - router {<router - id>|self}** parameters.
`inter-area-prefix`	Show inter - area prefix link states (LSA type 3)
`inter-area-router`	Show inter - area router link states (LSA type 4)
`intra-area-prefix`	Show intra - area prefix link states (LSA type 9
`link`	Show link states (LSA type 8)
`network`	Show network LSAs (LSA type 2).
`nssa-external`	Show NSSA external link states (LSA type 7).

Parameter	Description
`router`	Show router LSAs (LSA type 1).

Examples

Showing OSPFv3 AF link state database (LSDB) general information:

```
switch# show ospfv3-af ipv6 lsdb
Address-family : IPv6 Unicast
-----------------------------
OSPF Router with ID (50.50.50.50) (Process ID 1 VRF default)
============================================================
Router Link State Advertisements (Area 0.0.0.0)
----------------------------------------------------------------------------
-
ADV Router      Age     Seq#        Checksum    LSID         Link Count
Bits
----------------------------------------------------------------------------
-
40.40.40.40     930     0x80000004  0x2ea1      0            3
None
50.50.50.50     935     0x80000002  0x8b52      0            1           E
60.60.60.60     943     0x800003c5  0x9854      0            2
None
Network Link State Advertisements (Area 0.0.0.0)
------------------------------------------------------------------------
ADV Router      Age     Seq#        Checksum    LSID        Router Count
------------------------------------------------------------------------
60.60.60.60     944     0x80000001  0x7179      1360007168  2
50.50.50.50     935     0x80000001  0x516a      19          1
Inter Area Prefix Link State Advertisements (Area 0.0.0.0)
------------------------------------------------------------------
ADV Router      Age     Seq#        Checksum    LSID        Prefix
------------------------------------------------------------------
40.40.40.40     929     0x80000001  0x2498      131072      FEC0:3344::/32
50.50.50.50     928     0x80000001  0x5b2f      65536       111::/64
Inter Area Router Link State Advertisements (Area 0.0.0.0)
----------------------------------------------------------------------------
-----
ADV Router      Age     Seq#        Checksum    LSID        Destination
Router ID
----------------------------------------------------------------------------
-----
```

```
40.40.40.40     929     0x80000001  0x2498      1           33.33.33.33
AS External Link State Advertisements (Area 0.0.0.0)
------------------------------------------------------------------
ADV Router      Age     Seq#        Checksum    LSID        Prefix
------------------------------------------------------------------
40.40.40.40     264     0x80000001  0x24cc4     1           10::/64
40.40.40.40     675     0x80000001  0x5b00f     2           11::/64
NSSA External Link State Advertisements (Area 0.0.0.0)
------------------------------------------------------------------
ADV Router      Age     Seq#        Checksum    LSID        Prefix
------------------------------------------------------------------
3.3.3.3         264     0x80000001  0x24ac2     1           200::/64
Link-local Link State Advertisements (Area 0.0.0.0)
---------------------------------------------------------------------
ADV Router      Age     Seq#        Checksum    LSID        Interface
---------------------------------------------------------------------
50.50.50.50     264     0x80000001  0x653c4     19          1/1/1
Intra Area Prefix Link State Advertisements (Area 0.0.0.0)
----------------------------------------------------------------------------
-------------------
ADV Router      Age     Seq#        Checksum    LSID        Referenced LS
Type  Referenced LSID
----------------------------------------------------------------------------
-------------------
50.50.50.50     263     0x80000001  0x1da34     1
0x2001              0
50.50.50.50     264     0x80000001  0x2a45d     1
0x2002              19
```

Showing AS external link states:

```
switch# show ospfv3-af ipv6 lsdb as-external
Address-family : IPv6 Unicast
-----------------------------
OSPF Router with ID (60.60.60.60) (Process ID 1 VRF default)
========================================================
AS External Link State Advertisements (Area 0.0.0.0)
------------------------------------------------------------------
ADV Router      Age     Seq#        Checksum    LSID        Prefix
------------------------------------------------------------------
40.40.40.40     264     0x80000001  0x24cc4     1           10::/64
40.40.40.40     675     0x80000001  0x5b00f     2           11::/64
```

Showing the LSDB database summary:

```
switch# show ospfv3-af ipv6 lsdb database-summary
Address-family : IPv6 Unicast
```

```
-----------------------------
OSPF Router with ID (10.1.1.1) (Process ID 1 VRF default)
==========================================================
Area 0.0.0.0 database summary
--------------------------
LSA Type             Count
--------------------------
Router               2
Network              1
Inter Area Prefix    1
Inter Area Router    0
NSSA External        0
Link                 3
Intra Area Prefix    3
--------------------------
Total                10
Process 1 database summary
--------------------------
LSA Type             Count
--------------------------
Router               2
Network              1
Inter Area Prefix    1
Inter Area Router    0
AS External          2
NSSA External        0
Link                 3
Intra Area Prefix    3
--------------------------
Total                12
```

Showing inter-area prefix LSAs:

```
switch# show ospfv3-af ipv6 lsdb inter-area-prefix
Address-family : IPv6 Unicast
-----------------------------
OSPF Router with ID (6.6.6.6) (Process ID 1 VRF default)
========================================================
Inter Area Prefix Link State Advertisements (Area 0.0.0.0)
------------------------------------------------------------------
ADV Router      Age     Seq#        Checksum    LSID        Prefix
------------------------------------------------------------------
40.40.40.40     929     0x80000001  0x2498      131072      FEC0:3344::/32
50.50.50.50     928     0x80000001  0x5b2f      65536       111::/64
```

Showing network LSAs:

```
switch# show ospfv3-af ipv6 lsdb network
Address-family : IPv6 Unicast
-----------------------------
OSPF Router with ID (50.50.50.50) (Process ID 1 VRF default)
========================================================
Network Link State Advertisements (Area 0.0.0.0)
------------------------------------------------------------------------
ADV Router       Age      Seq#         Checksum     LSID         Router Count
------------------------------------------------------------------------
60.60.60.60      944      0x80000001   0x7179       1360007168   2
50.50.50.50      935      0x80000001   0x516a       19           1
```

Showing NSSA external link states:

```
switch# show ospfv3-af ipv6 lsdb nssa-external
Address-family : IPv6 Unicast
-----------------------------
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
========================================================
NSSA External Link State Advertisements (Area 0.0.0.0)
------------------------------------------------------------------
ADV Router       Age      Seq#         Checksum     LSID         Prefix
------------------------------------------------------------------
3.3.3.3          264      0x80000001   0x24ac2      1            200::/64
```

Showing router LSAs:

```
switch# show ospfv3-af ipv6 lsdb router
Address-family : IPv6 Unicast
-----------------------------
OSPF Router with ID (50.50.50.50) (Process ID 1 VRF default)
========================================================
Ruter Link State Advertisements (Area 0.0.0.0)
----------------------------------------------------------------------------
-
ADV Router       Age      Seq#         Checksum     LSID          Link Count
Bits
----------------------------------------------------------------------------
-
40.40.40.40      930      0x80000004   0x2ea1       0             3
None
50.50.50.50      935      0x80000002   0x8b52       0             1            E
60.60.60.60      943      0x800003c5   0x9854       0             2
None
```

Showing inter-area router LSAs:

```
switch# show ospfv3-af ipv6 lsdb inter-area-router
Address-family : IPv6 Unicast
-----------------------------
OSPF Router with ID (60.60.60.60) (Process ID 1 VRF default)
========================================================
Inter Area Router Link State Advertisements (Area 0.0.0.0)
----------------------------------------------------------------------------
-----
ADV Router       Age     Seq#         Checksum    LSID         Destination
Router ID
----------------------------------------------------------------------------
-----
40.40.40.40      929     0x80000001   0x2498      1            33.33.33.33
```

Showing link states:

```
switch# show ospfv3-af ipv6 lsdb link
Address-family : IPv6 Unicast
-----------------------------
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
========================================================
Link-local Link State Advertisements (Area 0.0.0.0)
---------------------------------------------------------------------
ADV Router       Age     Seq#         Checksum    LSID         Interface
---------------------------------------------------------------------
50.50.50.50      264     0x80000001   0x653c4     19           1/1/1
Link-local Link State Advertisements (Area 0.0.0.1)
---------------------------------------------------------------------
ADV Router       Age     Seq#         Checksum    LSID         Interface
---------------------------------------------------------------------
50.50.50.50      264     0x80000001   0xad324     20           1/1/2
```

Showing lintra-area prefix LSAs:

```
switch# show ospfv3-af ipv6 lsdb intra-area-prefix
Address-family : IPv6 Unicast
-----------------------------
OSPF Router with ID (0.0.0.0) (Process ID 1 VRF default)
========================================================
Intra Area Prefix Link State Advertisements (Area 0.0.0.0)
----------------------------------------------------------------------------
-------------------
ADV Router       Age     Seq#         Checksum    LSID         Referenced LS
Type  Referenced LSID
----------------------------------------------------------------------------
-------------------
50.50.50.50      263     0x80000001   0x1da34     1
```

```
0x2001                 0
50.50.50.50      264      0x80000001   0x2a45d      1
0x2002                 19
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show ospfv3-af ipv6 neighbors`

Syntax

```
show ospfv3-af [<PROCESS-ID>] ipv6 neighbors [<NEIGHBOR-ID>]
   [interface <INTERFACE-NAME>] | vlan <VLAN ID> | lag <LAG ID> | loopback
<LOOPBACK ID> | tunnel<tunnel id>][detail | summary]
   [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows information about OSPFv3 AF neighbors.

Parameter	Description
`<PROCESS-ID>`	Specifies an OSPFv3 AF process ID to show OSPFv3 AF neighbor information for the particular OSPFv3 AF process. Range: 1 to 65535.
`neighbors <NEIGHBOR-ID>`	Shows information about a particular neighbor, specified in IPv4 format (A.B.C.D).

Parameter	Description
`interface <INTERFACE-NAME>`	Shows neighbor information only for the specified interface.
`detail`	Shows detailed information for the neighbors.
`summary`	Shows summary information for the neighbors.
`lag <LAG ID>`	Specifies lag interface name.
`loopback <LOOPBACK ID>`	Specifies the vlan interface name.
`all-vrfs`	Shows neighbor information for all VRFs.
`vrf <VRF-NAME>`	Specifies the name of a VRF. Default: default.

Examples

Showing OSPFv3 AF neighbors information:

```
switch# show ospfv3-af ipv6 neighbors
Address-family : IPv6 Unicast
-----------------------------
OSPFv3 AF Process ID 1 VRF default
================================
Total Number of Neighbors: 1
Neighbor ID      Priority  State             Interface
-------------------------------------------------------
3.3.3.3          1         FULL/DR           1/1/1
  Neighbor address fe80::7272:cfff:fe79:7510
```

Showing OSPFv3 AF neighbors information for neighbor 3.3.3.3:

```
switch# show ospfv3-af ipv6 neighbors 3.3.3.3

Address-family : IPv6 Unicast
-----------------------------
VRF : default                          Process : 1
--------------------------------------------------------------
Router-Id           : 3.3.3.3          Area              : 0.0.0.0
Interface           : 1/1/1            Address           :
fe80::7272:cfff:fe79:7510
State               : FULL             Neighbor Priority : 1
Dead Timer Due      : 00:00:36         Options           : 0x13
Time since last state change : 00h:14m:45s
```

Showing detail OSPFv3 AF neighbors information for neighbor 3.3.3.3:

```
switch# show ospfv3-af ipv6 neighbors 3.3.3.3 detail
Address-family : IPv6 Unicast
-----------------------------
VRF : default                         Process : 1
--------------------------------------------------------------
Router-Id          : 3.3.3.3          Area               : 0.0.0.0
Interface          : 1/1/1            Address            :
fe80::7272:cfff:fe79:7510
State              : FULL             Neighbor Priority : 1
DR                 : 3.3.3.3          BDR                : 1.1.1.3
Dead Timer Due     : 00:00:36         Options            : 0x13
Retransmission Queue Length  : 0
Time Since Last State Change : 00h:14m:45s
```

Showing OSPFv3 AF neighbors information for interface 1/1/1:

```
switch# show ospfv3-af ipv6 neighbors 3.3.3.3 interface 1/1/1
Address-family : IPv6 Unicast
-----------------------------
VRF : default                         Process : 1
--------------------------------------------------------------
Router-Id          : 3.3.3.3          Area               : 0.0.0.0
Interface          : 1/1/1            Address            :
fe80::7272:cfff:fe79:7510
State              : FULL             Neighbor Priority : 1
Dead Timer Due     : 00:00:36         Options            : 0x13
Time Since Last State Change : 00h:14m:45s
```

Showing summary OSPFv3 AF neighbors information for a specific neighbor for all VRFs:

```
switch# show ospfv3-af ipv6 neighbors 3.3.3.3 summary all-vrfs
Address-family : IPv6 Unicast
-----------------------------
OSPFv3 AF Process ID 1 VRF default, Neighbor Summary
==================================================
Interface    Down Attempt Init TwoWay ExStart Exchange Loading Full Total
--------------------------------------------------------------------------
1/1/1        0    0       0    0      0       0        0       1    1
Total        0    0       0    0      0       0        0       1    1
OSPFv3 AF Process ID 1 VRF red, Neighbor Summary
==============================================
Interface    Down Attempt Init TwoWay ExStart Exchange Loading Full Total
--------------------------------------------------------------------------
1/1/2        0    0       0    0      0       0        0       1    1
Total        0    0       0    0      0       0        0       1    1
```

Showing OSPFv3 AF neighbors information for VRF red:

```
switch# show ospfv3-af ipv6 neighbors vrf red
Address-family : IPv6 Unicast
-----------------------------
OSPFv3 AF Process ID 2 VRF red
============================
Total Number of Neighbors: 1
Neighbor ID       Priority  State             Interface
-------------------------------------------------------
4.4.4.4           1         FULL/DR           1/1/2
  Neighbor address fe80::7272:cfff:fe79:7510
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ospfv3-af ipv6 routes

Syntax

```
show ospfv3-af [<PROCESS-ID>] ipv6 routes [<PREFIX/LENGTH>]
   [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows the OSPFv3 AF routing table information.

Parameter	Description
<PROCESS-ID>	Specifies an OSPFv3 AF process ID that shows information fro m the OSPFv3 AF routing table for the particular OSPFv3 AF p rocess. Range: 1 to 65535.
<PREFIX/LENGTH>	Specifies the IPv6 destination prefix showing information about a particular destination prefix. For example, 2010:bd9::/32.
`all-vrfs`	Select to show OSPFv3 AF routing table information for all VR Fs.
`vrf` *<VRF-NAME>*	Specifies the name of a VRF. Default: default.

Examples

Showing OSPFv3 AF routing table information:

```
switch# show ospfv3-af ipv6 routes
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2
Address-family : IPv6 Unicast
-----------------------------
OSPFv3 AF Process ID 1 VRF default, Routing Table
-------------------------------------------------
Total Number of OSPFv3 AF Routes : 3
111::/64            (i) area:0.0.0.0
     directly attached to interface 1/1/1, cost 1 distance 110
2000::/64           (i) area:0.0.0.0
     directly attached to interface tunnel1, cost 100 distance 110
fd00::/64           (i) area:0.0.0.1
     directly attached to interface vlan10, cost 1 distance 110
```

Showing OSPFv3 AF routing table information for VRF red:

```
switch# show ospfv3-af ipv6 routes vrf red
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2
Address-family : IPv6 Unicast
-----------------------------
OSPFv3 AF Process ID 2 VRF red, Routing Table
---------------------------------------------
Total Number of OSPFv3 AF Routes : 1
```

```
222::/64            (i) area:0.0.0.1
     directly attached to interface 1/1/2, cost 1 distance 110
```

Showing OSPFv3 AF routing table information for destination prefix fd00::/64:

```
switch# show ospfv3-af 1 ipv6 routes fd00::/64
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2
Address-family : IPv6 Unicast
-----------------------------
OSPFv3 AF Process ID 1 VRF default, Routing Table for prefixes fd00::/64
------------------------------------------------------------------------
Total Number of OSPFv3 AF Routes : 1
fd00::/64           (i) area:0.0.0.1
     directly attached to interface vlan10, cost 1 distance 110
```

Showing OSPFv3 AF routing table information for destination prefix 2000::/64:

```
switch# show ospfv3-af 1 ipv6 routes 2000::/64
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2
Address-family : IPv6 Unicast
-----------------------------
OSPFv3 AF Process ID 1 VRF default, Routing Table for prefixes 2000::/64
------------------------------------------------------------------------
Total Number of OSPFv3 AF Routes : 1
2000::/64           (i) area:0.0.0.0
     directly attached to interface tunnel1, cost 100 distance 110
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ospfv3-af ipv6 statistics

Syntax

```
show ospfv3-af ipv6 [<PROCESS-ID>] statistics
     [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows OSPFv3 AF statistics.

Parameter	Description
<PROCESS-ID>	Specifies an OSPFv3 AF process ID that shows information on the OSPFv3 AF SPF statistics for the particular OSPFv3 AF process. Range: 1 to 65535.
`all-vrfs`	Select to show OSPFv3 AF SPF statistics information for all VRFs.
`vrf` *<VRF-NAME>*	Specifies the name of a VRF. Default: default.

Examples

Showing OSPFv3 AF statistics information:

```
switch# show ospfv3-af ipv6 statistics
Address-family : IPv6 Unicast
-----------------------------
OSPFv3 AF Process ID 1 VRF default, Statistics (cleared 3h 2m 21s ago)
---------------------------------------------------------------------
Unknown Interface Drops         : 0
Unknown Virtual Interface Drops : 0
Bad IPv6 Header Length Drops    : 0
Wrong OSPFv3 Version Drops      : 0
Bad Source IPv6 Drops           : 0
Resource Failure Drops          : 0
Bad Header Length Drops         : 0
Total Drops                     : 0
```

Showing OSPFv3 AF statistics information for VRF red:

```
switch# show ospfv3-af 2 ipv6 statistics vrf red
Address-family : IPv6 Unicast
-----------------------------
OSPFv3 AF Process ID 2 VRF red, Statistics (cleared 3h 2m 30s ago)
```

```
------------------------------------------------------------------
Unknown Interface Drops          : 0
Unknown Virtual Interface Drops : 0
Bad IPv6 Header Length Drops     : 0
Wrong OSPFv3 Version Drops       : 0
Bad Source IPv6 Drops            : 0
Resource Failure Drops           : 0
Bad Header Length Drops          : 0
Total Drops                      : 0
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

show ospfv3-af ipv6 statistics interface

Syntax

```
show ospfv3-af ipv6 [<PROCESS-ID>] statistics interface [<INTERFACE-NAME>|
vlan <VLAN-ID> | lag <LAG-ID> | loopback <LOOPBACK-ID> | tunnel <TUNNEL-ID>]
   [all-vrfs | vrf <VRF-NAME>]
```

Description

Shows the OSPFv3 AF statistics for the enabled OSPFv3 AF interfaces.

Parameter	Description
`<PROCESS-ID>`	Specifies an OSPFv3 AF process ID to show OSPF - enabled interface statistics information on the specified OSPFv3 AF process. Range: 1 to 65535.
`<INTERFACE-NAME>`	Selects to show information only for the specified interface.
`vlan <VLAN-ID>`	Specifies the vlan interface name.
`lag <LAG ID>`	Specifies lag interface name.
`loopback <LOOPBACK ID>`	Specifies the vlan interface name.
`tunnel <TUNNEL-ID>`	Specifies the tunnel interface name.
`all-vrfs`	Select to show OSPF - enabled interface statistics information for all VRFs.
`vrf <VRF-NAME>`	Specifies the name of a VRF. Default: default.

Example

Showing enabled OSPFv3 AF interfaces information for interface 1/1/1:

```
switch# show ospfv3-af ipv6 statistics interface 1/1/1
Address-family : IPv6 Unicast
-----------------------------
OSPFv3 AF Process ID 1 VRF default, Interface 1/1/1 Statistics  (cleared 3h
22m 18s ago)
============================================================================
==========
Tx Hello packets      : 1213            Rx Hello packets      : 1213
Tx Hello bytes        : 48516           Rx Hello bytes        : 48516
Tx DD packets         : 3               Rx DD packets         : 2
Tx DD bytes           : 164             Rx DD bytes           : 136
Tx LS request packets : 1               Rx LS request packets : 1
Tx LS request bytes   : 64              Rx LS request bytes   : 64
```

```
Tx LS update packets   : 22              Rx LS update packets   : 22
Tx LS update bytes     : 1520            Rx LS update bytes     : 1844
Tx LS ack packets      : 21              Rx LS ack packets      : 20
Tx LS ack bytes        : 976             Rx LS ack bytes        : 800
Total IPsec packets processed : 10
Total IPsec bytes processed   : 6400
Total Number of State Changes : 3
Number of LSAs                : 2
LSA Checksum Sum              : 21758
Total OSPFv3-AF Packets Discarded: 0
----------------------------------
Reason                            Packets Dropped
--------------------------------------------------
Invalid Type                     0
Invalid Length                   0
Invalid Version                  0
Bad or Unknown Source            0
Area Mismatch                    0
Self-originated                  0
Duplicate Router ID              0
Interface Standby                0
Total Hello Packets Dropped      0
  Hello Interval Mismatch        0
  Dead Interval Mismatch         0
  Options Mismatch               0
  MTU Mismatch                   0
  Neighbor Ignored               0
Resource Failures                0
Bad LSA Length                   0
Others                           0
IPsec Authentication Errors      0
Total LSAs Ignored : 0
-----------------------
Bad Type             : 0
Bad Length           : 0
Invalid Data         : 0
Invalid Checksum     : 0
```

Showing enabled OSPFv3 AF interfaces information for tunnel interface:

```
switch# show ospfv3-af ipv6 statistics interface tunnel1
Address-family : IPv6 Unicast
-----------------------------
OSPFv3 AF Process ID 1 VRF default, Interface tunnel1 Statistics (cleared
0h16m32s ago)
```

```
============================================================================
=========
Tx Hello packets      : 100                Rx Hello packets      : 75
Tx Hello bytes        : 3900               Rx Hello bytes        : 3000
Tx DD packets         : 2                  Rx DD packets         : 2
Tx DD bytes           : 136                Rx DD bytes           : 116
Tx LS request packets : 1                  Rx LS request packets : 1
Tx LS request bytes   : 52                 Rx LS request bytes   : 64
Tx LS update packets  : 3                  Rx LS update packets  : 5
Tx LS update bytes    : 388                Rx LS update bytes    : 464
Tx LS ack packets     : 4                  Rx LS ack packets     : 3
Tx LS ack bytes       : 244                Rx LS ack bytes       : 208
Total IPsec packets processed : 0
Total IPsec bytes processed   : 0
Total Number of State Changes : 2
Number of LSAs                : 2
LSA Checksum Sum              : 76646
Total OSPFv3-AF Packets Discarded: 0
----------------------------------
Reason                           Packets Dropped
------------------------------------------------
Invalid Type                     0
Invalid Length                   0
Invalid Version                  0
Bad or Unknown Source            0
Area Mismatch                    0
Self-originated                  0
Duplicate Router ID              0
Interface Standby                0
Total Hello Packets Dropped      0
  Hello Interval Mismatch        0
  Dead Interval Mismatch         0
  Options Mismatch               0
  MTU Mismatch                   0
  Neighbor Ignored               0
Resource Failures                0
Bad LSA Length                   0
Others                           0
IPsec Authentication Errors      0
IPsec ESP Errors                 0
Total LSAs Ignored : 0
-----------------------
Bad Type           : 0
```

```
Bad Length          : 0
Invalid Data        : 0
Invalid Checksum    : 0
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show ospfv3-af ipv6 virtual-links

Syntax

```
show ospfv3-af [<PROCESS-ID>] ipv6 virtual-links [brief]
   [all-vrfs | vrf <VRF-NAME>]
```

Description

Displays the current state and parameters of the OSPFv3 AF virtual links.

Parameter	Description
`<PROCESS-ID>`	Enter an OSPFv3 AF process ID to display information on the OSPFv3 AF virtual links for the particular OSPFv3 AF process. Range: 1 to 65535.
`brief`	Select to display brief overview information for the OSPFv3 AF virtual links.
`all-vrfs`	Select to display OSPFv3 AF virtual links information for all VRFs.

Parameter	Description
`vrf <VRF-NAME>`	Specifies the name of a VRF. Default: default.

Examples

Show OSPFv3 AF virtual links information:

```
switch# show ospfv3-af ipv6 virtual-links
Address-family : IPv6 Unicast
-----------------------------
Virtual link to router 40.40.40.40 is up
----------------------------------------
VRF                  : default                      Process             :
21
Transit area         : 0.0.0.1                      Authentication      :
No
Hello Interval       : 10                           Dead Interval       :
40
Transit Delay        : 1                            Retransmit Interval : 5
Number of Link LSAs : 0                             Checksum Sum        : 0
Authentication       : No
Number of State Changes : 4
```

Show OSPFv3 AF virtual links information with IPsec AH authentication:

```
switch# show ospfv3-af ipv6 virtual-links
Address-family : IPv6 Unicast
-----------------------------
Virtual link to router 40.40.40.40 is up
----------------------------------------
VRF                  : default                      Process             :
21
Transit area         : 0.0.0.1                      Authentication      :
No
Hello Interval       : 10                           Dead Interval       :
40
Transit Delay        : 1                            Retransmit Interval : 5
Number of Link LSAs : 0                             Checksum Sum        : 0
AH Authentication    : MD5  SPI: 1234
Number of State Changes : 4
```

Show OSPFv3 AF virtual links information with IPsec ESP authentication:

```
switch# show ospfv3-af ipv6 virtual-links
Address-family : IPv6 Unicast
-----------------------------
```

```
Virtual link to router 40.40.40.40 is up
----------------------------------------
VRF                   : default                         Process           :
21
Transit area          : 0.0.0.1                         Authentication    :
No
Hello Interval        : 10                              Dead Interval     :
40
Transit Delay         : 1                               Retransmit Interval : 5
Number of Link LSAs : 0                                 Checksum Sum        : 0
ESP Authentication  : SHA1 Encryption: NULL  SPI: 1234
Number of State Changes : 4
```

Show brief overview information for OSPFv3 AF virtual links:

```
switch# show ospfv3-af ipv6 virtual-links brief
Address-family : IPv6 Unicast
-----------------------------
OSPFv3 AF Process ID 1 VRF default
==============================
Total Number of Virtual Links: 1
Neighbor-ID      Transit-Area     Status
--------------------------------------
2.1.1.2          0.0.0.1          down
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show running-config`

Syntax

```
show running-config
```

Description

This command shows the configurations for all the protocols along with OSPFv3 AF.

Example

```
switch# show running-config
Current configuration:
!
!Version ArubaOS-CX genericx86-64.01.01.000X
!Schema version 0.1.8
vrf blue
vrf green
vrf red
led locator on
!
!
!
!
!
!
!
vlan 1,10
interface 1/1/1
      no shutdown
      ip address 1.1.1.1/24
      ipv6 address 111::1/64
      ip ospf 1 area 0.0.0.0
      ospfv3-af 1 ipv4 area 0.0.0.0
      ospfv3-af ipv4 authentication ipsec spi 300 sha1 ciphertext AQBapX1/
Ow3jK2zC6WCOv3mGwmvDQ02L9WKRTxVPtSyXpUNqBQAAACoBmj2c
interface 1/1/2
   no shutdown
   vrf attach red
   ip address 2.2.2.2/24
   ipv6 address 222::2/64
   ip ospf 2 area 0.0.0.1
   ospfv3-af 1 ipv4 area 0.0.0.1
   ospfv3-af ipv4 authentication ipsec spi 300 md5 ciphertext AQBapYH+i/
6HV2w+yyJCb45XhHPFfLNIt876Mh9EMRyf0XgwBAAAAGE/tKs=
interface 1/1/3
   no shutdown
```

```
   ip address 3.3.3.1/24
   ipv6 address 333::3/64
   ospfv3-af 1 ipv6 area 0.0.0.0 instance 1
   no ospfv3-af ipv6 passive
   ospfv3-af 1 ipv4 area 0.0.0.0 instance 65
interface 1/1/4
   no shutdown
   ipv6 address 1000::1/64
interface tunnel 1 mode ip 6in6
   no shutdown
   source ipv6 1000::1
   destination ipv6 1000::2
   ipv6 address 2000::1/64
   ospfv3-af 1 ipv4 area 0.0.0.0
   ospfv3-af ipv4 passive
   ospfv3-af ipv4 hello-interval 22
   ospfv3-af ipv4 dead-interval 88
   ospfv3-af ipv4 transit-delay 44
   ospfv3-af ipv4 priority 88
   ospfv3-af ipv4 cost 8
   ospfv3-af ipv4 network point-to-point
   ospfv3-af ipv4 shutdown
interface vlan10
   ip address 10.1.1.1/24
   ipv6 address fd00::1/64
   ip ospf 1 area 0.0.0.0
   ospfv3-af 1 ipv4 area 0.0.0.1
   ospfv3-af ipv4 encryption ipsec spi 401 sha1 ciphertext
AQBapd+HuD8Y4kMJkh5dFzG6OTBvW4iY+ueK4xpETxZPsZ1FDgAAAE3pkLmvMCtdFk2zz/LS
aes ciphertext
AQBapbPnk0Kl9giahf2dWGRHcU+6mTFiWzfxeWUXuWGYQ4teIgAAAOG7kcVrkrKV754nX+Li1ZDY
PHG5xsapOnyZJMv7cfHEctA=
   ospfv3-af 1 ipv6 area 0.0.0.1
   ospfv3-af ipv4 encryption ipsec spi 401 sha1 ciphertext
AQBapd+HuD8Y4kMJkh5dFzG6OTBvW4iY+ueK4xpETxZPsZ1FDgAAAE3pkLmvMCtdFk2zz/LS
   aes ciphertext
AQBapbPnk0Kl9giahf2dWGRHcU+6mTFiWzfxeWUXuWGYQ4teIgAAAOG7kcVrkrKV754nX+Li1ZDY
PHG5xsapOnyZJMv7cfHEctA=
interface loopback 2
   ipv6 address 99::1/64
   ospfv3-af 1 ipv4 area 0.0.0.0
router ospfv3-af 1
   router-id 1.1.1.1
```

```
   reference-bandwidth 4000
   address-family ipv4 unicast
    router-id 2.2.2.1
    area 0.0.0.0
    area 0.0.0.1
    area 0.0.0.2
    area 0.0.0.3
    distribute-list prefix list2 in
    redistribute static
    reference-bandwidth 4000
    active-backbone stub-default-route
    exit-address-family
    address-family ipv6 unicast
    router-id 2.2.2.2
    area 0.0.0.0
    area 0.0.0.1
    area 0.0.0.2
    area 0.0.0.3
    distribute-list prefix list2 in
    passive-interface default
    redistribute static
    reference-bandwidth 4000
    active-backbone stub-default-route
    area 0.0.0.1 virtual-link 1.1.1.1
        hello-interval 15
        dead-interval 60
        authentication ipsec spi 256 md5 ciphertext AQBapYH+i/
6HV2w+yyJCb45XhHPFfLNIt876Mh9EMRyf0XgwBAAAAGE/tKs=
        exit
    area 0.0.0.1 authentication ipsec spi 256 md5 ciphertext AQBapYH+i/
6HV2w+yyJCb45XhHPFfLNIt876Mh9EMRyf0XgwBAAAAGE/tKs=
    area 0.0.0.2 virtual-link 2.2.2.2
        encryption ipsec spi 3456 md5 ciphertext AQBapVuqYeTJuT8/
BoIlC2z4MxuT3VVmiduKtDwIP1jbCCxZCAAAAEe+Xod/lZX8 null
        exit
    area 0.0.0.3 virtual-link 3.3.3.3
        encryption ipsec spi 5678 sha1 ciphertext AQBapVuqYeTJuT8/
BoIlC2z4MxuT3VVmiduKtDwIP1jbCCxZCAAAAEe+Xod/lZX8 des
ciphertext AQBapVuqYeTJuT8/BoIlC2z4MxuT3VVmiduKtDwIP1jbCCxZCAAAAEe+Xod/lZX8
        exit
    exit-address-family
router ospfv3-af 2
    router-id 1.1.1.3
```

```
    distance 88
    max-metric router-lsa
    throttle spf start-time 100 hold-time 1000 max-wait-time 10000
    address-family ipv4 unicast
    router-id 1.1.1.3
    distance 88
    area 0.0.0.0
    area 0.0.0.1
    passive-interface default
    redistribute ospfv3-af 1 route-map ospfv3_af4_routes
    max-metric router-lsa
    throttle spf start-time 100 hold-time 1000 max-wait-time 10000
    area 0.0.0.1 encryption ipsec spi 400 md5 ciphertext AQBapfwg9p/
z+rRS1mhk9jXzbOFBd82j2OSvEbAbI2f4KL+ADAAAAL3hhFDJlUbBwu1fvA==
des ciphertext
AQBapfTQ8jUL+zbNyX9ZKOF8Lc5iR6dF543pwK2egyCWSWkkEgAAAE0etJaie2d/
i2XF8pKkRpQvpw==
    exit-address-family
    address-family ipv6 unicast
    router-id 1.1.1.4
    distance 88
    area 0.0.0.0
    area 0.0.0.1
    passive-interface default
    redistribute ospfv3-af 1 route-map ospfv3_af6_routes
    max-metric router-lsa
    throttle spf start-time 100 hold-time 1000 max-wait-time 10000
    area 0.0.0.1 encryption ipsec spi 400 md5 ciphertext AQBapfwg9p/
z+rRS1mhk9jXzbOFBd82j2OSvEbAbI2f4KL+ADAAAAL3hhFDJlUbBwu1fvA==
des ciphertext
AQBapfTQ8jUL+zbNyX9ZKOF8Lc5iR6dF543pwK2egyCWSWkkEgAAAE0etJaie2d/
i2XF8pKkRpQvpw==
    exit-address-family
router ospfv3-af 1 vrf red
    reference-bandwidth 40000
    graceful-restart restart-interval 88
    no graceful-restart helper
    distance intra-area 34 inter-area 66 external 3
    address-family ipv4 unicast
    area 0.0.0.1
    redistribute ospf 2
    redistribute ospfv3-af 2
    reference-bandwidth 40000
```

```
    graceful-restart restart-interval 88
    no graceful-restart helper
    distance intra-area 34 inter-area 66 external 3
    exit-address-family
    address-family ipv6 unicast
    area 0.0.0.1
    redistribute ospfv3 2
    redistribute ospfv3-af 2
    reference-bandwidth 40000
    graceful-restart restart-interval 88
    max-metric router-lsa include-stub
    graceful-restart restart-interval 88
    no graceful-restart helper
    graceful-restart ignore-lost-interface
    distance intra-area 34 inter-area 66
    address-family ipv4 unicast
    router-id 1.1.1.8
    area 0.0.0.1
    max-metric router-lsa include-stub
    graceful-restart restart-interval 88
    no graceful-restart helper
    graceful-restart ignore-lost-interface
    distance intra-area 34 inter-area 66
    exit-address-family
    address-family ipv6 unicast
    router-id 1.1.1.9
    area 0.0.0.1
    graceful-restart restart-interval 88
    no graceful-restart helper strict-lsa-check
    max-metric router-lsa include-stub
    distance intra-area 34 inter-area 66
    exit-address-family
router ospfv3-af 1 vrf blue
    graceful-restart restart-interval 99
    no graceful-restart helper
    max-metric router-lsa on-startup 100
    address-family ipv4 unicast
    graceful-restart restart-interval 99
    no graceful-restart helper
    area 0.0.0.0
    max-metric router-lsa on-startup 100
    exit-address-family
    address-family ipv6 unicast
```

```
    graceful-restart restart-interval 99
    no graceful-restart helper
    area 0.0.0.0
    max-metric router-lsa on-startup 100
    exit-address-family
router ospfv3-af 1 vrf green
    max-metric router-lsa include-stub on-startup 100
    address-family ipv4 unicast
    area 0.0.0.0
    max-metric router-lsa include-stub on-startup 100
    exit-address-family
    address-family ipv6 unicast
    area 0.0.0.0
    max-metric router-lsa include-stub on-startup 100
    exit-address-family
switch#
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show running-config all`

Syntax

```
show running-config all
```

Description

Displays the current running configuration, including default values for all protocols such as OSPFv3 AF.

Examples

Showing running-config without default values:

```
switch# show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.05.0001BC-26-g94067ba
!export-password: default
led locator on
vrf blue
vrf green
vrf red
!
!
!
!
ssh server vrf mgmt
vlan 1
interface mgmt
    no shutdown
    ip dhcp
interface 1/1/1
    no shutdown
    ip address 1.1.1.1/24
    ipv6 address 111::1/64
    ipv6 address 99::1/64
    ospfv3-af 1 ipv4 area 0.0.0.0
interface 1/1/2
    no shutdown
interface loopback 2
    ipv6 address 929::1/64
    ospfv3-af 1 ipv4 area 0.0.0.0
interface loopback 3
    ipv6 address 123::1/64
    ospfv3-af 1 ipv4 area 0.0.0.0
    ospfv3-af 1 ipv6 area 0.0.0.0
!
!
!
!
!
router ospfv3-af 1 vrf blue
    max-metric router-lsa on-startup 100
    address-family ipv4 unicast
```

```
        distribute-list prefix list2 in
        max-metric router-lsa on-startup 100
        active-backbone stub-default-route
        area 0.0.0.0
    exit-address-family
    address-family ipv6 unicast
        distribute-list prefix list2 in
        max-metric router-lsa on-startup 100
        active-backbone stub-default-route
        area 0.0.0.0
    exit-address-family
router ospfv3-af 1
    router-id 1.1.1.1
    reference-bandwidth 4000
    address-family ipv4 unicast
        router-id 1.1.1.2
        reference-bandwidth 4000
        redistribute host-routes
        redistribute static
        area 0.0.0.0
        area 0.0.0.1
    exit-address-family
    address-family ipv6 unicast
        router-id 1.1.1.3
        reference-bandwidth 4000
        redistribute host-routes
        redistribute static
        area 0.0.0.0
        area 0.0.0.1
        area 0.0.0.1 virtual-link 1.1.1.2
            dead-interval 60
            hello-interval 15
            authentication ipsec spi 256 md5 ciphertext
            AQBapYH+i/6HV2w+yyJCb45XhHPFfLNIt876Mh9EMRyf0XgwBAAAAGE/tKs=
            exit
        area 0.0.0.2
        area 0.0.0.2 virtual-link 2.2.2.2
            encryption ipsec spi 3456 md5 ciphertext
            AQBapVuqYeTJuT8/BoIlC2z4MxuT3VVmiduKtDwIP1jbCCxZCAAAAEe+Xod/
lZX8 null
            exit
        area 0.0.0.3
        area 0.0.0.3 virtual-link 3.3.3.3
```

```
            encryption ipsec spi 5678 sha1 ciphertext
            AQBapVuqYeTJuT8/BoIlC2z4MxuT3VVmiduKtDwIP1jbCCxZCAAAAEe+Xod/
lZX8 des
            ciphertext AQBapVuqYeTJuT8/
BoIlC2z4MxuT3VVmiduKtDwIP1jbCCxZCAAAAEe+
            Xod/lZX8
            exit
    exit-address-family
router ospfv3-af 2
    router-id 1.1.1.3
    max-metric router-lsa
    address-family ipv4 unicast
        router-id 1.1.1.4
        max-metric router-lsa
        area 0.0.0.0
        area 0.0.0.1
        area 0.0.0.1 encryption ipsec spi 400 md5 ciphertext
        AQBapfwg9p/
z+rRS1mhk9jXzbOFBd82j2OSvEbAbI2f4KL+ADAAAAL3hhFDJlUbBwu1fvA==des
        ciphertext
AQBapfTQ8jUL+zbNyX9ZKOF8Lc5iR6dF543pwK2egyCWSWkkEgAAAE0etJaie2d/
        i2XF8pKkRpQvpw==
    exit-address-family
    address-family ipv6 unicast
        router-id 1.1.1.5
        max-metric router-lsa
        area 0.0.0.0
        area 0.0.0.1
        area 0.0.0.1 encryption ipsec spi 400 md5 ciphertext
        AQBapfwg9p/
z+rRS1mhk9jXzbOFBd82j2OSvEbAbI2f4KL+ADAAAAL3hhFDJlUbBwu1fvA==des
        ciphertext
AQBapfTQ8jUL+zbNyX9ZKOF8Lc5iR6dF543pwK2egyCWSWkkEgAAAE0etJaie2d/
        i2XF8pKkRpQvpw==
    exit-address-family
router ospfv3-af 1 vrf green
    max-metric router-lsa include-stub on-startup 100
    address-family ipv4 unicast
        max-metric router-lsa include-stub on-startup 100
        area 0.0.0.0
    exit-address-family
    address-family ipv6 unicast
        max-metric router-lsa include-stub on-startup 100
```

```
        area 0.0.0.0
    exit-address-family
router ospfv3-af 1 vrf red
    reference-bandwidth 40000
    address-family ipv4 unicast
        reference-bandwidth 40000
        redistribute ospfv3 2
        area 0.0.0.1
    exit-address-family
    address-family ipv6 unicast
        reference-bandwidth 40000
        redistribute ospfv3 2
        area 0.0.0.1
    exit-address-family
router ospfv3-af 2 vrf red
    router-id 1.1.1.7
    max-metric router-lsa include-stub
    address-family ipv4 unicast
        router-id 1.1.1.7
        max-metric router-lsa include-stub
        area 0.0.0.1
    exit-address-family
    address-family ipv6 unicast
        router-id 1.1.1.7
        max-metric router-lsa include-stub
        area 0.0.0.1
    exit-address-family
https-server vrf mgmt
```

Showing running-config with default values:

```
switch# show running-config all
Current configuration:
!
!
!
interface 1/1/1
    no shutdown
    mtu 1500
    qos trust none
    apply qos schedule-profile factory-default
    lldp transmit
    lldp receive
    lldp med poe
    no lldp med poe priority-override
```

```
lldp dot3 poe
lldp dot3 macphy
lldp med network-policy
lldp med capability
lldp med topology-trap
cdp
sflow
routing
vrf attach default
ip mtu 1500
ip address 1.1.1.1/24
ipv6 address 111::1/64 tag 0
ipv6 address 99::1/64 tag 0
no ip urpf-check
ipv6 nd mtu 1500
no ip proxy-arp
ipv6 nd dad attempts 1
ipv6 nd hop-limit 64
ipv6 nd ns-interval 1000
ipv6 nd ra min-interval 200
ipv6 nd ra max-interval 600
ipv6 nd ra lifetime 1800
ipv6 nd ra reachable-time 0
no ipv6 address autoconfig
ipv6 nd ra retrans-timer 0
no ipv6 nd ra managed-config-flag
no ipv6 nd ra other-config-flag
ipv6 nd suppress-ra
ipv6 nd router-preference medium
no ip irdp
ip irdp multicast
ip irdp minadvertinterval 450
ip irdp maxadvertinterval 600
ip irdp holdtime 1800
ip irdp preference 0
arp timeout 1800
ipv6 neighbor timeout 1800
power-over-ethernet
no power-over-ethernet pre-std-detect
power-over-ethernet priority low
power-over-ethernet allocate-by usage
power-over-ethernet assigned-class 4
ospfv3-af 1 ipv4 area 0.0.0.0
```

```
    ospfv3-af ipv4 dead-interval 40
    ospfv3-af ipv4 hello-interval 10
    ospfv3-af ipv4 retransmit-interval 5
    ospfv3-af ipv4 transit-delay 1
    ospfv3-af ipv4 priority 1
    ospfv3-af ipv4 cost 4
    ospfv3-af ipv4 network broadcast
!
!
!
interface loopback 2
    vrf attach default
    ipv6 address 929::1/64 tag 0
    no ip urpf-check
    arp timeout 1800
    ipv6 neighbor timeout 1800
    ospfv3-af 1 ipv4 area 0.0.0.0
    ospfv3-af ipv4 dead-interval 40
    ospfv3-af ipv4 hello-interval 10
    ospfv3-af ipv4 retransmit-interval 5
    ospfv3-af ipv4 transit-delay 1
    ospfv3-af ipv4 priority 1
    ospfv3-af ipv4 network broadcast
!
!
!
interface loopback 3
    vrf attach default
    ipv6 address 123::1/64 tag 0
    no ip urpf-check
    arp timeout 1800
    ipv6 neighbor timeout 1800
    ospfv3-af 1 ipv4 area 0.0.0.0
    ospfv3-af ipv4 dead-interval 40
    ospfv3-af ipv4 hello-interval 10
    ospfv3-af ipv4 retransmit-interval 5
    ospfv3-af ipv4 transit-delay 1
    ospfv3-af ipv4 priority 1
    ospfv3-af ipv4 network broadcast
    ospfv3-af 1 ipv6 area 0.0.0.0
    ospfv3-af ipv6 dead-interval 40
    ospfv3-af ipv6 hello-interval 10
    ospfv3-af ipv6 retransmit-interval 5
```

```
    ospfv3-af ipv6 transit-delay 1
    ospfv3-af ipv6 priority 1
    ospfv3-af ipv6 network broadcast
!
!
!
router ospfv3-af 1 vrf blue
    enable
    reference-bandwidth 100000
    max-metric router-lsa on-startup 100
    timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
    timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
    timers lsa-arrival 1000
    graceful-restart restart-interval 120
    distance 110
    address-family ipv4 unicast
        enable
        reference-bandwidth 100000
        max-metric router-lsa on-startup 100
        maximum-paths 4
        timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
        timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
        timers lsa-arrival 1000
        graceful-restart restart-interval 120
        default-metric 25
        distance 110
        active-backbone stub-default-route
        area 0.0.0.0
        area 0.0.0.0 default-metric 1
    exit-address-family
    address-family ipv6 unicast
        enable
        reference-bandwidth 100000
        max-metric router-lsa on-startup 100
        maximum-paths 4
        timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
        timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
        timers lsa-arrival 1000
        graceful-restart restart-interval 120
        default-metric 25
        distance 110
        active-backbone stub-default-route
        area 0.0.0.0
```

```
        area 0.0.0.0 default-metric 1
    exit-address-family
router ospfv3-af 1 vrf default
    enable
    router-id 1.1.1.1
    reference-bandwidth 4000
    timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
    timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
    timers lsa-arrival 1000
    graceful-restart restart-interval 120
    distance 110
    address-family ipv4 unicast
        enable
        router-id 1.1.1.1
        reference-bandwidth 4000
        maximum-paths 4
        timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
        timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
        timers lsa-arrival 1000
        graceful-restart restart-interval 120
        default-metric 25
        distance 110
        redistribute static
        no active-backbone stub-default-route
        area 0.0.0.0
        area 0.0.0.0 default-metric 1
        area 0.0.0.1
        area 0.0.0.1 default-metric 1
        area 0.0.0.2
        area 0.0.0.2 default-metric 1
        area 0.0.0.3
        area 0.0.0.3 default-metric 1
    exit-address-family
    address-family ipv6 unicast
        enable
        router-id 1.1.1.1
        reference-bandwidth 4000
        maximum-paths 4
        timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
        timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
        timers lsa-arrival 1000
        graceful-restart restart-interval 120
        default-metric 25
```

```
        distance 110
        redistribute static
        no active-backbone stub-default-route
        area 0.0.0.0
        area 0.0.0.0 default-metric 1
        area 0.0.0.1
        area 0.0.0.1 default-metric 1
        area 0.0.0.1 virtual-link 1.1.1.2
            dead-interval 60
            hello-interval 15
            retransmit-interval 5
            transit-delay 1
            authentication ipsec spi 256 md5 ciphertext
            AQBapYH+i/6HV2w+yyJCb45XhHPFfLNIt876Mh9EMRyf0XgwBAAAAGE/tKs=
            exit
        area 0.0.0.2
        area 0.0.0.2 default-metric 1
        area 0.0.0.2 virtual-link 2.2.2.2
            dead-interval 40
            hello-interval 10
            retransmit-interval 5
            transit-delay 1
            encryption ipsec spi 3456 md5 ciphertext
            AQBapVuqYeTJuT8/BoIlC2z4MxuT3VVmiduKtDwIP1jbCCxZCAAAAEe+Xod/
lZX8 null
            exit
        area 0.0.0.3
        area 0.0.0.3 default-metric 1
        area 0.0.0.3 virtual-link 3.3.3.3
        dead-interval 40
        hello-interval 10
        retransmit-interval 5
        transit-delay 1
        encryption ipsec spi 5678 sha1 ciphertext
        AQBapVuqYeTJuT8/BoIlC2z4MxuT3VVmiduKtDwIP1jbCCxZCAAAAEe+Xod/lZX8 des
        ciphertext AQBapVuqYeTJuT8/
BoIlC2z4MxuT3VVmiduKtDwIP1jbCCxZCAAAAEe+Xod/lZX8
        exit
    exit-address-family
router ospfv3-af 2 vrf default
    enable
    router-id 1.1.1.3
    reference-bandwidth 100000
```

```
    max-metric router-lsa
    timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
    timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
    timers lsa-arrival 1000
    graceful-restart restart-interval 120
    distance 110
    address-family ipv4 unicast
        enable
        router-id 1.1.1.3
        reference-bandwidth 100000
        max-metric router-lsa
        maximum-paths 4
        timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
        timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
        timers lsa-arrival 1000
        graceful-restart restart-interval 120
        default-metric 25
        distance 110
        no active-backbone stub-default-route
        area 0.0.0.0
        area 0.0.0.0 default-metric 1
        area 0.0.0.1
        area 0.0.0.1 encryption ipsec spi 400 md5 ciphertext
        AQBapfwg9p/
z+rRS1mhk9jXzbOFBd82j2OSvEbAbI2f4KL+ADAAAAL3hhFDJlUbBwu1fvA==des
        ciphertext
AQBapfTQ8jUL+zbNyX9ZKOF8Lc5iR6dF543pwK2egyCWSWkkEgAAAE0etJaie2d/
        i2XF8pKkRpQvpw==
        area 0.0.0.1 default-metric 1
    exit-address-family
    address-family ipv6 unicast
        enable
        router-id 1.1.1.3
        reference-bandwidth 100000
        max-metric router-lsa
        maximum-paths 4
        timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
        timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
        timers lsa-arrival 1000
        graceful-restart restart-interval 120
        default-metric 25
        distance 110
        no active-backbone stub-default-route
```

```
        area 0.0.0.0
        area 0.0.0.0 default-metric 1
        area 0.0.0.1
        area 0.0.0.1 encryption ipsec spi 400 md5 ciphertext
        AQBapfwg9p/
z+rRS1mhk9jXzbOFBd82j2OSvEbAbI2f4KL+ADAAAAL3hhFDJlUbBwu1fvA==des
        ciphertext
AQBapfTQ8jUL+zbNyX9ZKOF8Lc5iR6dF543pwK2egyCWSWkkEgAAAE0etJaie2d/
        i2XF8pKkRpQvpw==
        area 0.0.0.1 default-metric 1
    exit-address-family
router ospfv3-af 1 vrf green
    enable
    reference-bandwidth 100000
    max-metric router-lsa include-stub on-startup 100
    timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
    timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
    timers lsa-arrival 1000
    graceful-restart restart-interval 120
    distance 110
    address-family ipv4 unicast
        enable
        reference-bandwidth 100000
        max-metric router-lsa include-stub on-startup 100
        maximum-paths 4
        timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
        timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
        timers lsa-arrival 1000
        graceful-restart restart-interval 120
        default-metric 25
        distance 110
        no active-backbone stub-default-route
        area 0.0.0.0
        area 0.0.0.0 default-metric 1
    exit-address-family
    address-family ipv6 unicast
        enable
        reference-bandwidth 100000
        max-metric router-lsa include-stub on-startup 100
        maximum-paths 4
        timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
        timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
        timers lsa-arrival 1000
```

```
        graceful-restart restart-interval 120
        default-metric 25
        distance 110
        no active-backbone stub-default-route
        area 0.0.0.0
        area 0.0.0.0 default-metric 1
    exit-address-family
router ospfv3-af 1 vrf red
    enable
    reference-bandwidth 40000
    timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
    timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
    timers lsa-arrival 1000
    graceful-restart restart-interval 120
    distance 110
    address-family ipv4 unicast
        enable
        reference-bandwidth 40000
        maximum-paths 4
        timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
        timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
        timers lsa-arrival 1000
        graceful-restart restart-interval 120
        default-metric 25
        distance 110
        redistribute ospfv3 2
        no active-backbone stub-default-route
        area 0.0.0.1
        area 0.0.0.1 default-metric 1
    exit-address-family
    address-family ipv6 unicast
        enable
        reference-bandwidth 40000
        maximum-paths 4
        timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
        timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
        timers lsa-arrival 1000
        graceful-restart restart-interval 120
        default-metric 25
        distance 110
        redistribute ospfv3 2
        no active-backbone stub-default-route
        area 0.0.0.1
```

```
        area 0.0.0.1 default-metric 1
    exit-address-family
router ospfv3-af 2 vrf red
        enable
        router-id 1.1.1.7
        reference-bandwidth 100000
        max-metric router-lsa include-stub
        timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
        timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
        timers lsa-arrival 1000
        graceful-restart restart-interval 120
    distance 110
    address-family ipv4 unicast
        enable
        router-id 1.1.1.7
        reference-bandwidth 100000
        max-metric router-lsa include-stub
        maximum-paths 4
        timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
        timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
        timers lsa-arrival 1000
        graceful-restart restart-interval 120
        default-metric 25
        distance 110
        no active-backbone stub-default-route
        area 0.0.0.1
        area 0.0.0.1 default-metric 1
    exit-address-family
    address-family ipv6 unicast
        enable
        router-id 1.1.1.7
        reference-bandwidth 100000
        max-metric router-lsa include-stub
        maximum-paths 4
        timers throttle spf start-time 200 hold-time 1000 max-wait-time 5000
        timers throttle lsa start-time 5000 hold-time 0 max-wait-time 0
        timers lsa-arrival 1000
        graceful-restart restart-interval 120
        default-metric 25
        distance 110
        no active-backbone stub-default-route
        area 0.0.0.1
        area 0.0.0.1 default-metric 1
```

```
    exit-address-family
!
!
!
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manage r (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show tech ospfv3-af`

Syntax

```
show tech ospfv3-af [local-file]
```

Description

Shows the output of the following OSPFv3 AF commands:

- **`show ospfv3-af all vrfs`**
- **`show ospfv3-af interface all-vrfs`**
- **`show ospfv3-af neighbors detail all-vrfs`**
- **`show ospfv3-af neighbors summary all-vrfs`**
- **`show ospfv3-af route all-vrfs`**
- **`show ospfv3-af statistics all-vrfs`**
- **`show ospfv3-af statistics interface all-vrfs`**

- `show ospfv3-af border-routers all-vrfs`
- `show ospfv3-af lsdb all-vrfs database-summary`
- `show ospfv3-af virtual-links all-vrfs`

Parameter	Description
`local-file`	Captures the command output into a local file. Optional.

Examples

```
switch(config-if)# show tech ospfv3-af
====================================================
Show Tech executed on Tue Sep 12 09:42:42 2017
====================================================
====================================================
[Begin] Feature ospfv3-af
====================================================
*********************************
Command : show ospfv3-af ipv6 neighbor summary all-vrfs
*********************************
Address-family : IPv6 Unicast
-----------------------------
OSPFv3 AF Process ID 1 VRF default, Neighbor Summary
==================================================
Interface     Down Attempt Init TwoWay ExStart Exchange Loading Full Total
--------------------------------------------------------------------------
1/1/1         0    0       0    0      0       0        0       1    1
Total         0    0       0    0      0       0        0       1    1
OSPFv3 AF Process ID 1 VRF red, Neighbor Summary
==============================================
Interface     Down Attempt Init TwoWay ExStart Exchange Loading Full Total
--------------------------------------------------------------------------
1/1/2         0    0       0    0      0       0        0       1    1
Total         0    0       0    0      0       0        0       1    1
*********************************
Command : show ospfv3-af ipv6 statistics all-vrfs
*********************************
Address-family : IPv6 Unicast
-----------------------------
OSPFv3 AF Process ID 1 VRF default, Statistics (cleared 0h 4m 46s ago)
--------------------------------------------------------------------
Unknown Interface Drops          : 0
```

```
Unknown Virtual Interface Drops : 0
Bad IPv6 Header Length Drops    : 0
Wrong OSPFv3 Version Drops      : 0
Bad Source IPv6 Drops           : 0
Resource Failure Drops          : 0
Bad Header Length Drops         : 0
Total Drops                     : 0
OSPFv3 AF Process ID 2 VRF red, Statistics (cleared 0h 4m 45s ago)
-----------------------------------------------------------------
Unknown Interface Drops         : 0
Unknown Virtual Interface Drops : 0
Bad IPv6 Header Length Drops    : 0
Wrong OSPFv3 Version Drops      : 0
Bad Source IPv6 Drops           : 0
Resource Failure Drops          : 0
Bad Header Length Drops         : 0
Total Drops                     : 0
*********************************
Command : show ospfv3-af ipv6 neighbor detail all-vrfs
*********************************
Address-family : IPv6 Unicast
-----------------------------
Neighbor 3.3.3.3, Interface address fe80::7272:cfff:fec6:6e6c
---------------------------------------------------------------
Process ID 1 VRF default, in Area 0.0.0.0 via Interface 1/1/1
    State is FULL, Neighbor priority is 1
    DR is 3.3.3.3, BDR is 1.1.1.3
    Options is 0x13
    Dead timer due in 00:00:34
    Retransmission queue length 0
Neighbor 4.4.4.4, Interface address fe80::7272:cfff:fec6:6e6c
---------------------------------------------------------------
Process ID 2 VRF red, in Area 0.0.0.1 via Interface 1/1/2
    State is FULL, Neighbor priority is 1
    DR is 4.4.4.4, BDR is 1.1.1.7
    Options is 0x13
    Dead timer due in 00:00:35
    Retransmission queue length 0
*********************************
Command : show ospfv3-af ipv6 route all-vrfs
*********************************
Address-family : IPv6 Unicast
-----------------------------
```

```
Codes: i - Intra-area route, I - Inter-area route
       E1 - External type-1, E2 - External type-2
OSPFv3 AF Process ID 1 VRF default, Routing Table
-----------------------------------------------
Total Number of OSPFv3 Routes : 2
111::/64           (i) area:0.0.0.0
     directly attached to interface 1/1/1, cost 1 distance 110
fd00::/64          (i) area:0.0.0.1
     directly attached to interface vlan10, cost 1 distance 110
OSPFv3 AF Process ID 2 VRF red, Routing Table
-------------------------------------------
Total Number of OSPFv3 Routes : 1
222::/64           (i) area:0.0.0.1
     directly attached to interface 1/1/2, cost 1 distance 110
*********************************
Command : show ospfv3-af ipv6 all-vrfs
*********************************
Address-family : IPv6 Unicast
-----------------------------
Routing Process 1 with ID: 1.1.1.3 VRF default
-----------------------------------------------
Graceful-restart is configured
Restart Interval: 120, State: inactive
Last Graceful Restart Exit Status: none
Maximum Paths to Destination: 4
Number of external LSAs 0, checksum sum 0
Number of areas is: 2, 2 normal, 0 stub
Number of active areas is: 2, 2 normal, 0 stub
Area (0.0.0.0) (Active)
  Interfaces in this Area: 1 Active Interfaces: 1
  Passive Interfaces: 0 Loopback Interfaces: 0
  SPF calculation has run 5 times
  Number of LSAs: 5, checksum sum 138261
Area (0.0.0.1) (Active)
  Interfaces in this Area: 1 Active Interfaces: 1
  Passive Interfaces: 0 Loopback Interfaces: 0
  SPF calculation has run 5 times
  Area ranges:
    fd00::1/64, inter-area, no-advertise
  Number of LSAs: 4, checksum sum 85028
Routing Process 2 with ID: 1.1.1.7 VRF red
-------------------------------------------
Graceful-restart is configured
```

```
Restart Interval: 120, State: inactive
Last Graceful Restart Exit Status: none
Maximum Paths to Destination: 4
Number of external LSAs 0, checksum sum 0
Number of areas is: 1, 1 normal, 0 stub
Number of active areas is: 1, 1 normal, 0 stub
Area (0.0.0.1) (Active)
  Interfaces in this Area: 1 Active Interfaces: 1
  Passive Interfaces: 0 Loopback Interfaces: 0
  SPF calculation has run 4 times
  Number of LSAs: 4, checksum sum 149947
*********************************
Command : show ospfv3-af ipv6 interface all-vrfs
*********************************
Address-family : IPv6 Unicast
-----------------------------
Interface 1/1/1 is up, Line Protocol is up
-------------------------------------------
IPv6 address fe80::7272:cfff:fea7:c8fa
Process ID 1 VRF default, Area 0.0.0.0
    State Backup-dr, Status up, Network type Broadcast, cost 1
    Transit delay 1 sec, Router priority 1
    Designated Router ID: 3.3.3.3
    Backup Designated Router ID: 1.1.1.3
    Timer Intervals: Hello 10, Dead 40, Retransmit 5
    Number of Link LSAs: 2, checksum sum 60947
Interface 1/1/2 is up, Line Protocol is up
-------------------------------------------
IPv6 address fe80::7272:cfff:fea7:c8fa
Process ID 2 VRF red, Area 0.0.0.1
    State Backup-dr, Status up, Network type Broadcast, cost 1
    Transit delay 1 sec, Router priority 1
    Designated Router ID: 4.4.4.4
    Backup Designated Router ID: 1.1.1.7
    Timer Intervals: Hello 10, Dead 40, Retransmit 5
    Number of Link LSAs: 2, checksum sum 75181
*********************************
Command : show ospfv3-af ipv6 statistics interface all-vrfs
*********************************
Address-family : IPv6 Unicast
-----------------------------
OSPFv3 AF Process ID 1 VRF default, Interface 1/1/1 Statistics (cleared 0h
4m 45s ago)
```

```
=============================================================================
========
Tx Hello packets       : 28                 Rx Hello packets       : 28
Tx Hello bytes         : 1116               Rx Hello bytes         : 1116
Tx DD packets          : 3                  Rx DD packets          : 2
Tx DD bytes            : 164                Rx DD bytes            : 136
Tx LS request packets  : 1                  Rx LS request packets  : 1
Tx LS request bytes    : 64                 Rx LS request bytes    : 64
Tx LS update packets   : 4                  Rx LS update packets   : 4
Tx LS update bytes     : 368                Rx LS update bytes     : 452
Tx LS ack packets      : 3                  Rx LS ack packets      : 2
Tx LS ack bytes        : 208                Rx LS ack bytes        : 152
Total IPsec packets processed : 10
Total IPsec bytes processed   : 6400
Total Number of State Changes : 3
Number of LSAs                : 2
LSA Checksum Sum              : 60947
Total OSPFv3-AF Packets Discarded: 0
----------------------------------
Reason                           Packets Dropped
------------------------------------------------
Invalid Type                     0
Invalid Length                   0
Invalid Version                  0
Bad or Unknown Source            0
Area Mismatch                    0
Self-originated                  0
Duplicate Router ID              0
Interface Standby                0
Total Hello Packets Dropped      0
  Hello Interval Mismatch        0
  Dead Interval Mismatch         0
  Options Mismatch               0
  MTU Mismatch                   0
  Neighbor Ignored               0
Resource Failures                0
Bad LSA Length                   0
Others                           0
IPsec Authentication Errors      0
Total LSAs Ignored : 0
-----------------------
Bad Type            : 0
Bad Length          : 0
```

```
Invalid Data       : 0
Invalid Checksum   : 0
OSPFv3 AF Process ID 2 VRF red, Interface 1/1/2 Statistics (cleared 0h 4m
45s ago)
============================================================================
====
Tx Hello packets      : 28                Rx Hello packets      : 28
Tx Hello bytes        : 1116              Rx Hello bytes        : 1116
Tx DD packets         : 3                 Rx DD packets         : 2
Tx DD bytes           : 144               Rx DD bytes           : 136
Tx LS request packets : 1                 Rx LS request packets : 1
Tx LS request bytes   : 64                Rx LS request bytes   : 52
Tx LS update packets  : 4                 Rx LS update packets  : 4
Tx LS update bytes    : 332               Rx LS update bytes    : 452
Tx LS ack packets     : 3                 Rx LS ack packets     : 2
Tx LS ack bytes       : 208               Rx LS ack bytes       : 132
Total IPsec packets processed : 10
Total IPsec bytes processed   : 6400
Total Number of State Changes : 3
Number of LSAs                : 2
LSA Checksum Sum              : 75181
Total OSPFv3-AF Packets Discarded: 0
----------------------------------
Reason                           Packets Dropped
------------------------------------------------
Invalid Type                     0
Invalid Length                   0
Invalid Version                  0
Bad or Unknown Source            0
Area Mismatch                    0
Self-originated                  0
Duplicate Router ID              0
Interface Standby                0
Total Hello Packets Dropped      0
  Hello Interval Mismatch        0
  Dead Interval Mismatch         0
  Options Mismatch               0
  MTU Mismatch                   0
  Neighbor Ignored               0
Resource Failures                0
Bad LSA Length                   0
Others                           0
IPsec Authentication Errors      0
```

```
Total LSAs Ignored : 0
-----------------------
Bad Type            : 0
Bad Length          : 0
Invalid Data        : 0
Invalid Checksum    : 0
*********************************
Command : show ospfv3-af ipv6 virtual-links all-vrfs
*********************************
Address-family : IPv6 Unicast
-----------------------------
Virtual link to router 40.40.40.40 is up
-----------------------------------------
Process ID 21 VRF default, Transit area 0.0.0.1
Transit delay 1 sec
Timer Intervals: hello 10, dead 40, retransmit 5
No authentication
Number of Link LSAs: 0, checksum sum 0
4 state changes
====================================================
[End] Feature ospfv3-af
====================================================
====================================================
Show Tech commands executed successfully
====================================================
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manage r (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

summary-address

Syntax

```
summary-address  <PREFIX/LENGHT> [no-advertise | tag <TAG-VALUE>]
no summary-address <PREFIX/LENGHT> [no-advertise | tag <TAG-VALUE>]
```

Description

Summarizes the external routes with the matching address and mask. When advertising this route, its metric is set to the lowest cost path from among the routes that were summarized.

The **no** form of this command disables route summarization.

> **NOTE**
> This command only works for an ASBR (Autonomous System Boundary Router).

Parameter	Description
<PREFIX/LENGHT>	Specifies the IP prefix/mask for advertising the aggregate route.
`no-advertise`	Do not advertise the aggregate route. Suppress routes that m atch the specified prefix/mask pair.
`tag` *<TAG-VALUE>*	Specify the tag for the aggregate route. The summary prefix wi ll be advertised along with the tag value in External LSAs. Rang e: 0 to 4294967295

Examples

Setting OSPFv3 AF route summarization for IPv4:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-ipv4-uc)# summary-address 10.1.0.0/16
switch(config-ospfv3-ipv4-uc)# summary-address 10.1.0.0/16 no-advertise
switch(config-ospfv3-ipv4-uc)# summary-address 10.1.0.0/16 tag 10
```

Disabling route summarization for IPv4:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-ipv4-uc)# no summary-address 10.1.0.0/16
switch(config-ospfv3-ipv4-uc)# no summary-address 10.1.0.0/16 no-advertise
```

```
switch(config-ospfv3-ipv4-uc)# no summary-address 10.1.0.0/16 tag 10
```

Setting OSPFv3 AF route summarization for IPv6:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-ipv6-uc)# summary-address fd00::1/64
switch(config-ospfv3-ipv6-uc)# summary-address fd00::1/64 no-advertise
switch(config-ospfv3-ipv6-uc)# summary-address fd00::1/64 tag 10
```

Disabling route summarization for IPv6:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-ipv6-uc)# no summary-address fd00::1/64
switch(config-ospfv3-ipv6-uc)# no summary-address fd00::1/64 no-advertise
switch(config-ospfv3-ipv6-uc)# no summary-address fd00::1/64 tag 10
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-ipv4-uc<PROCESS-ID>` `config-ospfv3-ipv6-uc<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

timers lsa-arrival

Syntax

```
timers lsa-arrival <DELAY>
no timers lsa-arrival
```

Description

Configures the minimum delay between receiving the same LSA from a peer. The same LSA is an LSA that contains the same LSA ID number, LSA type, and advertising router ID. If an instance of the same LSA arrives before the delay expires, the LSA is dropped. Generally, the LSA arrival timer should be set to a value less than or equal to the start-time value for the command **timers throttle lsa** on the neighbor.

The **no** form of this command sets the LSA timers to default values.

Parameter	Description
<DELAY>	Specifies the delay in milliseconds. Range: 0 to 600000. Default : 1000.

Examples

Setting the LSA arrival timer:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# timers lsa-arrival 10
```

Setting the LSA arrival timer to default:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# no timers lsa-arrival
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420	`config-ospfv3-af-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

Platforms	Command context	Authority
6200		

timers throttle lsa

Syntax

```
timers throttle lsa start-time <START-TIME> hold-time <HOLD-TIME> max-wait-
time <WAIT-TIME>
no timers throttle lsa
```

Description

Configures the timers for LSA generation.

The **no** form of this command sets the LSA timers to default values.

Parameter	Description
`start-time <START-TIME>`	Specifies the initial wait time in milliseconds after which LSAs a re generated. When set to 0, the LSAs are generated without a ny delay. Range: 0 to 600000. Default: 5000.
`hold-time <HOLD-TIME>`	Specifies the amount of time, in milliseconds, between regene ration of an LSA. The hold time doubles each time the same L SA must be regenerated, until **max - wait - time** is reached. W hen set to 0, LSA regeneration time is not increased. Range: 0 to 600000. Default: 0.
`max-wait-time <WAIT-TIME>`	Specifies the maximum wait time, in milliseconds, for regenerati on of the same LSA. When set to 0, LSA regeneration time is n ot increased. Range: 0 to 600000. Default: 0.

Examples

Setting the LSA timers:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# timers throttle lsa start-time 100 hold-time
1000 max-wait-time 10000
```

Setting LSA timers to default values:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# no timers throttle lsa
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-af-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`timers throttle spf`

Syntax

```
timers throttle spf start-time <milliseconds> hold-time <milliseconds>
max-wait-time <milliseconds>
no timers throttle spf
```

Description

Configures timers for SPF calculation. There are three timers:

- **start-time**: Is the initial delay before an SPF calculation is started. Default is 200 milliseconds.
- **hold-time**: Is the progressive backoff time to wait before next scheduled SPF calculation. Default is 1000 milliseconds. If a route change event occurs during this period, the value doubles until it reaches the max-wait-time.
- **max-wait-time:** Is the maximum time to wait before the next scheduled SPF calculation. Default is 5000 milliseconds. This is used to limit the SPF hold timer and also defines the time to be considered for which the OSPF LSDB has to be stable, after which the SPF throttle mechanism is reset.

The **no** form of this command sets all the configured non-default timers to default value.

Parameter	Description
`<START-TIME>`	Time in milliseconds to set timer for initial SPF delay. Default: 200. Range: 1 - 600000 Required
`<HOLD-TINME>`	Time in milliseconds to set the minimum hold time between two consecutive SPF calculations. Default: 1000. Range: 1 - 600000 Required
`<WAIT-TIME>`	Time in milliseconds to set the maximum wait time between tw o consecutive SPF calculations. Default: 5000. Range: 1 - 600000 Required

Examples

Setting non-default timer values for SPF throttling:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# timers throttle spf start-time 100 hold-time
1000 max-wait-time 10000
```

Setting default timer values for SPF throttling after configuring non-default values:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# no throttle spf
switch(config-ospfv3-af-1)# no timers throttle spf start-time 100 hold-time
1000 max-wait-time 10000
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-ospfv3-af-<PROCESS-ID>`	Administrators or local user group members with execution righ ts for this command.

`transit-delay`

Syntax

```
transit-delay <seconds>
no transit-delay
```

Description

Sets the time delay in Link state transmission for virtual links.

The **no** form of this command sets the delay in Link state transmission to the default of 1 second for virtual links.

Parameter	Description
`<seconds>`	Specifies the time delay for the transit delay, in seconds. Range: 1 to 3600. Default: 1.

Examples

Setting OSPFv3 AF virtual links transit delay:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
switch(config-ospfv3-ipv6-uc)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# transit-delay 30
```

Setting OSPFv3 AF virtual links transit delay to default:

```
switch(config)# router ospfv3-af 1
switch(config-ospfv3-af-1)# address-family ipv6 unicast
```

```
switch(config-ospfv3-ipv6-uc)# area 100 virtual-link  100.0.1.1
switch(config-router-vlink6)# no transit-delay
```

Command History

Release	Modification
10.15	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config-router-vlink`	Administrators or local user group members with execution righ ts for this command.

Equal Cost Multipath (ECMP)

> **NOTE**
> ECMP is not supported on the HPE Aruba Networking 4100i Switch Series.

> **NOTE**
> ECMP is not supported on the HPE Aruba Networking 6000 and 6100 Switch Series.

Equal Cost Multipath (ECMP) allows a router to load balance traffic destined to some network that is reachable through multiple equal cost route nexthops. ECMP functionality is dependent on L3 routing being enabled on the router.

Subtopics

Overview

The nexthop chosen out of the ECMP group of nexthops for a given packet is typically determined by its header information, usually by hashing source and destination IP addresses as well as potentially source and destination L4 ports. The hashing algorithm used is dependent on the type of switch in use, and user configuration as well as whether the packet is flowing through the slow-path or fast-path.

By default all of the following fields are enabled and utilized for the load balancing decision.

- L3 Destination IP address
- L3 Source IP address
- L4 Destination port
- L4 Source port

ECMP support is NOT dependent on address family. The ECMP feature applies to both IPv4 and IPv6 traffic flowing through the router.

> **NOTE**
> If two different routing protocols are (mis)configured to use the same Administrative Distance, then they have the opportunity to create ECMP groups together.

ECMP commands

Select a command from the list in the left navigation menu.

Subtopics

`show ip ecmp`

Syntax

```
show ip ecmp
```

Description

Displays the Equal Cost Multipath (ECMP) configuration.

Examples

Showing output for HPE Aruba Networking 5420, 6200, 6300,6400, 8320,8325 and 8360 Switch series:

```
switch# show ip ecmp
ECMP Configuration
---------------------
 ECMP Status         : Enabled
ECMP Load Balancing by
------------------------
 Source IP           : Enabled
 Destination IP      : Enabled
 Source Port         : Enabled
 Destination Port    : Enabled
```

```
switch# show ip ecmp
ECMP Configuration
---------------------
ECMP Status         : Enabled
Dynamic Load Balancing Mode
----------------------
 Mode                : Flow-based
 Inactivity-timeout : 256 us
ECMP Load Balancing by
------------------------
 Source IP           : Enabled
 Destination IP      : Enabled
 Source Port         : Enabled
 Destination Port    : Enabled
```

```
switch# show ip ecmp
ECMP Configuration
---------------------
ECMP Status         : Enabled
Dynamic Load Balancing Mode
----------------------
Mode                : Packet-based
ECMP Load Balancing by
------------------------
 Source IP           : Enabled
 Destination IP      : Enabled
 Source Port         : Enabled
 Destination Port    : Enabled
```

```
switch# show ip ecmp
ECMP Configuration
---------------------
ECMP Status         : Enabled
```

```
Dynamic Load Balancing Mode
----------------------
Mode                  : None
ECMP Load Balancing by
------------------------
Source IP             : Enabled
Destination IP        : Enabled
Source Port           : Enabled
Destination Port      : Enabled
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Virtual Router Redundancy Protocol (VRRP)

NOTE

VRRP is not supported on the HPE Aruba Networking 4100i Switch Series.

NOTE

VRRP is not supported on the HPE Aruba Networking 6000 and 6100 Switch Series.

This section describes how to configure and verify VRRP configuration and operational states.

- All VRRP configurations work in VRRP context.
- VRRP can be configured on physical ports, VLAN interfaces, and LAG interfaces.

- VRRP mandates the associated interface to be routing interface.

Subtopics

Overview

In many networks, edge devices are configured to send packets to a statically configured default router. If this router becomes unavailable, devices that use it as a first-hop router become isolated from the network.

VRRP uses dynamic failover to ensure the availability of an end node default router. The IP address used as the default route is assigned to a virtual router (VR). The VR includes:

- An Active router assigned to forward traffic designated for the virtual route
- One or more prioritized Standby routers (If a Standby is forwarding traffic for the VR, it has replaced the Owner as the Active router.)

This redundancy provides a backup for gateway IP addresses (first-hop routers). If a VR Active router becomes unavailable, the traffic it supports will be transferred to a Standby router without major delays or operator intervention. This operation can eliminate single-point-of-failure problems and provide dynamic failover (and failback) support. As long as one physical router in a VR configuration is available, IP addresses assigned to the VR are always available. Edge devices can send packets to these IP addresses without interruption.

Advantages to using VRRP include:

- Minimizing failover time and bandwidth overhead if a primary router becomes unavailable
- Minimizing service disruptions during a failover
- Providing backup for a load-balanced routing solution

- Avoiding the need to make configuration changes in the end nodes if a gateway router fails
- Eliminating the need for router discovery protocols to support failover operation

Terminology

Owner router: The Owner router is automatically configured with the highest VRRP router priority in the VR (255). It operates as the Active router for the VR unless it becomes unavailable to the network. All Virtual Router members can be configured so virtual IP is not the same as physical (or real) IP. Such virtual address can be called pure virtual IP address.

Active router: The Owner or Standby router that is the physical forwarding agent for routed traffic using the VR as a gateway. There can be only one router operating as the Active for a network. If the router configured as the Owner for a VR is available to the network, it will also be the Active. If the Owner fails or loses availability to the network, the highest-priority Standby becomes the Active.

Standby: A router configured in a VR as a Standby to the Owner/Active for the same VR. There must be a minimum of one Standby in a VR to support VRRP operation if the Owner/Active fails. Every Standby is created with a configurable priority (default: 100). This priority determines the precedence for becoming the Active of the VR if the Owner or another Standby operating as Active becomes unavailable.

When the current Active router is no longer available, the Standby router with highest priority will become the current Active. When a router with higher priority becomes available, it is switched to Active. The user can override this behavior by disabling preemption mode.

Virtual Router (VR): Consists of one Owner/Active router and one or more Standby routers that belong to the same network. The Owner is the router that owns any IP addresses associated with the VR. The VR has one virtual IP address (multiple virtual IP addresses for multinetted interfaces) that correspond to a real IP address on the Owner. A VR includes the following:

- A VR identification (VRID) configured on all VRRP routers in the same network. For a multinetted interface, the VRID is configured on all routers in the same subnet.
- The same VIP configured on each instance of the same VR.
- The status of either Owner or Standby configured on each instance of the same VR. There can be one Owner and one or more Standbys configured on each VR.
- The priority level configured on each instance of the VR. On the Owner router, the highest priority setting of 255 is automatically fixed. On Standbys, the default priority setting is 100 and configurable.
- A VR MAC address that is not configurable.

Virtual IP address (VIP): The VIP associated with a VR must be a real IP address already configured in the associated interface on the Owner router.

- If the VIP is an IPv6 address, a link-local address must be configured before adding a global IPv6 address.

- The Owner and all Standby routers belonging to the VR have this IP address configured in their VRID contexts as the VIP.
- A subnetted interface allows multiple VIPs. If there are fewer IP addresses in an interface and a user wants VRRP support on multiple subnets, configure a separate VR instance for each IP address in the interface.

VRID: The identifier for a VR configured on an interface. A VRID can be used for only one VR in any interface on a router. It can be used again for a different VR in a different interface.

VRRP operation

VRRP supports router redundancy through a prioritized election process among routers configured as members of the same virtual router (VR).

A VR includes two or more member routers configured with:

- A virtual IP address that is also configured as a real IP address on one of the routers
- A virtual router MAC address

The router that owns the IP address is configured to operate as the Owner of the VR for traffic-forwarding purposes. By default, this router has the highest VRRP priority in the VR. Any other routers in the VR have a lower priority and are configured to operate as Standbys if the Owner router becomes unavailable.

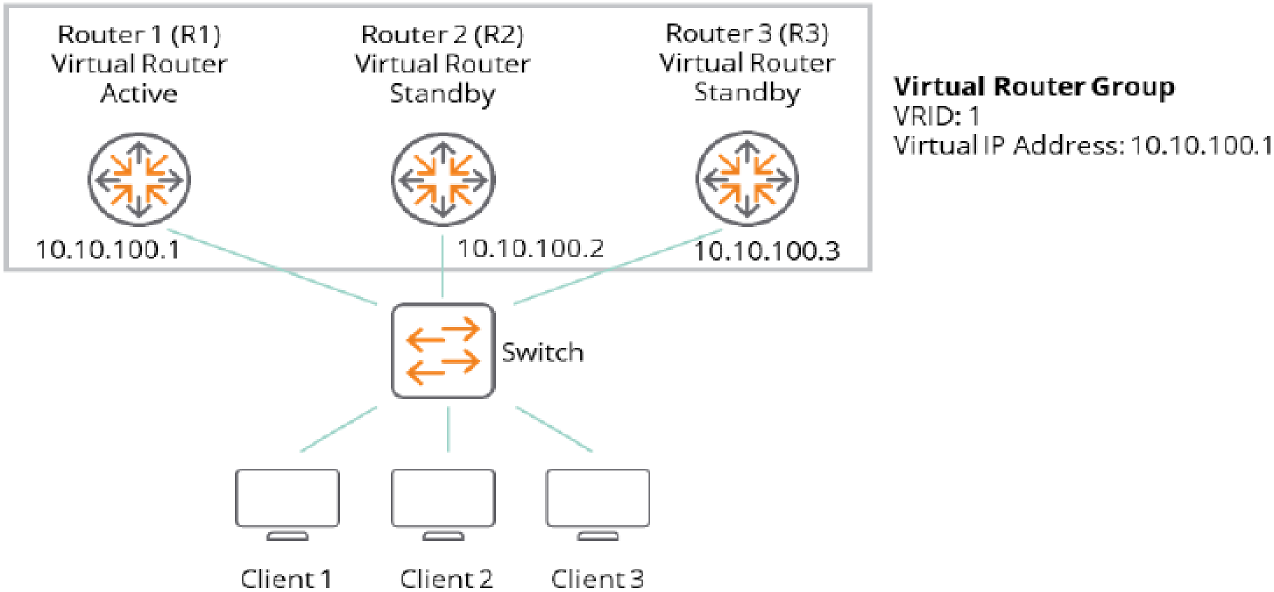
The Owner normally operates as the Active for a VR. If the Owner becomes unavailable, a failover to a Standby router belonging to the same VR occurs. This Standby becomes the Active. If the Owner recovers, a failback occurs, and Active status reverts to the Owner.

NOTE

- Using more than one Standby provides additional redundancy. If the Owner and the highest-priority Standby fail, another lower-priority Standby can take over as Active.
- In a VRRP owner scenario, duplicate address detection (DAD) should be disabled for VRRP to work.

The image shown here depicts a basic VLAN topology. In this example, Routers 1, 2 and 3 form a VRRP group. The IP address of the group is the same address configured for the SVI interface of R1 (10.10.100.1).

Figure 1. Basic VLAN Topology



Because the virtual IP address uses the IP address of the SVI interface of Router R1, Router R1 is the Active (also known as IP address Owner).

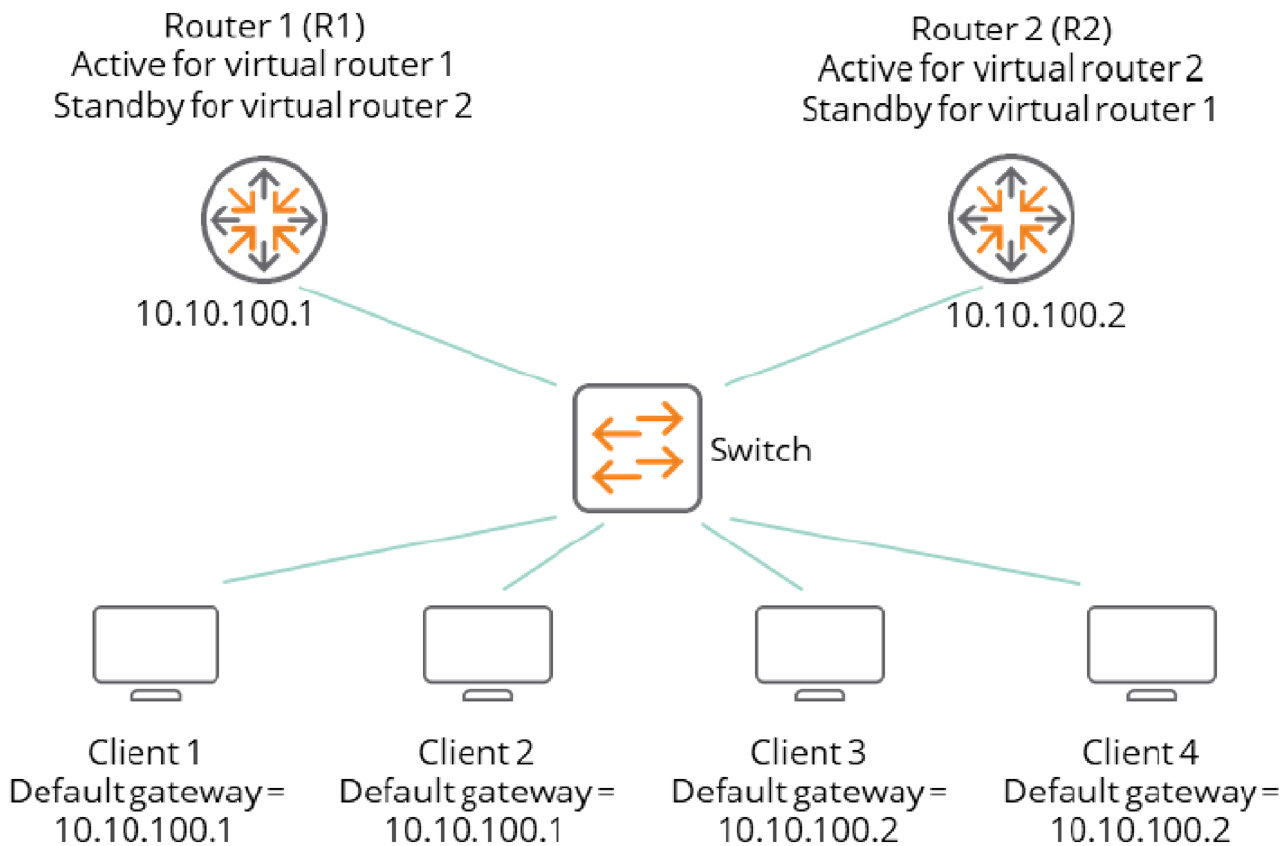
- As the Active, R1 owns the virtual IP address of the VRRP group and forwards packets sent to this IP address.
- Clients 1 through 3 are configured with the default gateway IP address of 10.10.100.1.
- Routers R2 and R3 function as Standbys. If the Active fails, the Standby router with highest priority becomes the Active and takes over the virtual IP address to provide uninterrupted service for the clients connected to the L2 Switch. When Router R1 recovers, it becomes the Active router again.

Multiple VRRP groups

A user can configure multiple VRRP groups on a router (or L3) interface (Physical, LAG, or SVI interfaces). In a topology in which multiple VRRP groups are configured on a router interface, the interface can act as an Active for one VRRP group and a Standby for one or more other VRRP groups.

This image displays a topology in which VRRP is configured so that Routers R1 and R2 share the traffic to and from clients 1 through 4. Routers R1 and R2 act as Standbys to each other if either router fails.

Figure 1. Routers share the traffic to and from clients 1 through 4.



This topology contains two virtual IP addresses for two VRRP groups that overlap. For VRRP group 1, Router R1 is the owner of IP address 10.10.100.1 and is the Active. Router R2 is the Standby to Router R1. Clients 1 and 2 are configured with the default gateway IP address of 10.10.100.1.

For VRRP group 2, Router R2 is the owner of IP address 10.10.100.2 and is the Active. Router R1 is the Standby to Router R2. Clients 3 and 4 are configured with the default gateway IP address of 10.0.0.2.

VRRP priority

In a Standby router VR configuration, the virtual router priority defaults to 100. The priority for the configured Owner is automatically set to the highest value of 255.

In a VR in which there are two or more Standby routers, priority settings can be reconfigured to define the order in which Standbys will be reassigned as Active if a failover occurs.

VRRP preemption

In a Standby router VR configuration, the virtual router priority defaults to 100. The priority for the configured Owner is automatically set to the highest value of 255.

When multiple Standby routers exist in a VR:

- If the current Active fails and the highest-priority Standby is not available, VRRP selects the next-highest priority Standby to operate as Active.
- If the highest-priority Standby later becomes available, it preempts the lower-priority Standby and takes over the Active function.

To remove this preemptive ability on a VR, disable it with the `no preempt` command.

> **NOTE**
> Preemption applies only to VRRP routers configured as Standbys.

Virtual Router MAC address

When a Virtual Router (VR) instance is configured, the protocol automatically assigns a MAC address based on the standard MAC prefix for VRRP packets plus the VRID number (as described in RFC 3768). The first five octets form the standard MAC prefix for VRRP and the last octet is the configured VRID. For example: `00-00-5E-00-01-<VRid>`

VRRP and ARP

The current Active for a VR responds to ARP requests for the virtual IP addresses with the VR-assigned MAC address. The virtual MAC address is also used as source MAC address for periodic advertisements sent by the current Active.

The VRRP router responds to ARP requests for non-virtual IP addresses with the system MAC address. Non-virtual IP addresses are not configured as virtual IP addresses for any VR on the interface.

VRRP tracking

VRRP supports object tracking. This functionality allows monitoring of the state of a configured object. The state determines the priority of the VRRP router in a VRRP group. A track object can be created using the `track <OBJECT-ID>` command and the object can be associated with an interface. A change in interface state will then affect the priority of a VRRP group.

High availability

VRRP supports high availability through stateful restarts and stateful switchovers.

- A stateful restart occurs when the VRRP process (or daemon) fails and is restarted.
- A stateful switchover occurs when the active management module (AMM) switches to the standby management module (SMM).

VRRP and Neighbor Discovery for IPv6

Neighbor Discovery (ND) is the IPv6 equivalent of the IPv4 ARP for layer 2 address resolution. It uses IPv6 ICMP messages to do the following:

- Determine the link-layer address of neighbors
- Verify that a neighbor is reachable
- Track neighbor (local) routers

Neighbor Discovery enables functions such as:

- Router and neighbor solicitations and discovery
- Detecting address changes for devices
- Identifying a replacement for a router or router path that has become unavailable
- Duplicate address detection (DAD)
- Router Advertisement processing
- Neighbor reachability
- Resolution of destination addresses
- Changes to link-layer addresses

An instance of Neighbor Discovery is triggered on a device when a new or changed IPv6 address is detected. VRRPv3 provides a faster failover to a Standby router by not using standard ND procedures. A failover to a Standby router can occur in approximately three seconds without any interaction with hosts and a minimum of VRRPv3 traffic.

Duplicate address detection (DAD)

Duplicate address detection verifies that a configured unicast IPv6 address is unique before it is assigned to an interface. When the Owner router fails, the Standby VRRP router assumes the Active role. When the Owner router becomes operational, DAD will fail because a Standby VRRP router is in the Active role that responds to the DAD request. To avoid this, on virtual routers that are in Owner mode (priority = 255) DAD needs to be disabled for the interface on which the Owner VR is configured.

Guidelines and limitations

- VRRP must be enabled at the global level using the `router vrrp enable` command. It must also be enabled at the interface level at which you configure VRRP.
- A maximum of 8 IPv4 and 8 IPv6 VRRP groups are supported on an interface.
- A maximum of 256 VRRP groups is supported on a router. The groups can be IPv4 or IPv6.
- Supported virtual router identification (VRID) range is 1-255.
- Proxy-ARP and Active-Gateway are not supported in conjunction with VRRP.

VRRP commands

Select a command from the list in the left navigation menu.

Subtopics

`address`

Syntax

```
address <IP-ADDR> [ primary | secondary ]
no address <IP-ADDR> [ primary | secondary ]
```

Description

Configures a primary or secondary IPv4 or IPv6 address for the VRRP group. To use secondary IP addresses in a VRRP group, you must first configure a primary IP address on the same group. A maximum of 16 IP addresses per IPv4 VRRP group and 8 IPv6 addresses per IPv6 VRRP group are supported.

> **NOTE**
> Do not configure an IPv4 VRRP group using addresses from the /30, /31, and /32 subnets of the interface IP address.

16 Virtual IP addresses per IPv4 VR and 8 Virtual IP addresses per IPv6 VR are supported.

> **NOTE**
> The total number of VIPs supported by a switch is:
> - 1024 VIPs for IPv4 VRs
> - 512 VIPs for IPv6 VRs

The **no** form of this command deletes a primary or secondary IPv4 or IPv6 address from the VRRP group.

Parameter	Description
`<IP-ADDR>`	Configures the IPv4 or IPv6 address.
`primary`	Configures a primary address.
`secondary`	Configures a secondary address.

Examples

```
switch(config)# system internal-vlan-range 4041-4050
switch(config)# router vrrp enable
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# address 10.0.0.1 primary
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv6
switch(config-if-vrrp)# address fe80::1 primary
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# no address 10.0.0.1 primary
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ipv6
switch(config-if-vrrp)# no address fe80::1 primary
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`authentication`

Syntax

```
authentication {text | md5} [{plaintext | ciphertext} <KEY>]
no authentication
```

Description

This command enables authentication mode and the authentication key for VRRP groups on an interface. VRRP members or routers of the same VRRP group must use the same authentication mode and authentication key.

The no form of this command disables authentication mode and the authentication key for VRRP groups on an interface.

IPv4 VRRPv3 and IPv6 VRRPv3 do not support VRRP packet authentication. Authentication mode and key configuration take effect only in VRRPv2 (IPv4 only - RFC2338).

In VRRPv3, authentication mode and authentication key settings do not take effect because VRRP Authentication was removed from RFC5798.

VRRP provides the following authentication modes as described in RFC2338:

Simple authentication

The sender fills an authentication key into the VRRP packet and the receiver compares the received authentication key with its local authentication key.

If the two authentication keys match, the received VRRP packet is legitimate. Otherwise, the received packet is illegitimate and is discarded.

> **NOTE**
> Authentication key text is sent in the clear and can be seen in a packet trace. This makes MD5 authentication more secure than text.

MD5 authentication

The sender computes a digest for the packet that will be sent using the authentication key and MD5 algorithm, and saves the result in the VRRP packet.

The receiver performs the same operation with the authentication key and MD5 algorithm, and compares the result with the content in the authentication header. If the results match, the received VRRP packet is legitimate. Otherwise, the received packet is illegitimate and is discarded.

Parameter	Description
`text`	Configures the simple authentication type.
`md5`	Configures the MD5 (message - digest) authentication type.
`plaintext`	Specifies that the key is provided as plaintext.
`ciphertext`	Specifies that the key is provided as ciphertext.
	Specifies the key in the chosen format.

Parameter	Description
<KEY>	

NOTE

When the key is not provided on the command line, plaintext key prompting occurs upon pressing Enter. The entered key characters are masked with asterisks.

Examples

Enabling VRRP authentication using MD5 with a provided plaintext key:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# version 2
switch(config-if-vrrp)# authentication md5 plaintext testvrrpkey
```

Enabling VRRP authentication using MD5 with a prompted plaintext key:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# version 2
switch(config-if-vrrp)# authentication md5
Enter the authentication key: *************
Re-Enter the authentication key: *************
```

Enabling VRRP authentication using MD5 with a provided ciphertext key:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# version 2
switch(config-if-vrrp)# authentication md5 ciphertext AQBapfciFZ/
P...biBAAAAOjc0a8=
```

Disabling VRRP authentication:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# version 2
switch(config-if-vrrp)# no authentication
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if-vrrp`	Administrators or local user group members with execution righ ts for this command.

`preempt`

Syntax

```
preempt
no preempt
```

Description

Enables the preempt option. The default value is enabled. In default mode, a Standby router with a higher priority than another Standby that is operating as Active will take over the Active function.

Applies to VRRP Standby routers only and is used to minimize network disruption caused by unnecessary preemption of the Active operation among Standby routers.

The **no** form of this command disables the preempt option, thus preventing the higher-priority Standby from taking over the Active operation from a lower-priority Standby. This command does not prevent an owner router from resuming the Active function after recovering from being unavailable.

Examples

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# preempt
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# no preempt
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`preempt delay minimum`

Syntax

```
preempt delay minimum <DELAY-IN-SECONDS>
no preempt delay minimum <DELAY-IN-SECONDS>
```

Description

Sets the time in seconds (1-3600) that the router will wait before taking control of the virtual IP and starting to route packets.

The **no** form of this command sets the preempt delay for the VRRP group to the default preempt delay of 0 seconds.

The VRRP Preempt Delay Timer (PDT) allows admin users to configure a period of time before the VR takes control of the virtual IP address. It does not transition to the Active state until the timer period expires.

The timer value configured should be long enough to allow upper layer protocol to converge. The PDT is applied during initialization and down/up events of the router.

Parameter	Description
<DELAY-IN-SECONDS>	Selects the time in seconds (1 - 3600). Default is 0 seconds.

Examples

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# preempt delay minimum 30
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# no preempt delay
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`priority`

Syntax

```
priority <1-254>
no priority
```

Description

Sets the priority for the VRRP group.

The **no** form of this command sets the priority for the VRRP group as default priority.

- The default value for non-Owner virtual routers is 100.
- The default value for Owner virtual router is 255, which cannot be changed.

Examples

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# priority 150
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# no priority
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`router vrrp {enable | disable}`

Syntax

```
router vrrp {enable | disable}
no router vrrp {enable | disable}
```

Description

Enables or disables VRRP protocol globally. You must globally enable the VRRP feature for VRRP virtual router.

no router vrrp enable disables VRRP protocol globally but does not remove all VRRP configurations.

no router vrrp disable enables VRRP protocol globally.

Example

Enabling VRRP protocol globally:

```
switch(config)# router vrrp enable
```

Disable VRRP protocol globally:

```
switch(config)# router vrrp disable
```

Disable VRRP protocol globally:

```
switch(config)# no router vrrp enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`no router vrrp`

Syntax

```
no router vrrp
```

Description

Removes VRRP configuration and VRRP global protocol. If **auto-confirm** is enabled or VRRP is not configured on any interface, this command will not ask for user confirmation.

Examples

Removing VRRP configuration:

```
switch(config)# no router vrrp
All VRRP configuration will be deleted.
Do you want to continue (y/n)?
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`show track`

Syntax

```
show track [brief | <OBJECT-ID>]
```

Description

Shows all or specific track object information.

Parameter	Description
`brief`	Displays brief information about all or specific track objects
`<OBJECT-ID>`	Displays information about a specified track object (1 - 128)

Examples

```
switch# show track
Track 1
  interface 1/1/1
  Interface is DOWN
switch# show track brief
Track  Interface        State
1      1/1/1            Down
2      None             Down
3      1/1/2            Up
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show track brief`

Syntax

```
show track brief
```

Description

Shows brief information for all track objects.

Examples

Showing brief information for all track objects:

```
switch# show track brief
Track  Interface        State
1      1/1/1            Down
2      None             Down
3      1/1/2            Up
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show vrrp`

Syntax

```
show vrrp [brief | detail | interface <INTERFACE-NAME> | interface <LAG-
NAME> | interface <VLAN-NAME> | ip | ipv6 | statistics | statistics
interface <INTERFACE-NAME> | statistics interface <LAG-NAME> | statistics
interface <VLAN-NAME>]
```

Description

Shows all VRRP virtual routers information.

Parameter	Description
`brief`	Displays brief output of all VRRP virtual routers Keywords used in displayed information: • Grp: VRRP virtual router group ID. • A - F: Address Family. • Pri: Priority. • Time: Uptime of VRRP virtual router since it moved out of I NIT state. • Pre: Preempt mode (Y is enabled, N if not enabled).
`detail`	Displays detailed output of all VRRP virtual routers
`interface <INTERFACE-NAME>`	Displays VRRP information for a specific interface

Parameter	Description
`interface <LAG-NAME>`	Displays VRRP information for a specific LAG interface
`interface <VLAN-NAME>`	Displays VRRP information for a specific VLAN interface
`ip`	Displays the IP address family
`ipv6`	Displays IPv6 address family
`statistics`	Displays VRRP statistics information for all interfaces
`statistics interface <INTERFACE-NAME>`	Displays VRRP statistics information for a specific interface
`statistics interface <LAG-NAME>`	Displays VRRP statistics information for a specific LAG interfac e
`statistics interface <VLAN-NAME>`	Displays VRRP statistics information for a specific VLAN interfa ce

Examples

```
switch# show vrrp
VRRP is enabled
Interface 1/1/1 - Group 1 - Address-Family IPv
  State is ACTIVE
  State duration 56 mins 57.826 secs
  Virtual IP address is 10.0.0.1
  Virtual MAC address is 00:00:5e:00:01:01
  Advertisement interval is 1000 msec
  Preemption enabled
  Priority is 100
  Active Router is 10.0.0.2 (local), priority is 100
  Active Advertisement interval is 1000 msec
  Active Down interval is unknown
  Tracked object ID is 1, and state Down
Interface 1/1/2 - Group 1 - Address-Family IPv4
  State is INIT (Interface Down)
  State duration 45 mins 28.313 secs
  Virtual IP address is no address
  Virtual MAC address is 00:00:5e:00:01:01
```

```
  Advertisement interval is 1000 msec
  Preemption enabled
  Priority is 100
  Active Router is unknown, priority is unknown
  Active Advertisement interval is unknown
  Active Down interval is unknown
Interface 1/1/2 - Group 1 - Address-Family IPv6
  State is INIT (Group Disabled)
  State duration 20 mins 19.794 secs
  Virtual IP address is no address
  Virtual secondary IP addresses:
    2201:13::110:4
  Virtual MAC address is 00:00:5e:00:02:01
  Advertisement interval is 1000 msec
  Preemption enabled
  Priority is 100
  Active Router is unknown, priority is unknown
  Active Advertisement interval is unknown
  Active Down interval is unknown
switch# show vrrp brief
VRRP is enabled
 Interface   Grp  A-F   Pri   Time Owner Pre State   Active addr/Group addr
  1/1/1       1   IPv4  100    0    N    Y   ACTIVE  10.0.0.2(local)
10.0.0.1
  1/1/2       1   IPv4  100    0    N    Y   INIT    AF-UNDEFINED no address
  1/1/2       1   IPv6  100    0    N    Y   INIT    AF-UNDEFINED no address
switch# show vrrp detail
VRRP is enabled
Interface 1/1/1 - VRRPv2 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0
Interface 1/1/1 - VRRPv3 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0
Interface 1/1/1 - Group 1 - Address-Family IPv4
  State is ACTIVE
  State duration 1 mins 35.486 secs
  Virtual IP address is 10.0.0.1
  Virtual MAC address is 00:00:5e:00:01:01
  Advertisement interval is 1000 msec
  Version 3
```

```
  Preemption enabled
  Priority is 100
  Active Router is 10.0.0.2 (local), priority is 100
  Active Advertisement interval is 1000 msec
  Active Down interval is unknown
  Tracked object ID is 1, and state Down
  VRRPv3 Advertisements: sent 3931 (errors 0) - rcvd 0
  VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
  Group Discarded Packets: 3537
    IP address Owner conflicts: 0
    IP address configuration mismatch : 3537
    Advert Interval errors : 0
    Adverts received in Init state: 0
    Invalid group other reason: 0
  Group State transition:
    Init to active: 0
    Init to standby: 2 (Last change Mon Jun 16 11:19:36.316 UTC)
    Standby to active: 2 (Last change Mon Jun 16 11:19:39.926 UTC)
    Active to standby: 0
    Active to init: 1 (Last change Mon Jun 16 11:17:49.978 UTC)
    Standby to init: 0
Interface 1/1/2 - VRRPv2 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0
Interface 1/1/2 - VRRPv3 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0
Interface 1/1/2 - Group 1 - Address-Family IPv4
  State is INIT (Interface Down)
  State duration 49 mins 23.507 secs
  Virtual IP address is no address
  Virtual MAC address is 00:00:5e:00:01:01
  Advertisement interval is 1000 msec
  Version 3
  Preemption enabled
  Priority is 100
  Active Router is unknown, priority is unknown
  Active Advertisement interval is unknown
  Active Down interval is unknown
  VRRPv3 Advertisements: sent 0 (errors 0) - rcvd 0
  VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
```

```
  Group Discarded Packets: 0
    IP address Owner conflicts: 0
    IP address configuration mismatch : 0
    Advert Interval errors: 0
    Adverts received in Init state: 0
    Invalid group other reason: 0
  Group State transition:
    Init to active: 0
    Init to standby: 0
    Standby to active: 0
    Active to standby: 0
    Active to init: 0
    Standby to init: 0
Interface 1/1/2 - Group 1 - Address-Family IPv6
  State is INIT (Interface Down)
  State duration 24 mins 14.988 secs
  Virtual IP address is no address
  Virtual secondary IP addresses:
    2201:13::110:4
  Virtual MAC address is 00:00:5e:00:02:01
  Advertisement interval is 1000 msec
  Preemption enabled
  Priority is 100
  Active Router is unknown, priority is unknown
  Active Advertisement interval is unknown
  Active Down interval is unknown
  VRRPv3 Advertisements: sent 0 (errors 0) - rcvd 0
  VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
  Group Discarded Packets: 0
    VRRPv2 incompatibility: 0
    IP address Owner conflicts: 0
    IP address configuration mismatch : 0
    Advert Interval errors : 0
    Adverts received in Init state: 0
    Invalid group other reason: 0
  Group State transition:
    Init to active: 0
    Init to standby: 0
    Standby to active: 0
    Active to standby: 0
    Active to init: 0
    Standby to init: 0
switch# show vrrp interface 1/1/1
```

```
VRRP is enabled
Interface 1/1/1 - Group 1 - Address-Family IPv4
  State is ACTIVE
  State duration 11 mins 21.617 secs
  Virtual IP address is 10.0.0.1
  Virtual MAC address is 00:00:5e:00:01:01
  Advertisement interval is 1000 msec
  Version 3
  Preemption enabled
  Priority is 100
  Active Router is 10.0.0.2 (local), priority is 100
  Active Advertisement interval is 1000 msec
  Active Down interval is unknown
switch# show vrrp interface lag10
VRRP is enabled
Interface lag10 - Group 1 - Address-Family IPv4
  State is ACTIVE
  State duration 11 mins 21.617 secs
  Virtual IP address is 10.0.0.1
  Virtual MAC address is 00:00:5e:00:01:01
  Advertisement interval is 1000 msec
  Version 3
  Preemption enabled
  Priority is 100
  Active Router is 10.0.0.2 (local), priority is 100
  Active Advertisement interval is 1000 msec
  Active Down interval is unknown
switch# show vrrp interface vlan100
VRRP is enabled
Interface vlan100 - Group 1 - Address-Family IPv4
  State is ACTIVE
  State duration 11 mins 21.617 secs
  Virtual IP address is 10.0.0.1
  Virtual MAC address is 00:00:5e:00:01:01
  Advertisement interval is 1000 msec
  Version 3
  Preemption enabled
  Priority is 100
  Active Router is 10.0.0.2 (local), priority is 100
  Active Advertisement interval is 1000 msec
  Active Down interval is unknown
switch# show vrrp statistics
VRRP is enabled
```

```
Interface 1/1/1 - VRRPv2 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0
Interface 1/1/1 - VRRPv3 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0
VRRP Statistics for interface 1/1/1 - Group 1 - Address-Family IPv4
  State is ACTIVE
  State duration 6 mins 55.006 secs
  VRRPv3 Advertisements: sent 4288 (errors 0) - rcvd 0
  VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
  Group Discarded Packets: 3856
    IP address Owner conflicts: 0
    IP address configuration mismatch : 0
    Advert Interval errors : 0
    Adverts received in Init state: 0
    Invalid group other reason: 0
  Group State transition:
    Init to active: 0
    Init to standby: 2 (Last change Mon Jun 16 11:19:36.316 UTC)
    Standby to active: 2 (Last change Mon Jun 16 11:19:39.926 UTC)
    Active to standby: 0
    Active to init: 1 (Last change Mon Jun 16 11:17:49.978 UTC)
    Standby to init: 0
Interface 1/1/2 - VRRPv2 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0
Interface 1/1/2 - VRRPv3 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0
VRRP Statistics for Interface 1/1/2 - Group 1 - Address-Family IPv4
  State is INIT (No Primary Group Address)
  State duration 54 mins 43.027 secs
  VRRPv3 Advertisements: sent 0 (errors 0) - rcvd 0
  VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
  Group Discarded Packets: 0
    IP address Owner conflicts: 0
    Invalid address count: 0
    IP address configuration mismatch : 0
```

```
    Advert Interval errors : 0
    Adverts received in Init state: 0
    Invalid group other reason: 0
  Group State transition:
    Init to active: 0
    Init to standby: 0
    Standby to active: 0
    Active to standby: 0
    Active to init: 0
    Standby to init: 0
VRRP Statistics for Interface 1/1/2 - Group 1 - Address-Family IPv6
  State is INIT (Interface Down)
  State duration 29 mins 34.508 secs
  VRRPv3 Advertisements: sent 0 (errors 0) - rcvd 0
  VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
  Group Discarded Packets: 0
    IP address Owner conflicts: 0
    IP address configuration mismatch : 0
    Advert Interval errors: 0
    Adverts received in Init state: 0
    Invalid group other reason: 0
  Group State transition:
    Init to active: 0
    Init to standby: 0
    Standby to active: 0
    Active to standby: 0
    Active to init: 0
    Standby to init: 0
switch# show vrrp statistics interface 1/1/1
VRRP is enabled
Interface 1/1/1 - VRRPv2 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0
Interface 1/1/1 - VRRPv3 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0
VRRP Statistics for interface 1/1/1 - Group 1 - Address-Family IPv4
  State is ACTIVE
  State duration 6 mins 55.006 secs
  VRRPv3 Advertisements: sent 4288 (errors 0) - rcvd 0
  VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
```

```
  Group Discarded Packets: 3856
    IP address Owner conflicts: 0
    IP address configuration mismatch : 0
    Advert Interval errors : 0
    Adverts received in Init state: 0
    Invalid group other reason: 0
  Group State transition:
    Init to active: 0
    Init to standby: 2 (Last change Mon Jun 16 11:19:36.316 UTC)
    Standby to active: 2 (Last change Mon Jun 16 11:19:39.926 UTC)
    Active to standby: 0
    Active to init: 1 (Last change Mon Jun 16 11:17:49.978 UTC)
    Standby to init: 0
switch# show vrrp statistics interface lag10
VRRP is enabled
Interface lag10 - VRRPv2 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0
Interface lag10 - VRRPv3 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0
VRRP Statistics for interface lag10 - Group 1 - Address-Family IPv4
  State is ACTIVE
  State duration 6 mins 55.006 secs
  VRRPv3 Advertisements: sent 4288 (errors 0) - rcvd 0
  VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
  Group Discarded Packets: 3856
    IP address Owner conflicts: 0
    IP address configuration mismatch : 0
    Advert Interval errors : 0
    Adverts received in Init state: 0
    Invalid group other reason: 0
  Group State transition:
    Init to active: 0
    Init to standby: 2 (Last change Mon Jun 16 11:19:36.316 UTC)
    Standby to active: 2 (Last change Mon Jun 16 11:19:39.926 UTC)
    Active to standby: 0
    Active to init: 1 (Last change Mon Jun 16 11:17:49.978 UTC)
    Standby to init: 0
switch# show vrrp statistics interface vlan100
VRRP is enabled
```

```
Interface vlan100 - VRRPv2 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0
Interface vlan100 - VRRPv3 Statistics
   Invalid group ID packets received : 0
   Invalid version packets received : 0
   Invalid checksum packets received : 0
VRRP Statistics for interface vlan100 - Group 1 - Address-Family IPv4
  State is ACTIVE
  State duration 6 mins 55.006 secs
  VRRPv3 Advertisements: sent 4288 (errors 0) - rcvd 0
  VRRPv2 Advertisements: sent 0 (errors 0) - rcvd 0
  Group Discarded Packets: 3856
    IP address Owner conflicts: 0
    IP address configuration mismatch : 0
    Advert Interval errors : 0
    Adverts received in Init state: 0
    Invalid group other reason: 0
  Group State transition:
    Init to active: 0
    Init to standby: 2 (Last change Mon Jun 16 11:19:36.316 UTC)
    Standby to active: 2 (Last change Mon Jun 16 11:19:39.926 UTC)
    Active to standby: 0
    Active to init: 1 (Last change Mon Jun 16 11:17:49.978 UTC)
    Standby to init: 0
```

Command History

Release	Modification
10.14	Replaced the **ipv4** parameter with the **ip** parameter. The **ipv4** parameter is dep recated.
10.08	Updated command output for inclusive language
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`shutdown`

Syntax

```
shutdown
no shutdown
```

Description

Enables standby VRRP groups on the interface to route the traffic sent to virtual router's MAC address.

It is recommended to enable this mode on VLANs where mc-lags are configured.

Disabled by default.

The **no** form of this command disables standby VRRP groups on the interface to route the traffic sent to virtual router's MAC address.

> **NOTE**
> Only supported on SVI interfaces.

Examples

Enabling standby VRRP groups on interface VLAN 10 to route traffic sent to the router's MAC address:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# vrrp dual-active-forwarding
```

Disabling the ability of standby VRRP groups on interface VLAN 10 to route traffic sent to the router's MAC address:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# no vrrp dual-active-forwarding
```

Command History

Release	Modification
10.12.1000	Command introduced.

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`timers advertise`

Syntax

```
timers advertise <ADVERTISE-IN-MILLISECONDS>
no timers advertise
```

Description

Sets the advertisement interval in ms (100-40950). The default value is 1000. Advertisement interval can be configured in multiples of 1,000 ms.

The **no** form of this command sets the advertisement interval in ms to the default value of 1000.

> **NOTE**
> This release does not support sub-second timer for VRRPv3.

Examples

Setting the advertisement interval in ms to 2000:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# timers advertise 2000
```

Setting the advertisement interval in ms to the default value of 1000:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# no timers advertise
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`track (VRRP group)`

Syntax

```
track <OBJECT-ID>
no track <OBJECT-ID>
```

Description

Sets the track object ID (1-128) for the group. The track object is first configured globally for the interface and then attached to the VRRP virtual router.

> **NOTE**
> The track object must not track the same interface for which a VRRP group is configured.

The **no** form of this command removes the track object ID from the group.

Examples

Setting the track object ID for the group:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# track 1
```

Removing the track object ID from the group:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# no track 1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`track (VRRP virtual router)`

Syntax

```
track <OBJECT-ID>
no track <OBJECT-ID>
```

Description

Configures a track object that can be associated with an interface. A change in interface state will then affect the priority of a VRRP group. By default, no interface is associated to a track object, so state is down.

The **no** form of this command deletes a tracked object for an interface. If it is not associated with a VRRP virtual router, a track object cannot be deleted.

> **NOTE**
>
> Track cannot be configured by using port with no routing.
>
> When all tracked interfaces go down on a virtual router, priority is automatically set to zero instead of its configured value. Owner virtual routers always use a default priority of 255.

Parameter	Description
`<OBJECT-ID>`	Specify the track object ID value. Range: 1 to 128.

Examples

Configuring a tracked object:

```
switch(config)# track 1
```

Deleting a tracked object:

```
switch(config)# no track 1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`track by`

Syntax

```
track by <OBJECT-ID>
no track by <OBJECT-ID>
```

Description

Specifies an interface to be tracked when changes in the state of the interface affect the priority of a VRRP group. Once track is associated with an interface, the track state reflects the interface forwarding state.

The **no** form of this command removes an interface from tracking, affecting VRRP states of any interfaces associated with VRRP groups.

> **NOTE**
> The VLAN interface 1 is always tracked.

Parameter	Description
<OBJECT-ID>	Specifies the track object ID value. Range: 1 to 128.

Example

Specifying an interface to be tracked:

```
switch (config)# interface 1/1/1
switch (config-if)# track by 1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	`config`	Administrators or local user group members with execution righ ts for this command.

Platforms	Command context	Authority
6200		

`version`

Syntax

```
version <VERSION-NUMBER>
```

Description

Sets the protocol version for the VRRP group. Version change is allowed only for the IPv4 address-family. The default value is 2, which supports IPv4 with minimum 1 second advertisement interval. Value 3 supports IPv4 and IPv6 with minimum 1 second advertisement interval.

Parameter	Description
`<VERSION-NUMBER>`	Specifies the VRRP protocol version. Possible values: 2 or 3. Th e default value is 2, which supports IPv4 with a minimum 1 seco nd advertisement interval.

Example

Setting the protocol version for the VRRP group to 3:

```
switch(config)# interface 1/1/1
switch(config-if)# vrrp 1 address-family ip
switch(config-if-vrrp)# version 3
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`vrrp`

Syntax

```
vrrp <VRID> address-family {ip | ipv6}
no vrrp <VRID> address-family {ip | ipv6}
```

Description

Creates a VRRP group and establishes VRRP group configuration context.

- A maximum of 16 VRRP groups, including both IPv4 and IPv6, are supported on an interface.
- A maximum of 256 VRRP groups is supported on a router. The groups can be IPv4 or IPv6 on a first come first serve basis.

The **no** form of this command deletes a VRRP group.

Parameter	Description
`<VRID>`	Selects the VRRP router ID value. Range: 1 to 255.
`address-family [IP| IPv6]`	Specifies which address family to use, IP or IPv6.

Examples

```
switch(config)# interface 1/1/1
switch(config-if)# routing
switch(config-if-vrrp)# vrrp 1 address-family ip
switch(config-if-vrrp)#
switch(config-if)# no vrrp 1 address-family ip
```

Command History

Release	Modification
10.14	Replaced the **ipv4** parameter with the **ip** parameter. The **ipv4** parameter is dep recated.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

Policy Based Routing (PBR)

> **NOTE**
>
> PBR is not supported on the 4100i Switch Series.

> **NOTE**
>
> PBR is not supported on the 6000 and 6100 Switch Series.

Policy Based routing (PBR) lets you manipulate the path of a packet based on the various attributes of the packet. Packets that can be manipulated by PBR are packets that are already routing through the system at Layer 3, with a destination IP address that is on a network other than the packet ingress Layer 3 interface. PBR is an extension of the existing classifier policy system where traffic to be manipulated is matched by a classifier **class**, and policy **actions** to be executed on the matching traffic. Matching traffic with the same destination can be routed over different paths so that different types of traffic, such as VoIP or traffic with special security requirements, can be better managed.

> **NOTE**
>
> PBR's ability to influence a packet path is limited to the current router in that the next router to which the traffic is redirected makes an independent decision on where to forward traffic to next, if anywhere. Packets not matched by a PBR policy entry are not affected and take routes specified in the system route table.

Subtopics

PBR actions

PBR can alter a packet path through the following PBR actions:

- **interface null:** equivalent to the policy **drop** policing action. Any packets matching the class criteria for that policy entry will be dropped and not routed any further.
- **nexthop:** allows for overriding the routing table's longest prefix match next-hop router for matching packets. If no such routing table entry exists for matching packets, (default or not) this action still affects matching packets.
- **default-nexthop:** allows for specifying a next-hop router for matching packets when there is no longest prefix match for those packets in the routing table. Such a default-nexthop overrides a system default route if already configured and also applies if there is no system default route.

> **NOTE**
>
> - Next-hop and default-nexthop facilitate routing matching packets where they otherwise might not be, due to the absence of routing table entries.
> - Unlike next-hop, default-nexthop only applies if there is no destination lookup match in the main routing table for matching packets.

PBR policy action and action list

Entries in a policy use a class to specify the criteria on what packets to match and a set of one or more policy actions to take on the matching packets such as dscp or mirror.

PBR is configured for use as another such policy action in a policy entry by using the `pbr` keyword followed by the name of a user configured PBR action list. This list may contain up to eight PBR action entries which are listed according to priority by sequence number. The PBR actions section lists the different action entry choices.

In a given action list applied to an interface, zero entries may be available (that is, reachable). Of the entries that are available, only one can be active for that action list. Due to real-time changes in the network operating environment, list entry availability can change at any moment. At any given time, the highest priority available entry in the list is selected as the active PBR action for that list. If the active entry becomes unavailable, the next highest available entry is automatically selected and replaces that now-unavailable entry as the active entry. If a higher priority entry (other than the current active entry) becomes available, this entry is selected and replaces the current active entry, as the new active entry (even though that current entry is still available). If no entries are available or the only available entry becomes unavailable, routing occurs based on the routing table and is not influenced by PBR.

NOTE

The **interface null** action is always considered available. If there are no higher priority PBR actions available, it will be selected as the active entry when configured in a PBR action list, and applied to an interface.

PBR action list maximum entries

There is a limit of eight entries per action list and a maximum of 16 action lists in the system. Only one entry from an action list can be 'active' at any given moment, so the maximum number of unique active entries in the system is also 16.

For example, a given next-hop specified in an action list applied to an interface counts against the limit once. If that same next-hop address is entered in a different action list and applied to another interface in the same VRF, this next-hop is not unique in the VRF and does not affect the limit further. If one of those action lists (that specify the address) is applied to an interface in a different VRF, the entry will affect the limit since addresses are unique to each VRF.

Another example is if two PBR action lists with eight unique next-hop entries (for a total of 16 unique next-hops) are configured as PBR actions in a policy. If that policy is applied to 32 route-only ports (each of which is in a different VRF), the entry supply will be exhausted.

NOTE

- PBR action list maximum capacity is 16.

IP versions in an action list

Configuring PBR actions of different IP versions in the same PBR action list is not supported. This limitation applies to the next-hop and default-nexthop actions specifically. If an action of one IP version is already configured, configuring an action with a different IP version in the same action list will be blocked. This limitation applies regardless of whether the action list is part of a policy, applied to an interface or whether any of the entries are selected or not.

Furthermore, use cases similar to the following are not supported:

- Applying a PBR action list with entries of one IP version (for example IPv4 next-hop) to an interface of the other IP version (for example a route-only port with an IPv6 address only).
- Creating a policy entry with a class of one IP version and a PBR action list with entries of the other IP version.

Configuration mismatches while not prevented are not supported and behavior is not defined. At best, traffic will not reach its intended destination.

Specifying valid next-hop and default-nexthop addresses

PBR supports all types of directly connected hosts as nexthop or default-nexthop actions.

This means that while you may specify any IPv4 or IPv6 address as a nexthop or default-nexthop, afrer application only those reachable through a directly connected network will be installed as an active PBR nexthop. Even if a specified address is reachable from the router, if it is not on a directly connected subnet it will not be installed.

Hardware path PBR versus software path PBR

Traffic to be routed that cannot be handled by the switching ASIC (for example IPv4 packets with IP options), is handled by the switch operating system kernel routing software.

The characteristics of network usage determine the proportion of traffic handled by software path PBR relative to the hardware path. Under normal conditions, software path PBR is likely to range from a fraction of a percent of all traffic to almost none.

Packets with IP options are sent to the hardware ASIC as well as the CPU. Since the software path PBR actions are applied to the packets in the CPU, a default entry is programmed in the hardware to take no action on those packets. This hardware entry count can be seen in diagnostics with the command `diag-dump policy basic`.

Hardware versus software path for default-nexthop action

When a policy containing an entry with PBR is applied to an interface and the active action for that entry action list is 'default-nexthop', packets that match the class criteria and have no explicit route table hit (that is, no destination address longest prefix match, a.k.a 'route-miss'), they are forwarded to the specified active PBR default-nexthop. This is true for packets that follow the hardware and software paths through the switch.

There is a difference between PBR hardware and software path behavior when a route table hit occurs for class-matched packets (when the policy entry is a PBR default-nexthop).

Hardware path match criteria for a PBR policy entry with default-nexthop is extended to include the route table miss along with the class qualifiers, resulting in a **policy entry hit** for that policy entry. Conversely when there is a route table hit, the result is a policy entry miss in hardware path. When a policy entry miss occurs, policy processing moves on to the next entry in the policy and takes whatever action is specified, if any exist. This can include a different PBR routing action including interface null or no PBR action at all, for example.

In software path, the class match criteria is the only criteria required to achieve a policy entry hit. When that occurs, policy processing will stop. When there is a class match and a route table hit (with PBR action default-nexthop), the packet is forwarded according to that route table entry, not the PBR default-nexthop entry, nor is it influenced by any subsequent policy entries.

This difference in behavior is due to a limitation in the software path matching and routing engine, relative to hardware.

Table 1. default - nexthop behaviors

Routing packet	System route miss	System route hit
Hardware	Take PBR default - nexthop route.	Policy entry miss, continue policy processing with next entry, if pres ent. No further matches result in system default route being used, if present.
Software	Take PBR default - nexthop route.	Policy entry hit is overridden by system route hit. System route ent ry used, policy processing stops.

Software path and system default route

AOS-CX default behavior (for packets with destination networks that lack entries in the system route table with a reachable next-hop address), sends packets to the system CPU for special handling, if any are specified (for example, sending an ICMP unreachable reply message).

A side effect of this CPU routing is that it has higher priority than an applied PBR next-hop action. Even though a policy is applied with an entry that matches the traffic and specifies a PBR next-hop action which is reachable, the traffic will still be routed to the system CPU (due to the absence of a reachable next-hop in the system route table), and not through the desired PBR hardware path. With traffic routing to the system CPU, it will be properly routed by PBR software path, but it will also be rate limited by the control plane policing feature. Loss can occur at higher traffic rates.

CAUTION

The workaround for this issue is to create a default next-hop route in the system with a reachable next-hop router/host. This will result in a route hit, or reachable next-hop detected for the traffic with no further need to route traffic to the CPU. The PBR hardware path next-hop action will then occur, as desired.

PBR, ECMP, and routing protocols

When a policy with PBR actions is applied to an interface, the highest priority action from the action list is applied to matching traffic on that interface and overrides any static, ECMP or dynamically installed routes in the system.

PBR, VSX, and VLAN ACLs

Multichassis Link Aggregation Group (MCLAG) with VSX is a high-availability feature where a switch containing LAG members is connected to multiple switches to allow for node-level redundancy on that link. If one of the other switches goes down the LAG remains up and can continue to carry all the LAG traffic, bandwidth permitting.

A LAG (and an MCLAG) can be a member of a VLAN and a Layer 3 VSI (virtual switch interface), which can be created for that VLAN for the purposes of routing a policy with Layer 3 specific actions (that is, PBR), which can therefore be applied to that interface to influence routing decisions for matching Layer 3 packets on the MCLAG; like any other route-only port or VSI. Under such a configuration, it is likely that it is desirable to use VLAN ACLs applied to the VLAN the MCLAG is a member of.

NOTE

There is a limit with this particular combination when the VLAN ACL specifies both IPv4 and IPv6 entries and the PBR policy has entries with IPv4 and IPv6 classes. This configuration will exhaust the switching ASIC resources and will fail to apply. To achieve a similar configuration, you may use port ACLs for IPv4 traffic (applying the ACL to all ports individually in the VLAN) while preserving the rest of the configuration.

PBR software path, VSX, and VRRP

Due to the dynamic nature of the VSX and VRRP protocols, applying a policy with PBR to VSX or VRRP Layer 3 kernel interfaces and then making further changes to those protocols may not result in expected behavior. Apply policies with PBR to VSX and VRRP Layer 3 interfaces after all changes have been made. If further changes are necessary the policy should be removed first and reapplied when configuration updates are complete.

PBR and next-hop router reachability

When a policy with PBR actions is applied to an interface, all nexthop and default-nexthop entries in the associated action list are continuously monitored for reachability. A probe occurs every 5 seconds for each entry. These probes guarantee that the reaction to the loss of reachability or detection of the new reachability of any such entry will be 5 seconds or less.

For example, if there are two reachable nexthops in an applied action list and the active nexthop (highest priority by lowest sequence number) entry becomes unreachable, the switchover to the lower priority nexthop will occur within 5 seconds. If there is only one nexthop or default-nexthop entry and no entries of any other type in the action list, routing decisions will return to the system routing table within 5 seconds.

Conversely if an applied action list has a single unreachable nexthop and that nexthop at some point becomes reachable, the switch back to routing to that nexthop will occur within 5 seconds of the new reachability of the nexthop on the network. Or in another scenario an action list has multiple nexthops and the active nexthop is not the highest priority entry. When the nexthop with the higher priority becomes available, the promotion of that now reachable nexthop to active will occur within 5 seconds of it's new reachability on the network.

PBR nexthop over VXLAN L3VNI

In general PBR does not support remote (recursive) nexthops. However, PBR supports configuring remote non-connected hosts as PBR nexthops or default-nexthop actions that are learned over EVPN-distributed subnets across a routed VXLAN L3VNI tunnel. These nexthops are not guaranteed to exist, but are supported as a PBR nexthop or default-nexthop action as long as there is a resolved EVPN-BGP route that matches the PBR nexthop address. As a result, hosts learned over EVPN-distributed subnets across a routed VXLAN L3VNI tunnel will not be probed for reachability.

NOTE

For more information on VXLAN or EVPN, refer to the VXLAN Guide.

A use-case of PBR over VxLAN L3VNI is to route packets to a dedicated machine over a tunnel such as a firewall or load balancer. The incoming packet will hit a VTEP and PBR policy will send the packet into the tunnel (ingress). On the other end, the packet will exit the tunnel and PBR policy will route the packet to the correct nexthop (egress). It is important to note that

NOTE

In order to configure the end-to-end solution, a policy needs to be configured on both VTEPs, ingress and egress.

Ingress policy

In order to route traffic to a nexthop learned over VXLAN L3VNI, a policy needs to be applied to the ingress VTEP's L3 interface associated with the arriving traffic that needs to be redirected.

In the following example the policy is applied to **vlan 20**, (where **nexthop 100.1.1.4** is learned over VxLAN L3VNI):

```
class ip class1
    10 match any any any
pbr-action-list pbr1
    10 nexthop 100.1.1.4
policy p1
    10 class ip class1 action pbr pbr1
interface vlan 20
    vrf attach red
    ip address 200.1.1.2/16
    active-gateway ip mac aa:ee:ff:00:00:20
    active-gateway ip 200.1.1.1
    apply policy p1 routed-in
```

Egress policy

A policy must be applied to the egress VTEP's L3VNI that is receiving the traffic from the VXLAN tunnel.

In the following example the policy is applied to VNI 10000:

```
class ip class1
    10 match any any any
pbr-action-list pbr1
    10 nexthop 100.1.1.4
policy p1
    10 class ip class1 action pbr pbr1
interface vxlan 1
    source ip 1.1.1.1
    no shutdown
    vni 10
        vlan 10
    vni 10000
        vrf red
        routing
        apply policy p1 routed-in
```

L2VNI vs L3VNI

There is an important distinction between PBR nexthops learned over L2VNI and L3VNI. Any L2VNI is treated the same as a locally connected nexthop. It will be probed periodically for reachability. If this L2VNI nexthop is determined to be unreachable, then PBR will take action on the next item in the action list. If the nexthop becomes reachable again and it has a higher precedence than the currently configured action, then it will take over. See PBR policy action and action list for more details.

For recursive nexthops (learned over L3VNI), the connected nexthop to reach the recursive nexthop is constantly probed for reachability. However, the recursive PBR nexthop itself will not be periodically probed. They are not guaranteed to be reachable. The PBR action list will use the nexthop as long as there is a EVPN-BGP route that matches.

CLI errors

Table 1. Error messages and triggering events

Message	Event
`Failed to create action list`	A PBR action list create operation was unable to cre ate the OVSDB row.
`PBR action list '%s' doesn't exis t`	A PBR action list update or delete operation was una ble to locate the list OVSDB row. OR

Message	Event
	A PBR action list entry create, update, or delete operation was unable to locate its parent action list OVSDB row.
`Unable to automatically set sequence number`	The automatically generated sequence number for a new PBR action list entry is higher than the maximum permitted sequence number.
`Unable to add PBR action list entry`	A PBR action list entry create failed to create an entry row in OVSDB.
`Configuration of IPv4 and IPv6 addresses in the same PBR action list is not supported`	An attempt to create both IPv4 and IPv6 addresses in the same action list was made.
`Invalid sequence number`	A PBR action list entry delete did not specify the entry sequence number.
`Cannot add entry because its automatic sequence number would exceed the maximum`	The automatic sequence number exceeded the maximum.
`Action list entry does not exist`	A PBR action list entry delete could not locate the OVSDB entry row.
`Action list name required`	A PBR action list delete was passed an invalid list name.
`Invalid action list name length`	A PBR action list name was longer than the permitted length.

PBR commands

Select a command from the list in the left navigation menu.

Subtopics

- apply policy
- pbr-action-list
- pbr-action-list copy
- pbr-action-list resequence
- pbr-action-list reset
- policy

[show pbr](#)
[show pbr-action-list](#)
[show running-config current-context](#)

`apply policy`

Syntax

```
apply policy <POLICY-NAME> routed-in
no apply policy <POLICY-NAME> routed-in
```

Description

Applies a classifier policy containing a PBR action to an interface. A policy with PBR actions is only applicable to L3/routing interfaces.

The **no** form of this command removes a classifier policy containing a PBR action from an interface.

```
config-if
```

Parameter	Description
<POLICY-NAME>	Specifies name of the policy.

Restrictions

- Layer 3 LAGs are not supported on 5420 or 6200 series switches. PBR policies cannot be applied to Layer 3 LAGs.

Usage

To use route-only ports (ROPs) as Layer 3 interfaces, an internal VLAN range must be configured first. A policy with PBR actions can be applied to ROPs.

Example

Applying a policy to an interface:

```
switch(config)# interface 1/1/10.10
switch(config-if)# routing
switch(config-if)# apply policy pbr_policy routed-in
switch(config-if)# exit
```

```
switch(config)# interface 1/1/10.10
switch(config-if)# apply policy my_policy in
switch(config-if)# exit
```

Command History

Release	Modification
10.08	PBR support added for 6200 Switch Series.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`pbr-action-list`

Syntax

```
pbr-action-list <ACTION-LIST-NAME>
    [<SEQUENCE-NUMBER>]
      {nexthop | default-nexthop} <NEXT-HOP-IP-ADDR>
       interface null
    no [<SEQUENCE-NUMBER>]
      {nexthop | default-nexthop} <IP-ADDR>
       interface null
no pbr-action-list <ACTION-LIST-NAME>
```

Description

Creates a PBR action list or modifies its entries.

The **no** form of this command can be used to delete an action list or an individual action list entry.

Parameter	Description
<ACTION-LIST-NAME>	Specifies the action list name. An action list name can be 1 to 6 4 alphanumeric characters.
<SEQUENCE-NUMBER>	Specifies list entry sequence number. Range: 1 - 4294967295 {nexthop | default - nexthop} Selects a regular next - hop (**nexthop**) or a default next - hop (**default - nexthop**). These parameters specify the address of a next - hop router to forward traffic matched by a class under different conditions.
`nexthop`	Sets the next hop for routing the packet.
`default-nexthop`	Sets the next hop for routing the packet when there is no explic it route for its destination.
<NEXTHOP-IP-ADDR>	Specifies IPv4 or IPv6 address of the next - hop router.
`interface null`	Selects keyword interface: **null**.
`null`	Specifies to drop matching traffic.

Restrictions

Reachability of next-hop routers in the list is not guaranteed. Such reachability can change at any time due to the dynamic nature of the network environment.

Usage

- `interface null`
- `interface tunnel`
- `nexthop`
- `default-nexthop`

Each action list may contain up to four entries of three different entry types:

- `interface null`

- `nexthop`
- `default-nexthop`

List entries have a unique sequence number which, if not user specified, are automatically assigned beginning at 10 and continuing at intervals of 10 for each subsequent new list entry, for example 20, 30, and 40. Sequence numbers of any value can be specified manually, a different interval may be set, and new entries can be added to (or removed from) any location in the list at any time.

Specifying an existing sequence number causes the existing list entry to be replaced by the new details. The list entry with the lowest sequence number has the highest priority entry in the list. The sequence numbers may be renumbered with the pbr-action-list resequence command.

Only one next-hop router or interface from the list is used per packet matched. This router or interface is defined as the highest priority list entry that is reachable or available at the time of the traffic match. If the highest priority list entry next-hop router is reachable - that list entry is chosen, the search is stopped, and the traffic is forwarded to the next-hop router or interface for the entry. If the highest priority list entry next-hop router is not reachable, the next highest priority list entry reachability is determined and used if reachable, otherwise the process continues down the list. If none of the routers in the list are reachable, the packet may be dropped (through the null interface entry if configured) or forwarded according to a system route table entry.

NOTE

An action list that contains a next-hop of one IP version cannot also contain an entry of another IP version. For example, an action list must contain only IPv4 or IPv6 next-hop addresses.

Examples

The list name is included in the context prompt for easy current-list identification. Any list name over 10 characters will be truncated at 10 characters and terminated with the tilde character (~) to indicate a reduced list name display. This reduction affects the prompt display of the list name only:

```
switch(config)# pbr-action-list eighteenchars
switch(config-pbr-action-list-eighteench~)#
```

The following example creates an action list with two IPv4 next-hops, a default IPv4 next-hop, and a null interface. The example uses default sequence numbering for its list entries.

```
switch(config)# pbr-action-list test1
switch(config-pbr-action-list-test1)# nexthop 1.1.1.1
switch(config-pbr-action-list-test1)# nexthop 2.2.2.2
switch(config-pbr-action-list-test1)# default-nexthop 9.9.9.9
switch(config-pbr-action-list-test1)# interface null
switch(config-pbr-action-list-test1)# end
switch(config)# show pbr-action-list test1
           Name
  Sequence Type                            Address/Interface
-------------------------------------------------------------------
           test1
```

```
        10  nexthop                          1.1.1.1
        20  nexthop                          2.2.2.2
        30  default-nexthop                  9.9.9.9
        40  interface                        null
```

The following example creates an action list with an IPv4 next-hop and a null interface with manual sequence numbers for its entries.

```
switch(config)# pbr-action-list test2
switch(config-pbr-action-list-test2)# 6 default-nexthop 4.4.4.4
switch(config-pbr-action-list-test2)# 1 interface null
switch(config-pbr-action-list-test2)# exit
switch(config)# show pbr-action-list test2
           Name
  Sequence Type                             Address/Interface
-------------------------------------------------------------------
           test2
         1  interface                       null
         6  default-nexthop                 4.4.4.4
```

The following example creates an action list with two IPv4 tunnel interfaces, with default sequence numbering.

```
switch(config)# pbr-action-list test3
switch(config-pbr-action-list-test3)# interface tunnel10
switch(config-pbr-action-list-test3)# interface tunnel15
switch(config-pbr-action-list-test3)# end
switch(config)# show pbr-action-list test3
           Name
  Sequence Type                             Address/Interface
-------------------------------------------------------------------
           test3
        10  interface                       tunnel10
        20  interface                       tunnel15
```

The following example creates an action list with two IPv6 next-hops and the null interface, with manual sequence numbers.

```
switch(config)# pbr-action-list test4
switch(config-pbr-action-list-test4)# 5 nexthop 2000:abcd::cccc:dddd
switch(config-pbr-action-list-test4)# 6 nexthop 1000:abcd::1234:5678
switch(config-pbr-action-list-test4)# 7 interface null
switch(config-pbr-action-list-test4)# end
switch(config)# show pbr-action-list test4
           Name
  Sequence Type                             Address/Interface
-------------------------------------------------------------------
           test4
         5  nexthop                         2000:abcd::cccc:dddd
```

```
        6   nexthop                                     1000:abcd::1234:5678
        7   interface                                   null
```

Command History

Release	Modification
10.08	PBR support added for 6200 Switch Series.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config` The `pbr-action-list <ACTION-LIST-NAME>` command takes you into the `config-pbr-action-list-<ACTION-LIST-NAME>` context where you modify entries for a PBR action list.	Administrators or local user group members with execution rights for this command.

`pbr-action-list copy`

Syntax

```
pbr-action-list <ACTION-LIST-NAME> copy <DESTINATION-ACTION-LIST-NAME>
```

Description

Copies an existing PBR action list.

Parameter	Description
<ACTION-LIST-NAME>	Specifies the action list name to be copied.
<DESTINATION-ACTION-LIST-NAME>	Specifies the name of the copied action list. A destination actio n list name can be 1 to 64 alphanumeric characters.

Examples

The following example copies test4 action list to test 5.

```
switch(config)# show pbr-action-list test4
           Name
  Sequence Type                                Address/Interface
-------------------------------------------------------------------
           test4
        5  nexthop                             2000:abcd::cccc:dddd
        6  nexthop                             1000:abcd::1234:5678
        7  interface                           null
switch(config)# pbr-action-list test4 copy test5
switch(config-pbr-action-list-test4)# show pbr-action-list test5
           Name
  Sequence Type                                Address/Interface
-------------------------------------------------------------------
           test4
        1  nexthop                             2000.abcd::cccc.dddd
       11  nexthop                             1000.abcd::1234.5678
       21  interface                           null
```

Command History

Release	Modification
10.08	PBR support added for 6200 Switch Series.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`pbr-action-list resequence`

Syntax

```
pbr-action-list <ACTION-LIST-NAME> resequence <STARTING-SEQUENCE-NUMBER>
            <INCREMENT>
```

Description

Renumbers the entries in an action list. The list entry with the lowest sequence number has the highest priority entry in the list.

Parameter	Description
<ACTION-LIST-NAME>	Specifies the action list name to have its entries resequenced.
<STARTING-SEQUENCE-NUMBER>	Specifies the starting sequence number. Range: 1 - 42949672 95
<INCREMENT>	Specifies the increment of the resequencing. Range: 1 - 42949 67295

Examples

The following command shows how a PBR action list is resequenced. In the following example, an action list named **test4** is resequenced so that instead of its entries starting at 5 and being numbered sequentially, its entries start now at 1 and they are numbered in increments of 10:

```
switch(config)# show pbr-action-list test4
            Name
  Sequence Type                              Address/Interface
------------------------------------------------------------------
            test4
         5  nexthop                          2000.abcd::cccc.dddd
         6  nexthop                          1000.abcd::1234.5678
         7  interface                        null
switch(config)# pbr-action-list test4 resequence 1 10
            Name
  Sequence Type                              Address/Interface
------------------------------------------------------------------
            test4
         1  nexthop                          2000.abcd::cccc.dddd
        11  nexthop                          1000.abcd::1234.5678
        21  interface                        null
```

Command History

Release	Modification
10.08	PBR support added for 6200 Switch Series.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`pbr-action-list reset`

Syntax

```
pbr-action-list <ACTION-LIST-NAME> reset
```

Description

Resets a specified PBR action list to its last successful configuration.

Parameter	Description
<ACTION-LIST-NAME>	Specifies the action list name to be reset.

Examples

```
switch(config)# pbr-action-list test reset
```

Command History

Release	Modification
10.08	PBR support added for 6200 Switch Series.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`policy`

Syntax

```
policy <POLICY-NAME>
    [<SEQUENCE-NUMBER>]
      class {ip|ipv6|mac} <CLASS-NAME>
      action {<REMARK-ACTIONS> | <POLICE-ACTIONS> | <OTHER-ACTIONS>}
      [{<REMARK-ACTIONS> | <POLICE-ACTIONS> | <OTHER-ACTIONS>}]
```

```
    [<SEQUENCE-NUMBER>]
    comment ...
no policy <POLICY-NAME>
```

Description

Creates, modifies, or deletes a classifier policy. A policy contains one or more policy entries ordered and prioritized by sequence numbers. Each entry has an IPv4/IPv6/MAC class and one or more policy actions associated with it. An applied policy processes a packet sequentially against policy entries in the list until the last entry in the list has been evaluated or the packet matches an entry. If a match occurs the related entry, actions are taken.

The **no** form of this command is used to delete a policy or an individual policy entry.

Parameter	Description
`<POLICY-NAME>`	Specifies the name of the policy.
`<SEQUENCE-NUMBER>`	Specifies a sequence number for the policy entry. Optional. Ran ge: 1 to 4294967295.
`comment`	Stores the remaining entered text as a policy entry comment.
`class {ip|ipv6|mac} <CLASS-NAME>`	Specifies a type of class, **ip** for IPv4, **ipv6** for IPv6 and **mac** for a MAC policy. And specifies a class name.
`<REMARK-ACTIONS>`	Remark actions can be any of the following options: **{pbr <ACT ION - LIST> | pcp <PRIORITY> | ip - precedence <IP - PRECE DENCE - VALUE> | dscp <DSCP - VALUE> | local - priority < LOCAL - PRIORITY - VALUE>}** where: pbr <ACTION - LIST> Specifies the PBR action list to be used. pcp <PCP - VALUE> Specifies Priority Code Point (PCP) value. Range: 0 to 7. ip - precedence <IP - PRECEDENCE - VALUE> Specifies the numeric IP precedence value. Range: 0 to 7. dscp <DSCP - VALUE>

Parameter	Description
	Specifies a Differentiated Services Code Point (DSCP) value. Enter either a numeric value (0 to 63) or a keyword as follows: • **AF11** - DSCP 10 (Assured Forwarding Class 1, low drop probability) • **AF12** - DSCP 12 (Assured Forwarding Class 1, medium drop probability) • **AF13** - DSCP 14 (Assured Forwarding Class 1, high drop probability) • **AF21** - DSCP 18 (Assured Forwarding Class 2, low drop probability) • **AF22** - DSCP 20 (Assured Forwarding Class 2, medium drop probability) • **AF23** - DSCP 22 (Assured Forwarding Class 2, high drop probability) • **AF31** - DSCP 26 (Assured Forwarding Class 3, low drop probability) • **AF32** - DSCP 28 (Assured Forwarding Class 3, medium drop probability) • **AF33** - DSCP 30 (Assured Forwarding Class 3, high drop probability) • **AF41** - DSCP 34 (Assured Forwarding Class 4, low drop probability) • *AF42* - DSCP 36 (Assured Forwarding Class 4, medium drop probability) • **AF43** - DSCP 38 (Assured Forwarding Class 4, high drop probability) • **CS0** - DSCP 0 (Class Selector 0: Default) • **CS1** - DSCP 8 (Class Selector 1: Scavenger) • **CS2** - DSCP 16 (Class Selector 2: OAM) • **CS3** - DSCP 24 (Class Selector 3: Signaling) • **CS4** - DSCP 32 (Class Selector 4: Real time) • **CS5** - DSCP 40 (Class Selector 5: Broadcast video)

Parameter	Description
	• **CS6** - DSCP 48 (Class Selector 6: Network control) • **CS7** - DSCP 56 (Class Selector 7) • **EF** - DSCP 46 (Expedited Forwarding) `local-priority <LOCAL-PRIORITY-VALUE>` Specifies a local priority value. Range: 0 to 7.

`<POLICE-ACTIONS>`	Police actions can be the following **{cir <RATE - BPS> cbs < BYTES> exceed}** where:
	`cir <RATE-BPS>`
	Specifies a Committed Information Rate value in Kilobits per se cond. Range: 1 to 4294967295.
	`cbs <BYTES>`
	Specifies a Committed Burst Size value in bytes. Range: 1 to 42 94967295.
	`exceed`
	Specifies action to take on packets that exceed the rate limit.

`<OTHER-ACTIONS>`	Other actions can be the following: `drop` Specifies drop traffic.

Restrictions

MAC classes are not applicable to policies containing PBR actions. Applying such policies to an interface are blocked.

Usage

- For Policy Based Routing, the policy action keyword is **pbr** which itself takes the name of a PBR action list as a parameter.
- A policy entry that contains a PBR action can contain other action types as well.
- An applied policy processes a packet sequentially against policy entries in the list until the last policy entry in the list has been evaluated or the packet matches an entry.

- Entering an existing **<POLICY-NAME>** value will cause the existing policy to be modified, with any new **<SEQUENCE-NUMBER>** value creating an additional policy entry, and any existing **<SEQUENCE-NUMBER>** value replacing the existing policy entry with the same sequence number.
- If no sequence number is specified, a new policy entry is appended to the end of the entry list with a sequence number equal to the highest policy entry currently in the list plus 10.

Examples

Create a policy with two PBR actions:

```
switch(config)# policy pbr_policy
switch (config-policy)# 10 class ip v4_class action pbr action_list1
switch (config-policy)# 20 class ipv6 v6_class action pbr action_list2
switch (config-policy)# exit
```

Command History

Release	Modification
10.08	PBR support added for 6200 Switch Series.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config` The `policy` command takes you into the `config-policy` context where you enter the policy entries.	Administrators or local user group members with execution rights for this command.

`show pbr`

Syntax

```
show pbr {interface <INTERFACE-NAME>|vrf <VRF-NAME>|summary}
```

Description

Shows a detailed view of Policy Based Routing (PBR) in the system.

Parameter	Description
<VRF-NAME>	Specifies name of a VRF.
<INTERFACE-NAME>	Specifies an interface. Format: **member/slot/port**.

Usage

Show commands can only reference the default VRF.

Examples

Showing PBR summary information when there is no active next-hop in the system:

```
switch# show pbr summary
VRF       Port    Policy    PBR       Seq  Type       Nexthop
---------------------------------------------------------------------
No active PBR nexthop found
---------------------------------------------------------------------
```

Showing PBR summary information when there are active next-hops in the system:

```
switch# show pbr summary
VRF       Port    Policy    PBR       Seq  Type       Nexthop
---------------------------------------------------------------------
default  1/1/1   policy_1  pbr_1     10   nexthop    1.1.1.1 (active)
         1/1/2   policy_2  pbr_2     20   nexthop    5.5.5.5 (active)
---------------------------------------------------------------------
```

Showing PBR summary information when displaying a policy with a **pbr-action-list** applied on a VxLAN L3VNI:

```
switch# configure terminal
switch(config)# interface vxlan 1
switch(config-vxlan-if)# vni 10000
switch(config-vni-10000)# apply policy p1 routed-in
switch(config-vni-10000)# show pbr summary
VRF
      Port
```

```
                 Policy
                           Class
                                      PBR
                                              Sequence  Type           Nexthop
------------------------------------------------------------------------------
---
red
      vni10000
                p1
                          c1
                                    pbr1
                                                    10  nexthop        11.2.1.4
(active)
------------------------------------------------------------------------------
---
```

Command History

Release	Modification
10.08	PBR support added for 6200 Switch Series.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show pbr-action-list

Syntax

```
show pbr-action-list [<ACTION-LIST-NAME>] [commands] [configuration] [vsx-peer]
```

Description

Shows the current PBR action list configuration. Action list entries are displayed in ascending order of their sequence number.

Parameter	Description
`<ACTION-LIST-NAME>`	Specifies the PBR action list name.
`commands`	Formats output as CLI commands.
`configuration`	Displays user - specified configuration.
`vsx-peer`	Displays VSX peer switch information.

Restrictions

If an action list entry is modified to an invalid value (for example through the REST interface), this command will indicate a mismatch for that action entry when run. In this event, use the **pbr-action-list <NAME> reset** command to restore it to the previous valid value.

Usage

- This command does not indicate whether the action list is configured in a policy or applied to an interface. Use the **show pbr** command for PBR status involving action lists.
- A single action list is shown by specifying its name or you can show all action lists by omitting a name argument.
- Using the additional commands keyword, you can change the tabulated output to a configuration style output for single or all list display.

Examples

Create two PBR action lists then run **show pbr-action-list** to display all configured action lists in the default configuration mode:

```
switch(config)# pbr-action-list v4_pbr
switch(config-pbr-action-list-v4_pbr)# 1 nexthop 1.1.1.1
switch(config-pbr-action-list-v4_pbr)# 5 default-nexthop 2.2.2.2
switch(config-pbr-action-list-v4_pbr)# 10 interface null
switch(config-pbr-action-list-v4_pbr)# exit
switch(config)#
switch(config)# pbr-action-list v6_pbr
switch(config-pbr-action-list-v6_pbr)# 20 nexthop 2000:abcd::cccc:dddd
```

```
switch(config-pbr-action-list-v6_pbr)# 40 default-nexthop
1000:abcd::1234:5678
switch(config-pbr-action-list-v6_pbr)# 60 interface null
switch(config-pbr-action-list-v6_pbr)# exit
switch#
switch# show pbr-action-list
           Name
           Additional PBR-Action-List Parameters
Sequence     Type                Nexthop
-----------------------------------------------------------------------------
-----------
             v4_pbr
1            nexthop             1.1.1.1
5            default-nexthop     2.2.2.2
10           interface           null
             v6_pbr
20           nexthop             2000:abcd::cccc:dddd
40           default-nexthop     1000:abcd::1234:5678
60           interface           null
```

Command History

Release	Modification
10.08	PBR support added for 6200 Switch Series.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show running-config current-context`

Syntax

```
show running-config current-context
```

Description

Displays the configuration of the PBR action list in the current configuration context, in commands mode.

Parameter	Description
`running-config`	Shows configuration currently running on switch.
`current-context`	Limits display to current config context only, in commands mod e.

Usage

Useful for reexamining entries previously entered into the action list after its entries have scrolled off the terminal due to other output or upon reentering the context of an existing action list.

Examples

Creating two PBR action lists and running **show running-configuration curent-context** to display the action list configuration in commands mode:

```
switch(config)# pbr-action-list v4_pbr
switch(config-pbr-action-list-v4_pbr)# 1 nexthop 1.1.1.1
switch(config-pbr-action-list-v4_pbr)# 5 default-nexthop 2.2.2.2
switch(config-pbr-action-list-v4_pbr)# 10 interface null
switch(config-pbr-action-list-v4_pbr)# exit
switch(config)#
switch(config)# pbr-action-list v6_pbr
switch(config-pbr-action-list-v6_pbr)# 20 nexthop 2000:abcd::cccc:dddd
switch(config-pbr-action-list-v6_pbr)# 40 default-nexthop
1000:abcd::1234:5678
switch(config-pbr-action-list-v6_pbr)# 60 interface null
switch(config-pbr-action-list-v6_pbr)#
switch(config-pbr-action-list-v6_pbr)# show running-config current-context
pbr-action-list v6_pbr
    20 nexthop 2000:abcd::cccc:dddd
    40 default-nexthop 1000:abcd::1234:5678
    60 interface null
```

Switching context back to the first actionl ist and running the same command:

```
switch(config-pbr-action-list-v6_pbr)# pbr-action-list v4_pbr
switch(config-pbr-action-list-v4_pbr)#
switch(config-pbr-action-list-v4_pbr)# show running-config current-context
```

```
pbr-action-list v4_pbr
    1 nexthop 1.1.1.1
    5 default-nexthop 2.2.2.2
    10 interface null
```

Removing action list entry number 5 and running the command again:

```
switch(config-pbr-action-list-v4_pbr)# no 5
switch(config-pbr-action-list-v4_pbr)# show running-config current-context
pbr-action-list v4_pbr
    1 nexthop 1.1.1.1
    10 interface null
```

Command History

Release	Modification
10.08	PBR support added for 6200 Switch Series.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

IP Directed Broadcast

> **NOTE**
> IP Directed Broadcast is only supported on the 5420 and 6200 Switch Series. It is not supported on the 4100i, 6000, or 6100 Switch Series.

IP Directed Broadcast is a feature by which remote administration tasks such as backups and wake-on-LAN (WOL) application can be achieved by sending directed broadcast packets for hosts/servers residing on a different subnet. The datagram is routed by normal mechanisms until it reaches a gateway attached to the destination hardware network, at which point it is broadcast. This class of broadcasting is also known as 'directed broadcasting'. IP Directed Broadcast packets are Layer 3 subnet broadcasts.

For example, for the broadcast IP address 192.168.1.255, the corresponding egress interface Subnet IP would be 192.168.1.x/24. Classless subnetting (CIDR) is also supported. For example, for the broadcast IP address 10.10.15.255, the broadcast address for the corresponding interface IP would be 10.10.15.x/21.

IP Directed Broadcast is supported on Route Only Port (ROP), Switched Virtual Interface (SVI), Layer 3 Link Aggregation Group (L3LAG) and Virtual Extensible Lan (VxLAN) interfaces, but is disabled by default. The feature is supported for the associated interface addresses for both the primary and secondary addresses. IP directed broadcast can be enabled on all the configured L3 interfaces. It is only supported for IPv4 addresses.

NOTE

- VxLAN is only supported on the 6300, 6400, 8100 and 8360 Switch Series.
- VxLAN interfaces require IP Directed Broadcast to be enabled on a Switched Virtual Interface associated with the VxLAN interface.
- IP Directed Broadcast is not supported with Dynamic VRF Route Leak.

For more information on this feature, see the related video on the **HPE Aruba Networking AirHeads Broadcasting Channel**.

Subtopics

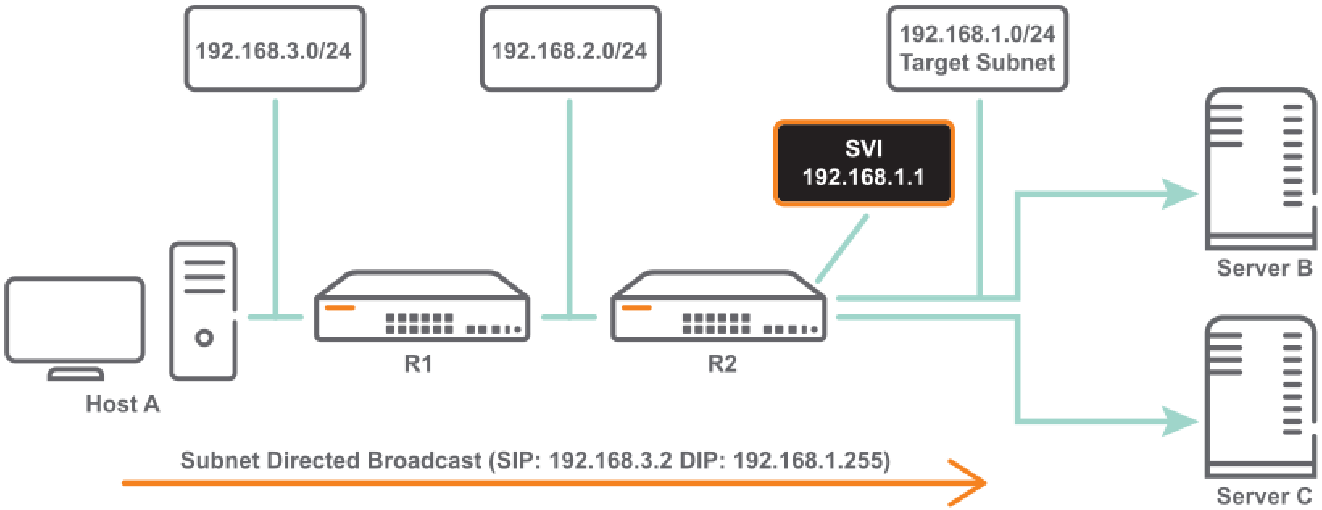
IP Directed Broadcast configuration example
IP Directed Broadcast commands

IP Directed Broadcast configuration example

The following are sample topology diagrams for an IP Directed Broadcast configuration. Figure 1 shows when the egress interface is an SVI, while figure 2 shows the diagram when the egress interface is an ROP.

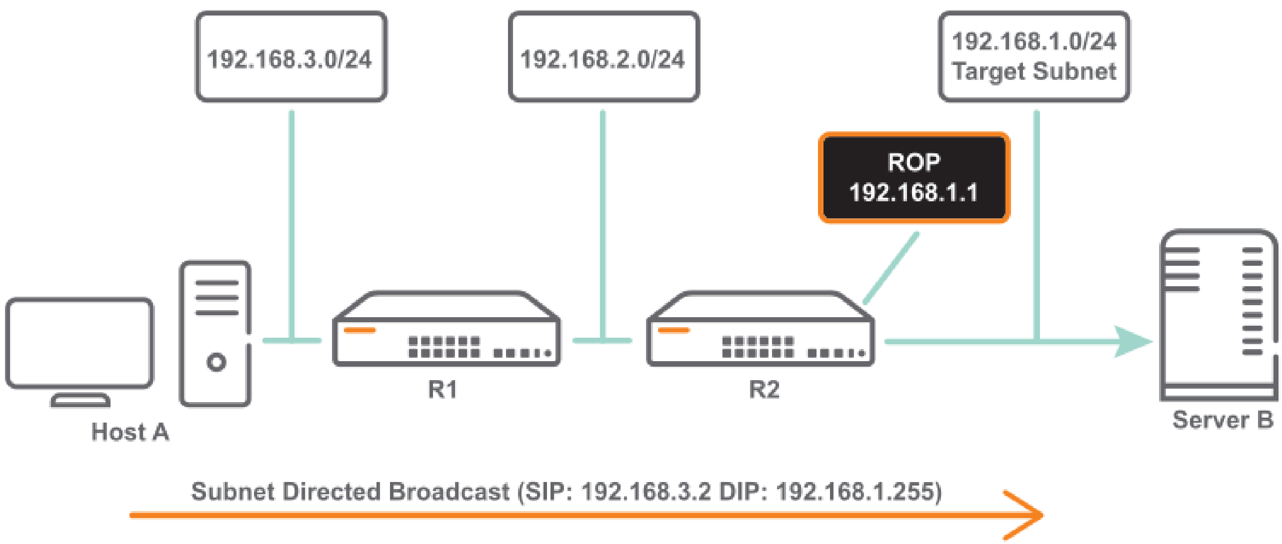
Figure 1. SVI: IP Directed Broadcast

d



Broadcast

Figure 2. ROP: IP Directed Broadcast



SVI: IP Directed Broadcast

Intermediate routers forward IP Directed Broadcast packets as Unicast. The IP directed broadcast packet is broadcast or flood in the target subnet (DA MAC: All 0xFFs) only after the last hop router.

Host A in subnet 192.168.3.0/24 wants to inject a IP Directed Broadcast (192.168.1.255) packet into Target Subnet 192.168.1.0/24. Router R1 forwards the IP Datagram with DIP 192.168.1.255 as a regular Unicast Datagram. Router R2 then floods the IP Datagram over egress ROP, SVI or VxLAN interface with Destination MAC as all 0xFFs.

At Ingress, Port Based ACLs (PACL) and VLAN Based ACLs (VACL) can be used to restrict/allow IP Directed Broadcast traffic. Existing Port based ACLs (PACL) can be used to allow or disallow certain IP Directed Broadcast Traffic.

An ACL can be configured using the `access-list ip <ACL-NAME>` command and then applied using the `apply access-list ip <ACL-NAME>` command as shown in the following output.

```
switch(config)# access-list ip ipdbacl
switch(config)# interface 1/1/1
switch(config-if)# apply access-list ipdbacl
  in   Inbound (ingress) traffic
  out  Outbound (egress) traffic
switch(config-if)# int lag 10
switch(config-lag-if)# apply access-list ipdbacl
  in   Inbound (ingress) traffic
  out  Outbound (egress) traffic
```

The following is an example of the `show running-config` command on an ROP interface.

```
switch(config)# interface 1/1/1
    no shutdown
    ip address 192.168.1.1/24
    ip directed-broadcast
```

The following is an example of the `show running-config` command on an SVI interface.

```
switch(config)# vlan 10
interface vlan10
    no shutdown
    ip address 192.168.1.1/24
    ip directed-broadcast
```

The following is an example of the `show running-config` command on an L3LAG interface.

```
switch(config)# interface lag 3
    no shutdown
    ip address 192.168.1.1/24
    ip directed-broadcast
```

NOTE

Currently egress ACL is supported only on ROP and LAG interfaces, and not on an SVI interface.

IP Directed Broadcast commands

Select a command from the list in the left navigation menu.

Subtopics

show ip directed-broadcast

`copy support-file feature`

Syntax

```
copy support-file feature l3
```

Description

Captures support logs to debug any IP Directed Broadcast issues.

Examples

Capturing the support logs into a local file:

```
switch# copy support-file feature l3 sftp
```

Command History

Release	Modification
10.09	Command introduced for the 6200 Switch Series.

Command Information

Platforms	Command context	Authority
5420 6200	Manager (`#`)	Administrators or local user group members with execution righ ts for this command.

`ip directed-broadcast`

Syntax

```
ip directed-broadcast
no ip directed-broadcast
```

Description

Turns on IP Directed Broadcast for the specified interface. The **no** form of this command turns it off.

This command is disabled by default.

Examples

Enabling and disabling IP Directed Broadcast on an physical interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip directed-broadcast
switch(config-if)# no ip directed-broadcast
```

Enabling and disabling IP Directed Broadcast on a VLAN interface:

```
switch(config)# interface vlan 100
switch(config-if-vlan)# ip directed-broadcast
switch(config-if-vlan)# no ip directed-broadcast
```

Command History

Release	Modification
10.09	Command introduced for the 6200 Switch Series.

Command Information

Platforms	Command context	Authority
5420 6200	`config-if`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show arp`

Syntax

```
show arp
```

Description

Shows IP directed broadcast verification.

> **NOTE**
>
> IP Directed Broadcast is not supported on subinterfaces on 8100, 6300, 6400, 8325, 8360 and 10000 Switch series.

Examples

Showing IP directed broadcast verification:

```
switch# show arp
IPv4 Address      MAC                Port         Physical Port
State
----------------------------------------------------------------------------
------
1.1.1.255         FF:FF:FF:FF:FF:FF  1/1/1        1/1/1
permanent
3.1.1.255         FF:FF:FF:FF:FF:FF  vlan10
permanent
Total Number Of ARP Entries Listed: 2.
----------------------------------------------------------------------------
------
```

Command History

Release	Modification
10.09	Command introduced for the 6200 Switch Series.

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show ip interface`

Syntax

```
show ip interface <INTERFACE-NAME>
```

Description

Displays the status of IP Directed Broadcast on the specified interface along with other interface related attributes.

Parameter	Description
`<INTERFACE-NAME>`	Specifies the interface to use as a source for displaying the status of the IP Directed Broadcast.

Examples

Displaying the IP Directed Broadcast status on the specified interface:

```
switch# show ip interface vlan30
Interface vlan30 is up
 Admin state is up
 Hardware: Ethernet, MAC Address: 94:f1:28:21:63:00
 IP MTU 1500
 IP Directed Broadcast is Enabled
 IPv4 address 192.168.3.1/24
 L3 Counters: Rx Disabled, Tx Disabled
 Statistics                      RX                     TX
Total
 ------------- -------------------- --------------------
--------------------
 L3 Packets                       0                      0
0
 L3 Bytes                         0                      0
0
```

Command History

Release	Modification
10.09	Command introduced for the 6200 Switch Series.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`show ip directed-broadcast`

Syntax

```
show ip directed-broadcast
```

Description

Displays the summary of the interfaces on which IP Directed Broadcast is enabled.

Examples

On the HPE Aruba Networking 6400 Switch Series, interface identification differs.

Displaying the summary of the interfaces on which IP Directed Broadcast is enabled:

```
switch# show ip directed-broadcast
 IPv4 Directed Broadcast Configuration
 Interface        Status
 ----------      --------
 1/1/1            Enabled
 vlan10           Enabled
 vlan30           Enabled
```

Displaying IP Directed Broadcast Host entries installed in Neighbor cache:

```
switch# show arp state permanent
IPv4 Address      MAC                Port          Physical Port     State
----------------------------------------------------------------------------
52.1.1.255        FF:FF:FF:FF:FF:FF  1/1/1         1/1/1             permanent
40.0.0.255        FF:FF:FF:FF:FF:FF  vlan20        vlan20            permanent
Total Number Of ARP Entries Listed- 2.
-----------------------------------------------------------------------------
---
```

Command History

Release	Modification
10.09	Command introduced for the 6200 Switch Series.

Command Information

Platforms	Command context	Authority
5420 6200	Operator (`>`) or Manager (`#`)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Key chain

> **NOTE**
> Key chain is not supported on the HPE Aruba Networking 6000 and 6100 Switch Series.

This chapter provides details for configuring and verifying the functions of a key chain. Key chain configurations are spread across `config` context, `keychain` context, and `keychain-key` context.

A key chain provides seamless authentication-key rollover. When authentication method is enabled, using key chain allows OSPF to overcome static key configuration to change the key periodically. Using key chain also reduces risks of keys being guessed by configuring rotating keys.

> **NOTE**
> A key chain must be used by an application like OSPF that communicates by using the keys with its peers.

Subtopics

Key chain commands

Select a command from the list in the left navigation menu.

Subtopics

`accept-lifetime`

Syntax

```
accept-lifetime [start-time <time> <month>/<day>/<year>] {duration
{<seconds> | infinite} | end-time <time> <month>/<day>/<year>}
```

Description

Configures the duration for which the key is valid for receiving packets.

The **no** form of this command configures the key packet receiving duration to the default value of an infinite time.

Parameter	Description
`start-time`	Time at which the key chain lifetime starts. Required. Format: HH:MM:SS
`end-time`	Time at which the key chain lifetime expires. Required. Format: HH:MM:SS
`day`	Day of the month. Required. Range: 1 - 31.
`month`	Month of the year. Required.

Parameter	Description
`year`	Year. Required. Range: 2020 - 2050
`duration`	Time in seconds. Optional. Range: 1 - 2147483646.
`infinite`	Specifies infinite time for the key. Optional.

Examples

Configuring the duration for which the key is valid for receiving packets:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# accept-lifetime start-time 10:10:10 10/25/2020
end-time 10:10:10 11/25/2020
switch(config-keychain-key)# accept-lifetime start-time 10:10:10 10/25/2020
duration 1000
switch(config-keychain-key)# accept-lifetime start-time 10:10:10 10/25/2020
duration infinite
switch(config-keychain-key)# accept-lifetime end-time 10:10:10 11/25/2020
switch(config-keychain-key)# accept-lifetime duration 1000
switch(config-keychain-key)# accept-lifetime duration infinite
```

Configuring the key packet receiving duration to the default value of an infinite time:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# no accept-lifetime
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-keychain-key`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`cryptographic-algorithm`

Syntax

```
recv-id cryptographic-algorithm {aes-cmac-128 |hmac-sha-1 | hmac-sha-256 |
hmac-sha-384 | hmac-sha-512 | md5}
no cryptographic-algorithm
```

Description

Configures the recv-id cryptographic algorithm for the key. The key will not be valid until the receive ID, the send ID, and send lifetime is configured for TCP-AO, Choose one of the authentication algorithms from the following parameters. The **no** form of this command configures the default cryptographic algorithm for a key, md5.

> **NOTE**
> TCP Authentication Option (TCP-AO) authentication supports only the aes-cmac-128 and hmac-sha-1 algorithms. If you are configuring TCP-AO, you must select one of these options.

Parameter	Description
`aes-cmac-128`	Sets the authentication algorithm for the key to AES - CMAC - 128. This parameter is only supported for TCP - AO.
`hmac-sha-1`	Sets the authentication algorithm for the key to SHA - 1. This p arameter is also supported for TCP - AO.
`hmac-sha-256`	Sets the authentication algorithm for the key to SHA - 256.
`hmac-sha-384`	Sets the authentication algorithm for the key to SHA - 384.
`hmac-sha-512`	Sets the authentication algorithm for the key to SHA - 512.
`md5`	Sets the authentication algorithm for the key to md5. Maximum length of the key string supported: 16 bytes (**config** - if contex t), 64 bytes (**config** - **keychain** - **key context**).

Examples

Set the authentication algorithm for the key to **SHA-384**:

```
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# recv-id cryptographic-algorithm hmac-sha-384
```

Set the authentication algorithm to the default, **md5**:

```
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# no recv-id cryptographic-algorithm
```

Command History

Release	Modification
10.11	The **aes - cmac - 128** parameter is introduced.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6200	`config-keychain-key`	Administrators or local user group members with execution righ ts for this command.

`key`

Syntax

```
key <KEY-ID>
```

Description

Creates the key for a key chain and enters the key chain key context. A maximum of 64 keys can be configured per key chain.

The **no** form of this command deletes the key from the key chain.

Parameter	Description
`<KEY-ID>`	ID of the key. Required. Range: 1 - 255.

Examples

Creating a key for a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
```

Deleting a key from a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# no key 1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-keychain`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`keychain`

Syntax

```
keychain <KEYCHAIN-NAME>
```

Description

Creates the key chain and enters the key chain context. A maximum of 64 key chains can be configured in the system.

The **no** form of this command removes the key chain if it is not used by any subscribers.

Parameter	Description
`<KEYCHAIN-NAME>`	Name of the key chain. Required.

Examples

Creating a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
```

Removing a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

key-string

Syntax

```
key-string [{ciphertext | plaintext} <PASSWORD>]
```

Description

Sets the key password. The password is internally stored in encrypted form. The key is not valid until its password has been set.

The **no** form of this command deletes the password used for the key.

Parameter	Description
`ciphertext`	Specifies that the key password is provided as ciphertext.
`plaintext`	Specifies that the key password is provided as plaintext.
<PASSWORD>	Specifies the key password.

> **NOTE**
> When the key password is not provided on the command line, plaintext password prompting occurs upon pressing Enter. The entered password characters are masked with asterisks.

Examples

Setting the key password with plaintext:

```
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# key-string plaintext F82#450bHP
```

Setting the key password with plaintext prompting:

```
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# key-string
Enter the key password: *************
```

```
Re-Enter the key password: *************
```

Setting the key password with ciphertext:

```
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# key-string ciphertext AQBpfciFZ/
P...biAAAOjc0a8=
```

Deleting the password for the key:

```
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# no key-string
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	`config-keychain-key`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

name

Syntax

```
name <KEY-NAME>
no name <KEY-NAME>
```

Description

Configures a name for a numbered key in a key chain.

The **no** form of this command removes the name of the key.

Parameter	Description
<KEY-NAME>	Specifies the name of the key in alphanumeric characters. Range: 1 - 64.

Examples

Creating a name for a key in a key chain called **abcdef123456**:

```
switch# configure terminal
switch(config)# keychain macsec_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# name abcdef123456
```

Removing the name of the key named **abcdef123456**:

```
switch# configure terminal
switch(config)# keychain macsec_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# no name abcdef123456
```

Command History

Release	Modification
10.11	Command added.

Command Information

Platforms	Command context	Authority
All platforms	`config-keychain-key`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

`recv-id`

Syntax

```
recv-id <0-255>
```

Description

Configures the receive ID for a keychain key. The receive ID has to be unique across keys in the keychain.

The **no** form of this command configures removes the recv-id value. The receive ID can not be changed for an active key of a keychain which is associated with BGP neighbor.

Parameter	Description
`<0-255>`	Set the receive ID corresponding to the keychain key. Supported values are 0 - 255.

Examples

Configuring the receive ID for the keychain key.

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# recv-id 1
```

Command History

Release	Modification
10.11	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config-keychain-key`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

send-lifetime

Syntax

```
send-lifetime [start-time <time> <month>/<day>/<year>] {duration {<seconds>
| infinite} | end-time <time> <month>/<day>/<year>}
```

Description

Configures the duration for which the key is valid for sending packets.

The **no** form of this command configures the key packet sending duration to the default value of an infinite time.

Parameter	Description
`start-time`	Time at which the key chain lifetime starts. Required. Format: HH:MM:SS
`end-time`	Time at which the key chain lifetime expires. Required. Format: HH:MM:SS
`day`	Day of the month. Required. Range: 1 - 31.
`month`	Month of the year. Required.
`year`	Year. Required. Range: 2020 - 2050
`duration`	Time in seconds. Optional. Range: 1 - 2147483646.
`infinite`	Specifies infinite time for the key. Optional.

Examples

Configuring the duration for which the key is valid for sending packets:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# send-lifetime start-time 10:10:10 10/25/2020
end-time 10:10:10 11/25/2020
switch(config-keychain-key)# send-lifetime start-time 10:10:10 10/25/2020
duration 1000
switch(config-keychain-key)# send-lifetime start-time 10:10:10 10/25/2020
duration infinite
switch(config-keychain-key)# send-lifetime end-time 10:10:10 11/25/2020
```

```
switch(config-keychain-key)# send-lifetime duration 1000
switch(config-keychain-key)# send-lifetime duration infinite
```

Configuring the key packet sending duration to the default value of an infinite time:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# no send-lifetime
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	`config-keychain-key`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`send-id`

Syntax

```
send-id <0-255>
```

Description

Configures the send ID for a keychain key. The send ID has to be unique across keys in the keychain.

The **no** form of this command configures removes the send-id value. The send id can not be changed for an active key of a keychain which is associated with BGP neighbor.

Parameter	Description
`<0-255>`	Set the send IDcorresponding to the keychain key. Supported values are 0 - 255.

Examples

Configuring the send ID for the keychain key.

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# send-id 218
```

Command History

Release	Modification
10.11	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	`config-keychain-key`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show capacities keychain`

Syntax

```
show capacities keychain
```

Description

Shows the maximum number of key chains and keys configurable in a key chain.

Example

```
switch# show capacities keychain
System Capacities: Filter Keychain
Capacities
Name
               Value
----------------------------------------------------------------------------
-------------------------------
```

```
Maximum number of keychains supported in the
system                                                        64
Maximum number of Keys supported in a single
Keychain                                                      64
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>)	Administrators or local user group members with execution righ ts for this command.

show keychain

Syntax

```
show keychain [<KEYCHAIN-NAME>]
```

Description

Shows information about configured and active keys of a named key chain or (if **keychain-name** is not specified) all configured key chains.

Parameter	Description
<KEYCHAIN-NAME>	Name of the key chain. Optional.

Example

```
switch# show keychain
Keychain Name : macsec_keys
```

```
  Number of Keys      : 1
  Active Send Key ID  :
  Active Recv Key IDs :
  Key ID : 1
    Key name          : abcdef123456
    Key string        :
AQBapYa+0qQDzcakbB1TopeX0AMYDDWDW015orkH5mY3qJDaBAAAADASiBQ=
    Send Key Validity : 00:00:00 01/01/2020 to Infinite
    Recv Key Validity : 00:00:00 01/01/2020 to Infinite
Keychain Name : ospf_keys
  Number of Keys      : 2
  Active Send Key ID  : 7
  Active Recv Key IDs : 7, 200
  Key ID              : 7
    Key name          : -
    Key string        :
AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    Crypto-Algorithm  : sha256
    Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
    Recv Key Validity : 00:00:01 10/1/2020 to infinite
  Key ID              : 200
    Key name          : -
    Key string        :
AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    Crypto-Algorithm  : sha512
    Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
    Recv Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
Keychain Name : bgp_keys
  Number of Keys      : 2
  Active Send Key ID  : 7
  Active Recv Key IDs : 7
  Key ID              : 7
    Key name          : -
    Key string        :
AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    Crypto-Algorithm  : md5
    Send Key Validity : 00:00:01 10/26/2020 to 23:59:01 10/1/2021
    Recv Key Validity : 00:00:01 10/22/2020 to infinite
  Key ID              : 8
    Key name          : -
    Key string        :
AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    Crypto-Algorithm  : sha384
```

```
    Send Key Validity : 00:00:01 10/1/2021 to 23:59:01 10/1/2021
    Recv Key Validity : 00:00:01 10/1/2021 to 23:59:01 10/1/2021
```
```
Keychain Name : ospf_keys
  Number of Keys       : 2
  Active Send Key ID   : 7
  Active Recv Key IDs : 7, 200
  Key ID               : 7
    Key name           : -
    Key string         :
AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    Crypto-Algorithm   : sha256
    Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
    Recv Key Validity : 00:00:01 10/1/2020 to infinite
  Key ID               : 200
    Key name           : -
    Key string         :
AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    Crypto-Algorithm   : sha512
    Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
    Recv Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (`>`)	Administrators or local user group members with execution righ ts for this command.

`show running-config keychain`

Syntax

```
show runnning-config keychain
```

Description

Shows the configurations for key chain protocol.

Example

```
switch# show running-config keychain
keychain ospf_keys
  key 1
    key-string ciphertext
AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    cryptographic-algorithm md5
    accept-lifetime start-time 10:10:10 10/25/2020 end-time 10:10:10
11/25/2020
    send-lifetime start-time 10:10:10 10/25/2020 end-time 10:10:10
11/25/2020
  key 45
    key-string ciphertext
AQBapZ1OHiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
    accept-lifetime start-time 10:10:10 10/25/2020 end-time 10:10:10
11/25/2020
  key 33
keychain macsec_keys
  key 1
    name abcdef123456
    key-string ciphertext
AQBapYa+0qQDzcakbB1TopeX0AMYDDWDW015orkH5mY3qJDaBAAAADASiBQ=
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>)	Administrators or local user group members with execution righ ts for this command.

IP Client Tracker

The client IP address tracking feature will learn and update the IP addresses of the access devices and clients connected to the switch. It can track addresses of directly connected clients, as well as clients connected to a downstream device such as a wireless access point.

IP Client tracker can be enabled for user-based tunneling (UBT) clients to learn the IP addresses on the required VLANs and interfaces.

Enabling IP Client Tracker based on UBT modes:

- If the UBT mode is Local VLAN, enable IP Client tracker for the UBT client VLAN defined using the `ubt-client-vlan` command.
- If the UBT mode is VLAN extend, enable IP Client tracker for the UBT client VLAN defined under role.

Subtopics

IP Client Tracker commands

IP Client Tracker commands

Select a command from the list in the left navigation menu.

Subtopics

client track ip
client track ip { enable | disable | auto }
client track ip client-limit
client track ip update-interval
client track ip update-method probe
show capacities
show client ip { count | port | vlan }

`client track ip`

Syntax

```
client track ip
```

Description

Enables client IP address tracking on the switch. The default is disabled on global and VLAN levels.

Admin users can enable client IP address tracking at the VLAN level.

> **NOTE**
>
> Tracking enabling will take effect only if the client IP address tracking is enabled at system and VLAN level.

The **no** form of the command disables client IP address tracking. If tracking is disabled at switch level, it will be stopped even if it is enabled at VLAN or port level.

Example

Enable client IP address tracking at switch level:

```
switch(config)# client track ip
```

Enable client IP address tracking on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# client track ip
```

Enable client IP address tracking on VLANs 10 to 100:

```
switch(config)# vlan 10-100
switch(config-vlan-<10-100>)# client track ip
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 5420 6000 6100 6200	`config`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

client track ip { enable | disable | auto }

Syntax

```
client track ip { enable | disable | auto }
```

Description

Enables client IP address tracking on the specified set of interfaces. Tracking will take effect only if client IP address tracking is enabled at both the system level and for the VLAN to which the port belongs. Default: auto.

The **no** form of the command disables client IP address tracking on the specified set of interfaces.

Parameter	Description
`enable`	Specifies that all client IP addresses will be tracked in the port.
`disable`	Specifies that client IP addresses will not be tracked in the port.
`auto`	Specifies the following: • For LLDP devices: Only the specified client IP address will be tracked in the port and other client IP addresses will not be tracked. • For non - LLDP devices: All client IP addresses will be trac ked in the port.

Example

Enable client IP address tracking on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# client track ip enable
```

Enable client IP address tracking on interfaces 1/1/1 to 1/1/5:

```
switch(config)# interface 1/1/1-1/1/5
switch(config-if-<1/1/1-1/1/5>)# client track ip enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 5420 6000 6100 6200	`config-if`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`client track ip client-limit`

Syntax

```
client track ip client-limit <CLIENT-LIMIT>
```

Description

Configures the maximum number of clients to be tracked on the specified set of interfaces.

For the HPE Aruba Networking 6000 and 6100 Switch Series the **no** form of the command resets the client limit to the default value of 128.

For the HPE Aruba Networking 5420 and 6200 Switch Series the **no** form of the command resets the client limit to the default value of 2048.

- 6300: 2048
- 6400: 4096

Parameter	Description
CLIENT-LIMIT	Specifies the maximum number of clients tracked on a port. Re quired. For the HPE Aruba Networking 6000 and 6100 Switch Series, Range: 1 - 128. Default: 128. For the HPE Aruba Netwo rking 5420 and 6200 Switch Series, Range: 1 - 2048. Default: 2048.

Example

Configure the maximum number of clients to be tracked on interface 1/1/5:

```
switch(config)# interface 1/1/5
switch(config-if)# client track ip client-limit 32
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
4100i 5420 6000 6100 6200	`config-if`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`client track ip update-interval`

Syntax

```
client track ip update-interval <INTERVAL>
```

Description

Configures how often client IP addresses are updated.

The **no** form of the command resets the update interval to the default of 1800 seconds.

Parameter	Description
INTERVAL	Specifies the update interval in seconds. Required. Range: 60 - 28000. Default: 1800.

Example

Configure the update interval for an interface:

```
switch(config)# interface 1/1/1
switch(config-if)# client track ip update-interval 600
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 5420 6000 6100 6200	`config-if`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`client track ip update-method probe`

Syntax

```
client track ip update-method probe
```

Description

Enables probing the client to update the IP address.

The probe is sent to all clients on the tracking list that have an IP address in the following scenarios:

1. IP packets are not received from the clients during the IP address update cycle.
2. There is no IP packet from a learned IP address. In this case, a probe will be sent for the IP address to confirm if it is still owned by that client.

The **no** form of the command disables probing.

Example

Disable probing to update the client IP address:

```
switch(config)# no client track ip update-method probe
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 5420 6000 6100 6200	`config`	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this c ommand from the operator context (>) only.

`show capacities`

Syntax

```
show capacities
```

Description

Shows the capacities configured on the switch.

Example

```
switch# show capacities
System Capacities:
Capacities
Name                                                                    Value
-----------------------------------------------------------------------------
------
Maximum number of Access Control Entries configurable in a
```

```
system             14336
Maximum number of Access Control Lists configurable in a
system                1024
Maximum number of class entries configurable in a
system                       1024
Maximum number of classes configurable in a
system                              512
Maximum number of entries in an Access Control
List                               1024
Maximum number of entries in a
class                                              1024
Maximum number of entries in a
policy                                             1024
Maximum number of classifier policies configurable in a
system                  512
Maximum number of policy entries configurable in a
system                      1024
Maximum number of clients supported for tracking the IP address in the
system  128
switch# show capacities client-track-ip-client-limit
System Capacities: Filter Client Track IP Client Limit
Capacities
Name                                                                  Value
----------------------------------------------------------------------------
-------
Maximum number of clients supported for tracking the IP address in the
system
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 5420 6000	Operator (>)	Administrators or local user group members with execution righ ts for this command.

Platforms	Command context	Authority
6100 6200		

show client ip { count | port | vlan }

Syntax

```
show client ip { count | port | vlan }
```

Description

Shows number of client IP addresses or information about client IP addresses tracked on ports and VLANs.

Parameter	Description
count	Displays number of clients tracked.
port	Displays client IP addresses tracked on the ports.
vlan	Displays client IP addresses tracked on the VLANs.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 5420 6000 6100 6200	Operator (>)	Administrators or local user group members with execution righ ts for this command.

Routing Information Protocol (RIP)

> **NOTE**
> Routing Information Protocol (RIP, RIPv2 and RIPng) is not supported on the HPE Aruba Networking 6000 and 6100 Switch Series.

Routing Information protocol (RIP, RIPv2, RIPng) is a distance-vector routing protocol that uses hop count as a routing metric. It is useful in small networks. RIP can be configured on L3 ports, LAG, VLAN and Loopback interfaces. All configurations work in the interface context and require the associated interface to be routing interface.

- **RIP/RIPv2:** IPv4 implementation of Routing Information protocol defined in RFC 2453.
- **RIPng:** IPv6 implementation of Routing Information protocol defined in RFC 2080.

Subtopics

Overview

RIP characteristics:

- Prevents routing loops by adding a limit to the number of hops allowed in a path from source to destination.

- Allows a maximum of 15 hops which limits the size of the network that RIP can support.
- Uses broadcast UDP data packets to exchange routing information.
- RIPv2 is a classless protocol that supports variable-length subnet mask (VLSM), classless inter-domain routing (CIDR) and route summarization.
- RIPng uses ipsec for message authentication.

RIP limitations:

- RIPv1 is not supported.
- Only poison reverse is supported for loop prevention.
- RIP/RIPng metric configuration support is not available.
- Redistribute RIP/RIPng is not supported in BGP/BGP+.

RIPv2 (IPv4) commands

Select a command from the list in the left navigation menu.

Subtopics

Configuration commands

Select a command from the list in the left navigation menu.

Subtopics

`router rip`

Syntax

```
router rip <PROCESS-ID> [vrf <VRF-NAME>]
no router rip <PROCESS-ID> [vrf <VRF-NAME>]
```

Description

Creates RIP process if not already created and enters the **router rip <PROCESS-ID>** context for the VRF mentioned. If no VRF is mentioned, a default is used. Only one RIP process is allowed per VRF.

The **no** form of this command deletes the RIP instance for the VRF. If no VRF is mentioned the default is deleted.

Parameter	Description
<PROCESS-ID>	Specifies name of the RIP process ID. Range: 1 - 63.
vrf <VRF-NAME>	Specifies VRF name.

Examples

Creating RIP process and naming the VRF:

```
switch(config)# router rip 2 vrf red
```

Deleting RIP process:

```
switch(config)# no router rip 2 vrf red
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config`	Administrators or local user group members with execution righ ts for this command.

Interface commands

Select a command from the list in the left navigation menu.

Subtopics

- **ip rip**
- **ip rip all-ip enable**
- **ip rip all-ip disable**
- **ip rip all-ip send disable**
- **ip rip all-ip receive disable**

`ip rip`

Syntax

```
ip rip <PROCESS-ID> {all-ip | ip-address}
no ip rip <PROCESS-ID> {all-ip | ip-address}
```

Description

Enables RIP process on an interface.

The **no** form of this command deletes the RIP process from an interface.

Parameter	Description
`ip rip <PROCESS-ID>`	Specifies RIP process ID. Range: 1 - 63.
`all-ip`	Specifies RIP for all IP addresses configured on the interface.

Parameter	Description
`ip-address`	Specifies IP address for RIP on the interface.

Usage

- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.
- If **ip rip** 1 **all-ip** is configured and a new IP address is added to the interface, RIP configurations will not be applicable for the newly added IP address.

Examples

Configuring RIP for all IP addresses configured on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 all-ip
```

Deleting RIP for all IP addresses configured on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip 1 all-ip
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ip rip all-ip enable`

Syntax

```
ip rip all-ip enable
no ip rip all-ip enable
```

Description

Enables RIP process for all RIP enabled IP addresses configured on interface.

The **no** form of this command disables RIP process on the interface.

Usage

- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

Examples

Enabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip all-ip enable
```

Disabling RIP process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip all-ip enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

ip rip all-ip disable

Syntax

```
ip rip all-ip disable
no ip rip all-ip disable
```

Description

Disables RIP process for all RIP enabled IP addresses configured on the interface.

The **no** form of this command enables RIP process for all RIP enabled IP addresses configured on the interface.

Usage

- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

Examples

Disabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip all-ip enable
```

Enabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip all-ip enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ip rip all-ip send disable`

Syntax

```
ip rip all-ip send disable
no ip rip all-ip send disable
```

Description

Disables interface from sending RIP packets for all RIP enabled IP addresses.

The **no** form of this command enables interface to send RIP packets for all RIP enabled IP addresses.

Usage

- Default settings allow an interface to send RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

Examples

Disabling interface from sending RIP packets for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip all-ip send disable
```

Enabling interface to send RIP packets for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip all-ip send disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`ip rip all-ip receive disable`

Syntax

```
ip rip all-ip receive disable
no ip rip all-ip receive disable
```

Description

Disables interface from receiving RIP packets for all enabled IP addresses.

The **no** form of this command enables interface to receive RIP packets for all RIP enabled IP addresses.

Usage

- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

Examples

Disabling interface from receiving RIP packets for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip all-ip receive disable
```

Enabling interface to receive RIP packets for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip all-ip receive disable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

RIPv2 (IPv4) commands

Select a command from the list in the left navigation menu.

Subtopics

- **enable**
- **disable**
- **distance**
- **maximum-paths**
- **redistribute**
- **timers update**

`enable`

Syntax

```
enable
no enable
```

Description

Enables RIP process if disabled. By default RIP process is enabled.

The **no** form of this command disables the RIP process.

Examples

Enabling RIP process when disabled:

```
switch(config)# router rip 1
switch(config-rip-1)# enable
```

Disabling RIP process when enabled:

```
switch(config)# router rip 1
switch(config-rip-1)# no enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`disable`

Syntax

```
disable
no disable
```

Description

Disables RIP process.

The **no** form of this command enables the RIP process.

Examples

Disabling RIP process:

```
switch(config)# router rip 1
switch(config-rip-1)# disable
```

Enabling RIP process:

```
switch(config)# router rip 1
switch(config-rip-1)# no disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`distance`

Syntax

```
distance <DISTANCE>
no distance
```

Description

Configures administrative distance for RIP. Administrative distance is used as criteria to select the best route when multiple protocols have the same route.

The **no** form of this command sets the RIP administrative distance to the default. Default: 120.

Parameter	Description
`<DISTANCE>`	Specifies RIP administrative distance. Range: 1 to 255.

Examples

Configuring administrative distance for RIP:

```
switch(config)# router rip 1
switch(config-rip-1)# distance 100
```

Setting administrative distance for RIP to default values:

```
switch(config)# router rip 1
switch(config-rip-1)# no distance
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`maximum-paths`

Syntax

```
maximum-paths <MAX-VALUE>
no maximum-paths
```

Description

Sets the maximum number of ECMP routes that RIP can support.

The **no** form of this command sets the maximum number of ECMP routes to the default value of 4.

Parameter	Description
`<MAX-VALUE>`	Sets the number of RIP ECMP routes. Range: 1 - 8.

Examples

Setting maximum number of RIP ECMP routes:

```
switch(config)# router rip 1
switch (config-rip-1)# maximum-paths 8
```

Setting maximum number of RIP ECMP routes to default:

```
switch(config)# router rip 1
switch (config-rip-1)# no maximum-paths
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config`	Administrators or local user group members with execution righ ts for this command.

redistribute

Syntax

```
redistribute {bgp | connected | ospf <PROCESS-ID> | static}
no redistribute {bgp | connected | ospf <PROCESS-ID> | static}
```

Description

Redistributes routes originating from other protocols into RIP.

The **no** form of this command disables redistribution of routes originating from other protocols into RIP.

Parameter	Description
`bgp`	Specifies BGP routes to redistribute into RIP.
`connected`	Specifies connected routes (directly attached subnet or host) to redistribute into RIP.
`ospf <PROCESS-ID>`	Specifies the OSPF route to redistribute into RIP. Range: <1 - 65535>
`static`	Specifies static route to redistribute into RIP.

> **NOTE**
> Parameters **ospf** and **bgp** are not supported on 6100 switches.

Examples

Redistributing BGP routes into RIP:

```
switch(config)# router rip 1
switch(config-rip-1)# redistribute bgp
```

Disabling BGP routes that originate from other protocols and redistribute into RIP:

```
switch(config)# router rip 1
switch(config-rip-1)# no redistribute bgp
```

Command History

Release	Modification
10.14	Supported process ID range expanded from 1 - 63 to 1 - 65535.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`timers update`

Syntax

```
timers update <INTERVAL> timeout <DURATION> garbage-collection <PERIOD>
no timers
```

Description

Configures RIP timers with specific values.

The **no** form of this command sets all RIP timers to default values.

Parameter	Description
`timers update <INTERVAL>`	Specifies frequency at which RIP sends updates to all of its pee rs. Range: 1 to 2147484. Default: 30.
`timeout <DURATION>`	Specifies timeout duration from the point of the last refresh a fter a route is received from a peer timeout and is marked as ex pired. Range: 1 to 255. Default: 180.
`garbage-collection <PERIOD>`	Specifies amount of time route remains in routing table after ro ute expiration. Range: 1 to 255. Default: 120.

Examples

Configuring RIP timers with specific values:

```
switch(config)# router rip 1
switch(config-rip-1)# timers update 40 timeout 200 garbage-collection 150
```

Configuring RIP timers with default values:

```
switch(config)# router rip 1
switch(config-rip-1)# no timers
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config`	Administrators or local user group members with execution righ ts for this command.

RIPv2 (IPv4) commands

Select a command from the list in the left navigation menu.

Subtopics

[clear ip rip statistics]

`clear ip rip statistics`

Syntax

```
clear ip rip [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]
```

Description

Clears RIP event statistics.

Parameter	Description
`<PROCESS-ID>`	Specifies RIP process ID. Range: 1 - 63
`all-vrfs`	Clears statistics for all VRFs.
`vrf`	Selects VRF to clear statistics for.
`<VRF-NAME>`	Specifies VRF name.

Examples

Clearing RIP event statistics:

```
switch# clear ip rip statistics
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	Operator (>) or Manage r (#)	Administrators or local user group members with execution righ ts for this command.

RIPng interface commands

Select a command from the list in the left navigation menu.

Subtopics

- **enable**
- **disable**
- **send disable**
- **receive disable**

`enable`

Syntax

```
enable
no enable
```

Description

Enables RIP process for RIP enabled IP address configured on interface.

The **no** form of this command disables RIP process on interface.

Usage

- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

Examples

Enabling RIP process for RIP enabled IP address:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# enable
```

Disabling RIP process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
```

```
switch(config-if-rip)# no enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`disable`

Syntax

```
disable
no disable
```

Description

Disables RIP process for RIP enabled IP addresses configured on interface.

The **no** form of this command enables RIP process on interface.

Examples

Disabling RIP process for RIP enabled IP addresses configured on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# disable
```

Enabling RIP process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
```

```
switch(config-if-rip)# no disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`send disable`

Syntax

```
send disable
no send disable
```

Description

Disables an interface from sending RIP packets for a specific IP address.

The **no** form of this command enables interface for sending RIP packets for a specific IP address.

Usage

- Default settings allow an interface to send and receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

Examples

Disabling interface from sending RIP packets for a specific IP address :

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# send disable
```

Enabling interface to send RIP packets for a specific IP address:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# no send disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`receive disable`

Syntax

```
receive disable
no receive disable
```

Description

Disables interface from receiving RIP packets for a specific IP address.

The **no** form of this command enables interface for receiving RIP packets for a specific IP address.

Usage

- Default settings allow an interface to receive RIP packets.

- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

Examples

Disabling interface from receiving RIP packets for a specific IP address:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# receive disable
```

Enabling interface for receiving RIP packets for a specific IP address:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# no receive disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

RIPv2 (IPv4) commands

Select a command from the list in the left navigation menu.

Subtopics

show capacities rip
show capacities-status rip
show ip rip

- **show ip rip interface**
- **show ip rip neighbors**
- **show ip rip routes**
- **show ip rip statistics**
- **show ip rip statistics interface**
- **show running-config**

show capacities rip

Syntax

```
show capacities rip
```

Description

Displays maximum number of RIP interfaces, routes and process.

Examples

Displaying maximum number of RIP interfaces, routes and process:

```
switch# show capacities rip
System Capacities: Filter RIP
Capacities Name
Value
----------------------------------------------------------------------------
------
Maximum number of RIP interfaces configurable in the system
32
Maximum number of RIP processes supported across each VRF
1
Maximum number of routes in RIP supported across all VRFs
2540
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	Operator (`>`) or Manager (`#`)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

`show capacities-status rip`

Syntax

```
show capacities-status rip
```

Description

Displays number of RIP interfaces, routes and process configured in the system.

Examples

Displaying number of RIP interfaces, routes and process:

```
switch# show capacities-status rip
System Capacities Status: Filter RIP
Capacities Name                                                  Value Maximum
----------------------------------------------------------------------
Number of RIP interfaces configured in the system                   0      32
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100	Operator (`>`) or Manager (`#`)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Platforms	Command context	Authority
6200		

show ip rip

Syntax

```
show ip rip [<PROCESS-ID>] [all-vrfs | vrf <VRF-NAME>]
```

Description

Displays general RIP configuration.

Parameter	Description
`<PROCESS-ID>`	Specifies RIP process ID. Range: 1 - 63.
`all vrfs`	Displays general RIP information for all VRFs.
`vrf`	Selects VRF to display general RIP information for.
`<VRF-NAME>`	Specifies VRF name.

Usage

- Parameters display general RIP information for a specific RIP process.
- Parameters display general RIP information for a specific or all VRFs.
- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

Displaying general RIP configuration for all VRFs:

```
switch# show ip rip 34 all-vrfs
VRF : Default                          Process-ID : 34
---------------------------------------------------------------------
```

```
RIP Version              : RIPv2     Protocol Status : Enabled
Update Time              : 60  sec   Timeout Time    : 240 sec
Garbage Collection Time : 250 sec   ECMP            : 6
Distance                 : 100       Redistribution  : static,
                                                       ospf 1
VRF : vrf_1                          Process-ID : 34
-----------------------------------------------------------------------
RIP Version              : RIPv2     Protocol Status : Enabled
Update Time              : 30  sec   Timeout Time    : 180 sec
Garbage Collection Time : 120 sec   ECMP            : 4
Distance                 : 120       Redistribution  : None
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	Operator (`>`) or Manager (`#`)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

`show ip rip interface`

Syntax

```
show ip rip [<PROCESS-ID>] interface [<INTERFACE-NAME>] [brief] [all-vrfs |
vrf <VRF-NAME>]
```

Description

Displays information about RIP enabled interfaces.

Parameter	Description
<PROCESS-ID>	Specifies RIP process ID. Range: 1 - 63.
<INTERFACE-NAME>	Specifies interface.
`brief`	Shows brief overview information for the RIP interface.
`all-vrfs`	Displays interface information for all VRFs.
`vrf`	Selects specific VRF.
<VRF-NAME>	Specifies VRF.

Usage

- Parameters display general RIP information for a specific RIP process.
- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

```
switch# show ip rip interface
Interface 1/1/1 is up, IP Address is 10.10.10.1/24
-----------------------------------------------------------------------
VRF             : Default            Process-ID     : 1
Status          : Oper Up            Mode           : Send and Receive
MTU             : 500                Version        : RIPv2
Poision Reverse : Enabled
Interface 1/1/2 is up, IP Address is 20.10.10.1/24
-----------------------------------------------------------------------
VRF             : Default            Process-ID     : 1
Status          : Admin Down         Mode           : Receive
MTU             : 500                Version        : RIPv2
Poision Reverse : Enabled
Interface 1/1/3 is up, IP Address is 30.10.10.1/24
-----------------------------------------------------------------------
```

```
VRF                : Default              Process-ID      : 1
Status             : Admin Down           Mode            : Send
MTU                : 500                  Version         : RIPv2
Poision Reverse : Enabled
switch# show ip rip interface brief
VRF : default   Process-ID : 1
-------------------------------
Total Number of Interfaces: 2
Interface      IP-Address/Mask    Status    MTU
--------------------------------------------------
1/1/1          10.10.10.1/24       up        500
1/1/2          20.10.10.1/24       up        500
1/1/3          30.10.10.1/24       up        500
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	Operator (`>`) or Manager (`#`)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

`show ip rip neighbors`

Syntax

```
show ip rip [<PROCESS-ID>] neighbors [<IP-ADDRESS>] [all-vrfs | vrf <VRF-NAME>]
```

Description

Displays information about RIP neighbors.

Parameter	Description
<PROCESS-ID>	Specifies RIP process ID. Range: 1 - 63.
<IP-ADDRESS>	Specifies IP address of a specific neighbor to display informatio n on.
`all-vrfs`	Displays neighbor information for all VRFs.
`vrf`	Selects VRF to display neighbor information.
<VRF-NAME>	Specifies VRF name.

Usage

- Parameters display RIP neighbor information for a specific RIP process.
- Parameters display RIP neighbor information for a specific neighbor.
- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

Displaying RIP neighbor information for all VRFs:

```
switch# show ip rip neighbors all-vrfs
VRF : default             Process-ID : 1
----------------------------------------
Total Number of Neighbors: 1
Peer-Address    Type    Last-Update  Rcvd-Bad-Pkts    Rcvd-Bad-Routes
------------------------------------------------------------------
1.1.1.2         RIPv2   0:0:7        4                5
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	Operator (`>`) or Manager (`#`)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

`show ip rip routes`

Syntax

```
show ip rip [<PROCESS-ID>] routes [<PREFIX/LENGTH>] [all-vrfs | vrf <VRF-NAME>]
```

Description

Displays RIP routing table for a specific RIP process.

Parameter	Description
`<PROCESS-ID>`	Specifies RIP process ID to display information for a specific RIP process. Range: 1 - 63.
`<PREFIX/LENGTH>`	Specifies the network prefix.
`all-vrfs`	Displays RIP routing information for all VRFs.
`vrf`	Selects VRF to display RIP routing information.
`<VRF-NAME>`	Specifies VRF name.

Usage

- <PREFIX/LENGTH> is an optional parameter that displays RIP routing table information for a specific subnet.
- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

Displaying RIP routing table for all VRFs:

```
switch# show ip rip routes all-vrfs
VRF : default                   Process-ID : 1
----------------------------------------
Total Number of Routes : 6
Prefix              Metric     Interface      Nexthop
--------------------------------------------------------
10.1.0.0/16           2          1/1/1         30.1.1.2
20.1.2.0/24           3          1/1/1         30.1.1.2
30.1.1.0/24           1          1/1/1
VRF : vrf_1                     Process-ID : 34
----------------------------------------
Prefix               Metric     Interface      Nexthop
--------------------------------------------------------
20.1.0.0/16           10         1/1/2         50.1.1.2
40.1.2.0/24           14         1/1/2         50.1.1.2
50.1.1.0/24            1         1/1/2
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	Operator (>) or Manage r (#)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this co mmand from the auditor context (auditor>) only.

show ip rip statistics

Syntax

```
show ip rip [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]
```

Description

Displays RIP statistics.

Parameter	Description
`<PROCESS-ID>`	Specifies RIP process ID. Range: 1 - 63.
`all-vrfs`	Displays statistics information for all VRFs.
`vrf`	Selects VRF to display RIP statistics information for.
`<VRF-NAME>`	Specifies VRF name.

Usage

- Parameters can display information for all VRFs or a specific VRF.
- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

Displaying RIP statistics for all VRFs:

```
switch# show ip rip statistics all-vrfs
VRF : default  Process-ID : 1
-------------------------------------
Global Route Changes : 50
Global Queries       : 2
Last Cleared         : 0h 30m 28s ago
VRF : vrf_1    Process-ID : 34
-------------------------------------
Global Route Changes : 20
Global Queries       : 0
Last Cleared         : 0h 30m 28s ago
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6100 6200	Operator (`>`) or Manager (`#`)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

`show ip rip statistics interface`

Syntax

```
show ip rip [<PROCESS-ID>] statistics interface [<INTERFACE-NAME>] [all-vrfs | vrf <VRF-NAME>]
```

Description

Displays RIP statistics for RIP enabled interfaces.

Parameter	Description
`<PROCESS-ID>`	Specifies RIP process ID. Range: 1 - 63.
`<INTERFACE-NAME>`	Specifies name of interface.
`all-vrfs`	Displays RIP interface statistics for all VRFs.
`vrf`	Selects VRF to display RIP interface statistics.

Parameter	Description
<VRF-NAME>	Specifies VRF name.

Usage

- Parameters can display information for all VRFs or a specific VRF.
- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

Displaying RIP statistics for a RIP enabled interface:

```
switch# show ip rip statistics interface 1/1/1
VRF : default  Process-ID : 1      interface 1/1/1
---------------------------------------------
IP-Address    Trigger-Updates    Rcvd-Bad-Packets    Rcvd-Bad-Routes
----------------------------------------------------------------
10.1.1.1      15                 3                   4
Last Cleared : 0h 30m 28s ago
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	Operator (>) or Manager (#)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

show running-config

Syntax

```
show running config
```

Description

Displays all running configurations for all protocols including RIP.

Examples

Displaying all running configurations for all protocols including RIP:

```
switch# show running-config
Current configuration:
!
!Version Halon 0.1.0 (Build: genericx86-64-Halon-0.1.0-
master-20170309054955-dev)
!Schema version 0.1.8
lldp enable
timezone set utc
vrf blue
vrf green
vrf red
led base-loc_fdc on
led base-loc on
led base-hlth_fdc fast_blink
led base-pwr_fdc on
!
!
!
!
!
!
aaa authentication login default local
aaa authorization commands default none
!
!
!
!
router ospf 1 vrf red
router rip 1
    maximum-paths 5
    distance 1
router rip 1 vrf red
```

```
    default-information originate always
    maximum-paths 7
    distance 5
    redistribute ospf 1
    timers update 40 timeout 200 garbage-collection 120
vlan 1
    no shutdown
interface lag 44
    no shutdown
    ip address 33.1.1.1/24
    ip rip 1 33.1.1.1
        send disable
interface 1/1/1
    no shutdown
    ip address 33.44.1.1/24
    ip address 44.44.1.1/24 secondary
    ip rip 1 33.44.1.1
        send disable
    ip rip 1 44.44.1.1
        send disable
interface 1/1/2
interface loopback 2
    ip address 55.55.55.55/32
    ip rip 1 55.55.55.55
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	Operator (>) or Manager (#)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

RIPng (IPv6) commands

Select a command category from the list in the left navigation menu.

Subtopics

- **Configuration commands**
- **Interface commands**
- **Routing commands**
- **RIPng clear commands**
- **Interface commands**
- **RIPng show commands**

Configuration commands

Select a command from the list in the left navigation menu.

Subtopics

- **router ripng**

`router ripng`

Syntax

```
router ripng <PROCESS-ID> [vrf <VRF-NAME>]
no router ripng <PROCESS-ID> [vrf <VRF-NAME>]
```

Description

Creates RIPng process if not already created and enters the **router ripng <PROCESS-ID>** context for the VRF mentioned. If no VRF is mentioned, a default is used. Only one RIPng process is allowed per VRF.

The **no** form of this command deletes the RIPng instance for the VRF. If no VRF is mentioned the default is deleted.

Parameter	Description
<PROCESS-ID>	Specifies name of the RIPng process ID. Range: <1 - 63>

Parameter	Description
`vrf`	Sets VRF name for RIPng process.
`<VRF-NAME>`	VRF name for VRF.

Examples

Creating RIPng process and naming the VRF:

```
switch(config)# router ripng 2 vrf red
```

Deleting RIPng process:

```
switch(config)# no router ripng 2 vrf red
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config`	Administrators or local user group members with execution righ ts for this command.

Interface commands

Select a command from the list in the left navigation menu.

Subtopics

- **ipv6 ripng**

`ipv6 ripng`

Syntax

```
ipv6 ripng <PROCESS-ID>
no ipv6 ripng <PROCESS-ID>
```

Description

Enables RIPng process on interface and creates a new context.

The **no** form of this command deletes RIPng process on interface.

Parameter	Description
`<PROCESS-ID>`	Specifies RIPng process ID. Range: 1 - 63.

Examples

Enabling RIPng process on an interface:

```
switch(config)# interface 1/1/1
switch (config-if)# ipv6 ripng 1
switch (config-if-ripng)#
```

Deleting RIPng process on an interface:

```
switch(config)# interface 1/1/1
switch (config-if)# no ipv6 ripng 1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

Routing commands

Select a command from the list in the left navigation menu.

Subtopics

- **enable**
- **disable**
- **distance**
- **maximum-paths**
- **redistribute**
- **timers update**

`enable`

Syntax

```
enable
no enable
```

Description

Enables RIPng process if disabled. By default RIPng process is enabled.

The **no** form of this command disables the RIPng process.

Examples

Enabling RIPng process when disabled:

```
switch(config)# router ripng 1
switch(config-ripng-1)# enable
```

Disabling RIPng process when enabled:

```
switch(config)# router ripng 1
switch(config-ripng-1)# no enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`disable`

Syntax

```
disable
no disable
```

Description

Disables RIPng process.

The **no** form of this command enables the RIPng process.

Examples

Disabling RIPng process:

```
switch(config)# router ripng 1
switch(config-ripng-1)# disable
```

Enabling RIPng process:

```
switch(config)# router ripng 1
switch(config-ripng-1)# no disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`distance`

Syntax

```
distance <DISTANCE>
no distance
```

Description

Configures administrative distance for RIPng. Administrative distance is used as criteria to select the best route when multiple protocols have the same route.

The **no** form of this command sets the RIPng administrative distance to the default. Default: 120.

Parameter	Description
`<DISTANCE>`	Specifies RIPng administrative distance. Range: 1 to 255.

Examples

Configuring administrative distance for RIPng:

```
switch(config)# router ripng 1
switch(config-ripng-1)# distance 100
```

Setting administrative distance for RIPng to default values:

```
switch(config)# router ripng 1
switch(config-ripng-1)# no distance
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`maximum-paths`

Syntax

```
maximum-paths <MAX-VALUE>
no maximum-paths
```

Description

Sets the maximum number of ECMP routes that RIPng can support.

The **no** form of this command sets the maximum number of ECMP routes to the default value of 4.

Parameter	Description
<MAX-VALUE>	Sets the number of RIPng ECMP routes. Range: 1 - 8.

Examples

Setting maximum number of RIPng ECMP routes:

```
switch(config)# router ripng 1
switch (config-ripng-1)# maximum-paths 8
```

Setting maximum number of RIPng ECMP routes to default:

```
switch(config)# router ripng 1
switch (config-ripng-1)# no maximum-paths
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`redistribute`

Syntax

```
redistribute {bgp | connected | ospfv3 <PROCESS-ID> | static}
no redistribute {bgp | connected | ospfv3 <PROCESS-ID> | static}
```

Description

Redistributes routes originating from other protocols into RIPng.

The **no** form of this command disables redistribution of routes originating from other protocols into RIPng.

Parameter	Description
`bgp`	Specifies BGP routes to redistribute into RIPng.
`connected`	Specifies connected routes (directly attached subnet or host) to redistribute into RIPng.
`ospfv3 <PROCESS-ID>`	Specifies the OSPFv3 route to redistribute into RIPng. Range: < 1 - 65535>
`static`	Specifies static route to redistribute into RIPng.

> **NOTE**
> Parameters **ospfv3** and **bgp** are not supported on 6100 switches.

Examples

Redistributing BGP routes into RIPng:

```
switch(config)# router ripng 1
switch(config-ripng-1)# redistribute bgp
```

Disabling BGP routes that originate from other protocols and redistribute into RIPng:

```
switch(config)# router ripng 1
switch(config-ripng-1)# no redistribute bgp
```

Command History

Release	Modification
10.14	Supported process ID range expanded from 1 - 63 to 1 - 65535.

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6200	`config`	Administrators or local user group members with execution righ ts for this command.

`timers update`

Syntax

```
timers update <INTERVAL> timeout <DURATION> garbage-collection <PERIOD>
no timers
```

Description

Configures RIPng timers with specific values.

The **no** form of this command sets all RIPng timers to default values.

Parameter	Description
`timers update <INTERVAL>`	Specifies frequency at which RIPng sends updates to all of its peers. Range: 1 to 2147484. Default: 30.
`timeout <DURATION>`	Specifies timeout duration from the point of the last refresh a fter a route is received from a peer timeout and is marked as ex pired. Range: 1 to 255. Default: 180.
`garbage-collection <PERIOD>`	Specifies amount of time route remains in routing table after ro ute expiration. Range: 1 to 255. Default: 120.

Examples

Configuring RIPng timers with specific values:

```
switch(config)# router ripng 1
switch(config-ripng-1)# timers update 40 timeout 200 garbage-collection 150
```

Configuring RIPng timers with default values:

```
switch(config)# router ripng 1
switch(config-ripng-1)# no timers
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config`	Administrators or local user group members with execution righ ts for this command.

RIPng clear commands

Select a command from the list in the left navigation menu.

Subtopics

clear ipv6 ripng statistics

`clear ipv6 ripng statistics`

Syntax

```
clear ipv6 ripng [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]
```

Description

Clears RIPng event statistics.

Parameter	Description
<PROCESS-ID>	Specifies RIPng process ID. Range: 1 - 63
`all-vrfs`	Clears statistics for all VRFs.
`vrf`	Selects VRF to clear statistics for.
<VRF-NAME>	Specifies VRF name.

Examples

Clearing RIPng event statistics:

```
switch# clear ipv6 ripng statistics
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	Operator (`>`) or Manager (`#`)	Administrators or local user group members with execution rights for this command.

Interface commands

Select a command from the list in the left navigation menu.

Subtopics

- enable
- disable
- send disable
- receive disable

`enable`

Syntax

```
enable
no enable
```

Description

Enables RIPng process on interface.

The **no** form of this command disables RIPng process on interface.

Examples

Enabling RIPng process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch(config-if-ripng)# enable
```

Disabling RIPng process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch(config-if-ripng)# no enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`disable`

Syntax

```
disable
no disable
```

Description

Disables RIPng process on interface.

The **no** form of this command enables RIPng process on interface.

Examples

Disabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch(config-if-ripng)# disable
```

Enabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch(config-if-ripng)# no disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`send disable`

Syntax

```
send disable
no send disable
```

Description

Disables interface from sending RIPng packets. An interface can send RIPng packets by default.

The **no** form of this command enables interface to send RIPng packets, if disabled.

Examples

Disabling interface from sending RIPng packets:

```
switch(config)# interface 1/1/1
switch (config-if)# ipv6 ripng 1
switch (config-if-ripng)# send disable
```

Enabling interface to send RIPng packets:

```
switch(config)# interface 1/1/1
switch (config-if)# ipv6 ripng 1
switch (config-if-ripng)# no send disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

`receive disable`

Syntax

```
receive disable
no receive disable
```

Description

Disables interface from receiving RIPng packets for all enabled IP addresses. An interface can receive RIPng packets by default.

The **no** form of this command enables interface to receive RIPng packets, if disabled.

Examples

Disabling interface from receiving RIPng packets:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch (config-if-ripng)# receive disable
```

Enabling interface to receive RIPng packets when disabled:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch (config-if-ripng)# no receive disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	`config-if`	Administrators or local user group members with execution righ ts for this command.

RIPng show commands

Select a command from the list in the left navigation menu.

Subtopics

- **show capacities ripng**
- **show capacities-status ripng**
- **show ipv6 ripng**
- **show ipv6 ripng interface**
- **show ipv6 ripng neighbors**
- **show ipv6 ripng routes**
- **show ipv6 ripng statistics**
- **show ipv6 ripng statistics interface**
- **show running-config**

`show capacities ripng`

Syntax

```
show capacities ripng
```

Description

Displays the maximum number of RIPng interfaces, routes and process.

Examples

Displaying maximum number of RIPng interfaces, routes and process:

```
switch# show capacities ripng
System Capacities: Filter RIPng
```

```
Capacities Name                                                       Value
---------------------------------------------------------------------------
Maximum number of RIPng interfaces configurable in the system         32
Maximum number of RIPng processes supported across each VRF           1
Maximum number of routes in RIPng supported across all VRFs           2540
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	Operator (`>`) or Manage r (`#`)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this co mmand from the auditor context (auditor>) only.

`show capacities-status ripng`

Syntax

```
show capacities-status ripng
```

Description

Displays number of RIPng interfaces, routes and process configured in the system.

Examples

Displaying number of RIPng interfaces, routes and process:

```
switch# show capacities-status ripng
System Capacities Status: Filter RIPng
Capacities
Name
      Value  Maximum
---------------------------------------------------------------------------
-------------------------------
```

```
Number of RIPng interfaces configured in the
system                                                         0        32
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	Operator (`>`) or Manage r (`#`)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this co mmand from the auditor context (auditor>) only.

`show ipv6 ripng`

Syntax

```
show ipv6 ripng [<PROCESS-ID>] [all-vrfs | vrf <VRF-NAME>]
```

Description

Displays general RIPng configuration.

Parameter	Description
`<PROCESS-ID>`	Specifies RIPng process ID. Range: 1 - 63.
`all vrfs`	Displays general RIPng information for all VRFs.
`vrf`	Selects VRF to display general RIPng information for.
	Specifies VRF name.

Parameter	Description
<VRF-NAME>	

Usage

- Parameters display general RIPng information for a specific RIPng process.
- Parameters display general RIPng information for a specific or all VRFs.
- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

Displaying general RIPng configuration for all VRFs:

```
switch# show ipv6 ripng 34 all-vrfs
VRF : Default                    Process-ID : 34
---------------------------------------------------------------------
Protocol Status         : Enabled   ECMP            : 6
Update Time             : 60  sec   Timeout Time    : 240 sec
Garbage Collection Time : 250 sec   Distance        : 100
Redistribution          : static,
                          ospfv3 1
VRF : vrf_1                      Process-ID : 34
---------------------------------------------------------------------
Protocol Status         : Enabled   ECMP            : 4
Update Time             : 30  sec   Timeout Time    : 180 sec
Garbage Collection Time : 120 sec   Distance        : 120
Redistribution          : None
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100	Operator (`>`) or Manager (`#`)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Platforms	Command context	Authority
6200		

`show ipv6 ripng interface`

Syntax

```
show ipv6 ripng [<PROCESS-ID>] interface [<INTERFACE-NAME>] [brief] [all-
vrfs | vrf <VRF-NAME>]
```

Description

Displays information about RIPng enabled interfaces.

Parameter	Description
`<PROCESS-ID>`	Specifies RIPng process ID. Range: 1 - 63.
`<INTERFACE-NAME>`	Specifies interface.
`brief`	Shows brief overview information for RIPng interface.
`all-vrfs`	Displays interface information for all VRFs.
`vrf`	Selects specific VRF.
`<VRF-NAME>`	Specifies VRF.

Usage

- Parameters display general RIPng information for a specific RIPng process.
- Parameters display general RIPng information for a specific or all VRFs.

- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

```
switch# show ipv6 ripng interface
Interface 1/1/1 is up, IPv6 Address is fe80::7272:cfff:fe70:67a
-----------------------------------------------------------------------
VRF              : Default            Process-ID      : 1
Status           : Oper Up            Mode            : Send and Receive
MTU              : 500                Poision Reverse : Enabled
Interface 1/1/2 is up, IPv6 Address is fe80::7272:cfff:fe70:67a
-----------------------------------------------------------------------
VRF              : Default            Process-ID      : 1
Status           : Admin Down         Mode            : Receive
MTU              : 500                Poision Reverse : Enabled
Interface 1/1/3 is up, IPv6 Address is fe80::7272:cfff:fe70:67a
-----------------------------------------------------------------------
VRF              : Default            Process-ID      : 1
Status           : Admin Down         Mode            : Send
MTU              : 500                Poision Reverse : Enabled
switch# show ipv6 ripng interface brief
VRF : default   Process-ID : 1
-------------------------------
Total Number of Interfaces: 2
Interface     IPv6-Address                  Status    MTU
-----------------------------------------------------------
1/1/1         fe80::7272:cfff:fe70:67a      up        500
1/1/2         fe80::7272:cfff:fe71:67a      up        500
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	Operator (`>`) or Manage r (`#`)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this co mmand from the auditor context (auditor>) only.

`show ipv6 ripng neighbors`

Syntax

```
show ipv6 ripng [<PROCESS-ID>] neighbors [<LINK-LOCAL-ADDRESS>] [all-vrfs |
vrf <VRF-NAME>]
```

Description

Displays information about RIPng neighbors.

Parameter	Description
`<PROCESS-ID>`	Specifies RIPng process ID. Range: 1 - 63.
`neighbors`	Specifies neighbor IP address.
`<LINK-LOCAL-ADDRESS>`	Specifies link - local address.
`all-vrfs`	Displays neighbor information for all VRFs.
`vrf`	Selects VRF to display neighbor information.
`<VRF-NAME>`	Specifies VRF name.

Usage

- Parameters display RIPng neighbor information for a specific RIPng process.
- Parameters display RIPng neighbor information for a specific neighbor.
- Parameters display general RIPng information for a specific or all VRFs.
- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

Displaying RIPng neighbor information for all VRFs:

```
switch# show ipv6 ripng neighbors all-vrfs
VRF : default             Process-ID : 1
----------------------------------------
Total Number of Neighbors: 1
Peer-Address      Type     Last-Update  Rcvd-Bad-Pkts     Rcvd-Bad-Routes
----------------------------------------------------------------------
fe80::7272:cfff:fe70:86ae
                  RIPng    0:0:7        4                 5
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	Operator (`>`) or Manager (`#`)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

`show ipv6 ripng routes`

Syntax

```
show ipv6 ripng [<PROCESS-ID>] routes [<PREFIX/LENGTH>] [all-vrfs | vrf
<VRF-NAME>]
```

Description

Displays RIPng routing table for a specific RIPng process.

Parameter	Description
<PROCESS-ID>	Specifies RIPng process ID to display information for a specific RIPng process. Range: 1 - 63.
<PREFIX/LENGTH>	Specifies the network prefix.
`all-vrfs`	Displays RIPng routing information for all VRFs.
`vrf`	Selects VRF to display RIPng routing information.
<VRF-NAME>	Specifies VRF name.

Usage

- <PREFIX/LENGTH> is an optional parameter that displays RIPng routing table information for a specific subnet.
- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

Displaying RIPng routing table for all VRFs:

```
switch# show ipv6 ripng routes all-vrfs
VRF : default              Process-ID : 1
------------------------------------------
Prefix              Metric     Interface      Nexthop
-----------------------------------------------------------------------
2001:DB8:10::/64      2         1/1/1         FE80::2E0:E6FF:FE1B:8242
2002:DB8:10::/64      3         1/1/1         FE80::2E0:E6FF:FE1B:8242
2003:DB8:10::/64      1         1/1/1
VRF : vrf_1                Process-ID : 34
-------------------------------------------
Prefix               Metric     Interface      Nexthop
------------------------------------------------------------------------
3001:DB8:10::/64      10         1/1/2         FE80::2E0:E6FF:FE1B:8232
```

```
3002:DB8:10::/64       14          1/1/2          FE80::2E0:E6FF:FE1B:8232
3003:DB8:10::/64        1          1/1/2
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	Operator (`>`) or Manager (`#`)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

`show ipv6 ripng statistics`

Syntax

```
show ipv6 ripng [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]
```

Description

Displays RIPng statistics.

Parameter	Description
`<PROCESS-ID>`	Specifies RIPng process ID. Range: 1 - 63.
`all-vrfs`	Displays statistics information for all VRFs.
`vrf`	Selects VRF and displays RIPng statistics for it.
	Specifies VRF name.

Parameter	Description
<VRF-NAME>	

Usage

- Parameters can display information for all VRFs or a specific VRF.
- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

Displaying RIPng statistics for all VRFs:

```
switch# show ipv6 ripng statistics all-vrfs
VRF : default  Process-ID : 1
-------------------------------------
Global Route Changes : 50
Global Queries       : 2
Last Cleared         : 0h 30m 28s ago
VRF : vrf_1    Process-ID : 34
-------------------------------------
Global Route Changes : 20
Global Queries       : 0
Last Cleared         : 0h 30m 28s ago
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	Operator (>) or Manage r (#)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this co mmand from the auditor context (auditor>) only.

`show ipv6 ripng statistics interface`

Syntax

```
show ipv6 ripng [<PROCESS-ID>] statistics interface [<INTERFACE-NAME>] [all-
vrfs | vrf <VRF-NAME>]
```

Description

Displays RIPng statistics for RIPng enabled interfaces.

Parameter	Description
`<PROCESS-ID>`	Specifies RIPng process ID. Range: 1 - 63.
`<INTERFACE-NAME>`	Specifies name of interface.
`all-vrfs`	Displays RIPng interface statistics for all VRFs.
`vrf`	Selects VRF to display RIPng interface statistics.
`<VRF-NAME>`	Specifies VRF name.

Usage

- Parameters can display information for all VRFs or a specific VRF.
- If a VRF is not mentioned, information for the default VRF is displayed.

Examples

Displaying RIPng statistics for a RIPng enabled interface:

```
switch# show ipv6 ripng statistics interface 1/1/1
VRF : default  Process-ID : 1      interface 1/1/1
--------------------------------------------------
IPv6-Address          Trigger-Updates    Rcvd-Bad-Packets    Rcvd-Bad-Routes
----------------------------------------------------------------------------
-
```

```
fe80::7272:cfff:fe70:86ae
                     15                   3                   4
Last Cleared: 0h 30m 28s ago
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	Operator (`>`) or Manager (`#`)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

`show running-config`

Syntax

```
show running config
```

Description

Displays all running configurations for all protocols including RIPng.

Examples

Displaying all running configurations for all protocols including RIPng:

```
switch# show running-config
Current configuration:
!
!Version Halon 0.1.0 (Build: genericx86-64-Halon-0.1.0-
master-20170309054955-dev)
!Schema version 0.1.8
lldp enable
timezone set utc
vrf green
```

```
vrf red
led base-loc_fdc on
led base-loc on
led base-hlth_fdc fast_blink
led base-pwr_fdc on
!
!
!
!
!
!
aaa authentication login default local
aaa authorization commands default none
!
!
!
!
router ospfv3 1
router ripng 1
    maximum-paths 5
    distance 1
    redistribute ospfv3 1
    timers update 40 timeout 200 garbage-collection 150
router ripng 1 vrf red
    default-information originate always
    maximum-paths 7
    distance 5
vlan 1
    no shutdown
interface lag 44
    no shutdown
    ipv6 address link-local
    ipv6 ripng 1
        send disable
interface 1
    no shutdown
    ipv6 address link-local
    ipv6 ripng 1
        receive disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
5420 6100 6200	Operator (`>`) or Manager (`#`)	Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Feature and Proactive Troubleshooting

AOS-CX switches include an intelligent, intuitive and distributed troubleshooting framework that makes network troubleshooting simpler, inexpensive and efficient. Use the CLI commands in this section to troubleshoot individual features of AOS-CX software, or issue the **troubleshoot proactive** command to set up proactive system troubleshooting to automatically troubleshoot issues based on any of the following.

1. Events
2. Log messages
3. CoPP or other database changes
4. CPU levels exceeding a threshold
5. Memory exceeding a threshold
6. Daemon crashes

> **NOTE**
> Feature-specific troubleshooting information can also be viewed via the **Deepshoot >Troubleshoot** page of the Switch WebUI. For more details, refer to the latest WebUI Reference Guide for your switch model.

Subtopics

troubleshoot l3

`troubleshoot l3`

Syntax

```
troubleshoot l3
   arp [{health[verbose]}|operations]
   config
   dhcp-relay  [config|{health[verbose]}|operations]
   dhcp-server [config|{health[verbose]}|operations]
   dhcpv6-relay  [config|{health[verbose]}|operations]
   dhcpv6-server [config|{health[verbose]}|operations]
   external-storage health [verbose]]
   health [verbose]
   operations
   ospf [config|{health[verbose]}|operations]
   ospfv3 [config|{health[verbose]}|operations]
```

Description

Troubleshoot issues with Layer-3 features and protocols.

Parameter	Description
`arp health [{health[verbose]}|operations]`	Troubleshoot the Address Resolution Protocol (ARP). By defaul t, the **troubleshoot l3 arp** command displays both feature heal th and feature operations troubleshooting information. Include t he optional **health** or **operations** parameters to display inform ation for that category only.
`config`	Issue the **troubleshoot l3 Config** command to display the resul ts of configuration validation checks for all L3 features. Include the **config** parameter after a specific L3 component to display c onfiguration validation information for that component only.
`dhcp-relay [config|{health[verbose]}|operations]`	By default, the **troubleshoot l3 dhcpv-relay** command displays feature configuration, health, and operations troubleshooting in formation. Include the optional **Config**, **health** , or **operations** p arameter to display information for that category only. Include the optional **verbose** keyword after the **health** param eter to display detailed diagnostic progress and confirm whethe r additional errors were suppressed.

Parameter	Description
`dhcp-server [config|{health[verbose]}|operations]`	By default, the **troubleshoot l3 dhcpv-server** command displays feature configuration, health, and operations troubleshooting information. Include the optional **Config**, **health** , or **operations** parameter to display information for that category only. Include the optional **verbose** keyword after the **health** parameter to display detailed diagnostic progress and confirm whether additional errors were suppressed.
`dhcpv6-relay [config|{health[verbose]}|operations]`	By default, the **troubleshoot l3 dhcpv6-relay** command displays feature configuration, health, and operations troubleshooting information. Include the optional **Config**, **health** , or **operations** parameter to display information for that category only. Include the optional **verbose** keyword after the **health** parameter to display detailed diagnostic progress and confirm whether other errors were suppressed.
`dhcpv6-server [config|{health[verbose]}|operations]`	Troubleshoot DHCPv6 server settings. By default, the **troubleshoot l3 dhcpv6-server** command displays feature configuration, health, and operations troubleshooting information. Include the optional **Config**, **health** , or **operations** parameter to display information for that category only. Include the optional **verbose** keyword after the **health** parameter to display detailed diagnostic progress and confirm whether additional errors were suppressed.
`external-storage [config|{health[verbose]}]`	Troubleshoot issues with external storage. By default, the **troubleshoot l3 external-storage** command displays feature health, and configuration troubleshooting information. Include the optional **health** or **Config** parameter to display information for that category only.
`health [verbose]`	The **troubleshoot l3 health** command displays basic health information for all L3 features, including error messages, message IDs, severity levels, and when the messages were reported. Include the **health** parameter after an L3 feature name (for example, **troubleshoot l3 ospf health**) to display health troubleshooting information for that feature only. Include the optional **verbose** keyword after the **health** parameter to display detailed diagnostic progress and confirm whether additional errors were suppressed.
`operations`	Display comprehensive L3 feature diagnostics with a detailed analysis.

Parameter	Description
`ospf [config|{health[verbose]}|operations]`	Troubleshoot issues with Open Shortest Path First version 2 (OSPF). By default, the **troubleshoot l3 ospf** command displays feature configuration, health, and operations troubleshooting information. Include the optional **Config**, **health, or operations** parameter to display information for that category only.
`ospfv3 [config|{health[verbose]}|operations]`	Troubleshoot issues with Open Shortest Path First version 3 (OSPFv3). By default, the **troubleshoot l3 ospfv3** command displays feature configuration, health, and operations troubleshooting information. Include the optional **Config**, **health** , or **operations** parameter to display information for that category only.

Usage

Filter the output of a troubleshooting command by including a **parameters "<query><word>"** option after the **health**, **config**, or **operations** parameters for some troubleshooting l3 commands. The "**<query><word>**" string must be enclosed in double quotation marks ("). More than one set of filter parameters can be used in a single command, but all filter parameters must be enclosed within a single set of quotation marks, for example, "interface <interface> vrf <vrf - name>".

Table 1. Filter examples for L3 Troubleshooting Commands

Command	Filter options
troubleshooting l3 dhcp-relay config|health|operations	• parameters "interface <interface>": Troubleshoot issues with the specified interface.
troubleshooting l3 dhcp-server config|health|operations	• parameters "--pool <pool>" Troubleshoot issues with the specified address pool. • parameters "--vrf <vrf>": Troubleshoot issues with the specified VRF.
troubleshooting l3 dhcpv6-relay config|health|operations	• parameters "interface <interface>": Troubleshoot issues with the specified interface.
troubleshooting l3 dhcpv6-server config|health|operations	• parameters "--pool <pool>" Troubleshoot issues with the specified address pool. • parameters "--vrf <vrf>": Troubleshoot issues with the specified VRF.

Command	Filter options
troubleshooting l3 ospf config|health|operations	• parameters "neighbor <router - id>"`: Troubleshoot issues with the specified neighbor. • parameters "neighbor <router - id> interface <interface>": Troubleshoot neighbor issues on a specific interface. • parameters "neighbor <router - id> interface <interface> vrf <vrf - name>": Troubleshoot neighbor issues on a specific interface in a VRF. • parameters "interface <interface>": Troubleshoot issues with the specified interface. • parameters "interface <interface> vrf <vrf - name>": Troubleshoot interface issues in a specific VRF. • parameters "route <ipv6 - prefix>": Troubleshoot issues with the specified IPv6 route. • parameters "interface vlan{vlan_id}": Troubleshoot issues on a specific VLAN. • parameters "area <area-id>": Troubleshoot issues within a specific OSPF area.
troubleshooting l3 ospfv3 config|health|operations	• parameters "neighbor <router - id>"`: Troubleshoot issues with the specified neighbor. • parameters "neighbor <router - id> interface <interface>": Troubleshoot neighbor issues on a specific interface. • parameters "neighbor <router - id> interface <interface> vrf <vrf - name>": Troubleshoot neighbor issues on a specific interface in a VRF. • parameters "interface <interface>": Troubleshoot issues with the specified interface. • parameters "interface <interface> vrf <vrf - name>": Troubleshoot interface issues in a specific VRF. • parameters "route <ipv6 - prefix>": Troubleshoot issues with the specified IPv6 route. • parameters "interface vlan{vlan_id}": Troubleshoot issues on a specific VLAN.

Command	Filter options
	• parameters "area <area-id>": Troubleshoot issues within a specific OSPF area.

Example

Displaying display the results of configuration validation checks for an OSPFv3 configuration

```
switch# troubleshoot l3 ospfv3 config parameters "neighbor 1.1.1.1
interface 1/1/1"
--------------------------------------------------
Configuration troubleshoot
--------------------------------------------------
ospfv3 : error: RCVD_BAD_UNKNOWN.1/1/1
         Possible errors:
         1. DUT and Peer Not in Same Network
         error: RCVD_DEAD_MISMATCH1/1/1
         error: 1/1/1 TX_RX_MISMATCH
         error: Neighbor not found. Troubleshoot interface for any issues
with the ospfv3 interface.
====================================================
Troubleshoot OSPFv3 executed successfully
```

The following command displays troubleshooting information for the health of the DHCP-relay feature.

```

DocTeam-8325# troubleshoot l3 dhcp-relay health
--------------------------------------------------
Health troubleshoot
--------------------------------------------------
Component: dhcp-relay
----------------------------------------------------------------------------
------------------------------------------------
Message ID
Message
       Last-Reported(UTC)   Severity
----------------------------------------------------------------------------
------------------------------------------------
1710        Atleast one helper-address should be configured for DHCPv4
Relay to be effective.  2026-03-27 02:38:32  ERROR

====================================================
Troubleshoot dhcp-relay executed successfully
====================================================
```

Command History

Release	Modification
10.18	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	`manager`	Administrators or local user group members with execution righ ts for this command.

Support and Other Resources

Access HPE Aruba Networking support and updates, and view warranty and regulatory information

Subtopics

- Accessing HPE Aruba Networking Support
- Accessing Updates
- Warranty Information
- Regulatory Information
- Documentation Feedback

Accessing HPE Aruba Networking Support

HPE Aruba Networking Support Services	**https://www.hpe.com/us/en/networking/hpe - aruba - networking - support - services.html**
AOS - CX Switch Software Documentation Portal	**https://arubanetworking.hpe.com/techdocs/ArubaDocPortal/content/new-portal/aoscx.html**
HPE Aruba Networking Support Portal	**https://networkingsupport.hpe.com/home**

North America telephone	1 - 800 - 943 - 4526 (US & Canada Toll - Free Number) +1 - 650 - 750 - 0350 (Backup—Toll Number)
International telephone	**https://www.hpe.com/psnow/doc/a50011948enw**

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	**https://developer.arubanetworks.com/hpe - aruba - networking - aoscx/docs/about**
Airheads social forums and Knowledge Base	**https://community.arubanetworks.com/**
AOS - CX Software Technical Update channel on YouTube.	Videos on new features introduced in this release : **https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP - Q_UL3CskS**
HPE Aruba Networking Hardware Documentation and Translations Portal	
HPE Aruba Networking software	**https://networkingsupport.hpe.com/downloads https://networkingsupport.hpe.com/downloads**
Software licensing and Feature Packs	**https://licensemanagement.hpe.com/**

End - of - Life information	**https://networkingsupport.hpe.com/end - of - life**

Accessing Updates

You can access updates from the HPE Aruba Networking Support Portal at **https://networkingsupport.hpe.com**.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

https://networkingsupport.hpe./notifications/subscriptions (requires an active HPE Aruba Networking Support Portal account to manage subscriptions). Security notices are viewable without an HPE Aruba Networking Support Portal account.

Warranty Information

To view warranty information for your product, go to **https://www.arubanetworks.com/support-services/product-warranties/**.

Regulatory Information

To view the regulatory information for your product, view the Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products, available at **https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts**

Additional regulatory information

HPE Aruba Networking is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see **https://www.arubanetworks.com/company/about-us/environmental-citizenship/**.

Documentation Feedback

HPE Aruba Networking is committed to providing documentation that meets your needs. To help us improve the documentation, send any errors, suggestions, or comments to Documentation Feedback (**docsfeedback-switching@hpe.com**). When submitting your feedback, include the document title, part number, edition, and publication date located on the front cover of the document. For online help content, include the product name, product version, help edition, and publication date located on the legal notices page.